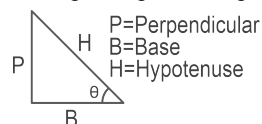


Trigonometry

It is that branch of Mathematics which deals with the measurements of the sides and angles of a triangle and also the problems associated with angles.

Important Formulae

In a right angled triangle, as shown below



$$\begin{aligned}\bullet \sin \theta &= \frac{P}{H} & \bullet \operatorname{cosec} \theta &= \frac{H}{P} \\ \bullet \cos \theta &= \frac{B}{H} & \bullet \sec \theta &= \frac{H}{B} \\ \bullet \tan \theta &= \frac{P}{B} & \bullet \cot \theta &= \frac{B}{P}\end{aligned}$$

Note that

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}, \therefore \operatorname{cosec} \theta \times \sin \theta = 1$$

$$\text{Similarly, } \sec \theta = \frac{1}{\cos \theta}$$

$$\therefore \sec \theta \times \cos \theta = 1$$

$$\text{And, } \cot \theta = \frac{1}{\tan \theta}$$

$$\therefore \cot \theta \times \tan \theta = 1$$

$$\text{Also, } \tan \theta = \frac{\sin \theta}{\cos \theta}$$

- $\sin \theta = \cos (90^\circ - \theta)$
- $\operatorname{cosec} \theta = \sec (90^\circ - \theta)$
- $\tan \theta = \cot (90^\circ - \theta)$

Conversion of trigonometric ratios

- $\sin (-x) = -\sin x$
- $\operatorname{cosec} (-x) = -\operatorname{cosec} x$
- $\cos (-x) = \cos x$
- $\sec (-x) = \sec x$
- $\tan (-x) = -\tan x$
- $\cot (-x) = -\cot x$

Signs of the trigonometric ratios in different quadrants :

2nd quadrant Sine and cosecant are positive	1st quadrant All positive
3rd quadrant tangent and cotangent are positive	4th quadrant cosine and secant are positive

- x axis $\pm \theta \rightarrow$ no change
- y axis $\pm \theta \rightarrow$ changes like
- $\sin \theta \leftrightarrow \cos \theta$
- $\tan \theta \leftrightarrow \cot \theta$
- $\operatorname{cosec} \theta \leftrightarrow \sec \theta$

Important Identities

- $\sin^2 \theta + \cos^2 \theta = 1$
- $\sec^2 \theta - \tan^2 \theta = 1$
- $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$

We can derive all other identities from these three basic identities.

Values of Trigonometric functions for different values of θ .

	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
$\cot \theta$	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
$\operatorname{cosec} \theta$	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

The value of $\sin \theta$ or $\cos \theta$ never exceeds 1, whereas the value of $\sec \theta$ or $\operatorname{cosec} \theta$ is always greater than or equal to 1.

Range of all trigonometric functions

	Minimum	Maximum
$\sin x, \cos x$ (odd powers)	-1	+1
$\sin x, \cos x$ (even powers)	0	+1
$\tan x, \cot x$ (odd powers)	$-\infty$	$+\infty$
$\tan x, \cot x$ (even powers)	0	$+\infty$
$\sec x, \operatorname{cosec} x$ (odd powers)	$-\infty$	$+\infty$
$\sec x, \operatorname{cosec} x$ (even powers)	+1	$+\infty$

Maximum and Minimum value of the trigonometric expressions of the different form :

- $a \sin x \pm b \cos x$

$$\text{Maximum value} = \sqrt{a^2 + b^2}$$

$$\text{Minimum value} = -\sqrt{a^2 + b^2}$$

- $a \sin^2 x + b \cos^2 x$

$$\text{If, } a > b \text{ Maximum value} = a$$

$$\text{Minimum value} = b$$

$$\text{If } a < b \text{ Maximum value} = b$$

$$\text{Minimum value} = a$$

- $a \operatorname{cosec}^2 x + b \sec^2 x$

$$\text{Maximum value} = \infty$$

$$\text{Minimum value} = (\sqrt{a} + \sqrt{b})^2$$

- $a \sin^2 x + b \operatorname{cosec}^2 x$

- $a \tan^2 x + b \cot^2 x$

- $a \cos^2 x + b \sec^2 x$

Maximum and Minimum value of these three trigonometric expression is ∞ and $2\sqrt{ab}$ respectively.

- $\sin^{2n} x + \cos^{2m} x$

$$\text{Maximum value} = 1$$

$$\text{Minimum value} = (\text{use } x = 45^\circ)$$

Example : Find the minimum and

maximum value of $\sin^2 x + \cos^4 x$?

$$\text{Maximum value of } \sin^2 x + \cos^4 x = 1$$

$$\text{Minimum value} = \sin^2 45^\circ + \cos^4 45^\circ =$$

$$\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$\bullet (\sin x \cos x)^n$$

$$\text{Maximum value} = \infty$$

$$\text{Minimum value} = \left(\frac{1}{2}\right)^n$$

Example : Find the minimum value of $\sin^4 x \cos^4 x$?

$$\text{Minimum value} = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

Compound Formulae

- $\sin (A + B)$
 $= \sin A \cos B + \cos A \sin B$
- $\sin (A - B)$
 $= \sin A \cos B - \cos A \sin B$
- $\cos (A + B)$
 $= \cos A \cos B - \sin A \sin B$
- $\cos (A - B)$
 $= \cos A \cos B + \sin A \sin B$
- $\sin 2A = 2 \sin A \cos A$
 $= \frac{2 \tan A}{1 + \tan^2 A}$
- $\cos 2A = \cos^2 A - \sin^2 A$
 $= 1 - 2 \sin^2 A$
- $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$
 $= 2 \cos^2 A - 1$
- $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
- $\sin 3A = 3 \sin A - 4 \sin^3 A$
- $\cos 3A = 4 \cos^3 A - 3 \cos A$
- $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$
- $\sin \theta \cdot \sin 2\theta \cdot \sin 4\theta$
 $= \frac{1}{4} \sin 3\theta$
- $\cos \theta \cdot \cos 2\theta \cdot \cos 4\theta$
 $= \frac{1}{4} \cos 3\theta$
- $\tan \theta \cdot \tan 2\theta \cdot \tan 4\theta = \tan 3\theta$
- $\tan(A + B) = \frac{\tan A + \tan B}{1 - (\tan A \times \tan B)}$
- $\tan(A - B) = \frac{\tan A - \tan B}{1 + (\tan A \times \tan B)}$
- $\cot(A + B) = \frac{\cot B \cot A - 1}{\cot B + \cot A}$
- $\cot(A - B) = \frac{\cot B \cot A + 1}{\cot B - \cot A}$

Transformations of sums or differences into products

- $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$
- $\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$

$$\bullet \cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$\bullet \cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$$

Concepts

$$\bullet a \sin X + b \cos X = m$$

$$a \cos X - b \sin X = n$$

$$\text{Then, } a^2 + b^2 = m^2 + n^2$$

Example : if $3 \sin \theta + 5 \cos \theta = 5$. Find the value of $3 \cos \theta - 5 \sin \theta$?

$$3 \cos \theta - 5 \sin \theta = \sqrt{5^2 + 3^2 - 5^2} = \pm 3$$

$$\bullet a \sec X + b \tan X = m$$

$$a \tan X - b \sec X = n$$

$$\text{Then, } a^2 - b^2 = m^2 - n^2$$

If $\sec x$ and $\tan x$ are replaced by $\operatorname{cosec} x$ and $\cot x$ respectively. Then the outcome will be the same.

Example : if $29 \sec \theta - 21 \tan \theta = 20$. Find the value of $29 \tan \theta - 21 \sec \theta$?

$$\text{Sol:- } = 29 \tan \theta - 21 \sec \theta$$

$$= \sqrt{20^2 - 29^2 + 21^2} = 0$$

$$\bullet \sec x + \tan x = \frac{1}{\sec x - \tan x}$$

$$\bullet \operatorname{cosec} x + \cot x = \frac{1}{\operatorname{cosec} x - \cot x}$$

Some Important results

(i). If $A + B + C = 180^\circ$. then,

$$\bullet \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$\bullet \cot A \cot B + \cot B \cot C + \cot C \cot A = 1$$

(ii). If $A + B + C = 90^\circ$. then,

$$\bullet \tan A \tan B + \tan B \tan C + \tan C \tan A = 1$$

$$\bullet \cot A + \cot B + \cot C = \cot A \cot B \cot C$$

(iii). If $A + B = 45^\circ$. then

$$\bullet (1 + \tan A)(1 + \tan B) = 2$$

$$\bullet (1 - \cot A)(1 - \cot B) = 2$$

Variety Questions

Q.1. Find the value of $\tan 72^\circ - \tan 27^\circ - \tan 72^\circ \tan 27^\circ$.

SSC CGL 25/09/2024 (2nd Shift)

- (a) -1 (b) 1 (c) -2 (d) 0

Q.2. Simplify $\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta}$.

SSC CGL 24/09/2024 (3rd Shift)

- (a) $\cos \frac{\theta}{2}$ (b) $\sin \frac{\theta}{2}$ (c) $\cot \frac{\theta}{2}$ (d) $\tan \frac{\theta}{2}$

Q.3. The value of the expression

$$\frac{1 - \sin(2t)}{1 + \sin(2t)} \times \frac{\cos(t) + \sin(t)}{\cos(t) - \sin(t)}$$
 is

SSC CGL 19/09/2024 (2nd Shift)

- (a) $\frac{1 - 2 \tan(t)}{1 + 2 \tan(t)}$ (b) $\frac{1 - \tan(t)}{1 + \tan(t)}$
(c) $\frac{1 + 2 \tan(t)}{1 - 2 \tan(t)}$ (d) $\frac{1 + \tan(t)}{1 - \tan(t)}$

Q.4. $\sin A \sin B =$ _____.

SSC CGL 18/09/2024 (3rd Shift)

(a) $\frac{1}{2} \{(\sin(A+B) + \sin(A-B))\}$

(b) $\frac{1}{2} \{(\sin(A+B) - \sin(A-B))\}$

(c) $\frac{1}{2} \{(\cos(A+B) - \cos(A-B))\}$

(d) $\frac{1}{2} \{(\cos(A-B) - \cos(A+B))\}$

Q.5. If $a \sin^3 x + b \cos^3 x = \sin x \cos x$ and $a \sin x = b \cos x$, then find the value of $a^2 + b^2$, provided that x is neither 0° nor 90° .

SSC CGL 11/09/2024 (1st Shift)

- (a) 0 (b) 1 (c) $a^2 - b^2$ (d) $a^2 + b^2$

Q.6. If $p \cos A = 2q \sin A$ and $2p \operatorname{cosec} A - q \sec A = 3$, then the value of $p^2 + 4q^2$ is:

SSC CGL 10/09/2024 (2nd Shift)

- (a) 2 (b) 4 (c) 1 (d) 3

Q.7. If $\tan A + \sin A = x$, $\tan A - \sin A = y$, what is the value of $x^2 - y^2$?

SSC CHSL 11/07/2024 (4th Shift)

- (a) $4\sqrt{xy}$ (b) $2\sqrt{\frac{x}{y}}$ (c) $\sqrt{xy}$ (d) $4\sqrt{\frac{x}{y}}$

Q.8. What is the value of

$$\cos^{-1}\left(\frac{7}{25}\right) - \cos^{-1}\left(\frac{15}{17}\right) ?$$

SSC CHSL 11/07/2024 (2nd Shift)

- (a) $\cos^{-1}\left(\frac{304}{425}\right)$ (b) $\cot^{-1}\left(\frac{297}{425}\right)$
(c) $\tan^{-1}\left(\frac{297}{425}\right)$ (d) $\sin^{-1}\left(\frac{304}{425}\right)$

Q.9. If $f(p) = \sin^{p+2} x + \cos^{p+2} x$, then the value of $6f(2) - 4f(4) + 10f(0)$ is :

SSC CHSL 10/07/2024 (2nd Shift)

- (a) 12 (b) 10 (c) 11 (d) 14

Q.10. If $\tan A = a \tan B$ and $\sin A = b \sin B$, then the value of $\cos^2 A$ is:

SSC CHSL 05/07/2024 (1st Shift)

- (a) $\frac{b^2 + 1}{a^2 + 1}$ (b) $\frac{b^2 - 1}{a^2 - 1}$
(c) $\frac{a^2 + 1}{b^2 + 1}$ (d) $\frac{a^2 - 1}{b^2 - 1}$

Q.11. If $\sin(X - Y) = \sin X \cos Y - \cos X \sin Y$, then the value of $\frac{\sin(A - B)}{\cos A \cos B} +$

$$\frac{\sin(B - C)}{\cos B \cos C}$$
 is :

SSC CHSL 05/07/2024 (2nd Shift)

- (a) $\frac{\cos(A+C)}{\cos A \cos C}$ (b) $\frac{\sin(C-A)}{\cos A \cos C}$
(c) $\frac{\cos(A+C)}{\sin A \sin C}$ (d) $\frac{\sin(A-C)}{\cos A \cos C}$

Q.12. $\frac{3 \cos^3 \theta - 2 \cos \theta}{\sin \theta - 3 \sin^3 \theta}$ is equal to :

SSC CHSL 03/07/2024 (3rd Shift)

- (a) $\tan^2 \theta$ (b) $\cot^2 \theta$ (c) $\tan \theta$ (d) $\cot \theta$

Q.13. If $\cot A = 7$, then the value of

$$\frac{5 \cos A + 4 \sin A}{\cos^3 A + 7 \sin^3 A + 6 \sin A}$$
 is :

SSC CHSL 01/07/2024 (3rd Shift)

- (a) 2 (b) 0 (c) 3 (d) 1

Q.14. What is the value of $\frac{[(\sin x + \sin y)(\sin y - \sin x)]}{[(\cos x + \cos y)(\cos y - \cos x)]}$

SSC CPO 29/06/2024 (3rd Shift)

- (a) 0 (b) 1 (c) -1 (d) 2

Q.15. Simplify $\frac{\cos 13^\circ + \sin 13^\circ}{\cos 13^\circ - \sin 13^\circ}$

SSC CPO 28/06/2024 (1st Shift)

- (a) $\cot 58^\circ$ (b) $\tan 32^\circ$
(c) $\tan 58^\circ$ (d) $\cos 26^\circ$

Q.16. If $(\sin x + \sin y) = a$ and $(\cos x + \cos y) = b$, then find the value of $(\sin x \sin y + \cos x \cos y)$.

SSC CPO 27/06/2024 (2nd Shift)

- (a) $\frac{a^2 + b^2 - 1}{2}$ (b) $\frac{a^2 - b^2 - 2}{2}$
(c) $\frac{a^2 + b^2 - 2}{2}$ (d) $\frac{a^2 + b^2 + 2}{2}$

Q.17. If $\tan(t) + \cot(t) = 1$, then one of the values of the expression $\frac{1}{\sin(t) + \cos(t)}$ is _____.

SSC CPO 27/06/2024 (1st Shift)

- (a) $\frac{1}{\sqrt{3}}$ (b) $\frac{2}{\sqrt{3}}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{3}$

Q.18. If $3 \sec^4 \theta + 8 = 10 \sec^2 \theta$, then the value of $\tan \theta$ can be:

Graduate Level 26/06/2024 (Shift - 4)

- (a) $2\sqrt{3}$ (b) $1\sqrt{3}$ (c) $2\frac{1}{\sqrt{3}}$ (d) $1\frac{1}{\sqrt{3}}$

Q.19. If $\cot^4 A + \cot^2 A = 1$, then find the value of $\cos^4 A - 3 \cos^2 A$.

SSC CHSL Tier II 02/11/2023

- (a) -1 (b) -2 (c) 2 (d) 1

Q.20. If $\cot^2 \theta = 1 - e^2$, then the value of $\operatorname{cosec} \theta + \cot^3 \theta \sec \theta$ is:

SSC CPO 05/10/2023 (3rd Shift)

- (a) $(2 - e^2)^{\frac{3}{2}}$ (b) $(1 - e^2)$
(c) $(1 - e^2)^{\frac{3}{2}}$ (d) $(2 - e^2)^{\frac{1}{2}}$

Q.21. Find the least value of $16 \operatorname{cosec}^2 \theta + 25 \sin^2 \theta$.

SSC CPO 04/10/2023 (3rd Shift)

- (a) 35 (b) 38 (c) 42 (d) 40

Q.22. If $\operatorname{cosec} A + \cot A = a\sqrt{b}$, then find the value of $\frac{(a^2 b - 1)}{(a^2 b + 1)}$.

SSC CHSL 17/08/2023 (2nd Shift)

- (a) $\cos A$ (b) $\tan A$ (c) $\frac{1}{\sin A}$ (d) $\frac{1}{\cot A}$

Q.23. If $\sin \theta + \tan \theta = p$ and $\tan \theta - \sin \theta = q$, then which of the relation satisfies for given values of p and q ?

SSC CHSL 11/08/2023 (4th Shift)

(a) $p^2 - q^2 = 4\sqrt{pq}$ (b) $p^2 - q^2 = 8\sqrt{pq}$
 (c) $p^2 + q^2 = 4\sqrt{pq}$ (d) $p^2 + q^2 = 4pq$

Q.24. If θ is an acute angle and $\sin\theta \cos\theta = 2 \cos^3\theta - \frac{1}{4} \cos\theta$, then the value of $\sin\theta$ is:

SSC CHSL 10/08/2023 (2nd Shift)

(a) $\frac{\sqrt{15}-1}{8}$ (b) $\frac{\sqrt{15}-1}{4}$
 (c) $\frac{\sqrt{15}+1}{4}$ (d) $\frac{\sqrt{15}-1}{2}$

Q.25. Simplify the following: $7(\cot 11^\circ \cot 13^\circ \cot 19^\circ \cot 29^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ)$

SSC CHSL 09/08/2023 (3rd Shift)

(a) 7 (b) -7 (c) -1 (d) 1

Q.26. The given expression is equal to :

$$\frac{\sin^4 A + \cos^4 A}{1 - 2\sin^2 A \cos^2 A}$$

SSC CHSL 09/08/2023 (2nd Shift)

(a) 1 (b) -1 (c) 0 (d) 2

Q.27. If $\cot A + \cos A = p$ and $\cot A - \cos A = q$, then which of the following relation is correct ?

SSC CHSL 03/08/2023 (4th Shift)

(a) $\sqrt{\frac{1}{16pq}} = p^2 + q^2$ (b) $\frac{1}{4pq} = p^2 + q^2$
 (c) $\sqrt{16pq} = p^2 - q^2$ (d) $4pq = p^2 + q^2$

Q.28. The value of the expression $4(\sin^6 A + \cos^6 A) - 6(\sin^4 A + \cos^4 A) + 8$ is :

SSC CHSL 02/08/2023 (4th Shift)

(a) 4 (b) 8 (c) 7 (d) 6

Q.29. Evaluate the following:

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + 2\cos 8\theta}}}$$

SSC CGL 27/07/2023 (3rd shift)

(a) $2\cos\theta$ (b) $2\cos 2\theta$ (c) $\sin 2\theta$ (d) $\cos 2\theta$

Q.30. Find the value of $\sqrt{\frac{1 - \sin 3\theta}{1 + \sin 3\theta}}$

SSC CGL 21/07/2023 (2nd shift)

(a) $\sec 3\theta - \tan 3\theta$ (b) $(\sec 3\theta - \tan 3\theta)^3$
 (c) $(\sec 3\theta - \tan 3\theta)^2$ (d) $\sec 3\theta + \tan 3\theta$

Q.31. The solution of the equation $\cos^2\theta + \sin\theta + 1 = 0$ lies between :

Graduate Level 30/06/2023 (shift 1)

(a) $(\frac{3\pi}{4}, \frac{5\pi}{4})$ (b) $(\frac{5\pi}{4}, \frac{7\pi}{4})$
 (c) $(\frac{\pi}{4}, \frac{3\pi}{4})$ (d) $(-\frac{\pi}{4}, \frac{\pi}{4})$

Q.32. What is the value of $(1 + \cot A + \tan A)(\sin A - \cos A)(\frac{\sin A \cos A}{\sin^3 A - \cos^3 A})$?

SSC CHSL Tier II (26/06/2023)

(a) 1 (b) -1 (c) 0 (d) 2

Q.33. If $\sin m + \sin n = p$, $\cos m + \cos n = q$, then find the value of $\sin m \times \sin n + \cos m \times \cos n$.

SSC CHSL 20/03/2023 (2nd Shift)

(a) $p^2 + q^2 - 2$ (b) $\frac{(p^2 + q^2 - 2)}{2}$
 (c) $p + q - pq$ (d) $p + q + pq$

Q.34. If $a \cot \theta + b \operatorname{cosec} \theta = p$ and $b \cot \theta + a \operatorname{cosec} \theta = q$ then $p^2 - q^2$ is equal to _____.

SSC CGL Tier II (07/03/2023)

(a) $b^2 - a^2$ (b) $a^2 - b^2$ (c) $b - a$ (d) $a^2 + b^2$

Q.35. If A is an acute angle, the simplified form of

$$\frac{\cos(\pi - A) \cdot \cot(\frac{\pi}{2} + A) \cos(-A)}{\tan(\pi + A) \tan(\frac{3\pi}{2} + A) \sin(2\pi - A)}$$

SSC CGL 13/12/2022 (4th Shift)

(a) $\cos^2 A$ (b) $\sin A$ (c) $\sin^2 A$ (d) $\cos A$

Q.36. What is the value of the expression $\cos 2A \cos 2B + \sin^2(A - B) - \sin^2(A + B)$?

SSC CGL 12/12/2022 (1st Shift)

(a) $\sin(2A - 2B)$ (b) $\sin(2A + 2B)$
 (c) $\cos(2A + 2B)$ (d) $\cos(2A - 2B)$

Q.37. If $\frac{k - k \cot^2 30^\circ}{1 + \cot^2 30^\circ} = \sin^2 60^\circ +$

$4 \tan^2 45^\circ - \operatorname{cosec}^2 60^\circ$, then the value of k (correct to two decimal places) is:

SSC CGL 03/12/2022 (3rd Shift)

(a) 5.55 (b) -6.83 (c) -5.58 (d) 6.83

Q.38. If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, find the value of $(\cos \theta - \sin \theta)$.

SSC CPO 10/11/2022 (Afternoon)

(a) $\sqrt{2} \sin \theta$ (b) $\sqrt{2} \cos \theta$
 (c) $\frac{1}{\sqrt{2}} \sin \theta$ (d) $\frac{1}{2} \cos \theta$

Q.39. The value of $\sin 73^\circ + \cos 137^\circ$ is:

SSC CHSL 10/06/2022 (Evening)

(a) $\sin 13^\circ$ (b) $\cos 13^\circ$

(c) $\cos 18^\circ$ (d) $\sin 18^\circ$

Q.40. The value of $\sin 18^\circ$ is given as $\frac{\sqrt{5}-1}{4}$. Using this value, find the value of $\cos 18^\circ$.

SSC CHSL 30/05/2022 (Afternoon)

(a) $\frac{\sqrt{10+2\sqrt{5}}}{4}$ (b) $\frac{\sqrt{5}+1}{4}$
 (c) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (d) $\frac{\sqrt{10-2\sqrt{5}}}{4}$

Q.41. The value of $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$ is:

SSC CHSL 27/05/2022 (Morning)

(a) $\tan 3\theta$ (b) $\cot 3\theta$ (c) $\tan \theta$ (d) $\cot \theta$

Q.42. The value of $\cot^2 46^\circ - \sec^2 44^\circ + (\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ)$

SSC CHSL 10/08/2021 (Evening)

(a) 23.5 (b) 20.5 (c) 22.5 (d) 21.5

Q.43. If A lies between 45° and 540° , and $\sin A = 0.5$, what is the value of $\frac{A}{3}$ in degree ?

SSC CHSL 15/04/2021 (Morning)

(a) 170° (b) 175° (c) 165° (d) 160°

Q.44. If $\cos 27^\circ = x$, then the value of $\tan 63^\circ$ is:

SSC CHSL 26/10/2020 (Evening)

(a) $\frac{\sqrt{1+x^2}}{x}$ (b) $\frac{x}{\sqrt{1+x^2}}$
 (c) $\frac{\sqrt{1-x^2}}{x}$ (d) $\frac{x}{\sqrt{1-x^2}}$

Q.45. If $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ and $x \sin \theta = y \cos \theta$, then the value of $x^2 + y^2$ is:

SSC CHSL 20/10/2020 (Morning)

(a) 0 (b) 4 (c) 1 (d) 2

Q.46. If $\cos^2 \theta + \cos^4 \theta = 1$, then the value of $\sin \theta + \sin^2 \theta$ is:

SSC CHSL 16/10/2020 (Morning)

(a) 0 (b) $\frac{1}{2}$ (c) 2 (d) 1

Q.47. If $\sin \theta - \cos \theta = \frac{7}{17}$, then find the value of $\sin \theta + \cos \theta$.

SSC CHSL 14/10/2020 (Evening)

(a) $\frac{8}{17}$ (b) $\frac{23}{13}$ (c) $\frac{23}{17}$ (d) $\frac{8}{13}$

Q.48. If $\sec \theta$ and $\sin \theta$ ($0^\circ < \theta < 90^\circ$) are the roots of the equation $\sqrt{6}x^2 - kx + \sqrt{6} = 0$, then the value of k is:

SSC CHSL 14/10/2020 (Morning)

(a) $\sqrt{3}$ (b) $3\sqrt{2}$ (c) $2\sqrt{3}$ (d) $3\sqrt{3}$

Q.49. If $\cos A$, $\sin A$, $\cot A$ are in geometric progression, then the value of $\tan^6 A - \tan^2 A$ is:

SSC CHSL 13/10/2020 (Afternoon)

(a) $\frac{1}{2}$ (b) 3 (c) $\frac{1}{3}$ (d) 1

Q.50. Find the value of $\cos 225^\circ$

SSC CHSL 17/03/2020 (Afternoon)

(a) -0.7071 (b) -0.866
 (c) 0.7071 (d) 0.866

Q.51. If $0^\circ < A, B < 45^\circ$, $\cos(A+B) = \frac{24}{25}$

and $\sin(A-B) = \frac{15}{17}$, then $\tan 2A = ?$

SSC CGL 06/03/2020 (Evening)

(a) 0 (b) 1 (c) $\frac{416}{87}$ (d) $\frac{213}{4}$

Q.52. If $\cos^2 \theta - \sin^2 \theta = \tan^2 \theta$, then which of the following is true ?

SSC CPO 13/12/2019 (Evening)

- (a) $\cos\theta\cos\theta = 1$
 (b) $\cos\theta\cos\theta = \sqrt{2}$
 (c) $\cos^2\theta - \sin^2\theta = \cot^2\theta$
 (d) $\cos^2\theta - \sin^2\theta = \tan^2\theta$

Q.53. everything between 0° and 90° For α_i 's, ($i = 1, 2, 3, \dots, 20$) it is given that $\sin\alpha_1 + \sin\alpha_2 + \sin\alpha_3 + \dots + \sin\alpha_{20} = 20$. Find the value (in degrees) of $(\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_{20})$.

SSC CHSL 11/07/2019 (Morning)

- (a) 1800° (b) 900° (c) 0° (d) 20°

Q.54. For α and β both being acute angles, it is given that $\sin(\alpha + \beta) = 1$, $\cos(\alpha - \beta) = \frac{1}{2}$. The values of α and β are :

SSC CHSL 10/07/2019 (Evening)

- (a) $75^\circ, 15^\circ$ (b) $45^\circ, 15^\circ$
 (c) $75^\circ, 45^\circ$ (d) $60^\circ, 30^\circ$

Q.55. If $\theta = 9^\circ$, then what is the value of $\cot\theta \cdot \cot 2\theta \cdot \cot 3\theta \cdot \cot 4\theta \cdot \cot 5\theta \cdot \cot 6\theta \cdot \cot 7\theta \cdot \cot 8\theta \cdot \cot 9\theta$?

SSC CHSL 10/07/2019 (Evening)

- (a) $\sqrt{3} - 1$ (b) 1 (c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$

Q.56. Which among the following is an irrational quantity?

SSC CHSL 10/07/2019 (Afternoon)

- (a) $\tan 30^\circ \cdot \tan 60^\circ$ (b) $\sin 30^\circ$
 (c) $\tan 45^\circ$ (d) $\cos 30^\circ$

Q.57. Which among the following increases continuously in the range $0^\circ < \theta < 90^\circ$?

SSC CHSL 09/07/2019 (Evening)

- (a) $\cot\theta$ (b) $\operatorname{cosec}\theta$ (c) $\tan\theta$ (d) $\cos\theta$

Q.58. If $\sec\theta = 4x$ and $\tan\theta = \frac{4}{x}$, ($x \neq 0$)

then the value of $8(x^2 - \frac{1}{x^2})$ is :

SSC CHSL 05/07/2019 (Afternoon)

- (a) $\frac{1}{16}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{8}$

Q.59. If $\sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}} \times \sqrt{\frac{\operatorname{cosec}\theta - \cot\theta}{\operatorname{cosec}\theta + \cot\theta}} = \frac{1-r}{1+r}$, then the value of r is:

SSC CHSL 03/07/2019 (Afternoon)

- (a) $\sin\theta$ (b) $\operatorname{Cosec}\theta$ (c) $\sec\theta$ (d) $\cos\theta$

Q.60. If $\frac{(\sin\theta - \operatorname{cosec}\theta)(\cos\theta - \sec\theta)}{\tan^2\theta - \sin^2\theta} = r^3$, then $r =$?

SSC CHSL 03/07/2019 (Morning)

- (a) $\sin\theta\cos\theta$ (b) $\tan\theta$
 (c) $\cot\theta$ (d) $\operatorname{Cosec}\theta\sec\theta$

Q.61. If $3 - 2\sin^2\theta - 3\cos\theta = 0$, $0^\circ < \theta < 90^\circ$, then what is the value of $(2\operatorname{Cosec}\theta + \tan\theta)$?

SSC CHSL 02/07/2019 (Evening)

- (a) $7\sqrt{3}$ (b) $5\sqrt{3}$ (c) $\frac{5\sqrt{3}}{3}$ (d) $\frac{7\sqrt{3}}{3}$

Q.62. A simplified of

$$\left(\frac{\sin\theta}{1 + \cos\theta} + \frac{1 + \cos\theta}{\sin\theta}\right) \left(\frac{1}{\tan\theta + \cot\theta}\right) \text{ is :}$$

SSC CHSL 02/07/2019 (Evening)

- (a) $\cos\theta$ (b) $2\sin\theta$ (c) $\sin\theta$ (d) $2\cos\theta$

Q.63. The value of

$$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\sin 56^\circ \cdot \sec 34^\circ + \cos 25^\circ \cdot \operatorname{cosec} 65^\circ} \text{ is:}$$

SSC CHSL 02/07/2019 (Evening)

- (a) $\frac{1}{2}$ (b) 4 (c) 2 (d) 1

Q.64. $\frac{(\sec\theta + \tan\theta)(1 - \sin\theta)}{\operatorname{cosec}\theta(1 + \cos\theta)(\operatorname{cosec}\theta - \cot\theta)}$ is equal to:

SSC CHSL 02/07/2019 (Morning)

- (a) $\sin\theta$ (b) $\sec\theta$ (c) $\cos\theta$ (d) $\operatorname{Cosec}\theta$

Q.65. If $3\cos^2 A + 7\sin^2 A = 4$, then what is the value of $\cot A$, given that A is an acute angle?

SSC CGL 13/06/2019 (Evening)

- (a) 1 (b) $\sqrt{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{\sqrt{3}}$

Q.66. If $\operatorname{Cosec}\theta = \frac{13}{12}$, then $\sin\theta + \cos\theta - \tan\theta$ is equal to:

SSC CGL 11/06/2019 (Afternoon)

- (a) $\frac{91}{65}$ (b) $\frac{139}{65}$ (c) $\frac{71}{65}$ (d) $-\frac{71}{65}$

Q.67. If $\sec 4\theta = \operatorname{Cosec}(\theta + 20^\circ)$, then θ is equal to:

SSC CGL 10/06/2019 (Afternoon)

- (a) 22° (b) 18° (c) 14° (d) 20°

Q.68. If $3\sin\theta = 2\cos^2\theta$, $0^\circ < \theta < 90^\circ$, then the value of $(\tan^2\theta + \sec^2\theta - \operatorname{cosec}^2\theta)$ is:

SSC CGL 10/06/2019 (Morning)

- (a) -2 (b) $-\frac{7}{3}$ (c) $\frac{7}{3}$ (d) 2

Q.69. If $\frac{\tan\theta + \sin\theta}{\tan\theta - \sin\theta} = \frac{k+1}{k-1}$, then $k =$?

SSC CGL 10/06/2019 (Morning)

- (a) $\operatorname{cosec}\theta$ (b) $\sec\theta$ (c) $\cos\theta$ (d) $\sin\theta$

Q.70. If $\frac{1}{\operatorname{cosec}\theta - 1} + \frac{1}{\operatorname{cosec}\theta + 1} = 2\sec\theta$, $0^\circ < \theta < 90^\circ$, then the value of $(\cot\theta + \cos\theta)$ is:

SSC CGL 07/06/2019 (Evening)

- (a) $\frac{1 + \sqrt{2}}{2}$ (b) $\frac{2 + \sqrt{2}}{2}$
 (c) $\frac{2 + \sqrt{3}}{\sqrt{2}}$ (d) $1 + \sqrt{2}$

Q.71. If $\sin\theta = \frac{a}{\sqrt{a^2 + b^2}}$, $0^\circ < \theta < 90^\circ$,

then the value of $\sec\theta + \tan\theta$ is :

SSC CGL 07/06/2019 (Afternoon)

- (a) $\frac{\sqrt{a^2 + b^2} + a}{b}$ (b) $\frac{\sqrt{a^2 + b^2} + b}{2a}$
 (c) $\frac{\sqrt{a^2 + b^2} + a}{2b}$ (d) $\frac{\sqrt{a^2 + b^2} + b}{a}$

Q.72. If $\cos^2\theta - 3\cos\theta + 2 = \sin^2\theta$, $0^\circ < \theta < 90^\circ$, then the value of $2\operatorname{cosec}\theta + 4\cot\theta$ is :

SSC CGL 07/06/2019 (Afternoon)

- (a) $\frac{8\sqrt{3}}{3}$ (b) $\frac{4\sqrt{3}}{4}$ (c) $2\sqrt{3}$ (d) $4\sqrt{3}$

Q.73. The value of $\frac{1}{\sec x - \tan x} - \frac{1}{\cos x}$, $0^\circ < x < 90^\circ$, is equal to :

SSC CGL 07/06/2019 (Morning)

- (a) $2\sec x$ (b) $\tan x$ (c) $2\cos x$ (d) $\cot x$

Q.74. If $\tan^2\theta - 3\sec\theta + 3 = 0$, $0^\circ < x < 90^\circ$, then the value of $\sin\theta + \cot\theta$ is :

SSC CGL 07/06/2019 (Morning)

- (a) $\frac{5\sqrt{3}}{6}$ (b) $2\sqrt{3}$ (c) $\frac{5\sqrt{3}}{3}$ (d) $3\sqrt{3}$

Q.75. If $(1 + \tan^2\theta) + (1 + (\tan^2\theta)^{-1}) = k$, then $\sqrt{k} =$?

SSC CGL 06/06/2019 (Evening)

- (a) $\operatorname{cosec}\theta \cdot \sec\theta$ (b) $\operatorname{cosec}\theta \cdot \cos\theta$
 (c) $\sin\theta \cdot \cos\theta$ (d) $\sin\theta \cdot \sec\theta$

Q.76. The value of $\frac{1}{\sin\theta} - \frac{\cot^2\theta}{1 + \operatorname{cosec}\theta}$ is

SSC CGL 06/06/2019 (Evening)

- (a) 0 (b) 1 (c) 2 (d) -1

Q.77. $\left(\frac{1}{1 + \sin^2\theta} + \frac{1}{1 + \operatorname{cosec}^2\theta}\right) =$?

SSC CGL 06/06/2019 (Evening)

- (a) $\sin^2\theta$ (b) 1 (c) $\operatorname{cosec}^2\theta$ (d) 2

Q.78. If $\frac{\cos\theta}{1 - \sin\theta} + \frac{\cos\theta}{1 + \sin\theta} = 4$, $0^\circ < \theta < 90^\circ$, then the value of $(\tan\theta + \operatorname{cosec}\theta)$ is :

SSC CGL 06/06/2019 (Evening)

- (a) $\frac{5\sqrt{2}}{2}$ (b) $\frac{5\sqrt{3}}{3}$ (c) $\frac{4\sqrt{3}}{3}$ (d) $\frac{5\sqrt{2}}{3}$

Q.79. $\frac{\sin\theta - \cos\theta + 1}{\sin\theta + \cos\theta - 1} =$?

SSC CGL 06/06/2019 (Morning)

- (a) $\sec\theta \sin\theta$ (b) $\sec\theta \tan\theta$
 (c) $\sec\theta + \tan\theta$ (d) $\sec\theta - \tan\theta$

Q.80. The value of

$\sqrt{\sec^2\theta + \operatorname{cosec}^2\theta} \times \sqrt{\tan^2\theta - \sin^2\theta}$ is equal to :

SSC CGL 06/06/2019 (Morning)

- (a) $\operatorname{cosec}\theta \sec^2\theta$ (b) $\sin\theta \sec^2\theta$
 (c) $\sin\theta \cos^2\theta$ (d) $\operatorname{cosec}\theta \cos^2\theta$

Q.81. If $\sec\theta - \tan\theta = P$, then $\operatorname{cosec}\theta = ?$
SSC CGL 04/06/ 2019 (Evening)

- (a) $\frac{2P}{1-P^2}$ (b) $\frac{1-P^2}{1+P^2}$
(c) $\frac{P^2+1}{1-P^2}$ (d) $\frac{2P}{1+P^2}$

Q.82. If $0^\circ < \theta < 90^\circ$ and $\cos^2\theta = 3(\cot^2\theta - \cos^2\theta)$ then the value of $(\frac{1}{2} \sec\theta + \sin\theta)^{-1}$ is:

SSC CGL 04/06/2019 (Afternoon)

- (a) $\sqrt{3} + 2$ (b) $2(2 - \sqrt{3})$
(c) $2(\sqrt{3} - 1)$ (d) $\sqrt{3} + 1$

Q.83. $(\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta})^2 + 1, \theta \neq 45^\circ$ is equal to.

SSC CGL 04/06/2019 (Afternoon)

- (a) $\operatorname{cosec}^2\theta$ (b) $\sec^2\theta$ (c) $\cot^2\theta$ (d) $2\tan^2\theta$

Q.84. $(\operatorname{cosec} A - \sin A)^2 + (\sec A - \cos A)^2 - (\cot A - \tan A)^2$ is equal to:

SSC CPO 16/03/2019 (Evening)

- (a) 2 (b) 0 (c) 1 (d) -1

Q.85. Find the value of $1 + \frac{\tan^2 A}{1 + \sec A}$

SSC CPO 15/03/2019 (Evening)

- (a) $\operatorname{cosec} A$ (b) $\cos A$ (c) $\sec A$ (d) $\sin A$

Practice Questions

SSC CHSL 2024 Tier - 2

Q.86. Find the value of $(\sin^6 A + \cos^6 A + 3\sin^2 A \cdot \cos^2 A) - (\sin^4 A - \cos^4 A + 1) \operatorname{cosec}^2 A$.
SSC CHSL Tier II (18/11/2024)

- (a) 3 (b) -3 (c) 0 (d) -1

Q.87. Find the value of $3(\sec^2 A - 1)(\operatorname{cosec} A - 1)(\operatorname{cosec} A + 1) + 4(\operatorname{cosec}^2 A - \cot^2 A)$.

SSC CHSL Tier II (18/11/2024)

- (a) 12 (b) 3 (c) 10 (d) 7

SSC CGL 2024 Tier - 1

Q.88. The value of $\sin^2 30^\circ - \sin^2 40^\circ + \sin^2 45^\circ - \sin^2 55^\circ - \sin^2 35^\circ + \sin^2 45^\circ - \sin^2 50^\circ + \sin^2 60^\circ$ is:

SSC CGL 09/09/2024 (1st Shift)

- (a) 1 (b) 2 (c) 4 (d) 0

Q.89. The value of $\cos^2 29^\circ + \cos^2 61^\circ$ is:

SSC CGL 09/09/2024 (1st Shift)

- (a) $\frac{\sqrt{3}}{2}$ (b) 2 (c) 1 (d) 0

Q.90. If $\operatorname{cosec} \theta = \frac{5}{3}$, then evaluate

$$(\sec^2 \theta - 1) \times \cot^2 \theta \times (1 + \cot^2 \theta)$$

SSC CGL 09/09/2024 (1st Shift)

- (a) $\frac{25}{16}$ (b) $\frac{9}{16}$ (c) $\frac{25}{9}$ (d) $\frac{4}{5}$

Q.91. Simplify the following.
 $\frac{\sin^3 A - \cos^3 A}{\sin A - \cos A}$ where A is an acute angle.

SSC CGL 09/09/2024 (2nd Shift)

- (a) $1 + \sin A \cos A$ (b) $1 - 3 \sin A$
(c) $3 \cos A - 1$ (d) $\sin A + \cos A$

Q.92. If $\frac{\sec \alpha + \tan \alpha}{\sec \alpha - \tan \alpha} = \frac{7}{4}$, then find the value of $\operatorname{cosec} \alpha$.

SSC CGL 09/09/2024 (3rd Shift)

- (a) $\frac{1}{3}$ (b) $\frac{1}{11}$ (c) $\frac{11}{3}$ (d) $\frac{3}{11}$

Q.93. If A is an acute angle, which of the following is equal to $\frac{\sin A}{1 + \cos A}$?

SSC CGL 09/09/2024 (3rd Shift)

- (a) $\frac{1 - \cos A}{\sin A}$ (b) $\frac{1 + \cos A}{\sin A}$
(c) $\frac{1 - \sin A}{\cos A}$ (d) $\frac{1 + \sin A}{\cos A}$

Q.94. If $\frac{7\sin\theta + 4\cos\theta}{9\sin\theta - 2\cos\theta} = \frac{5}{4}$, then the value of $\frac{\tan^2\theta + 5}{\tan^2\theta - 5}$ is:

SSC CGL 09/09/2024 (3rd Shift)

- (a) $\frac{2121}{769}$ (b) $-\frac{2121}{769}$
(c) $\frac{769}{2121}$ (d) $-\frac{769}{2121}$

Q.95. $\frac{2 + \tan^2 2A + \cot^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A}$ is equal to__.

SSC CGL 10/09/2024 (1st Shift)

- (a) $\sec 2A + \tan 2A$ (b) $\sec 2A + \operatorname{cosec} 2A$
(c) $\cot 2A - \operatorname{cosec} 2A$ (d) $\sec 2A \cdot \operatorname{cosec} 2A$

Q.96. Let $0^\circ < t < 90^\circ$. Then which of the following is true?

SSC CGL 10/09/2024 (1st Shift)

- (a) $\sin(t) \neq \cos(t)$ when $t = 45^\circ$
(b) $\sin(t) > \cos(t)$ when $t < 45^\circ$
(c) $\sin(t) < \cos(t)$ when $t < 45^\circ$
(d) $\sin(t) < \cos(t)$ when $t > 45^\circ$

Q.97. If $2 \tan \theta = 3$, then $\frac{3 \sin \theta - 2 \cos \theta}{3 \sin \theta + 2 \cos \theta}$ is equal to:

SSC CGL 10/09/2024 (2nd Shift)

- (a) $\frac{6}{13}$ (b) $\frac{6}{7}$ (c) $\frac{5}{7}$ (d) $\frac{5}{13}$

Q.98. If $\cot \theta = \frac{4}{3}$, then evaluate $\frac{(\sec \theta (1 + \cot^2 \theta) (\operatorname{cosec}^2 \theta - \cot^2 \theta))}{\operatorname{cosec}^3 \theta}$.

SSC CGL 10/09/2024 (2nd Shift)

- (a) $\frac{3}{4}$ (b) $\frac{4}{5}$ (c) $\frac{4}{3}$ (d) $\frac{3}{5}$

Q.99. $\frac{3 \sin 58^\circ}{5 \cos 32^\circ} + \frac{2 \cos 62^\circ}{5 \sin 28^\circ}$ is equal to:

SSC CGL 10/09/2024 (3rd Shift)

- (a) 0 (b) 2 (c) 3 (d) 1

Q.100. If $7 \cos^2 \theta + 5 \sin^2 \theta - 6 = 0$, ($0^\circ < \theta < 90^\circ$), then what is the value of $1 + \sqrt{\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta}}$?

SSC CGL 10/09/2024 (3rd Shift)

- (a) $\sqrt{2} + 2$ (b) $\sqrt{2} + 1$ (c) $\sqrt{2} - 1$ (d) $\sqrt{2} - 2$

Q.101. Find the value of $\frac{\sec 35^\circ}{\operatorname{cosec} 55^\circ} - \frac{\cos 25^\circ}{\sin 65^\circ}$

SSC CGL 11/09/2024 (1st Shift)

- (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) 2

Q.102. Evaluate the value of $(\operatorname{cosec} 56^\circ \cos 34^\circ - \cos 59^\circ \operatorname{cosec} 31^\circ)$.

SSC CGL 11/09/2024 (2nd Shift)

- (a) 1 (b) 2 (c) -1 (d) 0

Q.103. Find the value of: $[\frac{4}{3} \tan^2 60^\circ + 3 \cos^2 30^\circ - 2 \sec^2 30^\circ - \frac{3}{4} \cot^2 60^\circ] \div [\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ]$

SSC CGL 11/09/2024 (2nd Shift)

- (a) $6\frac{2}{3}$ (b) $3\frac{1}{2}$ (c) $4\frac{1}{3}$ (d) $6\frac{3}{2}$

Q.104. Express $\sin 74^\circ + \tan 74^\circ$ in terms of trigonometric ratios of angles between 0° and 45° .

SSC CGL 11/09/2024 (3rd Shift)

- (a) $\sec 16^\circ + \cot 16^\circ$
(b) $\operatorname{cosec} 16^\circ + \sec 16^\circ$
(c) $\operatorname{cosec} 16^\circ + \cot 16^\circ$
(d) $\cos 16^\circ + \cot 16^\circ$

Q.105. If $p \sin A - \cos A = 1$, then $p^2 - (1 + p^2) \cos A$ equals:

SSC CGL 11/09/2024 (3rd Shift)

- (a) 1 (b) -1 (c) 2 (d) 0

Q.106. The value of

$$(\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta})^3 \frac{1}{\tan\theta} - \sec^2\theta$$
 is:

SSC CGL 12/09/2024 (1st Shift)

- (a) 2 (b) 1 (c) 0 (d) -1

Q.107. If $2 \operatorname{cosec}^2 \theta + 3 \cot^2 \theta = 17$, then the value of ' θ ' when $0^\circ \leq \theta \leq 90^\circ$ is:

SSC CGL 12/09/2024 (1st Shift)

- (a) 45° (b) 90° (c) 30° (d) 75°

Q.108. Solve the following:

$$\frac{1}{1 + \cos \theta} + \frac{1}{1 - \cos \theta} = ?$$

SSC CGL 12/09/2024 (2nd Shift)

- (a) $2 \operatorname{cosec}^2 \theta$ (b) $2 \sin^2 \theta$
(c) $2 \sec^2 \theta$ (d) $2 \tan^2 \theta$

Q.109. If $7 \tan \theta = 3$, then

$$\frac{7 \sin \theta + 5 \cos \theta}{7 \sin \theta - 2 \cos \theta} \text{ is:}$$

SSC CGL 12/09/2024 (2nd Shift)

- (a) 7 (b) 5 (c) 8 (d) 6

Q.110. If $\cot^4 \phi - \cot^2 \phi = 1$, then the value of $\cos^4 \phi + \cos^2 \phi$ is:

SSC CGL 12/09/2024 (2nd Shift)

- (a) 2 (b) 1 (c)
- $\frac{1}{\sqrt{3}}$
- (d)
- $\frac{3}{2}$

Q.111. If $\tan 4\theta \cdot \tan 6\theta = 1$, where 6θ is an acute angle, then find the value of $\cot 5\theta$.

SSC CGL 12/09/2024 (3rd Shift)

- (a)
- $\sqrt{3}$
- (b)
- $-\sqrt{3}$
- (c)
- -1
- (d) 1

Q.112. If $7 \cos A = 6$, then the numerical value of $\frac{\operatorname{cosec} A + \cot A}{\operatorname{cosec} A - \cot A}$ is:

SSC CGL 12/09/2024 (3rd Shift)

- (a) 1 (b) 0 (c) 13 (d) -13

Q.113. If $p(\sqrt{3} + \cot 30^\circ) = \tan^3 60^\circ - 2 \sin 60^\circ$, then the value of p is:

SSC CGL 13/09/2024 (1st Shift)

- (a) -1 (b)
- $2\sqrt{3}$
- (c) 1 (d)
- $\sqrt{3}$

Q.114. If $\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} = 4 \sec \theta$, ($0 < \theta < 90^\circ$), then the value of $(\cot \theta + \operatorname{cosec} \theta)$ is:

SSC CGL 13/09/2024 (1st Shift)

- (a)
- $\sqrt{3}$
- (b)
- $\sqrt{6}$
- (c)
- $\sqrt{2}$
- (d)
- $\sqrt{5}$

Q.115. If $\sin 31^\circ = \alpha$, then the value of $\cot 59^\circ$ is:

SSC CGL 13/09/2024 (1st Shift)

- (a) $\frac{\sqrt{1 - \alpha^2}}{2\alpha}$ (b) $\frac{\alpha}{\sqrt{1 - \alpha^2}}$
 (c) $\frac{\sqrt{1 + \alpha^2}}{\alpha}$ (d) $\frac{\alpha}{\sqrt{1 + \alpha^2}}$

Q.116. If $\frac{\tan(x)}{\sec(x)} = \frac{1}{2}$, then the value of $(\sin(x) + \cos(x))^2$ is ____.

SSC CGL 13/09/2024 (2nd Shift)

- (a) $\frac{2 + \sqrt{3}}{3}$ (b) $\frac{2 + \sqrt{3}}{2}$
 (c) $\frac{2 - \sqrt{3}}{2}$ (d) $\frac{2 - \sqrt{3}}{3}$

Q.117. If $\tan A = \frac{5}{7}$, then find the value of

$$\frac{7 \sin A - 3 \cos A}{7 \sin A + 3 \cos A} + 4.$$

SSC CGL 13/09/2024 (2nd Shift)

- (a)
- $4\frac{1}{4}$
- (b)
- $3\frac{5}{7}$
- (c)
- $3\frac{1}{5}$
- (d)
- $7\frac{1}{3}$

Q.118. If $\sin B = \frac{15}{17}$, what is the value of

$\cos B(\sec B - \tan B)$? Given that $0 < B < \frac{\pi}{2}$

SSC CGL 13/09/2024 (2nd Shift)

- (a)
- $\frac{1}{16}$
- (b)
- $\frac{2}{15}$
- (c)
- $\frac{8}{15}$
- (d)
- $\frac{2}{17}$

Q.119. The value of $\left(\frac{\operatorname{cosec} A}{\operatorname{cosec} A - 1} + \frac{\operatorname{cosec} A}{\operatorname{cosec} A + 1}\right) \left(\frac{\sec A}{\sec A - 1} + \frac{\sec A}{\sec A + 1}\right)^{-1}$, when $A = 60^\circ$ is:

SSC CGL 13/09/2024 (3rd Shift)

- (a) 4 (b) 3 (c) 1 (d) 2

Q.120. In a right angle $\triangle ABC$, $\angle B = 90^\circ$, if $\tan A = 1$, then $4 \sin A \cos A =$ _____.

SSC CGL 17/09/2024 (1st Shift)

- (a) 1 (b) 2 (c)
- $\frac{1}{2}$
- (d) 4

Q.121. If $\sec \theta = \frac{29}{20}$ where $0 < \theta < 90^\circ$, then what is the value of $3 \operatorname{cosec} \theta + 3 \cot \theta$?

SSC CGL 17/09/2024 (1st Shift)

- (a)
- $\frac{7}{3}$
- (b) 7 (c) 14 (d) 49

Q.122. If $\cos 24^\circ = \frac{m}{n}$, then the value of $(\operatorname{cosec} 24^\circ - \cos 66^\circ)$ is:

SSC CGL 17/09/2024 (1st Shift)

- (a) $\frac{m^2}{n\sqrt{m^2 - n^2}}$ (b) $\frac{n^2}{m\sqrt{m^2 - n^2}}$
 (c) $\frac{n^2}{m\sqrt{n^2 - m^2}}$ (d) $\frac{m^2}{n\sqrt{n^2 - m^2}}$

Q.123. Find the value of

$$\left[\frac{4 \cos(90 - \theta) \sin^3(90 + \theta) - 4 \sin(90 + \theta) \cos^3(90 - \theta)}{\cos\left(\frac{180 + 8\theta}{2}\right)} \right]$$

SSC CGL 17/09/2024 (2nd Shift)

- (a) -1 (b) 1 (c) 2 (d) 0

Q.124. If $\sin A + \cos A = \frac{1}{2\sqrt{2}}$, then find the value of $\sin^4 A + \cos^4 A$.

SSC CGL 17/09/2024 (2nd Shift)

- (a)
- $\frac{79}{128}$
- (b)
- $\frac{7}{13}$
- (c)
- $\frac{79}{126}$
- (d)
- $\frac{5}{13}$

Q.125. If $a = x \cos \theta + y \sin \theta$ and $b = x \sin \theta - y \cos \theta$, then $a^2 + b^2$ is equal to:

SSC CGL 17/09/2024 (2nd Shift)

- (a)
- $x + y$
- (b)
- $x^2 - y^2$
- (c)
- $x^2 + y^2$
- (d)
- $x - y$

Q.126. If $\operatorname{cosec} x = \frac{5}{4}$, then the value of

$$\frac{5 \sin^2 x + 8 \cot^2 x}{27 \tan^2 x - 125 \cos^2 x} \text{ is:}$$

SSC CGL 17/09/2024 (3rd Shift)

- (a)
- $\frac{33}{91}$
- (b)
- $\frac{77}{30}$
- (c)
- $\frac{55}{87}$
- (d)
- $\frac{99}{47}$

Q.127. If $\sin \theta + \cos \theta = \sqrt{3} \cos \theta$, then the value of $\cot \theta$ is:

SSC CGL 17/09/2024 (3rd Shift)

- (a) $\sqrt{3} - 1$ (b) $\frac{(\sqrt{3} + 1)}{2}$
 (c) $\frac{\sqrt{3} - 1}{2}$ (d) $\sqrt{3} + 1$

Q.128. If $\sec(t) = \frac{x}{y}$, then $\cot(t)$ is equal to:

SSC CGL 18/09/2024 (1st Shift)

- (a) $\frac{y}{\sqrt{x^2 - y^2}}$ (b) $\frac{x}{\sqrt{x^2 + y^2}}$
 (c) $\frac{x}{\sqrt{x^2 - y^2}}$ (d) $\frac{y}{\sqrt{x^2 + y^2}}$

Q.129. At what angle θ , the value of $\sin(\theta)$ and $\cos(\theta)$ will have the same value?

SSC CGL 18/09/2024 (1st Shift)

- (a)
- 45°
- (b)
- 30°
- (c)
- 60°
- (d)
- 75°

Q.130. If $3 \sin A + 4 \cos A = 5$, then the value of $\tan A$ is:

SSC CGL 18/09/2024 (1st Shift)

- (a)
- $\frac{3}{4}$
- (b)
- $\frac{3}{5}$
- (c)
- $\frac{5}{4}$
- (d)
- $\frac{4}{5}$

Q.131. Find the value of $\frac{\tan 55^\circ}{\cot 35^\circ}$

SSC CGL 18/09/2024 (2nd Shift)

- (a)
- $\cos 35^\circ$
- (b) 1 (c)
- $\sec 55^\circ$
- (d) 0

Q.132. Simplify, $\frac{\sin A}{1 + \cos A} + \cot A$, where

A is an acute angle.

SSC CGL 18/09/2024 (2nd Shift)

- (a) $\sin A$ (b) $\cos A$
 (c) $\sin A + \cos A$ (d) $\operatorname{cosec} A$

Q.133. If $\tan \theta = \frac{8}{15}$, and θ is an acute

angle, then the value of $\frac{\sqrt{1 - \sin \theta}}{\sqrt{1 + \sin \theta}}$ is:

SSC CGL 18/09/2024 (2nd Shift)

- (a)
- $\frac{2}{5}$
- (b)
- $\frac{3}{5}$
- (c)
- $\frac{1}{5}$
- (d)
- $\frac{4}{5}$

Q.134. The given expression is equivalent to:

$$\sqrt{\frac{\sec \theta + 1}{\sec \theta - 1}} + \sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}}$$

SSC CGL 18/09/2024 (3rd Shift)

- (a) $2 \sin \theta$ (b) $2 \operatorname{cosec} \theta$
 (c) $2 \tan \theta$ (d) $2 \tan \theta \sec \theta$

Q.135. What is the value of

$$\sqrt{\frac{9 \sin^2 26^\circ + 9 \sin^2 64^\circ + 36}{32 - 7 \cos^2 32^\circ - 7 \cos^2 58^\circ}}?$$

SSC CGL 18/09/2024 (3rd Shift)

- (a)
- $\frac{3\sqrt{5}}{5}$
- (b)
- $\frac{5\sqrt{3}}{3}$
- (c)
- $\frac{5\sqrt{5}}{3}$
- (d)
- $\frac{3\sqrt{3}}{5}$

Q.136. The value of the expression $\sin x + \operatorname{cosec} x = 2$, then the value of $\sin^7 x + \operatorname{cosec}^7 x$ is:

SSC CGL 19/09/2024 (1st Shift)

- (a) 0 (b) 1 (c) 2 (d) 4

Q.137. If $\sin A + \cos A = \sqrt{3}$, then find the value of $\tan A + \cot A + 2 \sin A \cos A$.

SSC CGL 19/09/2024 (1st Shift)

- (a) 6 (b) 2 (c) 1 (d) 3

Q.138. If $3\sin^2\theta + 5\cos^2\theta = 4$ and θ is an acute angle, then the value of $\tan\theta$ is:

SSC CGL 19/09/2024 (1st Shift)

- (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\frac{1}{2}$ (d) 1

Q.139. The maximum value of $(2\sin\theta + 3\cos\theta)$ is:

SSC CGL 19/09/2024 (2nd Shift)

- (a) $\sqrt{17}$ (b) $\sqrt{11}$ (c) 9 (d) $\sqrt{13}$

Q.140. Simplify $[1 - \sin^2 32^\circ + \frac{1}{1 + \tan^2 58^\circ}]$.

SSC CGL 19/09/2024 (2nd Shift)

- (a) $\frac{1}{2}$ (b) $\sin 32^\circ$ (c) 1 (d) $\sqrt{3}$

Q.141. If $\tan\theta = \frac{5}{8}$, then find the value of

$$\frac{(1 + \cos\theta)(1 - \cos\theta)}{(1 + \sin\theta)(1 - \sin\theta)}$$

SSC CGL 19/09/2024 (3rd Shift)

- (a) $\frac{5}{64}$ (b) $\frac{2}{25}$ (c) $\frac{25}{64}$ (d) $\frac{64}{25}$

Q.142. If $\tan\theta + \cot\theta = 2$, and θ is an acute angle, then the value of θ is:

SSC CGL 19/09/2024 (3rd Shift)

- (a) 30° (b) 15° (c) 60° (d) 45°

Q.143. If A is an acute angle, then

$$\sqrt{\frac{1 - \cos A}{1 + \cos A}} + \sqrt{\frac{1 + \cos A}{1 - \cos A}}$$
 is equal to :

SSC CGL 19/09/2024 (3rd Shift)

- (a) $2 \operatorname{cosec} A$ (b) $2 \cos A$
(c) $2 \sec A$ (d) $2 \sin A$

Q.144. The value of θ , when $\sqrt{3} \cos\theta + \sin\theta = 1$ ($1 \leq \theta \leq 90^\circ$), is:

SSC CGL 23/09/2024 (1st Shift)

- (a) 20° (b) 60° (c) 90° (d) 30°

Q.145. Let $x = r \cos(t)$, $y = r \sin(t) \cos(u)$, $z = r \sin(t) \sin(u)$. Then the value of $x^2 + y^2 + z^2$ is ____.

SSC CGL 23/09/2024 (1st Shift)

- (a) $r^2 \cos(u)$ (b) r^2 (c) $2r$ (d) $r^2 \sin(t)$

Q.146. If $4\tan\theta - 3 = 0$, then the value of $\frac{1 - \cos 2\theta}{1 + \cos 2\theta}$ is:

SSC CGL 23/09/2024 (1st Shift)

- (a) 1 (b) $\frac{7}{15}$ (c) $\frac{4}{3}$ (d) $\frac{9}{16}$

Q.147. If $\sin A - \cos A = \frac{\sqrt{3} - \sqrt{2}}{2}$, then the value of $\sin A \cdot \cos A$ is:

SSC CGL 23/09/2024 (2nd Shift)

- (a) $\frac{2\sqrt{3} - 1}{4}$ (b) $\frac{2\sqrt{2} - 1}{8}$
(c) $\frac{2\sqrt{6} - 1}{8}$ (d) $\frac{3\sqrt{2} - 1}{4}$

Q.148. If $\cos A + \cos^2 A = 1$ then $\sin^2 A + \sin^4 A$ is equal to:

SSC CGL 23/09/2024 (2nd Shift)

- (a) $\frac{\cos^2 A}{\cos A}$ (b) 0 (c) $\frac{\cos A}{\cos^2 A}$ (d) 1

Q.149. The value of $\frac{1}{\operatorname{cosec}\theta - \cot\theta} - \frac{1}{\sin\theta}$ is equal to:

SSC CGL 23/09/2024 (2nd Shift)

- (a) $\tan\theta$ (b) $\operatorname{cosec}\theta$ (c) $\cot\theta$ (d) $\sec\theta$

Q.150. The value of $\frac{\sin^2\theta - 2\sin^4\theta}{2\cos^4\theta + \cos^2\theta}$ at $\theta = 45^\circ$

SSC CGL 23/09/2024 (3rd Shift)

- (a) 1 (b) 0 (c) 3 (d) 2

Q.151. For what value of t is the value of $\sin^2(t)$, half of the value of $\tan(t)$?

SSC CGL 23/09/2024 (3rd Shift)

- (a) 60° (b) 45° (c) 30° (d) 22.5°

Q.152. If $\tan A = \frac{1}{\sqrt{10}}$, A is an acute angle, then the value of $\sin A + \operatorname{cosec} A$ is:

SSC CGL 23/09/2024 (3rd Shift)

- (a) $\frac{11}{\sqrt{10}}$ (b) $\frac{1}{\sqrt{10}}$ (c) $\frac{12}{\sqrt{11}}$ (d) $\frac{10}{\sqrt{11}}$

Q.153. If $\sec\theta = \sqrt{2}$, then $\operatorname{cosec}\theta$ is equal to:

SSC CGL 24/09/2024 (1st Shift)

- (a) 0 (b) $\sqrt{2}$ (c) -1 (d) 1

Q.154. If $2\sin^2\theta + 3\sin\theta - 2 = 0$, ($0 < \theta < 90^\circ$), then the value of θ is:

SSC CGL 24/09/2024 (1st Shift)

- (a) 15° (b) 30° (c) 60° (d) 45°

Q.155. If $\sin 3A = \cos(A - 26^\circ)$, then $A =$ _____, where $3A$ is an acute angle.

SSC CGL 24/09/2024 (1st Shift)

- (a) 29° (b) 52° (c) 26° (d) 78°

Q.156. The value of $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2$ is:

SSC CGL 24/09/2024 (2nd Shift)

- (a) $3 + \tan^2 A + \cot^2 A$ (b) $7 + \tan^2 A + \cot^2 A$
(c) $5 + \tan^2 A + \cot^2 A$ (d) $1 + \tan^2 A + \cot^2 A$

Q.157. If $3 \tan A = 4$ and A is an acute angle, then the value of $4\sin A + 3\cos A$ is:

SSC CGL 24/09/2024 (2nd Shift)

- (a) 1 (b) 3 (c) 4 (d) 5

Q.158. If $10 \sin^2\theta + 6 \cos^2\theta = 7$, $0 < \theta < 90^\circ$ then find the value of $\tan\theta$.

SSC CGL 24/09/2024 (3rd Shift)

- (a) $\sqrt{3}$ (b) 1 (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{2}$

Q.159. If $A = 15^\circ$, then What is the value of $\frac{11\sqrt{2} \cos 3A + 10 \sin 2A}{7\sqrt{3} \sin 4A}$?

SSC CGL 24/09/2024 (3rd Shift)

- (a) $\frac{11}{7}$ (b) $\frac{21}{32}$ (c) $\frac{32}{21}$ (d) $\frac{7}{11}$

Q.160. Evaluate $2 \frac{\tan 54^\circ}{\cot 36^\circ} - \frac{\cot 41^\circ}{\tan 49^\circ}$.

SSC CGL 25/09/2024 (1st Shift)

- (a) -1 (b) 1 (c) 0 (d) 2

Q.161. Find the value of $(\operatorname{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta)$.

SSC CGL 25/09/2024 (1st Shift)

- (a) 0 (b) $\frac{1}{2}$ (c) 1 (d) -1

Q.162. If $m + 5 = \sec^2 A \operatorname{cosec}^2 A (1 - \cos^2 A) (1 - \sin^2 A) + 5$, then find the value of m:

SSC CGL 25/09/2024 (1st Shift)

- (a) 10 (b) 0 (c) 1 (d) 5

Q.163. If $\cot^2\theta - 2\cos^2\theta = 0$, ($0^\circ < \theta < 90^\circ$), then the value of θ is:

SSC CGL 25/09/2024 (2nd Shift)

- (a) 45° (b) 60° (c) 90° (d) 30°

Q.164. If $x = p \sec A \cos B$, $y = q \sec A \sin B$ and $z = r \tan A$, what is the value of the following expression?

$$\frac{x^2}{p^2} + \frac{y^2}{q^2} - \frac{z^2}{r^2}$$

SSC CGL 25/09/2024 (2nd Shift)

- (a) 1 (b) $p^2 + q^2 - r^2$ (c) $p^2 - q^2 + r^2$ (d) 0

Q.165. Find the value of $3\sin 15^\circ - 4\sin^3 15^\circ$

SSC CGL 25/09/2024 (3rd Shift)

- (a) 2 (b) $\frac{1}{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$

Q.166. If $\sec\theta + \tan\theta = x$, then find $\sin\theta$.

SSC CGL 25/09/2024 (3rd Shift)

- (a) $\frac{x^2 + 1}{1 - x^2}$ (b) $\frac{x^2 - 1}{1 + 2x^2}$
(c) $\frac{x^2 - 1}{1 + x^2}$ (d) $\frac{1 - x^2}{1 + x^2}$

Q.167. If $(48^\circ + k)$ is an acute angle and $\sin(48^\circ + k) = \cos 13^\circ$, what is the value of k (in $^\circ$)?

SSC CGL 26/09/2024 (1st Shift)

- (a) 17 (b) 23 (c) 29 (d) 37

Q.168. Evaluate the given expression.

$$\frac{5}{1 + \cot^2\theta} + \frac{3}{1 + \tan^2\theta} + 2\cos^2\theta$$

SSC CGL 26/09/2024 (1st Shift)

- (a) 0 (b) 3 (c) 5 (d) 2

Q.169. If A is an acute angle and $\tan A + \cot A = 2$, find the value of $7\tan^8 A - 6\cot^8 A + 8\sec^2 A$.

SSC CGL 26/09/2024 (1st Shift)

- (a) 6 (b) 17 (c) 16 (d) 7

Q.170. If $p = \frac{\sin A}{1 + \cos A}$, then $\frac{\sin A}{1 - \cos A}$

= ?

SSC CGL 26/09/2024 (2nd Shift)

- (a) $\frac{1}{p}$ (b) $\frac{1}{1-p}$ (c) $\frac{1}{p+1}$ (d) $p-1$

Q.171. If $8\cot\theta = 7$, then the value of $\frac{1+\sin\theta}{\cos\theta}$ is :

SSC CGL 26/09/2024 (2nd Shift)

- (a) $\frac{8+\sqrt{113}}{7}$ (b) $\frac{1+\sqrt{113}}{8}$
(c) $\frac{7+\sqrt{113}}{8}$ (d) $\frac{1+\sqrt{113}}{7}$

Q.172. In $\triangle ABC$, right angled at B if $\tan C = \sqrt{3}$, then find $\frac{\sin^2 C + \cos^2 C}{1 + \cot^2 C}$

SSC CGL 26/09/2024 (2nd Shift)

- (a) $\frac{3}{4}$ (b) $\frac{16}{3}$ (c) $\frac{4}{15}$ (d) $\frac{3}{16}$

Q.173. The greatest value of $\sin^4\theta + \cos^4\theta$ is:

SSC CGL 26/09/2024 (3rd Shift)

- (a) 2 (b) 1 (c) 3 (d) 4

Q.174. If $x = a\sec\theta\cos\phi$, $y = b\sec\theta\sin\phi$ and $z = c\tan\theta$, then the value of $(\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2})$ is equal to:

SSC CGL 26/09/2024 (3rd Shift)

- (a) 2 (b) 1 (c) 3 (d) 0

Q.175. If $\cot\theta = \sqrt{11}$, then the value of $\frac{\csc^2\theta - \sec^2\theta}{\csc^2\theta + \sec^2\theta}$ is:

SSC CGL 26/09/2024 (3rd Shift)

- (a) $\frac{3}{5}$ (b) $\frac{5}{6}$ (c) $\frac{4}{5}$ (d) $\frac{6}{7}$

SSC CHSL 2024 Tier - 1

Q.176. In right-angled triangle ABC, $\angle B = 90^\circ$ and angle A and angle C are acute angles.

If $\csc A = 2\sqrt{2}$, then find the value of $\sin A \cdot \cos C + \cos A \cdot \sin C$.

SSC CHSL 01/07/2024 (1st Shift)

- (a) $\sqrt{2}$ (b) 1 (c) 0 (d) $2\sqrt{2}$

Q.177. If $\sin\theta = \frac{2xy}{x^2+y^2}$, then the value of $\tan\theta$ is:

SSC CHSL 01/07/2024 (2nd Shift)

- (a) $\frac{2xy}{x^2-y^2}$ (b) $\frac{x^2-y^2}{2xy}$
(c) $\frac{x^2-y^2}{x^2+y^2}$ (d) $\frac{x^2+y^2}{x^2-y^2}$

Q.178. $12\sin^2 A + 2\cos^2 A = 7$, where $0^\circ < A < 180^\circ$. Then $\cot A$ is:

SSC CHSL 01/07/2024 (4th Shift)

- (a) $\pm \frac{1}{\sqrt{2}}$ (b) ± 1 (c) $\pm \frac{1}{\sqrt{3}}$ (d) $\pm \frac{1}{2}$

Q.179. What is the value of $\cos(x+y) - \sin(x-y) + \tan(2z)$?

SSC CHSL 02/07/2024 (1st Shift)

- (a) $\cos x \cdot \cos y - \sin x \cdot \sin y - \sin x \cdot \cos y - \cos x \cdot \sin y + \frac{2\tan z}{1-\tan^2 z}$
(b) $\sin x \cdot \sin y + \frac{2\tan z}{1-\tan^2 z}$
(c) $\cos x \cdot \cos y - \sin x \cdot \sin y - \sin x \cdot \cos y + \cos x \cdot \sin y + \frac{2\tan z}{1-\tan^2 z}$
(d) $\cos x \cdot \cos y - \sin x \cdot \sin y - \cos x \cdot \sin y + \frac{\tan z}{1-\tan^2 z}$

Q.180. Given that $\sin\theta = \frac{15}{17}$, Evaluate

$\frac{15\cot\theta + 17\sin\theta}{8\tan\theta + 16\sec\theta}$, where $0^\circ < \theta < 90^\circ$.

SSC CHSL 02/07/2024 (2nd Shift)

- (a) $\frac{23}{49}$ (b) $\frac{49}{26}$ (c) $\frac{26}{23}$ (d) $\frac{49}{23}$

Q.181. If θ is an acute angle and $\tan\theta + \cot\theta = 2$, then find the value of $\tan^3\theta + \cot^3\theta + 6\tan^3\theta\cot^2\theta$.

SSC CHSL 02/07/2024 (3rd Shift)

- (a) 8 (b) 10 (c) 12 (d) 6

Q.182. If $\sqrt{13}\sin\theta = 2$, then the value of $\frac{3\tan\theta + \sqrt{13}\sin\theta}{\sqrt{13}\cos\theta - 3\tan\theta}$ is:

SSC CHSL 02/07/2024 (4th Shift)

- (a) 2 (b) 3 (c) 0 (d) 4

Q.183. $(\cot\theta + \tan\theta)(\csc\theta - \sin\theta)(\cos\theta - \sec\theta) =$ _____.

SSC CHSL 03/07/2024 (1st Shift)

- (a) -1 (b) 2 (c) 0 (d) 1

Q.184. Find the value of the following expression.

$\sin(30+\theta) - \cos(60-\theta)$

SSC CHSL 03/07/2024 (2nd Shift)

- (a) $\frac{\sqrt{3}}{2}$ (b) $2\sin(30+\theta)$ (c) 1 (d) 0

Q.185. If $\alpha + \beta = 45^\circ$ and $(\tan\alpha + 1)(\tan\beta + 1) = 2x$, then x is:

SSC CHSL 03/07/2024 (4th Shift)

- (a) -1 (b) 0 (c) 1 (d) 2

Q.186. What is the value of $\tan 570^\circ$?

SSC CHSL 04/07/2024 (1st Shift)

- (a) $-\frac{1}{\sqrt{3}}$ (b) $-\sqrt{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$

Q.187. What is the value of

$\frac{1}{\sin^4(90^\circ - 2\alpha)} + \frac{1}{\cos^2(90^\circ - 2\alpha) - 1}$?

SSC CHSL 04/07/2024 (2nd Shift)

- (a) $\sin^2 2\alpha + \tan^2 2\alpha$ (b) $\sec^2 2\alpha + \cot^2 2\alpha$
(c) $\cos^2 2\alpha \cot^2 2\alpha$ (d) $\sec^2 2\alpha \tan^2 2\alpha$

Q.188. If θ is an acute angle and $\cot\theta +$

$\tan\theta = 2$, then find the value of $\tan^{12}\theta + \cot^{12}\theta + 2\tan^5\theta\cot^7\theta$.

SSC CHSL 04/07/2024 (3rd Shift)

- (a) 2 (b) 3 (c) 4 (d) 1

Q.189. If θ be acute angle and $\cos\theta = \frac{24}{25}$, then the value of $\cot(90^\circ - \theta)$ is equal to:

SSC CHSL 05/07/2024 (3rd Shift)

- (a) $\frac{6}{25}$ (b) $\frac{7}{24}$ (c) $\frac{7}{25}$ (d) $\frac{5}{24}$

Q.190. $\left[\frac{\csc 31^\circ - \sec 59^\circ}{(\sin 18^\circ \div \cos 72^\circ)} \times \frac{\tan 26^\circ}{\cot 64^\circ} \right] + \left[\frac{\tan 30^\circ + \tan 15^\circ}{(1 - \tan 30^\circ \tan 15^\circ)} \right] =$ _____.

SSC CHSL 05/07/2024 (4th Shift)

- (a) -1 (b) 0 (c) 1 (d) 2

Q.191. If $\cot A = \frac{5}{12}$, then find the value

of $\frac{\sqrt{\sec A - \csc A}}{\sqrt{\sec A + \csc A}} - \sqrt{\frac{17}{7}}$.

SSC CHSL 08/07/2024 (1st Shift)

- (a) $\frac{24}{\sqrt{119}}$ (b) $\frac{10}{\sqrt{119}}$ (c) 0 (d) $-\frac{10}{\sqrt{119}}$

Q.192. If $\tan(t) = \frac{3}{7}$, t is an acute angle, then what is the value of $\csc(t)$?

SSC CHSL 08/07/2024 (2nd Shift)

- (a) $\frac{\sqrt{58}}{3}$ (b) $\frac{\sqrt{53}}{7}$ (c) $\frac{\sqrt{58}}{7}$ (d) $\frac{\sqrt{53}}{3}$

Q.193. If $\sin\theta = \frac{11}{12}$, then what will be the value of $(\tan\theta + \sec\theta)$?

SSC CHSL 08/07/2024 (3rd Shift)

- (a) $\frac{12}{\sqrt{23}}$ (b) $\frac{11}{\sqrt{23}}$ (c) $\sqrt{23}$ (d) 23

Q.194. If $\cos^4\theta - \sin^4\theta = \frac{4}{5}$, then find the value of $\sin 4\theta$.

SSC CHSL 08/07/2024 (4th Shift)

- (a) $\frac{24}{25}$ (b) $\frac{16}{25}$ (c) $\frac{21}{25}$ (d) $\frac{18}{25}$

Q.195. If $\cot\alpha = \frac{P}{Q}$, then find the value of

$\frac{P\cos\alpha - Q\sin\alpha}{P\cos\alpha + Q\sin\alpha} - \frac{P^2 - Q^2}{P^2 + Q^2} + 3$.

SSC CHSL 09/07/2024 (1st Shift)

- (a) 0 (b) 3 (c) $\frac{P^2}{P^2 + Q^2}$ (d) $\frac{P}{Q}$

Q.196. If $\sin\theta = \frac{7}{25}$, then $\cos\theta$ and $\tan\theta$ are respectively _____. (where θ is in the first quadrant)

SSC CHSL 09/07/2024 (3rd Shift)

- (a) $\frac{25}{7}$ and $\frac{7}{24}$ (b) $\frac{24}{25}$ and $\frac{24}{7}$
(c) $\frac{24}{25}$ and $\frac{7}{24}$ (d) $\frac{25}{24}$ and $\frac{24}{7}$

Q.197. What is the maximum value of $7 \cos A + 24 \sin A + 32$?

SSC CHSL 09/07/2024 (4th Shift)

- (a) 39 (b) 32 (c) 25 (d) 57

Q.198. If $\tan A = \sqrt{2} - 1$ then find the value of $\sin A \cos A + \tan A$.

SSC CHSL 10/07/2024 (1st Shift)

- (a) $\frac{24\sqrt{2} + 15}{4}$ (b) $\frac{24\sqrt{2} - 15}{4}$
(c) $\frac{5\sqrt{2} + 4}{4}$ (d) $\frac{5\sqrt{2} - 4}{4}$

Q.199. If $4 \sin^2 \theta - 4\sqrt{3} \sin \theta + 3 = 0$, then find the value of θ . ($0 < \theta < 90^\circ$)

SSC CHSL 10/07/2024 (3rd Shift)

- (a) 45° (b) 30° (c) 60° (d) 75°

Q.200. The value of $\left(\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta}\right) \left(\frac{1}{\tan \theta + \cot \theta}\right)$ is equal to :

SSC CHSL 11/07/2024 (1st Shift)

- (a) $\tan \theta$ (b) $2 \sin \theta$ (c) $2 \cos \theta$ (d) 1

Q.201. Simplify $\frac{\sin 60^\circ - \cos 60^\circ}{\sin 60^\circ + \cos 60^\circ}$

SSC CHSL 11/07/2024 (3rd Shift)

- (a) $2 + \sqrt{3}$ (b) $\sqrt{3} - 2$
(c) $\sqrt{3} - 1$ (d) $2 - \sqrt{3}$

SSC CPO 2024 Tier - 1

Q.202. If $\cot \theta = \frac{28}{45}$, then find the value of $(\sin \theta - \cos \theta)$.

SSC CPO 27/06/2024 (1st Shift)

- (a) $\frac{45}{78}$ (b) $\frac{7}{53}$ (c) $\frac{28}{53}$ (d) $\frac{17}{53}$

Q.203. If $\sec 6A = \operatorname{cosec} (A - 29^\circ)$, where $2A$ is an acute angle, then the measure of $\angle A =$ _____.

SSC CPO 27/06/2024 (1st Shift)

- (a) 18 (b) 17 (c) 21 (d) 19

Q.204. Simplify: $(1 + \tan^2 A) + \left(1 + \frac{1}{\tan^2 A}\right)$

SSC CPO 27/06/2024 (1st Shift)

- (a) $\frac{1}{\sin^2 A - \sin^4 A}$ (b) $\frac{1}{\sin^2 A - \cos^4 A}$
(c) $\frac{1}{\cos^2 A - \sin^4 A}$ (d) $\frac{1}{\cot^2 A - \tan^4 A}$

Q.205. If $\sin \alpha = \frac{5}{13}$, then the value of $\cos \alpha \cdot \operatorname{cosec} \alpha \cdot \cot \alpha$ is _____.

SSC CPO 27/06/2024 (2nd Shift)

- (a) $\frac{12}{5}$ (b) $\frac{144}{25}$ (c) $\frac{5}{12}$ (d) $\frac{25}{144}$

Q.206. Find the value of $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 3(\sin^2 A + \cos^2 A)$.

SSC CPO 27/06/2024 (2nd Shift)

- (a) -1 (b) 1 (c) 2 (d) -2

Q.207. Find the maximum value of $(19 \sin \theta + 6 \cot \theta \sin \theta)$.

SSC CPO 27/06/2024 (3rd Shift)

- (a) $\sqrt{397}$ (b) $\sqrt{197}$ (c) $\sqrt{297}$ (d) $\sqrt{497}$

Q.208. If $\tan \theta = \frac{3}{10}$ then find the value of $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$

SSC CPO 27/06/2024 (3rd Shift)

- (a) $\frac{100}{7}$ (b) $\frac{100}{3}$ (c) $\frac{100}{9}$ (d) $\frac{50}{3}$

Q.209. Which of the following is true when $x = \sin A + \cos A$; $y = \sec A + \operatorname{cosec} A$?

SSC CPO 27/06/2024 (3rd Shift)

- (a) $y(1 + x^2) = 2x$ (b) $y - 2x = x^2 y$
(c) $y + 2x = x^2 y$ (d) $y(1 - 2x^2) = x$

Q.210. Solve the equation for A (in degrees):
 $2 \cos^2 A + 3 \cos A - 2 = 0$, $0 < A < 90^\circ$

SSC CPO 28/06/2024 (1st Shift)

- (a) 30° (b) 80° (c) 60° (d) 45°

Q.211. Find the value of $\frac{1 - \cot^2 \theta}{\tan^2 \theta - 1}$

SSC CPO 28/06/2024 (1st Shift)

- (a) $\tan^2 \theta$ (b) $\cot \theta$ (c) $\tan \theta$ (d) $\cot^2 \theta$

Q.212. If $\frac{(\sec A + \tan A)}{\sec A - \tan A} = 2 \frac{51}{79}$

then the value of $\sin A$ is equal to:

SSC CPO 28/06/2024 (1st Shift)

- (a) $\frac{77}{144}$ (b) $\frac{61}{169}$ (c) $\frac{87}{169}$ (d) $\frac{65}{144}$

Q.213. If $4 \sin^2 \theta = 3(1 + \cos \theta)$, $0^\circ < \theta < 90^\circ$, then what is the value of $\sqrt{15} \tan \theta + \frac{4}{\sqrt{15}} \sin \theta + 2 \sec \theta$?

SSC CPO 28/06/2024 (2nd Shift)

- (a) 24 (b) $4\sqrt{15}$ (c) $\frac{8}{\sqrt{15}}$ (d) 8

Q.214. If $\tan \theta + \cot \theta = 6$, then the value of $\tan^2 \theta + \cot^2 \theta$ is _____.

SSC CPO 28/06/2024 (2nd Shift)

- (a) 24 (b) 34 (c) 20 (d) 22

Q.215. $(1 - \cos^2 A) \sec^2 A$ is equal to:

SSC CPO 28/06/2024 (2nd Shift)

- (a) $\cos^2 A \cot^2 A$ (b) $\tan^2 A$
(c) $\cot^2 A$ (d) $\operatorname{cosec}^2 A$

Q.216. If $\sin x + \cos x = \sqrt{2} \sin x$, then the value of $\sin x - \cos x$ is:

SSC CPO 28/06/2024 (2nd Shift)

- (a) $\sqrt{2} \sin x$ (b) $2 \sin x$
(c) $-\sqrt{2} \cos x$ (d) $\sqrt{2} \cos x$

Q.217. If A , B and C are the interior angles of $\triangle ABC$, then what is the value of $\left\{\tan \frac{A}{2} + \operatorname{cosec} \frac{B+C}{2}\right\} \left\{\tan \frac{A}{2} - \operatorname{cosec} \frac{B+C}{2}\right\}$?

SSC CPO 28/06/2024 (2nd Shift)

$\frac{B+C}{2}$ } ?

SSC CPO 28/06/2024 (3rd Shift)

- (a) 2 (b) -1 (c) -2 (d) 1

Q.218. Simplify $\frac{\sin^3 A + \cos^3 A}{\sin A + \cos A}$

SSC CPO 28/06/2024 (3rd Shift)

- (a) $1 - \tan A \sec A$ (b) $1 - \cos A \cot A$
(c) $1 - \sin A \tan A$ (d) $1 - \sin A \cos A$

Q.219. If $\sin(x + y) = 1$ and $\cos(x - y) = \frac{\sqrt{3}}{2}$, find the value of 'y'.

SSC CPO 28/06/2024 (3rd Shift)

- (a) 90° (b) 60° (c) 30° (d) 20°

Q.220. The value of $\left(\frac{2 \cos^3 \theta - \cos \theta}{\sin \theta - 2 \sin^3 \theta}\right)^2 + 1$, $\theta \neq 45^\circ$ is :

SSC CPO 28/06/2024 (3rd Shift)

- (a) $\sin^2 \theta$ (b) $\sec^2 \theta$ (c) $\operatorname{cosec}^2 \theta$ (d) $\cot^2 \theta$

Q.221. If $\sin^4 \theta + \cos^4 \theta = 2 \sin^2 \theta \cos^2 \theta$, $0^\circ < \theta < 90^\circ$, then find the value of $\cot \theta$.

SSC CPO 29/06/2024 (1st Shift)

- (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{3}$ (d) 1

Q.222. $\tan^4 A + \tan^2 A$ is equal to :

SSC CPO 29/06/2024 (1st Shift)

- (a) $\sec^2 A - \sec^4 A$ (b) $-(\sec^2 A + \sec^4 A)$
(c) $\sec^4 A - \sec^2 A$ (d) $\sec^4 A + \sec^2 A$

Q.223. Solve the following :

$(\tan \theta + \sec \theta + 1)(\cot \theta - \operatorname{cosec} \theta + 1) = ?$

SSC CPO 29/06/2024 (1st Shift)

- (a) 2 (b) 1 (c) 0 (d) -1

Q.224. The value of $(\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)(\tan \theta + \cot \theta)$ is

SSC CPO 29/06/2024 (1st Shift)

- (a) -3 (b) 3 (c) $\sec \theta \operatorname{cosec} \theta$ (d) 1

Q.225. Find the value of the following expression.

$(1 + \tan \theta + \sec \theta) \times (1 + \cot \theta - \operatorname{cosec} \theta)$

SSC CPO 29/06/2024 (2nd Shift)

- (a) 1 (b) $\cos \theta \sin \theta$ (c) 2 (d) $\tan \theta$

Q.226. Simplify $\frac{\tan 71^\circ + \tan 19^\circ}{1 - \tan 71^\circ \tan 19^\circ}$

SSC CPO 29/06/2024 (2nd Shift)

- (a) Not defined (b) 1 (c) $\sqrt{3}$ (d) 0

Q.227. If $\sin(x + y) = \cos\{3(x + y)\}$, then, the value of $\cot\{2(x + y)\}$ is _____.

SSC CPO 29/06/2024 (2nd Shift)

- (a) -1 (b) 1 (c) -2 (d) 2

Q.228. The value of $\frac{2 \cos 15^\circ \sin 15^\circ}{\cos^2 15^\circ - \sin^2 15^\circ}$ is:

SSC CPO 29/06/2024 (2nd Shift)

- (a) $\frac{1}{\sqrt{3}}$ (b) ∞ (c) $\frac{\sqrt{5} - 1}{\sqrt{5} + 1}$ (d) $\sqrt{3}$

Q.229. The value of $\frac{17}{\sec^2 \theta} + 10 \sin^2 \theta +$

$\frac{7}{1 + \cot^2 \theta}$ is :

SSC CPO 29/06/2024 (3rd Shift)

(a) 34 (b) 20 (c) 17 (d) 19

Q.230. The value of $\sin^2 15^\circ + \sin^2 25^\circ + \sin^2 35^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 65^\circ + \sin^2 75^\circ$ is:

SSC CPO 29/06/2024 (3rd Shift)

(a) 4 (b) 7 (c) $\frac{7}{2}$ (d) $\frac{7}{3}$

SSC Selection Post (Phase - XII)

Q.231. Evaluate $7\left(\frac{\csc 24^\circ}{\sec 66^\circ}\right)^3 + 8$

$\left(\frac{\cot 37^\circ}{\tan 53^\circ}\right)^4 - \left(2\frac{\sec 14^\circ}{\csc 76^\circ}\right)^2 + \left(-3\frac{\tan 82^\circ}{\cot 8^\circ}\right)^3$

Higher Secondary 20/06/2024 (Shift - 2)

(a) -16 (b) 40 (c) 10 (d) 2

Q.232. If $8 \tan A = 5$, what is the value of

$\frac{8 \sin A - 7 \cos A}{8 \sin A + 11 \cos A}$?

Higher Secondary 21/06/2024 (Shift - 2)

(a) $-\frac{1}{8}$ (b) $\frac{1}{19}$ (c) $\frac{1}{8}$ (d) $-\frac{1}{6}$

Q.233. Which of the following is NOT a correct formula ?

Higher Secondary 24/06/2024 (Shift - 2)

(a) $\cos^2 A \cot^2 A = \cot^2 A - \cos^2 A$

(b) $\frac{1 + \tan^2 A}{\tan^2 A} = \sin^2 A$

(c) $\sec A \sin A = \tan A$

(d) $\csc^2 A - \cot^2 A = 1$

Q.234. Evaluate the following.

$\frac{5 \cos^2 120^\circ + 4 \sec^2 30^\circ - \tan^2 135^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$

Higher Secondary 25/06/2024 (Shift - 2)

(a) $\frac{37}{12}$ (b) $\frac{57}{5}$ (c) $\frac{67}{12}$ (d) $\frac{27}{5}$

Q.235. Evaluate

$\left(\frac{\sin 23^\circ \cos 67^\circ + \cos 23^\circ \sin 67^\circ}{\csc^2 15^\circ - \tan^2 75^\circ}\right)^3$

Higher Secondary 25/06/2024 (Shift - 4)

(a) 0 (b) 1 (c) 8 (d) -1

Q.236. If A is an acute angle, what is the

value of $\csc A$, if $\cot A = \frac{2p}{p^2 - 1}$?

Higher Secondary 26/06/2024 (Shift - 2)

(a) $\frac{p^2 + 1}{p^2 - 1}$ (b) $\frac{p^2 - 1}{p^2 + 1}$

(c) $\frac{p^2}{p^2 - 1}$ (d) $\frac{2p}{p^2 + 1}$

Q.237. The value of $\csc \theta \left[\frac{1 + \cos \theta}{\sin \theta} + \right.$

$\left. \frac{\sin \theta}{1 + \cos \theta} \right] - 2 \cot^2 \theta$ is equal to:

Higher Secondary 26/06/2024 (Shift - 2)

(a) 0 (b) 1 (c) 2 (d) 3

Q.238. If $\sin A + \cos A = \frac{4}{3}$, then find the value of $\tan A + \cot A$.

Graduate Level 20/06/2024 (Shift - 3)

(a) $\frac{7}{18}$ (b) $\frac{3}{4}$ (c) $\frac{18}{7}$ (d) $\frac{4}{3}$

Q.239. The expression $\sin^2 \theta + \cos^2 \theta - 1 = 0$ is satisfied by how many values of θ ?

Graduate Level 21/06/2024 (Shift - 3)

(a) No value
(b) Only one value
(c) Infinitely many values
(d) Two values

Q.240. Find the value of $(\sin 75^\circ + \sin 15^\circ)$.

Graduate Level 21/06/2024 (Shift - 3)

(a) $\frac{\sqrt{6}}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\sqrt{6}$ (d) $\sqrt{3}$

Q.241. Find the value of $\frac{\cos 25^\circ - \sin 65^\circ}{\cos 25^\circ + \sin 65^\circ}$

Graduate Level 24/06/2024 (Shift - 3)

(a) -1 (b) 0 (c) 1 (d) $\frac{1}{2}$

Q.242. If $\sin(5x - 40^\circ) = \cos(5y + 40^\circ)$, then the value of $x + y$ is:

Graduate Level 24/06/2024 (Shift - 3)

(a) 18° (b) 15° (c) 40° (d) 20°

Q.243. The sides of a right-angled triangle, right-angled at B, are 6, 8 and 10 units. C is the vertex opposite to the side with length 8 units. What is the value of $\tan^2 A + \cos^2 C$?

Graduate Level 25/06/2024 (Shift - 3)

(a) $\frac{369}{400}$ (b) $\frac{125}{400}$ (c) $\frac{321}{400}$ (d) $\frac{325}{400}$

Q.244. If $\sin \theta + \csc \theta = \sqrt{5}$, then the value of $\sin^3 \theta + \csc^3 \theta$ is:

Graduate Level 26/06/2024 (Shift - 3)

(a) $\frac{1}{\sqrt{5}}$ (b) $\sqrt{5}$ (c) $2\sqrt{5}$ (d) 0

Q.245. If $\cos 2\theta = 0$, where θ is an acute angle, then find the value of $\sin(75^\circ - \theta)$.

Graduate Level 26/06/2024 (Shift - 3)

(a) $\sqrt{2} + 1$ (b) $\frac{1}{2}$ (c) $\sqrt{2} - 1$ (d) $\frac{1}{\sqrt{2}}$

Q.246. If $P \sin 60^\circ = Q \csc 45^\circ$, then the

value of $\frac{P^2 + Q^2}{P^2 - Q^2}$ is equal to:

Graduate Level 26/06/2024 (Shift - 4)

(a) $\frac{5}{11}$ (b) $\frac{11}{7}$ (c) $\frac{11}{5}$ (d) $\frac{7}{11}$

SSC CPO 2023 Tier - 1

Q.247. Find the value of $(\tan^2 \theta + \tan^4 \theta)$.

SSC CPO 03/10/2023 (1st Shift)

(a) $\sec^2 \theta - \sec^4 \theta$ (b) $\cot^2 \theta - \tan^2 \theta$

(c) $\sec^4 \theta + \sec^2 \theta$ (d) $\sec^4 \theta - \sec^2 \theta$

Q.248. If $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$, then $(\cos \theta + \sin \theta)$ is:

SSC CPO 03/10/2023 (1st Shift)

(a) $-\sqrt{2} \sin \theta$ (b) $\sqrt{2} \tan \theta$

(c) $\sqrt{2} \cos \theta$ (d) $-\sqrt{2} \cos \theta$

Q.249. The value of

$(1 + \sin^4 A - \cos^4 A) \operatorname{cosec}^2 A$ is:

SSC CPO 03/10/2023 (2nd Shift)

(a) -1 (b) 1 (c) -2 (d) 2

Q.250. If $\cos A = \frac{15}{17}$, $0 \leq A \leq 90^\circ$, then the value of $\cot(90^\circ - A)$ is :

SSC CPO 03/10/2023 (2nd Shift)

(a) $\frac{8}{15}$ (b) $\frac{7}{15}$ (c) $\frac{\sqrt{2}}{15}$ (d) $\frac{2\sqrt{2}}{15}$

Q.251. If $\sin 2A = \cos 15^\circ$, then the smallest positive value of A is:

SSC CPO 03/10/2023 (3rd Shift)

(a) 7.5° (b) 47.5° (c) 75° (d) 37.5°

Q.252. In $\triangle ABC$, $\angle B = 90^\circ$ and $AB : BC = 1 : 2$. The value of $\cos A + \tan C$ is:

SSC CPO 04/10/2023 (1st Shift)

(a) $\frac{1 + \sqrt{5}}{2\sqrt{5}}$ (b) $\frac{2\sqrt{5}}{2 + \sqrt{5}}$

(c) $\frac{5 + \sqrt{5}}{2\sqrt{5}}$ (d) $\frac{2 + \sqrt{5}}{2\sqrt{5}}$

Q.253. If $\sec A - \tan A = p$, then find the value of $\sec A$.

SSC CPO 04/10/2023 (2nd Shift)

(a) $\frac{p^2 - 1}{p^2 + 1}$ (b) $\frac{p^2 + 1}{p^2 - 1}$

(c) $\frac{p^2 + 1}{2p}$ (d) $\frac{p^2 + 1}{p}$

Q.254. Find the value of $\sqrt{\frac{1 - \tan A}{1 + \tan A}}$.

SSC CPO 04/10/2023 (2nd Shift)

(a) $\sqrt{\frac{1 + \sin A}{\cos A}}$ (b) $\sqrt{\frac{1 + \sin 2A}{\cos 2A}}$

(c) $\sqrt{\frac{1 - \sin A}{\cos A}}$ (d) $\sqrt{\frac{1 - \sin 2A}{\cos 2A}}$

Q.255. The value of $\sqrt{\frac{1 + \sin A}{1 - \sin A}}$ is :

SSC CPO 04/10/2023 (3rd Shift)

(a) $\csc A + \cot A$ (b) $\csc A - \cot A$

(c) $\sec A - \tan A$ (d) $\sec A + \tan A$

Q.256. If $\sin A = \frac{2}{3}$, then find the value of

$(7 - \tan A)(3 + \cos A)$.

SSC CPO 05/10/2023 (1st Shift)

(a) $\frac{61}{3} + \frac{17}{\sqrt{5}}$ (b) $\frac{61}{3} - \frac{17}{3\sqrt{5}}$

(c) $\frac{61}{3} + \frac{17}{3\sqrt{5}}$ (d) $\frac{61}{3\sqrt{5}} + \frac{17}{3}$

Q.257. If $\sin \beta = \frac{1}{3}$, $(\sec \beta - \tan \beta)^2$ is equal to :

SSC CPO 05/10/2023 (1st Shift)

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{3}{4}$

Q.258. Find the value of $\frac{\sin^2 39^\circ + \sin^2 (90^\circ - 39^\circ)}{\cos^2 35^\circ + \cos^2 (90^\circ - 35^\circ)} + 3 \tan 25^\circ \tan 65^\circ$.

SSC CPO 05/10/2023 (2nd Shift)

- (a) 1 (b) 3 (c) 4 (d) 2

Q.259. If $\tan \theta + \cot \theta = 2$, θ is an acute angle, then find the value of $2 \tan^{25} \theta + 3 \cot^{20} \theta + 5 \tan^{30} \theta \cot^{15} \theta$

SSC CPO 05/10/2023 (2nd Shift)

- (a) 12 (b) 10 (c) 8 (d) 6

Q.260. Find the value of $\sin (50^\circ + A) - \cos (40^\circ - A)$.

SSC CPO 05/10/2023 (2nd Shift)

- (a) 1 (b) 0 (c) -1 (d) 2

Q.261. If $\sin \theta + \cos \theta = \sqrt{2} \cos \theta$, then find $\frac{\sin \theta - \cos \theta}{\sin \theta}$.

SSC CPO 05/10/2023 (3rd Shift)

- (a) $-\sqrt{2}$ (b) -1 (c) 1 (d) $\sqrt{2}$

SSC CHSL 2023 Tier - 1

Q.262. $\frac{\cos \theta}{\sec \theta - 1} + \frac{\cos \theta}{\sec \theta + 1}$ is equal to :

SSC CHSL 02/08/2023 (1st Shift)

- (a) $2 \sec^2 \theta$ (b) $2 \cot^2 \theta$
(c) $2 \cos^2 \theta$ (d) $2 \sin^2 \theta$

Q.263. If $(\sin \theta - \cos \theta) = 0$, then the value of $\sin(\pi - \theta) + \sin(\frac{\pi}{2} - \theta)$ is:

SSC CHSL 02/08/2023 (2nd Shift)

- (a) 1 (b) 0 (c) $\sqrt{3}$ (d) $\sqrt{2}$

Q.264. If $\cos \theta + \sec \theta = \sqrt{3}$, then the value of $\cos^3 \theta + \sec^3 \theta$ is :

SSC CHSL 02/08/2023 (3rd Shift)

- (a) 0 (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{3}$ (d) $2\sqrt{3}$

Q.265. If $\cos A = \frac{63}{65}$, then find the value of $\tan A + \cot A$ (up to two places of decimal).

SSC CHSL 02/08/2023 (4th Shift)

- (a) 4.19 (b) 2.76 (c) 3.19 (d) 5.23

Q.266. If $\frac{\cos \theta}{(1 + \sin \theta)} + \frac{\cos \theta}{(1 - \sin \theta)} = 4$

and θ is acute, then the value of θ is :

SSC CHSL 03/08/2023 (1st Shift)

- (a) 60° (b) 15° (c) 45° (d) 30°

Q.267. If $\sin \alpha + \cos \alpha = \frac{2}{\sqrt{3}}$ then what

is $(\tan \alpha + \cot \alpha)$ equal to ?

SSC CHSL 03/08/2023 (1st Shift)

- (a) 1 (b) 2 (c) 6 (d) 0

Q.268. The value of

$(\operatorname{cosec}^2 B - 1)(\sec^2 B - 1)$ is :

SSC CHSL 03/08/2023 (2nd Shift)

- (a) 4 (b) 3 (c) 2 (d) 1

Q.269. If $\sin \theta + \cos \theta = \frac{1}{29}$, find the value

of $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$.

SSC CHSL 03/08/2023 (2nd Shift)

- (a) $\frac{1}{41}$ (b) $\frac{43}{29}$ (c) $\frac{41}{29}$ (d) $\frac{1}{43}$

Q.270. The given expression is equal to :

$\frac{(1 + \tan^2 A)}{\operatorname{cosec}^2 A \cdot \tan A}$

SSC CHSL 03/08/2023 (3rd Shift)

- (a) $\sec^2 A$ (b) $\sec A$ (c) $\tan A$ (d) $\tan^2 A$

Q.271. $\cos^4 A - \sin^4 A$ is equal to :

SSC CHSL 03/08/2023 (3rd Shift)

- (a) $2 \cos^2 A + 1$ (b) $1 - 2 \sin^2 A$
(c) $2 \sin^2 A - 1$ (d) $-(2 \sin^2 A + 1)$

Q.272. If $\operatorname{cosec} \theta = 1 \frac{7}{22}$, find the value of $\cot^2 \theta$.

SSC CHSL 03/08/2023 (4th Shift)

- (a) $\frac{49}{484}$ (b) $\frac{357}{484}$ (c) $\frac{225}{484}$ (d) $\frac{7}{22}$

Q.273. What is the value of the expression

$\sin A (1 + \frac{\sin A}{\cos A}) + \cos A (1 + \frac{\cos A}{\sin A})$?

SSC CHSL 04/08/2023 (1st Shift)

- (a) $\sec A + \operatorname{cosec} A$ (b) $\sin A + \cos A$
(c) $\sin A - \cos A$ (d) $\sec A - \operatorname{cosec} A$

Q.274. Simplify the expression:

$\frac{3 - \sin^2 A + \cos^2 A}{2 + 2 \cos^2 A}$

SSC CHSL 04/08/2023 (1st Shift)

- (a) 1 (b) 0 (c) -1 (d) 2

Q.275. If $\sec \theta + \tan \theta = \sqrt{3}$, then the positive value of $\sin \theta + \cos \theta$ is:

SSC CHSL 04/08/2023 (3rd Shift)

- (a) $\frac{(1 + \sqrt{3})}{2}$ (b) $\frac{2}{\sqrt{3}}$
(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{2}{(1 + \sqrt{3})}$

Q.276. If $2 \tan^2 A + 4 \cos^4 A = 3$, then the possible value of A is:

SSC CHSL 04/08/2023 (4th Shift)

- (a) 45° (b) 60° (c) 0° (d) 30°

Q.277. If $a = \cot A + \cos A$ and $b = \cot A - \cos A$, then find the value of $a^2 - b^2 - 4\sqrt{ab}$.

SSC CHSL 04/08/2023 (4th Shift)

- (a) 0 (b) -1 (c) 1 (d) -4

Q.278. The given expression is equal to :

$1 - \frac{\tan^2 \phi}{\sec^2 \phi}$

SSC CHSL 07/08/2023 (1st Shift)

- (a) $\sin^2 \phi \cos^2 \phi$ (b) $\sin^2 \phi \cot^2 \phi$
(c) $\cot^2 \phi \cos^2 \phi$ (d) $\tan^2 \phi \cos^2 \phi$

Q.279. Find the value of $\cot^2 B - \operatorname{cosec}^2 B$ for $0 < B < 90^\circ$.

SSC CHSL 07/08/2023 (1st Shift)

- (a) 1 (b) 2 (c) -1 (d) -2

Q.280. In ΔOPQ , right-angled at P, $OP = 7$ cm and $OQ - PQ = 1$ cm. Determine the values of $\sin Q + \cos Q$.

SSC CHSL 07/08/2023 (2nd Shift)

- (a) $\frac{13}{25}$ (b) $\frac{21}{25}$ (c) $\frac{31}{25}$ (d) $\frac{33}{25}$

Q.281. The given expression is equal to:

$\frac{2 \sec A}{\sec^2 A - 1}$

SSC CHSL 07/08/2023 (3rd Shift)

- (a) $2 \operatorname{cosec} A \cot A$ (b) $2 \cos A \cot A$
(c) $2 \operatorname{cosec} A \tan A$ (d) $2 \cos A \sin A$

Q.282. The given expression is equal to:

$\frac{\cos^2 A}{(1 + \sin A)}$

SSC CHSL 07/08/2023 (3rd Shift)

- (a) $1 - \sin A$ (b) $1 + \cos A$
(c) $\sin A$ (d) $\cos A$

Q.283. The value of $\sin^2 30^\circ + \sin^2 40^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 35^\circ + \sin^2 45^\circ + \sin^2 50^\circ + \sin^2 60^\circ$ is:

SSC CHSL 07/08/2023 (3rd Shift)

- (a) 2 (b) 4 (c) 3 (d) 1

Q.284. Simplify the given expression.

$\frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos 53^\circ \sin 37^\circ$

SSC CHSL 07/08/2023 (4th Shift)

- (a) 2 (b) -1 (c) -2 (d) 1

Q.285. The value of $\frac{1}{\sin \theta} - \frac{\cot^2 \theta}{1 + \operatorname{cosec} \theta}$ is

SSC CHSL 07/08/2023 (4th Shift)

- (a) 2 (b) 1 (c) -1 (d) 0

Q.286. The value of $\frac{\sin 58^\circ}{\cos 32^\circ} +$

$\frac{\sin 55^\circ \sec 35^\circ}{\tan 5^\circ \tan 45^\circ \tan 85^\circ}$ is equal to:

SSC CHSL 07/08/2023 (4th Shift)

- (a) 2 (b) 1 (c) 0 (d) 3

Q.287. If $\sec \theta = 2 \frac{4}{23}$ find the value of $\tan^2 \theta$.

SSC CHSL 08/08/2023 (1st Shift)

- (a) $2 \frac{16}{529}$ (b) $1 \frac{200}{529}$ (c) $3 \frac{384}{529}$ (d) $3 \frac{177}{529}$

Q.288. Find the value of

$$\frac{\cos^2 25^\circ - \sin^2 65^\circ}{\cos^2 25^\circ + \sin^2 65^\circ} :$$

SSC CHSL 08/08/2023 (1st Shift)

- (a) $\frac{1}{2}$ (b) 1 (c) -1 (d) 0

Q.289. Simplify the given equation.

$$\frac{\cot^3 A - 1}{\cot A - 1} = ?$$

SSC CHSL 08/08/2023 (2nd Shift)

- (a) $\operatorname{cosec}^2 A - \cot A$ (b) $\operatorname{cosec}^2 A + \cot A$
(c) $\cot^2 A + \operatorname{cosec} A$ (d) $\cot^2 A - \operatorname{cosec} A$

Q.290. If $\tan \theta + \sec \theta = a$, then what is $\sec \theta$ equal to:

SSC CHSL 08/08/2023 (2nd Shift)

- (a) $\frac{a+1}{2a^2}$ (b) $\frac{(a-1)}{2a^2}$
(c) $\frac{a^2-1}{2a}$ (d) $\frac{a^2+1}{2a}$

Q.291. Find the value of

$$\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} + 2.$$

SSC CHSL 08/08/2023 (3rd Shift)

- (a) $1 + 2\sec A$ (b) $2 - \sec A$
(c) $2 + \sec A$ (d) $2(1 + \sec A)$

Q.292. If $\sec \theta + \tan \theta = \frac{1}{\sqrt{3}}$, then the

positive value of $\cot \theta + \cos \theta$ is :

SSC CHSL 08/08/2023 (4th Shift)

- (a) $\frac{3\sqrt{3}}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{2}{3\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}$

Q.293. If $\frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} = \frac{4}{5}$, then the

value of $\frac{\operatorname{cosec}^2 \theta}{2 - \operatorname{cosec}^2 \theta}$ is:

SSC CHSL 09/08/2023 (1st Shift)

- (a) $\frac{16}{25}$ (b) $\frac{40}{41}$ (c) $\frac{41}{40}$ (d) $\frac{31}{30}$

Q.294. ΔPQR is right angled at Q. If PQ = 12 cm PR = 13 cm, find $\tan P + \cot R$.

SSC CHSL 09/08/2023 (1st Shift)

- (a) $\frac{9}{10}$ (b) 0 (c) $\frac{10}{12}$ (d) $\frac{12}{10}$

Q.295. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, then

the value of $A + B$ is:

SSC CHSL 09/08/2023 (1st Shift)

- (a) 60° (b) 15° (c) 30° (d) 45°

Q.296. If $\cot A = \frac{5}{12}$, find the value of the following expression.

$$\frac{5(1 - \cos^2 A)}{6(1 - \sin^2 A)}$$

SSC CHSL 09/08/2023 (2nd Shift)

- (a) $\frac{144}{5}$ (b) $\frac{24}{25}$ (c) $\frac{144}{25}$ (d) $\frac{24}{5}$

Q.297. If $\tan \theta \cdot \tan 2\theta = 1$ then the value of

$\cot 5\theta$ is:

SSC CHSL 09/08/2023 (4th Shift)

- (a) -1 (b) 1 (c) $-\sqrt{3}$ (d) $\sqrt{3}$

Q.298. If $4 \sin^2 A - 3 = 0$ and $0 \leq A \leq 90^\circ$, then $3 \sin A - 4 \sin^3 A$ is:

SSC CHSL 10/08/2023 (1st Shift)

- (a) $\frac{1}{2}$ (b) $\sqrt{\frac{3}{2}}$ (c) 0 (d) 1

Q.299. ΔRMS is right-angled at M. The length of the base, RM = 4 cm and the length of perpendicular MS = 3 cm. Find the value of $\sec R$.

SSC CHSL 10/08/2023 (3rd Shift)

- (a) $\frac{3}{4}$ (b) $\frac{5}{4}$ (c) $\frac{4}{5}$ (d) $\frac{2}{5}$

Q.300. Find the value of the following expression.

$$2(\tan^2 A \cos^2 A + \cos^2 A)$$

SSC CHSL 10/08/2023 (4th Shift)

- (a) 4 (b) 1 (c) 3 (d) 2

Q.301. If A, B, C are the acute angles,

such that $\tan(A + B - C) = \frac{1}{\sqrt{3}}$, $\cos(B +$

$C - A) = \frac{1}{2}$ and $\sin(C + A - B) = \frac{1}{\sqrt{2}}$. The

value of $A + B + C$ will be equal to:

SSC CHSL 10/08/2023 (4th Shift)

- (a) 90° (b) 110° (c) 170° (d) 135°

Q.302. Find the value of the following expression.

$$\frac{(1 + \sec \theta)}{(\sec \theta)} (1 - \cos \theta)$$

SSC CHSL 10/08/2023 (4th Shift)

- (a) $\cos^2 \theta$ (b) $\sec^2 \theta$ (c) $\cot^2 \theta$ (d) $\sin^2 \theta$

Q.303. Let ABC be a right-angled triangle where $\angle A = 90^\circ$ and $\angle C = 45^\circ$. Find the value of $\sec C + \sin C \sec C$.

SSC CHSL 11/08/2023 (1st Shift)

- (a) 1 (b) $1 - \sqrt{2}$ (c) $1 + \sqrt{2}$ (d) $\sqrt{2} - 1$

Q.304. If $(4 \sin \theta + 5 \cos \theta) = 3$, then the value of $(4 \cos \theta - 5 \sin \theta)$ is:

SSC CHSL 11/08/2023 (1st Shift)

- (a) $3\sqrt{2}$ (b) $4\sqrt{2}$ (c) $2\sqrt{3}$ (d) $2\sqrt{5}$

Q.305. Let ABC be a triangle right angled

at B. If $\tan A = \frac{12}{5}$, then find the values

of $\operatorname{cosec} A$ and $\sec A$, respectively.

SSC CHSL 11/08/2023 (3rd Shift)

- (a) $\frac{13}{10}, \frac{5}{13}$ (b) $\frac{13}{12}, \frac{13}{5}$
(c) $\frac{10}{13}, \frac{5}{13}$ (d) $\frac{12}{13}, \frac{5}{13}$

Q.306. If $\cos \theta - \sin \theta = 0$, then $(\sin^8 \theta + \cos^8 \theta)$ is :

SSC CHSL 11/08/2023 (3rd Shift)

- (a) $\frac{1}{8}$ (b) $\frac{1}{4}$ (c) $\frac{1}{6}$ (d) $\frac{1}{2}$

Q.307. Simplify the given expression

$$\frac{3(\sin^4 Z - \cos^4 Z + 1)}{\sin^2 Z}$$

SSC CHSL 11/08/2023 (4th Shift)

- (a) 9 (b) 2 (c) 4 (d) 6

Q.308. Find the value of $\sin 47^\circ \cos 33^\circ - \cos 43^\circ \sin 57^\circ$.

SSC CHSL 14/08/2023 (2nd Shift)

- (a) -1 (b) 1 (c) 0 (d) $\frac{1}{2}$

Q.309. The value of $\frac{1 + \sin A}{\cos A} + \frac{\cos A}{1 + \sin A}$

is:

SSC CHSL 14/08/2023 (3rd Shift)

- (a) $2 \sec A$ (b) $2 \operatorname{cosec} A$
(c) $\sec A$ (d) $\operatorname{cosec} A$

Q.310. Simplify the given equation. $(1 + \tan^2 A)(1 + \cot^2 A) = ?$

SSC CHSL 14/08/2023 (3rd Shift)

- (a) $\frac{1}{\cos^2 A(1 + \sin^2 A)}$
(b) $\frac{1}{\sin^2 A(1 - \sin^2 A)}$
(c) $\frac{1}{\sin^2 A + 1 + \operatorname{cosec}^2 A}$
(d) $\frac{1}{\sin^2 A(1 + \cos^2 A)}$

Q.311. The given expression is equal to:

$(\cot B - \tan B) \cdot \sin B \cdot \cos B$

SSC CHSL 14/08/2023 (4th Shift)

- (a) $1 - 2\sec^2 B$ (b) $1 - 2\cos^2 B$
(c) $2 \cos^2 B - 1$ (d) $2\sec^2 B - 1$

Q.312. $\cot^2 \theta - \operatorname{cosec}^2 \theta$ equals to :

SSC CHSL 17/08/2023 (1st Shift)

- (a) 0 (b) $\tan \theta$ (c) 1 (d) -1

Q.313. If $\sin \theta \cos \theta = \frac{1}{\sqrt{3}}$ then the value

of $(\sin^4 \theta + \cos^4 \theta)$ is:

SSC CHSL 17/08/2023 (3rd Shift)

- (a) 1 (b) $\frac{5}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{3}$

Q.314. Simplify the given expression.

$$\sqrt{\frac{1 + \cos P}{1 - \cos P}}$$

SSC CHSL 17/08/2023 (3rd Shift)

- (a) $\operatorname{cosec} P - \cot P$ (b) $\sec P - \tan P$
(c) $\sec P + \tan P$ (d) $\operatorname{cosec} P + \cot P$

Q.315. Find the value of $\tan(50^\circ + \theta) - \cot(40^\circ - \theta)$.

SSC CHSL 17/08/2023 (4th Shift)

- (a) $\frac{1}{2}$ (b) -1 (c) 0 (d) 1

Q.316. What is the value of $(1 - \sin^2 A) \operatorname{cosec}^2 A$?

Given A is an acute angle.

SSC CHSL 17/08/2023 (4th Shift)

(a) $\cot^2 A$ (b) $\cos^2 A$ (c) $\tan^2 A$ (d) $\sin^2 A$

Q.317. If $\tan 15^\circ = 2 - \sqrt{3}$, then the value of $\tan 15^\circ \cot 75^\circ + \tan 75^\circ \cot 15^\circ$ is:
SSC CHSL 17/08/2023 (4th Shift)
(a) 6 (b) 10 (c) 8 (d) 14

SSC CGL 2023 Tier - 1

Q.318. If $\left\{ \left(\frac{3\sin\theta - \cos\theta}{\cos\theta + \sin\theta} \right) \right\} = 1$, then the value of $\cot\theta$ is:
SSC CGL 14/07/2023 (1st shift)
(a) 3 (b) 0 (c) 1 (d) 2

Q.319. Evaluate the following: $\cos(36^\circ + A) \cdot \cos(36^\circ - A) + \cos(54^\circ + A) \cdot \cos(54^\circ - A)$
SSC CGL 14/07/2023 (2nd shift)
(a) $\sin 2A$ (b) $\cos A$ (c) $\sin A$ (d) $\cos 2A$

Q.320. If $\sin 23^\circ = \frac{a}{b}$, then the value of $\sec 23^\circ - \sin 67^\circ$ is _____
SSC CGL 14/07/2023 (3rd shift)

(a) $\frac{a^2}{\sqrt{b^2 - a^2}}$ (b) $\frac{b^2 - a^2}{ab}$
(c) $\frac{a^2}{b\sqrt{b^2 + a^2}}$ (d) $\frac{a^2}{b\sqrt{b^2 - a^2}}$

Q.321. If $\frac{\cos\beta}{\sec\alpha} = 15$ and $\frac{\sin\beta}{\sec\alpha} = 16$, then the value of $\sin^2\beta$ is _____.
SSC CGL 14/07/2023 (3rd shift)
(a) $\frac{256}{481}$ (b) $-\frac{256}{481}$ (c) $\frac{481}{256}$ (d) $-\frac{481}{256}$

Q.322. Simplify the following.
 $\frac{\sin^3 \alpha + \cos^3 \alpha}{\sin \alpha + \cos \alpha}$
SSC CGL 14/07/2023 (4th shift)
(a) $1 + \sin \alpha \cos \alpha$ (b) $\tan \alpha$
(c) $1 - \sin \alpha \cos \alpha$ (d) $\sec \alpha$

Q.323. If $\tan \frac{\pi}{6} + \sec \frac{\pi}{6} = x$, then find x .
SSC CGL 17/07/2023 (2nd shift)
(a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$ (c) $-\frac{1}{\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}$

Q.324. If $\cos x = -\frac{1}{2}$, x lies in third quadrant, then $\tan x = ?$
SSC CGL 17/07/2023 (3rd shift)
(a) $\sqrt{3}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{2}{\sqrt{3}}$ (d) $\frac{1}{\sqrt{3}}$

Q.325. Using $\cos(A + B) = \cos A \cos B - \sin A \sin B$, find the value of $\cos 75^\circ$.
SSC CGL 17/07/2023 (3rd shift)
(a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}+1}{4}$
(c) $\frac{\sqrt{6}-\sqrt{2}}{4}$ (d) $\frac{\sqrt{6}+\sqrt{2}}{4}$

Q.326. $\frac{\sin^4 \theta + \cos^4 \theta}{1 - 2\sin^2 \theta \cdot \cos^2 \theta} = \underline{\hspace{2cm}}$

SSC CGL 17/07/2023 (4th shift)

(a) 1 (b) 2 (c) -1 (d) 0

Q.327. Using trigonometric formulas, find the value of $\left(\frac{\sin(x-y)}{\sin(x+y)} \right) \left(\frac{\tan x + \tan y}{\tan x - \tan y} \right)$
SSC CGL 18/07/2023 (1st shift)
(a) -2 (b) 2 (c) 0 (d) 1

Q.328. If $\tan \frac{A}{2} = x$, then find x .

SSC CGL 18/07/2023 (1st shift)

(a) $\frac{\sqrt{1+\cos A}}{\sqrt{1-\cos A}}$ (b) $\frac{\sqrt{1-\sin A}}{\sqrt{1+\cos A}}$
(c) $\frac{\sqrt{1-\cos A}}{\sqrt{1+\cos A}}$ (d) $\frac{\sqrt{\cos A - 1}}{\sqrt{1+\cos A}}$

Q.329. The value of $\left(\frac{1}{\sin\theta} + \frac{1}{\tan\theta} \right) \left(\frac{1}{\sin\theta} - \frac{1}{\tan\theta} \right)$ is :

SSC CGL 18/07/2023 (2nd shift)

(a) 0 (b) 2 (c) 3 (d) 1

Q.330. Simplify the following: $\sin 2x + 2 \sin 4x + \sin 6x$

SSC CGL 18/07/2023 (2nd shift)

(a) $4 \cos^2 x \sin 4x$ (b) $4 \cos^2 x \sin x$
(c) $2 \cos^2 x \sin 4x$ (d) $4 \sin^2 x \sin 4x$

Q.331. If $\sin A = -\frac{3}{5}$, A lies in III quadrant, the value of $\sec A$ is:

SSC CGL 18/07/2023 (2nd shift)

(a) $-\frac{4}{5}$ (b) $-\frac{5}{4}$ (c) $-\frac{3}{4}$ (d) $\frac{3}{4}$

Q.332. Find the value of the following expression. $12(\sin^4 \theta + \cos^4 \theta) + 18(\sin^6 \theta + \cos^6 \theta) + 78\sin^2 \theta \cos^2 \theta$
SSC CGL 18/07/2023 (3rd shift)
(a) 30 (b) 40 (c) 10 (d) 20

Q.333. Using $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$, find the value of $\tan 15^\circ$.

SSC CGL 18/07/2023 (4th shift)

(a) $\sqrt{3} + 1$ (b) $\sqrt{3} - 1$ (c) $2 - \sqrt{3}$ (d) $2 + \sqrt{3}$

Q.334. If $\frac{\sin x - \cos x}{\sin x + \cos x} = \frac{2}{5}$, then the value of $\frac{1 + \cot^2 x}{1 - \cot^2 x}$ is:

SSC CGL 19/07/2023 (1st shift)

(a) 2.25 (b) 1.45 (c) 3.75 (d) 5.25

Q.335. Find the value of $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$

SSC CGL 19/07/2023 (1st shift)

(a) 4 (b) 0 (c) 2 (d) 1

Q.336. If $\left(\frac{\cos A}{1 - \sin A} \right) + \left(\frac{\cos A}{1 + \sin A} \right) = 4$ then what will be the value of A ? ($0^\circ < \theta < 90^\circ$)

SSC CGL 19/07/2023 (1st shift)

(a) 90° (b) 45° (c) 60° (d) 30°

Q.337. Using $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$, find the value of $\cot 15^\circ$.

SSC CGL 19/07/2023 (2nd shift)

(a) $2 + \sqrt{3}$ (b) $2 - \sqrt{3}$
(c) $\sqrt{3} - 1$ (d) $\sqrt{3} + 1$

Q.338. If $P \cos \alpha = 3$ and $4 \tan \alpha = Q$, then what is the relation between P and Q , which is independent of α ?

SSC CGL 19/07/2023 (2nd shift)

(a) $\frac{9}{P^2} + \frac{16}{Q^2} = 1$ (b) $\frac{9}{P^2} - \frac{16}{Q^2} = 1$
(c) $\frac{P^2}{9} - \frac{Q^2}{16} = 1$ (d) $\frac{P^2}{9} + \frac{Q^2}{16} = 1$

Q.339. For any acute angle θ , $\sin\theta + \sin^2\theta = 1$, then the value of $\cos^2\theta + \cos^4\theta$ = _____.

SSC CGL 19/07/2023 (3rd shift)

(a) 0 (b) 1 (c) 2 (d) -1

Q.340. If $\cos x + \sin x = \sqrt{2} \cos x$, what is the value of $(\cos x - \sin x)^2 + (\cos x + \sin x)^2$?

SSC CGL 19/07/2023 (4th shift)

(a) 2 (b) 1 (c) 0 (d) $\frac{1}{\sqrt{2}}$

Q.341. Solve the given expression.
$$\frac{\sec^2(90^\circ - \theta) - \cot^2 \theta}{2(\sin^2 35^\circ + \sin^2 55^\circ)}$$

SSC CGL 20/07/2023 (1st shift)

(a) 0.5 (b) 2 (c) 1.5 (d) 1

Q.342. $\operatorname{cosec} 2910^\circ + \sec 4260^\circ + \tan 2565^\circ + \cot 1755^\circ = ?$

SSC CGL 20/07/2023 (1st shift)

(a) 3 (b) 1 (c) 4 (d) 0

Q.343. The maximum value of $(\sin^{12}\theta + \cos^{20}\theta)$ for all the real values of θ is:

SSC CGL 20/07/2023 (3rd shift)

(a) 2 (b) 3 (c) 1 (d) 0

Q.344. What is the value of $\sin(-405^\circ)$?

SSC CGL 20/07/2023 (3rd shift)

(a) $\frac{\sqrt{5}}{2}$ (b) $\frac{1}{2}$ (c) $-\frac{1}{2}$ (d) $-\frac{1}{\sqrt{2}}$

Q.345. Simplify the following expression
$$\frac{\sin\theta - 2345\sin^3\theta}{2\cos^3\theta - \cos\theta}$$

SSC CGL 21/07/2023 (3rd shift)

(a) $\tan\theta$ (b) $\sin\theta$ (c) $\sec\theta$ (d) $\cos\theta$

Q.346. Simplify $\frac{1 + \sin t}{4 - 4\sin t} - \frac{1 - \sin t}{4 + 4\sin t}$

SSC CGL 21/07/2023 (4th shift)

(a) $4 \tan t \cdot \sin t$ (b) $\tan t \cdot \sec t$
(c) $\tan t - \sin t$ (d) $\tan t + \sin t$

Q.347. PQR is a triangle right angled at Q

and PQ : QR = 3 : 4. What is the value of $\sin P + \sin Q + \sin R$?

SSC CGL 21/07/2023 (4th shift)

- (a) $\frac{4}{5}$ (b) $\frac{3}{5}$ (c) $\frac{12}{5}$ (d) $\frac{2}{5}$

Q.348. If $2 \cot \theta = 3$, then find the value of $\frac{\sqrt{13} \sin \theta - 3 \tan \theta}{3 \tan \theta + \sqrt{13} \cos \theta}$

SSC CGL 24/07/2023 (1st shift)

- (a) $\frac{1}{\sqrt{13}}$ (b) $\frac{2}{\sqrt{13}}$ (c) 0 (d) $\frac{2}{3}$

Q.349. Using the identity $\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$ find the value of $\tan 15^\circ$ correct to three decimal places. [Use $\sqrt{3} = 1.732$]

SSC CGL 24/07/2023 (1st shift)

- (a) 0.268 (b) 0.27 (c) 0.267 (d) 0.269

Q.350. If $\operatorname{Cosec} \theta + \cot \theta = m$, find the value of $\frac{m^2 - 1}{m^2 + 1}$

SSC CGL 24/07/2023 (2nd shift)

- (a) 1 (b) 0 (c) $\cos \theta$ (d) $-\sin \theta$

Q.351. If $\sin \theta + \operatorname{cosec} \theta = \sqrt{5}$, then the value of $\sin^3 \theta + \operatorname{cosec}^3 \theta =$ _____.

SSC CGL 24/07/2023 (3rd shift)

- (a) 0 (b) $3\sqrt{5}$ (c) $\frac{1}{\sqrt{5}}$ (d) $2\sqrt{5}$

Q.352. If $a \cot \theta = b$, then what will be the value of $\frac{b \cos \theta - a \sin \theta}{b \cos \theta + a \sin \theta}$?

SSC CGL 24/07/2023 (4th shift)

- (a) $\frac{b^2 + a^2}{b^2 - a^2}$ (b) $b^2 + a^2$
(c) $\frac{b^2 - a^2}{b^2 + a^2}$ (d) 0

Q.353. Simplify the given expression.

$$\frac{1 + \sin^4 \theta + \cos^4 \theta}{\cos^2 \theta + \sin^4 \theta}$$

SSC CGL 24/07/2023 (4th shift)

- (a) 3 (b) 1 (c) 2 (d) 4

Q.354. Using $\operatorname{cosec}(\alpha + \beta) = \frac{\sec \alpha \times \sec \beta \times \operatorname{cosec} \alpha \times \operatorname{cosec} \beta}{\sec \alpha \times \operatorname{cosec} \beta + \operatorname{cosec} \alpha \times \sec \beta}$, find the value of $\operatorname{cosec} 75^\circ$.

SSC CGL 25/07/2023 (1st shift)

- (a) $\frac{\sqrt{6} + \sqrt{2}}{4}$ (b) $\frac{\sqrt{6} - \sqrt{2}}{4}$
(c) $\sqrt{6} - \sqrt{2}$ (d) $\sqrt{6} + \sqrt{2}$

Q.355. The value of $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$ is :

SSC CGL 25/07/2023 (3rd shift)

- (a) $\sec \theta + \tan \theta$ (b) $\operatorname{cosec} \theta - \cot \theta$
(c) $\operatorname{cosec} \theta + \cot \theta$ (d) $\sec \theta - \tan \theta$

Q.356. Find $(\frac{1}{\sin \theta}) - \sin \theta$.

SSC CGL 25/07/2023 (4th shift)

- (a) $\cos \theta \cot \theta$ (b) $\cos \theta \sec \theta$
(c) $\cos \theta \operatorname{cosec} \theta$ (d) $\cos \theta \tan \theta$

Q.357. If $\frac{21 \cos A + 3 \sin A}{3 \cos A + 4 \sin A} = 2$, then find the value of $\cot A$

SSC CGL 26/07/2023 (2nd shift)

- (a) $\frac{9}{11}$ (b) $\frac{11}{9}$ (c) $\frac{1}{3}$ (d) $\frac{11}{10}$

Q.358. $\tan 4384^\circ + \cot 6814^\circ = ?$

SSC CGL 26/07/2023 (3rd shift)

- (a) -1 (b) 2 (c) 0 (d) 1

Q.359. If $A = 22.5^\circ$ what is the value of $(10\sqrt{2} \sin 2A - 7\sqrt{2} \cos 2A + 9 \tan 2A)$?

SSC CGL 26/07/2023 (4th shift)

- (a) 12 (b) 15 (c) 10 (d) 6

Q.360. $\sin 600^\circ \cos 750^\circ + \sin 150^\circ \cos 240^\circ = ?$

SSC CGL 26/07/2023 (4th shift)

- (a) $\frac{2}{3}$ (b) 1 (c) $\frac{1}{2}$ (d) -1

Q.361. What is the values of $\tan \frac{\theta}{2}$?

SSC CGL 26/07/2023 (4th shift)

- (a) $\frac{\cos \theta}{1 - \sin \theta}$ (b) $\frac{\sin \theta}{1 - \cos \theta}$
(c) $\frac{\cos \theta}{1 - \cos \theta}$ (d) $\frac{\sin \theta}{1 + \cos \theta}$

Q.362. If $3x = \sec A$ and $\frac{3}{x} = \tan A$, then

$$9(x^2 - \frac{1}{x^2}) \text{ is } \underline{\hspace{2cm}}.$$

SSC CGL 27/07/2023 (2nd shift)

- (a) 3 (b) 9 (c) 1 (d) $\frac{1}{9}$

Q.363. If $\tan^4 \theta + \tan^2 \theta = 1$, what is the value of $11(\cos^4 \theta + \cos^2 \theta)$?

SSC CGL 27/07/2023 (3rd shift)

- (a) -11 (b) 8 (c) 0 (d) 11

SSC Selection Post (Phase - XI)

Q.364. The value of $\frac{32}{\sec^2 \theta} - \frac{20}{1 + \tan^2 \theta} +$

$12 \sin^2 \theta$ is :

Graduate Level 27/06/2023 (Shift - 4)

- (a) 20 (b) 32 (c) 24 (d) 12

Q.365. If $\sin \theta + \cos \theta = \frac{\sqrt{11}}{3}$, then what is $\tan \theta + \cot \theta$ equal to ?

Graduate Level 27/06/2023 (Shift - 4)

- (a) 5 (b) 9 (c) 7 (d) 11

Q.366. If $1 + \sin^2 \theta = a \cos^2 \theta$, then find $\sec^4 \theta - \tan^4 \theta$.

Graduate Level 30/06/2023 (Shift - 1)

- (a) a^2 (b) a^{-1} (c) a (d) a^2

Q.367. In a triangle PQR, if $QR = 3$ cm, $PR = 5$ cm and $\sin P = \frac{3}{5}$, then $\angle Q$ equals:

Higher Secondary 28/06/2023 (Shift - 2)

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

Q.368. The value of $\operatorname{cosec} 41^\circ - \sec 49^\circ$ is :

Higher Secondary 30/06/2023 (Shift - 2)

- (a) 1 (b) 0 (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

SSC CHSL 2022 Tier - 2

Q.369. If $\sin x + \sin^2 x = 1$, then what is the value of $\cos^8 x + 2 \cos^6 x + \cos^4 x$?

SSC CHSL Tier II (26/06/2023)

- (a) 2 (b) 0 (c) 4 (d) 1

SSC CHSL 2022 Tier - 1

Q.370. What is the value of

$$\left(\frac{1 + \sec^2 A}{1 + \cos^2 A} \right) \left(\frac{1 + \sin^2 A}{1 + \operatorname{cosec}^2 A} \right) ?$$

SSC CHSL 09/03/2023 (1st Shift)

- (a) $\cos^2 A$ (b) $\operatorname{cosec}^2 A$
(c) $\sec^2 A$ (d) $\tan^2 A$

Q.371. What is $\frac{\sqrt{1 - \sin A}}{\sqrt{1 + \sin A}}$?

SSC CHSL 09/03/2023 (2nd Shift)

- (a) $\sec^2 A$ (b) $\sec A + \tan A$
(c) $\tan^2 A$ (d) $\sec A - \tan A$

Q.372. What is the value of $(1 + \tan \theta - \sec \theta)(1 + \cot \theta + \operatorname{cosec} \theta)$?

SSC CHSL 09/03/2023 (3rd Shift)

- (a) -1 (b) 0 (c) 1 (d) 2

Q.373. What is the value of

$$\frac{\cos(50^\circ + A) - \sin(40^\circ - A)}{\cos 40^\circ \sec 40^\circ} ?$$

SSC CHSL 10/03/2023 (1st Shift)

- (a) 1 (b) 2 (c) 0.5 (d) 0

Q.374. If $\tan A + \sec A = K$, then what is the value of $\sin A$?

SSC CHSL 10/03/2023 (1st Shift)

- (a) $\frac{2K}{K^2 + 1}$ (b) $\frac{K^2 - 1}{2K}$
(c) $\frac{K^2 + 1}{K^2 - 1}$ (d) $\frac{K^2 - 1}{K^2 + 1}$

Q.375. What is $\frac{\operatorname{cosec} 90^\circ}{(1 - \cos^2 60^\circ)}$?

SSC CHSL 10/03/2023 (2nd Shift)

- (a) 0 (b) $\frac{4}{3}$ (c) $\frac{1}{4}$ (d) $\frac{3}{4}$

Q.376. What is the value of

$$\sqrt{\frac{\operatorname{cosec} A + 1}{\operatorname{cosec} A - 1}} + \sqrt{\frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1}} ?$$

SSC CHSL 13/03/2023 (1st Shift)

- (a) $2 \cos A$ (b) $\sec A$
(c) $\cos A$ (d) $2 \sec A$

Q.377. If $m = \sec \theta - \tan \theta$ and $n = \operatorname{cosec} \theta + \cot \theta$, then what is the value of $m + n$ ($m - 1$) ?

SSC CHSL 13/03/2023 (2nd Shift)

- (a) 0 (b) 2 (c) 1 (d) -1

Q.378. ΔABC is a right angled triangle. $\angle A = 90^\circ$, $AB = 4$ cm, $BC = 5$ cm. What is the value of $\cos B + \cot C$?

SSC CHSL 13/03/2023 (3rd Shift)

- (a) $\frac{17}{20}$ (b) $\frac{29}{20}$ (c) $\frac{23}{20}$ (d) $\frac{31}{20}$

Q.379. If ΔPQR is right angled at Q, then what is the value of $\cos (P + R)$?

SSC CHSL 13/03/2023 (4th Shift)

- (a) 0.5 (b) 1 (c) 0.2 (d) 0

Q.380. If $\sqrt{3} \tan^2 \theta - 4 \tan \theta + \sqrt{3} = 0$, then what is the value of $\tan^2 \theta + \cot^2 \theta$?

SSC CHSL 13/03/2023 (4th Shift)

- (a) $\frac{4}{3}$ (b) $\frac{10}{3}$ (c) 3 (d) $\frac{6}{5}$

Q.381. If $6 \cot \theta = 5$, then find the value of $\frac{(6 \cos \theta + \sin \theta)}{(6 \cos \theta - 4 \sin \theta)}$?

SSC CHSL 14/03/2023 (2nd Shift)

- (a) 5 (b) 1 (c) 6 (d) 0

Q.382. If $-\sin \theta + \operatorname{cosec} \theta = 6$, then what is the value of $\sin \theta + \operatorname{cosec} \theta$?

SSC CHSL 14/03/2023 (2nd Shift)

- (a) 6 (b) $\sqrt{40}$ (c) $\sqrt{34}$ (d) $\sqrt{38}$

Q.383. If $\sqrt{2} \sec^2 \theta - 4 \sec \theta + 2\sqrt{2} = 0$, then what is the value $\sin^2 \theta + \tan^2 \theta$?

SSC CHSL 14/03/2023 (3rd Shift)

- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{5}{2}$ (d) $\frac{3}{2}$

Q.384. What is the value of

$$\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} + \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} ?$$

SSC CHSL 14/03/2023 (4th Shift)

- (a) $\frac{1}{(\sin^2 \theta - \cos^2 \theta)}$
(b) $2(\sin^2 \theta - \cos^2 \theta)$
(c) $\frac{2}{(\sin^2 \theta - \cos^2 \theta)}$
(d) $\sin^2 \theta - \cos^2 \theta$

Q.385. Simplify the following

$$\text{expression:- } \frac{1 - \sin A}{\cos A} + \frac{\cos A}{1 - \sin A} ?$$

SSC CHSL 15/03/2023 (1st Shift)

- (a) $2 \sin A$ (b) $2 \cos A$ (c) $2 \sec A$ (d) $2 \tan A$

Q.386. If $\sec x + \tan x = 5$ and $\operatorname{cosec} y - \cot y = \frac{1}{3}$ then find the value of $(\sec x + \operatorname{cosec} y) - (\tan x - \cot y)$.

SSC CHSL 15/03/2023 (1st Shift)

- (a) 2.2 (b) 3.2 (c) 4.2 (d) 3.1

Q.387. If $A = a \operatorname{cosec} \theta + b \cot \theta$ and $B = a \cot \theta + b \operatorname{cosec} \theta$, then what is the value of $A^2 - B^2$?

SSC CHSL 15/03/2023 (4th Shift)

- (a) ab (b) $(a + b)$ (c) $(a^2 - b^2)$ (d) $(a - b)$

Q.388. If $\operatorname{cosec} \theta + \cot \theta = s$, then what is the value of $\cos \theta$?

SSC CHSL 16/03/2023 (2nd Shift)

- (a) $\frac{s^2 - 1}{s^2 + 1}$ (b) $\frac{s^2 + 1}{s^2 - 1}$
(c) $\frac{s^2 - 1}{2s}$ (d) $\frac{2s}{s^2 + 1}$

Q.389. Find the value of $\tan x$, if $\sec x + \tan x = \sqrt{3}$ and x lies between 0° and 90° .

SSC CHSL 16/03/2023 (3rd Shift)

- (a) $\sqrt{3}$ (b) 1 (c) $\frac{1}{\sqrt{3}}$ (d) 0

Q.390. Tan of complementary of an angle is same as which of the angles ?

SSC CHSL 17/03/2023 (1st Shift)

- (a) $\sin$ of that angle
(b) $\cot$ of that angle
(c) $\cos$ of that angle
(d) cosec of that angle

Q.391. If $\sec \theta + \tan \theta = 2$, then what is the value of $\cot \theta$?

SSC CHSL 17/03/2023 (2nd Shift)

- (a) $\frac{5}{3}$ (b) $\frac{3}{5}$ (c) $\frac{4}{3}$ (d) $\frac{3}{4}$

Q.392. What is the value of $(\tan 10^\circ \tan 80^\circ + \tan 20^\circ \tan 70^\circ + \tan 30^\circ \tan 60^\circ + \tan 40^\circ \tan 50^\circ)$?

SSC CHSL 17/03/2023 (4th Shift)

- (a) 4 (b) 3 (c) 5 (d) 2

Q.393. What is the value of $\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1}$?

SSC CHSL 20/03/2023 (2nd Shift)

- (a) $2 \sec \theta$ (b) $2 \operatorname{cosec} \theta$
(c) $2 \cot \theta$ (d) $\cot \theta + \operatorname{cosec} \theta$

Q.394. If $6 \sec \theta = 10$, then find the value of $\frac{(5 \operatorname{cosec} \theta - 3 \cot \theta)}{(4 \cos \theta + 3 \sin \theta)}$?

SSC CHSL 20/03/2023 (3rd Shift)

- (a) $\frac{2}{3}$ (b) $\frac{3}{2}$ (c) $\frac{5}{6}$ (d) $\frac{6}{5}$

Q.395. What is the value of $\operatorname{cosec} A(1 - \cos A)(\operatorname{cosec} A + \cot A)$?

SSC CHSL 21/03/2023 (2nd Shift)

- (a) 2 (b) 1 (c) 0 (d) 3

Q.396. If $\frac{(2 \cos^3 A - \cos A)}{(\sin A - 2 \sin^3 A)} = 1$, then what is the value of A ?

SSC CHSL 21/03/2023 (2nd Shift)

- (a) 90° (b) 30° (c) 45° (d) 60°

SSC CGL 2022 Tier - 2

Q.397. $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = ?$

SSC CGL Tier II (07/03/2023)

- (a) $5 + \tan^2 \theta + \cot^2 \theta$
(b) $7 + \tan^2 \theta - \cot^2 \theta$
(c) $7 + \tan^2 \theta + \cot^2 \theta$
(d) $5 + \tan^2 \theta - \cot^2 \theta$

SSC CGL 2022 Tier - 1

Q.398. The value of $\tan^2 \theta + \cot^2 \theta - \sec^2 \theta \operatorname{cosec}^2 \theta$ is:

SSC CGL 01/12/2022 (1st Shift)

- (a) 2 (b) -2 (c) 0 (d) $\frac{1}{2}$

Q.399. What is $\sin \alpha - \sin \beta$?

SSC CGL 01/12/2022 (2nd Shift)

- (a) $2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$
(b) $2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$
(c) $2 \cos \frac{\alpha - \beta}{2} \sin \frac{\alpha + \beta}{2}$
(d) $2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$

Q.400. What is the value of $\tan 240^\circ$?

SSC CGL 01/12/2022 (3rd Shift)

- (a) $\sqrt{2}$ (b) $(-\sqrt{3})$ (c) $\sqrt{3}$ (d) 3

Q.401. If $\tan^2 \theta = 1 - a^2$, then the value of $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta$ is:

SSC CGL 01/12/2022 (4th Shift)

- (a) $(2 - a^2)^{\frac{3}{2}}$ (b) $(a^2 - 1)^{\frac{3}{2}}$
(c) $(2 - a^2)^{\frac{3}{2}}$ (d) $a^{\frac{3}{2}}$

Q.402. If $\sin (A + B) = \cos (A + B)$, what is the value of $\tan A$?

SSC CGL 02/12/2022 (1st Shift)

- (a) $\frac{1 - \tan B}{1 + \tan B}$ (b) $\frac{1 + \tan B}{1 - \tan B}$
(c) $\frac{1 + \sec B}{1 - \sec B}$ (d) $\frac{1 - \operatorname{cosec} B}{1 + \operatorname{cosec} B}$

Q.403. In ΔABC , right-angled at B, $AB = 7$ cm and $AC - BC = 1$ cm. Find the value of $\sin C$.

SSC CGL 02/12/2022 (2nd Shift)

- (a) $\frac{3}{7}$ (b) $\frac{12}{13}$ (c) $\frac{3}{25}$ (d) $\frac{7}{25}$

Q.404. If $\frac{(1 - \cos \theta)}{\sin \theta} = \frac{1}{5}$, then what will

be the value of $\frac{(1 + \cos \theta)}{\sin \theta}$?

SSC CGL 02/12/2022 (4th Shift)

- (a) 5 (b) $\frac{2}{5}$ (c) $\frac{4}{5}$ (d) $\frac{1}{5}$

Q.405. Simplify: $\frac{\cos A}{1 + \tan A} - \frac{\sin A}{1 + \cot A}$

SSC CGL 03/12/2022 (1st Shift)

- (a) $\tan A$ (b) $\cos A - \sin A$
(c) $\cos A \cdot \sin A$ (d) $\cos A + \sin A$

Q.406. $\frac{1}{1 + \cos(90^\circ - \theta)} + \frac{1}{1 - \cos(90^\circ - \theta)} = \underline{\hspace{2cm}}$

SSC CGL 03/12/2022 (1st Shift)

- (a) $2\sec^2\theta$ (b) 1 (c) 0 (d) $2\tan^2\theta$

Q.407. $\sin^4\theta + \cos^4\theta$ in terms of $\sin\theta$ can be written as:

SSC CGL 03/12/2022 (2nd Shift)

- (a) $2\sin^4\theta + 2\sin^2\theta - 1$
(b) $2\sin^4\theta - 2\sin^2\theta$
(c) $2\sin^4\theta - 2\sin^2\theta - 1$
(d) $2\sin^4\theta - 2\sin^2\theta + 1$

Q.408. Simplify the following :

$$\frac{\cos x - \sqrt{3}\sin x}{2}$$

SSC CGL 03/12/2022 (3rd Shift)

- (a) $\cos(\frac{\pi}{3} - x)$ (b) $\sin(\frac{\pi}{3} + x)$
(c) $\cos(\frac{\pi}{3} + x)$ (d) $\sin(\frac{\pi}{3} - x)$

Q.409. The value of $\sqrt{\frac{1 + \cos A}{1 - \cos A}}$ is:

SSC CGL 03/12/2022 (4th Shift)

- (a) $\sec A - \tan A$ (b) $\operatorname{cosec} A + \cot A$
(c) $\sec A + \tan A$ (d) $\operatorname{cosec} A - \cot A$

Q.410. If $\sin\theta + \cos\theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$, then what is the value of $\tan\theta + \cot\theta$?

SSC CGL 05/12/2022 (1st Shift)

- (a) $8(\sqrt{3}-2)$ (b) $12(\sqrt{3}-2)$
(c) $12(\sqrt{3}+2)$ (d) $8(\sqrt{3}+2)$

Q.411. If $\sec 2\theta = \operatorname{cosec}(\theta - 36^\circ)$, where 2θ is an acute angle, find the value of θ .

SSC CGL 05/12/2022 (1st Shift)

- (a) 32° (b) 46° (c) 20° (d) 42°

Q.412. What will be the value of $\cos x \operatorname{cosec} x - \sin x \sec x$?

SSC CGL 05/12/2022 (2nd Shift)

- (a) $\cot \frac{2x}{2}$ (b) $\tan 2x$
(c) $\cot 2x$ (d) $2\cot 2x$

Q.413. $\cot^2 A - \cos^2 A$ is equal to :

SSC CGL 05/12/2022 (3rd Shift)

- (a) $\cos^2 A \cdot \sin^2 A$ (b) $\cos^2 A \cdot \cot^2 A$

- (c) 1 (d) 0

Q.414. If $\operatorname{bsin}\theta = a$, then $\sec\theta + \tan\theta = ?$

SSC CGL 06/12/2022 (1st Shift)

- (a) $\sqrt{\frac{b+a}{b-a}}$ (b) $\sqrt{\frac{1}{b+a}}$
(c) $\sqrt{\frac{1}{b-a}}$ (d) $\sqrt{\frac{b-a}{b+a}}$

Q.415. What will be the value of $\sin 10^\circ - \frac{4}{3} \sin^3 10^\circ$?

$$\frac{4}{3} \sin^3 10^\circ ?$$

SSC CGL 06/12/2022 (1st Shift)

- (a) $\frac{1}{3\sqrt{3}}$ (b) $\frac{1}{6}$ (c) $\frac{1}{2\sqrt{3}}$ (d) $\frac{\sqrt{3}}{6}$

Q.416. What is the value of $\cos(-\frac{17\pi}{3})$

SSC CGL 06/12/2022 (2nd Shift)

- (a) 1 (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) 0

Q.417. Solve the following to find its value in terms of trigonometric ratios.

(sin A + cos A)(1 - sin A cos A)

SSC CGL 06/12/2022 (3rd Shift)

- (a) $\sin^3 A + \cos^3 A$
(b) $\sin^2 A - \cos^2 A$
(c) $[\cos A - \sin A][\sin^2 A + \cos^2 A]$
(d) $\sin^3 A - \cos^3 A$

Q.418. If $\sec\theta - 2\cos\theta = \frac{7}{2}$, where θ is a

positive acute angle, then the value of $\sec\theta$ is:

SSC CGL 06/12/2022 (4th Shift)

- (a) 6 (b) 8 (c) 5 (d) 4

Q.419. Find the value of $\tan 3\theta$ if $\sec 3\theta = \operatorname{cosec}(4\theta - 15^\circ)$.

SSC CGL 06/12/2022 (4th Shift)

- (a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{3}$ (c) -1 (d) 1

Q.420. If $\tan A - \tan B - \tan C = \tan A \tan B \tan C$, what is the value of A in terms of B and C ?

SSC CGL 07/12/2022 (1st Shift)

- (a) $A = B + C$ (b) $A = 2B - 2C$
(c) $A = B - C$ (d) $A = \frac{B-C}{2}$

Q.421. If $\tan^2 \alpha = 3 + Q^2$, then $\sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha = ?$

SSC CGL 07/12/2022 (1st Shift)

- (a) $(3 + Q^2)^{\frac{3}{2}}$ (b) $(7 + Q^2)^{\frac{3}{2}}$
(c) $(5 - Q^2)^{\frac{3}{2}}$ (d) $(4 + Q^2)^{\frac{3}{2}}$

Q.422. If $1 + \sin\theta = m\cos\theta$, then what is the value of $\sin\theta$?

SSC CGL 07/12/2022 (2nd Shift)

- (a) $\frac{2m^2-1}{m^2+1}$ (b) $\frac{m^2-1}{m^2+1}$
(c) $\frac{m^2+1}{2m^2-1}$ (d) $\frac{m^2+1}{m^2-1}$

Q.423. Find the value of the following expression.

$$5(\sin^4\theta + \cos^4\theta) + 3(\sin^6\theta + \cos^6\theta) + 19\sin^2\theta\cos^2\theta$$

SSC CGL 07/12/2022 (3rd Shift)

- (a) 8 (b) 5 (c) 6 (d) 7

Q.424. If $\sin Y = x$, then what will be the

value of $\cos 2Y$ (where $0 \leq Y \leq 90^\circ$) ?

SSC CGL 08/12/2022 (1st Shift)

- (a) $(\sqrt{2}-1)x$ (b) $\sqrt{2}x$ (c) $1-2x$ (d) $1-2x^2$

Q.425. If $A + B = C$, then $\tan A \tan B \tan C = ?$

SSC CGL 08/12/2022 (1st Shift)

- (a) $\tan C + \tan A - \tan B$
(b) $\tan C + \tan A + \tan B$
(c) $\tan A - \tan B - \tan C$
(d) $\tan C - \tan A - \tan B$

Q.426. If $x = 8(\sin\theta + \cos\theta)$ and $y = 9$

$(\sin\theta - \cos\theta)$, then the value of $\frac{x^2}{8^2} +$

$$\frac{y^2}{9^2}$$
 is:

SSC CGL 08/12/2022 (1st Shift)

- (a) 4 (b) 6 (c) 8 (d) 2

Q.427. $\theta = 135^\circ$; $y = 15^\circ$. What is the value of $2\cos(\theta) \sin(y)$?

SSC CGL 08/12/2022 (3rd Shift)

- (a) $\frac{\sqrt{3}-1}{2}$ (b) $\frac{1-\sqrt{3}}{2}$
(c) $2-\sqrt{3}$ (d) $\sqrt{3}-2$

Q.428. If $\sec x - \cos x = 4$, then what will

be the value of $\frac{(1 + \cos^2 x)}{\cos x}$?

SSC CGL 08/12/2022 (3rd Shift)

- (a) $\frac{9}{4}$ (b) $\frac{1}{4}$ (c) $2\sqrt{5}$ (d) $\sqrt{5}$

Q.429. If $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$, then $x = ?$

SSC CGL 08/12/2022 (4th Shift)

- (a) B (b) A (c) $A+B$ (d) $A-B$

Q.430. $\cos^3 60^\circ - \cos^3 240^\circ - \cos^3 360^\circ =$

SSC CGL 09/12/2022 (1st Shift)

- (a) $-\frac{3}{4}$ (b) $-\frac{7}{5}$ (c) $-\frac{3}{5}$ (d) $-\frac{9}{7}$

Q.431. If $5 \sin^2 A + 3 \cos^2 A = 4$, $0 < A < 90^\circ$, then what is the value of $\tan A$?

SSC CGL 09/12/2022 (1st Shift)

- (a) 0 (b) 3 (c) 1 (d) 2

Q.432. If $\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta} = \frac{3}{2}$, then the

value of $\sin^4\theta - \cos^4\theta$ is

SSC CGL 09/12/2022 (2nd Shift)

- (a) $\frac{5}{12}$ (b) $\frac{12}{13}$ (c) $\frac{11}{12}$ (d) $\frac{5}{13}$

Q.433. If $\sin^2\theta \cos^2\theta = \frac{2}{9}$, then what will

be the value of $\operatorname{cosec}^2\theta + \sec^2\theta$?

SSC CGL 09/12/2022 (3rd Shift)

- (a) $\frac{7}{2}$ (b) $\frac{5}{2}$ (c) $\frac{9}{2}$ (d) $9\sqrt{2}$

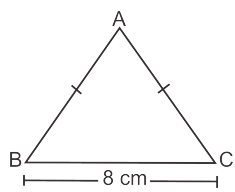
Q.434. If $2 \frac{\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$,

then $a, b = ?$

SSC CGL 09/12/2022 (3rd Shift)

- (a) $\frac{-3}{2}, \frac{-1}{2}$ (b) $\frac{3}{2}, \frac{1}{2}$ (c) $-3, -1$ (d) $3, 1$

Q.435. In the given figure, ABC is an isosceles triangle with $BC = 8$ cm and $AB = AC = 5$ cm. The value of $\tan C - \cot B$ is



SSC CGL 09/12/2022 (4th Shift)

- (a) $-\frac{5}{12}$ (b) $-\frac{7}{12}$ (c) $\frac{7}{12}$ (d) $\frac{5}{12}$

Q.436. If $b \cos \theta = a$, then $\operatorname{cosec} \theta + \cot \theta =$

SSC CGL 09/12/2022 (4th Shift)

- (a) $\sqrt{\frac{1}{b+a}}$ (b) $\sqrt{\frac{b-a}{b+a}}$
(c) $\sqrt{\frac{b+a}{b-a}}$ (d) $\sqrt{\frac{1}{b-a}}$

Q.437. If $\operatorname{cosec} \theta + \cot \theta = p$, then the value of $\frac{p^2 - 1}{p^2 + 1}$ is:

SSC CGL 12/12/2022 (1st Shift)

- (a) $\cos \theta$ (b) $\sin \theta$ (c) $\cot \theta$ (d) $\operatorname{cosec} \theta$

Q.438. If $\sin^2 \theta - 3 \sin \theta + 2 = 0$, then find the value of θ ($0^\circ \leq \theta \leq 90^\circ$).

SSC CGL 12/12/2022 (2nd Shift)

- (a) 45° (b) 0° (c) 60° (d) 90°

Q.439. If $\tan 3\theta \cdot \tan 7\theta = 1$, where 7θ is an acute angle, then find the value of $\cot 15\theta$.

SSC CGL 12/12/2022 (3rd Shift)

- (a) 1 (b) -1 (c) $-\sqrt{3}$ (d) $\sqrt{3}$

Q.440. If $2 \sin \theta + 2 \sin^2 \theta = 2$, then the value of $2 \cos^4 \theta + 2 \cos^2 \theta$ is:

SSC CGL 12/12/2022 (3rd Shift)

- (a) 4 (b) 2 (c) 1 (d) 0

Q.441. If $\cos \theta + \cos^2 \theta = 1$, find the value of $\sqrt{\sin^4 \theta + \cos^2 \theta}$

SSC CGL 12/12/2022 (4th Shift)

- (a) $\sqrt{2} \cos \theta$ (b) $2 \cos \theta$
(c) $\sqrt{2} \sin \theta$ (d) $2 \sin \theta$

Q.442. If $\sec^2 A + \tan^2 A = 3$, then what is the value of $\cot A$?

SSC CGL 13/12/2022 (1st Shift)

- (a) $\frac{1}{\sqrt{3}}$ (b) 0 (c) 1 (d) $\sqrt{3}$

Q.443. What is the value of $\operatorname{cosec} 15^\circ \sec 15^\circ$?

SSC CGL 13/12/2022 (2nd Shift)

- (a) 0.5 (b) 4 (c) 2 (d) 1

Q.444. $\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} = ?$

SSC CGL 13/12/2022 (2nd Shift)

- (a) $\tan A - \cot A$ (b) $\tan A + \cot A$
(c) $\sin A - \cos A$ (d) $\sin A + \cos A$

Q.445. If $\tan 40^\circ = a$, then find

$$\frac{\tan 320^\circ - \tan 310^\circ}{1 + \tan 320^\circ \tan 310^\circ}$$

SSC CGL 13/12/2022 (3rd Shift)

- (a) $\frac{1-a^2}{a}$ (b) $\frac{1+a^2}{2a}$
(c) $\frac{1-a^2}{2a}$ (d) $\frac{1+a^2}{a}$

Q.446. If $\tan(\alpha + \beta) = a$, $\tan(\alpha - \beta) = b$ then the value of $\tan 2\alpha$ is:

SSC CGL 13/12/2022 (4th Shift)

- (a) $\frac{a+b}{1-ab}$ (b) $\frac{a+b}{1+ab}$
(c) $\frac{a-b}{1+ab}$ (d) $\frac{a-b}{1-ab}$

SSC CPO 2022 Tier - 1

Q.447. If $x = \frac{2 \sin \theta}{(1 + \cos \theta + \sin \theta)}$ then the

value of $\frac{1 - \cos \theta + \sin \theta}{(1 + \sin \theta)}$ is:

SSC CPO 09/11/2022 (Morning)

- (a) $\frac{x}{(1+x)}$ (b) x (c) $\frac{1}{x}$ (d) $\frac{(1+x)}{x}$

Q.448. If $\left\{ \frac{\sec \theta - 1}{\sec \theta + 1} \right\}^n = \operatorname{cosec} \theta - \cot \theta$,

then $n = ?$

SSC CPO 09/11/2022 (Morning)

- (a) 1 (b) 0.5 (c) -1 (d) -0.5

Q.449. The value of $(2 \cos^2 \theta - 1)$

$$\left[\frac{1 + \tan \theta}{1 - \tan \theta} + \frac{1 - \tan \theta}{1 + \tan \theta} \right] \text{ is ;}$$

SSC CPO 09/11/2022 (Morning)

- (a) 2 (b) 0 (c) $\frac{\sqrt{3}}{2}$ (d) 1

Q.450. Simplify.

$$\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta}$$

SSC CPO 09/11/2022 (Afternoon)

- (a) $\cot \theta$ (b) $\cot 2\theta$ (c) $\tan \theta$ (d) $\tan 2\theta$

Q.451. The value of $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 1$ is:

SSC CPO 09/11/2022 (Afternoon)

- (a) 1 (b) 2 (c) 3 (d) 0

Q.452. Given that A and B are second quadrant angles, $\sin A = \frac{1}{3}$ and $\sin B = \frac{1}{5}$,

then find the value of $\cos(A - B)$.

SSC CPO 09/11/2022 (Evening)

- (a) $\frac{4\sqrt{3} + 1}{15}$ (b) $\frac{8\sqrt{3} - 1}{15}$
(c) $\frac{8\sqrt{3} + 1}{15}$ (d) $\frac{4\sqrt{3} - 1}{15}$

Q.453. What is the value of $\frac{\sin(A+B)}{\sin A \cos B}$?

SSC CPO 09/11/2022 (Evening)

- (a) $1 + \cot A \tan B$ (b) $1 + \tan A \cot B$
(c) $1 - \sin A \cos B$ (d) $1 - \cot A \tan B$

Q.454. The value of the expression $\cos^2 45^\circ + \cos^2 135^\circ + \cos^2 225^\circ + \cos^2 315^\circ$ is:

SSC CPO 10/11/2022 (Morning)

- (a) 2 (b) $\frac{1}{2}$ (c) $\frac{3}{2}$ (d) 1

Q.455. The value of $\frac{2 \cos^3 \theta - \cos \theta}{\sin \theta - 2 \sin^3 \theta}$ is:

SSC CPO 10/11/2022 (Morning)

- (a) $\sec \theta$ (b) $\sin \theta$ (c) $\cot \theta$ (d) $\tan \theta$

Q.456. If $\cos 48^\circ = \frac{m}{n}$, then $\sec 48^\circ - \cot 42^\circ$ is equal to:

SSC CPO 10/11/2022 (Morning)

- (a) $\frac{m - \sqrt{n^2 - m^2}}{m}$ (b) $\frac{m - \sqrt{n^2 - m^2}}{n}$
(c) $\frac{n - \sqrt{n^2 - m^2}}{n}$ (d) $\frac{n - \sqrt{n^2 - m^2}}{m}$

Q.457. Solve the following equation.

$$\sec^2 \theta \sqrt{1 - \sin^2 \theta} = \underline{\hspace{2cm}}$$

SSC CPO 10/11/2022 (Morning)

- (a) $\tan \theta$ (b) $\operatorname{cosec} \theta$ (c) $\sec \theta$ (d) 1

Q.458. If the value of $\operatorname{cosec} A + \cot A = m$, then the value of $\operatorname{cosec} A - \cot A$ is:

SSC CPO 10/11/2022 (Afternoon)

- (a) $\frac{1}{m}$ (b) m (c) $\sqrt{m}$ (d) m^2

Q.459. The value of $(\operatorname{cosec} A + \cot A)(1 - \cos A)$ is:

SSC CPO 10/11/2022 (Evening)

- (a) $\cos A$ (b) $\tan A$ (c) $\cot A$ (d) $\sin A$

Q.460. If $\cos A = \sin^2 A$, and $a \sin^{12} A + b \sin^{10} A + c \sin^8 A + \sin^6 A = 1$, then $a + b + c = ?$

SSC CPO 11/11/2022 (Morning)

- (a) 7 (b) 8 (c) 9 (d) 6

Q.461. If $4 \cos \theta + 3 \sin \theta = x$ and $4 \sin \theta - 3 \cos \theta = y$, find the value of $x^2 + y^2$.

SSC CPO 11/11/2022 (Morning)

- (a) 16 (b) 9 (c) 25 (d) 1

Q.462. Simplify. $\frac{\cos^4 \theta - \sin^4 \theta}{\sin^2 \theta}$

SSC CPO 11/11/2022 (Afternoon)

- (a) $1 - \tan^2 \theta$ (b) $\tan^2 \theta - 1$
(c) $\cot^2 \theta - 1$ (d) $1 - \cot^2 \theta$

Q.463. The value of $\frac{\sin A}{\cot A + \operatorname{cosec} A} -$

$$\frac{\sin A}{\cot A - \operatorname{cosec} A} + 1 \text{ is ?}$$

SSC CPO 11/11/2022 (Afternoon)

- (a) $\frac{1}{2}$ (b) 3 (c) 0 (d) 2

Q.464. If $1 + \sin^2 \theta - 3\sin \theta \cos \theta = 0$, then the value of $\cot \theta$ is:

SSC CPO 11/11/2022 (Evening)

- (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{3}$

Q.465. If $\cos^4 \alpha - \sin^4 \alpha = \frac{5}{6}$, then the value of $2\cos^2 \alpha - 1 =$ _____.

SSC CPO 11/11/2022 (Evening)

- (a) $\frac{11}{6}$ (b) $\frac{5}{6}$ (c) $\frac{6}{11}$ (d) $\frac{6}{5}$

SSC CGL 2021 Tier - 2

Q.466. Which of the following is equal to

$$\left[\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} \right] ?$$

SSC CGL Tier II (08/08/2022)

- (a) $\operatorname{cosec} \theta \cdot \sec \theta$ (b) $\sec \theta \cdot \tan \theta$
(c) $\operatorname{cosec} \theta \cdot \tan \theta$ (d) $\cot \theta \cdot \sec \theta$

Q.467. What is the value of $\frac{\cos 50^\circ}{\sin 40^\circ} + \frac{3\operatorname{cosec} 80^\circ}{\sec 10^\circ} - 2\cos 50^\circ \cdot \operatorname{cosec} 40^\circ$?

SSC CGL Tier II (08/08/2022)

- (a) 3 (b) 2 (c) 5 (d) 4

Q.468. If $\sqrt{3}\tan \theta = 3\sin \theta$, then what is the value of $\sin^2 \theta - \cos^2 \theta$?

SSC CGL Tier II (08/08/2022)

- (a) $\frac{1}{5}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{3}$

Q.469. Which of the following is equal to $\frac{1}{\tan \theta} + \tan \theta$?

SSC CGL Tier II (08/08/2022)

- (a) $\sec \theta \cdot \operatorname{cosec} \theta$ (b) 1
(c) $\frac{\operatorname{cosec} \theta}{\sec \theta}$ (d) $\tan^2 \theta$

SSC Selection Post (Phase - X)

Q.470. $[(\sin \theta \tan \theta + \cos \theta)^2 - 1]$ is equal to:

Graduate Level 02/08/2022 (Shift - 3)

- (a) $\tan^2 \theta$ (b) $\operatorname{cosec} \theta$ (c) $\sec^2 \theta$ (d) $\sec \theta$

Q.471. The value of

$$\sin^2 \frac{2\pi}{3} + \cos^2 \frac{5\pi}{6} - \tan^2 \frac{3\pi}{4}$$
 is:

Graduate Level 03/08/2022 (Shift - 3)

- (a) 2 (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) 4

Q.472. $\frac{2\sin \theta}{\sqrt{4 - 4\sin^2 \theta}} = ?$

Graduate Level 04/08/2022 (Shift - 4)

- (a) $\tan \theta$ (b) $2\tan \theta$ (c) $\cot \theta$ (d) $2\cot \theta$

Q.473. Simplify.

$$\frac{1}{\cos x} \sqrt{\frac{\cos(\pi + x) \cos(-x)}{\sin(\pi - x) \cos(-\frac{\pi}{2} + x)}}$$

Graduate Level 04/08/2022 (Shift - 4)

- (a) $\sec x$ (b) $\tan x$ (c) $\operatorname{cosec} x$ (d) $\cot x$

Q.474. If $\cos \theta = \sin(2\theta) \neq 0$, what is the value of $\cos^4 \theta + \sin^4 \theta + \cos^3 \theta + \sin^3 \theta + \sin^2 \theta + \cos^2 \theta + \sin \theta + \cos \theta$?

Graduate Level 05/08/2022 (Shift - 3)

- (a) $\frac{8 + 7\sqrt{3}}{18}$ (b) $\frac{18 + 7\sqrt{3}}{8}$
(c) $\frac{18 + 8\sqrt{3}}{7}$ (d) $\frac{7 + 18\sqrt{3}}{8}$

SSC CHSL 2021 Tier - 1

Q.475. If $\frac{x - x\operatorname{cosec}^2 30^\circ}{1 + \operatorname{cosec}^2 30^\circ} = \cos^2 60^\circ$

+ $4\cot^2 45^\circ - \sec^2 60^\circ$, then the value of x is _____.

SSC CHSL 25/05/2022 (Morning)

- (a) $\frac{5}{12}$ (b) $-\frac{5}{12}$ (c) $\frac{1}{5}$ (d) $-\frac{1}{12}$

Q.476. If $\sec \theta - \cos \theta = 14$ and $14\sec \theta = x$, then the value of x is _____.

SSC CHSL 25/05/2022 (Afternoon)

- (a) $\tan^2 \theta$ (b) $\sec^2 \theta$ (c) $2\sec \theta$ (d) $2\tan \theta$

Q.477. The value of $(1 + \tan 10^\circ)(1 + \tan 35^\circ)$ is:

SSC CHSL 25/05/2022 (Evening)

- (a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) 1 (d) 2

Q.478. The value of $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$ is:

SSC CHSL 27/05/2022 (Morning)

- (a) $\tan 3\theta$ (b) $\cot 3\theta$ (c) $\tan \theta$ (d) $\cot \theta$

Q.479. If $\sin x - 3\cos x = \sqrt{3}\cos x$, then the value of $\cot x$ is _____.

SSC CHSL 31/05/2022 (Morning)

- (a) $3 - \sqrt{3}$ (b) $3 + \sqrt{3}$ (c) $\sqrt{3}$ (d) $\frac{3 - \sqrt{3}}{6}$

Q.480. What is the value of

$$(4\sin^3 x - 3\sin x + \sin 3x) ?$$

SSC CHSL 31/05/2022 (Afternoon)

- (a) 1 (b) 0 (c) 4 (d) 3

Q.481. What is the simplified value of

$$\cos^2(90^\circ - \theta) - \left[\frac{\cos(90^\circ - \theta)\cos \theta}{\cot \theta} \right]$$

SSC CHSL 31/05/2022 (Evening)

- (a) 4 (b) 2 (c) 0 (d) 1

Q.482. The value of $\sin 2\theta (\tan \theta + \cot \theta)$ is:

SSC CHSL 01/06/2022 (Morning)

- (a) $\frac{2}{3}$ (b) 1 (c) $\frac{1}{2}$ (d) 2

Q.483. Using $2\cos A \cos B = \cos(A + B) + \cos(A - B)$, find the value of $\cos 75^\circ \cos 15^\circ$.

SSC CHSL 01/06/2022 (Afternoon)

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{3}}{4}$

Q.484. If $1 + \tan \theta = \sqrt{3}$, then $\sqrt{3}\cot \theta - 1 =$ _____

SSC CHSL 03/06/2022 (Afternoon)

- (a) $\frac{\sqrt{3} + 1}{2}$ (b) $\frac{\sqrt{3} - 1}{2}$
(c) $\frac{2\sqrt{3} - 1}{2}$ (d) $\frac{2\sqrt{3} + 1}{2}$

Q.485. If $\operatorname{cosec}^2 \theta + \cot^2 \theta = \frac{5}{3}$, then

what is the value of $\cot 2\theta$

SSC CHSL 07/06/2022 (Morning)

- (a) $\frac{1}{\sqrt{3}}$ (b) $-\frac{1}{\sqrt{3}}$ (c) $\frac{2}{\sqrt{3}}$ (d) $-\frac{2}{\sqrt{3}}$

Q.486. $(\operatorname{cosec} \theta - \cot \theta)^2 = ?$, $0^\circ < \theta < 90^\circ$

SSC CHSL 09/06/2022 (Evening)

- (a) $\frac{1 - \sin \theta}{1 + \cos \theta}$ (b) $\frac{1 - \cos \theta}{1 + \cos \theta}$
(c) $\frac{1 + \cos \theta}{1 - \cos \theta}$ (d) $\frac{1 + \cos \theta}{1 - \sin \theta}$

Q.487. Using the formula $\tan \frac{x}{2} =$

$$\frac{1 - \cos x}{\sin x}, \text{ find the value of } \tan 22.5^\circ.$$

SSC CHSL 10/06/2022 (Morning)

- (a) $\sqrt{2} - 1$ (b) $\sqrt{2} + 1$
(c) $\frac{\sqrt{3} + \sqrt{2}}{2}$ (d) $\frac{\sqrt{3} - \sqrt{2}}{2}$

SSC CGL 2021 Tier - 1

Q.488. If $4\sin^2 \theta = 3(1 + \cos \theta)$, $0^\circ < \theta < 90^\circ$, then what is the value of $(2\tan \theta + 4\sin \theta - \sec \theta)$?

SSC CGL 11/04/2022 (Morning)

- (a) $3\sqrt{15} - 4$ (b) $15\sqrt{3} - 4$
(c) $15\sqrt{3} + 3$ (d) $4\sqrt{15} - 3$

Q.489. If $3\sec^2 \theta + \tan \theta - 7 = 0$, $0^\circ < \theta < 90^\circ$, then what is the value of

$$\left(\frac{2\sin \theta + 3\cos \theta}{\operatorname{cosec} \theta + \sec \theta} \right) ?$$

SSC CGL 12/04/2022 (Morning)

- (a) 10 (b) $\frac{5}{2}$ (c) $\frac{5}{4}$ (d) $4\sqrt{2}$

Q.490. The value of $1 + \sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$,

if $0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL 12/04/2022 (Afternoon)

- (a) $1 - \sec \theta + \tan \theta$ (b) $1 - \sec \theta - \tan \theta$
(c) $1 + \sec \theta - \tan \theta$ (d) $1 + \sec \theta + \tan \theta$

Q.491. The value of $\left(\frac{1 - \cot \theta}{1 - \tan \theta} \right)^2 - 1$

when $0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL 12/04/2022 (Afternoon)

- (a) $\cos^2 \theta - 1$ (b) $\sec^2 \theta + 1$
(c) $\cot^2 \theta - 1$ (d) $\sin^2 \theta - 1$

Q.492. The expression $(\cos^6 \theta + \sin^6 \theta - 1)(\tan^2 \theta + \cot^2 \theta + 2) + 3$ is equal to:

SSC CGL 13/04/2022 (Afternoon)

- (a) 0 (b) 1 (c) 2 (d) -1

Q.493. If $\sin^2 \theta - \cos^2 \theta - 3 \sin \theta + 2 = 0$, $0^\circ < \theta < 90^\circ$, then what is the value of $1 + \sec \theta + \tan \theta$?

SSC CGL 13/04/2022 (Evening)

- (a) $-1 + \sqrt{3}$ (b) $-1 - \sqrt{3}$
(c) $1 + \sqrt{3}$ (d) $1 - \sqrt{3}$

Q.494. $(\sec \theta - \tan \theta)^2 (1 + \sin \theta)^2 \div \cos^2 \theta =$

SSC CGL 13/04/2022 (Evening)

- (a) $\cos^2 \theta$ (b) 1 (c) $\cot^2 \theta$ (d) -1

Q.495. If $2 \sin^2 \theta + 3 \cos \theta = 3$, $0^\circ < \theta < 90^\circ$, then the value of $(\sec^2 \theta + \cot^2 \theta)$ is

SSC CGL 18/04/2022 (Evening)

- (a) $3\frac{2}{3}$ (b) $3\frac{1}{3}$ (c) $4\frac{1}{3}$ (d) $4\frac{2}{3}$

Q.496. Simplify the following expression:

$$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} - \sin A$$

SSC CGL 19/04/2022 (Evening)

- (a) $1 + \cos A$ (b) $(1 + \sin A) \cos A$
(c) $1 + \sin A$ (d) $\cos A$

Q.497. Simplify the following expression:

$$\operatorname{cosec}^4 A (1 - \cos^4 A) - 2 \cot^2 A - 1$$

SSC CGL 20/04/2022 (Morning)

- (a) $\sin^2 A$ (b) $\operatorname{cosec}^2 A$ (c) 1 (d) 0

Q.498. If $6 \tan A (\tan A + 1) = 5 - \tan A$,

Given that $0 < A < \frac{\pi}{2}$ What is the value of

$(\sin A + \cos A)$?

SSC CGL 20/04/2022 (Morning)

- (a) $3\sqrt{5}$ (b) $\frac{5}{\sqrt{3}}$ (c) $5\sqrt{3}$ (d) $\frac{3}{\sqrt{5}}$

Q.499. The value of $\left(\frac{1 - \cot \theta}{1 - \tan \theta}\right)^2 + 1$, if

$0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL 20/04/2022 (Evening)

- (a) $\operatorname{cosec}^2 \theta$ (b) $\sin^2 \theta$ (c) $\cos^2 \theta$ (d) $\sec^2 \theta$

Q.500. If $\frac{\sin^2 \theta - 3 \sin \theta + 2}{\cos^2 \theta} = 1$, where 0°

$< \theta < 90^\circ$, then what is the value of $(\cos^2 \theta - \sin^3 \theta + \operatorname{cosec} 2\theta)$?

SSC CGL 21/04/2022 (Afternoon)

- (a) $\frac{-3 - 4\sqrt{3}}{6}$ (b) $\frac{-3 + 4\sqrt{3}}{6}$
(c) $\frac{3 + 4\sqrt{3}}{6}$ (d) $\frac{3 - 4\sqrt{3}}{6}$

SSC CGL 2020 Tier - 2

Q.501. $\frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}{\tan \theta + \cot \theta}$,

$0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL Tier II (29/01/2022)

- (a) $\tan \theta$ (b) $\sec \theta$ (c) $\operatorname{cosec} \theta$ (d) $\cot \theta$

Q.502. If $1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta = \frac{a}{b}$,

$0^\circ < \theta < 90^\circ$, then $\frac{a+b}{a-b} = ?$

SSC CGL Tier II (29/01/2022)

- (a) $\sec \theta$ (b) $\sin \theta$ (c) $\cos \theta$ (d) $\operatorname{cosec} \theta$

Q.503.

$$\frac{(1 + \sec \theta \operatorname{cosec} \theta)^2 (\sec \theta - \tan \theta)^2 (1 + \sin \theta)}{(\sin \theta + \sec \theta)^2 + (\cos \theta + \operatorname{cosec} \theta)^2}$$

, $0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL Tier II (29/01/2022)

- (a) $\cos \theta$ (b) $1 - \cos \theta$ (c) $\sin \theta$ (d) $1 - \sin \theta$

Q.504. $\left(\frac{\tan^3 \theta}{\sec^2 \theta} + \frac{\cot^3 \theta}{\operatorname{cosec}^2 \theta} + 2 \sin \theta \cos \theta\right)$

$\div (1 + \operatorname{cosec}^2 \theta + \tan^2 \theta)$, $0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL Tier II (29/01/2022)

- (a) $\operatorname{cosec} \theta \sec \theta$ (b) $\operatorname{cosec} \theta$
(c) $\sin \theta \cos \theta$ (d) $\sec \theta$

Q.505. The expression

$$\frac{\tan^6 \theta - \sec^6 \theta + 3 \sec^2 \theta \tan^2 \theta}{\tan^2 \theta + \cot^2 \theta + 2}, 0^\circ < \theta < 90^\circ$$

, is equal to:

SSC CGL Tier II (03/02/2022)

- (a) $\sec^2 \theta \operatorname{cosec}^2 \theta$ (b) $-\sec^2 \theta \operatorname{cosec}^2 \theta$
(c) $\cos^2 \theta \sin^2 \theta$ (d) $-\cos^2 \theta \sin^2 \theta$

Q.506. The expression

$$\frac{\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + \cot \theta + 1)(\operatorname{cosec} \theta - \cot \theta + 1) - 2}$$

, $0^\circ < \theta < 90^\circ$, is equal to

SSC CGL Tier II (03/02/2022)

- (a) $\frac{1}{2} \sin \theta$ (b) $2 \sin \theta$ (c) $\sec \theta$ (d) $2 \operatorname{cosec} \theta$

Q.507. The expression

$$\frac{(1 - \sin \theta + \cos \theta)^2 (1 - \cos \theta) \sec^3 \theta \operatorname{cosec}^2 \theta}{(\sec \theta - \tan \theta)(\tan \theta + \cot \theta)}$$

, $(0^\circ < \theta < 90^\circ)$, is equal to

SSC CGL Tier II (03/02/2022)

- (a) $\sin \theta$ (b) $2 \cos \theta$ (c) $\cot \theta$ (d) $2 \tan \theta$

Q.508. If $\frac{\sin^2 \theta}{\cos^2 \theta - 3 \cos \theta + 2} = 1$, θ lies in

the first quadrant, then the value of

$$\frac{\tan^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2}}{\tan \theta + \sin \theta}$$
 is:

SSC CGL Tier II (03/02/2022)

- (a) $\frac{2\sqrt{3}}{27}$ (b) $\frac{5\sqrt{3}}{27}$ (c) $\frac{2\sqrt{3}}{9}$ (d) $\frac{7\sqrt{3}}{54}$

Q.509. The expression

$$\frac{(1 - 2 \sin^2 \theta \cos^2 \theta)(\cot \theta + 1) \cos \theta}{(\sin^4 \theta + \cos^4 \theta)(1 + \tan \theta) \operatorname{cosec} \theta}$$

$0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL Tier II (03/02/2022)

- (a) $\cos^2 \theta$ (b) $-\sin^2 \theta$ (c) $\sec^2 \theta$ (d) $-\sec^2 \theta$

SSC MTS 2020 Tier - 1

Q.510. What is the average of the sixty terms given below?

$\cos^2 x, \cos^2 2x, \cos^2 3x, \dots, \cos^2 30x, \sin^2 x, \sin^2 2x, \sin^2 3x, \dots, \sin^2 30x$

SSC MTS 12/10/2021 (Afternoon)

- (a) $\cos^2 x$ (b) 0.5 (c) 1 (d) $\cos^2 x \sin^2 x$

SSC CGL 2020 Tier - 1

Q.511. $\frac{\operatorname{cosec} \theta}{\operatorname{cosec} \theta - 1} + \frac{\operatorname{cosec} \theta}{\operatorname{cosec} \theta + 1} - \tan^2 \theta$,

$0^\circ < \theta < 90^\circ$

SSC CGL 13/08/2021 (Morning)

- (a) $2 \sec^2 \theta$ (b) $\sec^2 \theta + 1$
(c) $\sec^2 \theta$ (d) $1 - \tan^2 \theta$

Q.512. If $2 \cos^2 \theta = 3 \sin \theta$, $0^\circ < \theta < 90^\circ$, then the value of $(\sec^2 \theta - \tan^2 \theta + \cos^2 \theta)$ is:

SSC CGL 13/08/2021 (Afternoon)

- (a) $\frac{7}{4}$ (b) $\frac{5}{4}$ (c) $\frac{9}{4}$ (d) $\frac{3}{4}$

Q.513. The value of $\frac{\tan(45^\circ - \alpha)}{\cot(45^\circ + \alpha)} -$

$$\frac{(\cos 19^\circ + \sin 71^\circ)(\sec 19^\circ + \operatorname{cosec} 71^\circ)}{\tan 12^\circ \tan 24^\circ \tan 66^\circ \tan 78^\circ}$$

SSC CGL 13/08/2021 (Evening)

- (a) -3 (b) 0 (c) -2 (d) 2

Q.514. $1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta$, $0^\circ < \theta < 90^\circ$, is equal to:

SSC CGL 16/08/2021 (Morning)

- (a) $\frac{1 - \sin \theta}{1 + \sin \theta}$ (b) $\frac{1 + \cos \theta}{1 - \cos \theta}$
(c) $\frac{1 - \cos \theta}{1 + \cos \theta}$ (d) $\frac{1 + \sin \theta}{1 - \sin \theta}$

Q.515. If $\sin\left(\frac{2A+B}{2}\right) = \cos\left(\frac{2A-B}{2}\right) =$

$= \frac{\sqrt{3}}{2}$, $0^\circ < \frac{2A+B}{2}$ and $\frac{2A-B}{2} < 90^\circ$

then find the value of $\sin[3(A-B)]$.

SSC CGL 16/08/2021 (Evening)

- (a) 1 (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}$

Q.516. $(\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta})$

$$\left(\frac{\sin \theta (1 + \cos \theta)}{1 + \cos \theta - \sin^2 \theta}\right), 0^\circ < \theta < 90^\circ$$
 is

equal to:

SSC CGL 17/08/2021 (Morning)

- (a) $\cot \theta$ (b) $\operatorname{cosec}^2 \theta$ (c) $\sec^2 \theta$ (d) $\tan \theta$

Q.517. For $0^\circ < \theta < 90^\circ$,

$$\frac{1}{\cos \theta} + \frac{1}{\tan \theta - \sec \theta}$$
 is equal to:

SSC CGL 17/08/2021 (Afternoon)

- (a) $-\sec \theta$ (b) $\tan \theta$ (c) $\sec \theta$ (d) $-\tan \theta$

Q.518. If $4 \sin^2(2x - 10^\circ) = 3$, $0 \leq (2x - 10) \leq 90$, then find the value of

$$\frac{\sin^4(x - 5^\circ) + \cos^4(x - 5^\circ)}{1 - 2 \sin^2(3x - 15^\circ) \cos^2(3x - 15^\circ)}$$

SSC CGL 17/08/2021 (Evening)

- (a) 1 (b) $\frac{5}{8}$ (c) $-\frac{5}{8}$ (d) -1

Q.519. The value of

$$\frac{\sqrt{2}\tan(60^\circ - \theta)\tan(30^\circ + \theta)}{\sin^2(45^\circ + \theta) + \sin^2(45^\circ - \theta)}$$
 is:

SSC CGL 17/08/2021 (Evening)

- (a) $\frac{1}{\sqrt{2}}$ (b) 1 (c) 2 (d) $\sqrt{2}$

Q.520. In ΔPQR , $\angle Q = 90^\circ$. If $\tan R = \frac{1}{3}$,

then what is the value of

$$\frac{\sec P(\cos R + \sin P)}{\operatorname{cosec} R(\sin R - \operatorname{cosec} P)}?$$

SSC CGL 18/08/2021 (Afternoon)

- (a) $-\frac{2}{7}$ (b) $\frac{18}{7}$ (c) $\frac{2}{7}$ (d) $-\frac{18}{7}$

Q.521. Find the value of

$$(\sec \theta + \tan \theta)^2 + \frac{1 + \operatorname{cosec} \theta}{1 - \operatorname{cosec} \theta}$$

where $0^\circ < \theta < 90^\circ$

SSC CGL 18/08/2021 (Evening)

- (a) 0 (b) -2 (c) 1 (d) 2

Q.522. If $\sin^6 \theta + \cos^6 \theta = \frac{1}{3}$, $0^\circ < \theta$ < 90° , then what is the value of $\sin \theta \cdot \cos \theta$

SSC CGL 18/08/2021 (Evening)

- (a) $\frac{\sqrt{6}}{6}$ (b) $\frac{\sqrt{2}}{3}$ (c) $\frac{\sqrt{2}}{\sqrt{3}}$ (d) $\frac{2}{9}$

Q.523. If $2\cos^2 \theta - 5\cos \theta + 2 = 0$, $0^\circ < \theta < 90^\circ$, then the value of $(\sec \theta + \tan \theta)$ is:

SSC CGL 20/08/2021 (Afternoon)

- (a) $1 - \sqrt{3}$ (b) $2 - \sqrt{3}$
(c) $1 + \sqrt{3}$ (d) $2 + \sqrt{3}$

Q.524. The value of $\sec^4 \theta (1 - \sin^4 \theta) - 2\tan^2 \theta$ is:

SSC CGL 20/08/2021 (Evening)

- (a) 1 (b) 0 (c) -1 (d) $-\frac{1}{2}$

Q.525. If $\sin^2 \theta - \cos^2 \theta - 3\sin \theta + 2 = 0$, $0^\circ < \theta < 90^\circ$, then what is the value of

$$\frac{1}{\sqrt{\sec \theta - \tan \theta}}$$
 is:

SSC CGL 23/08/2021 (Morning)

- (a) $\sqrt[4]{3}$ (b) $\sqrt[2]{2}$ (c) $\sqrt[2]{3}$ (d) $\sqrt[4]{2}$

Q.526. If $0^\circ < \theta < 90^\circ$,

$$\sqrt{\frac{\sec^2 \theta + \operatorname{cosec}^2 \theta}{\tan^2 \theta - \sin^2 \theta}}$$
 is equal to:

SSC CGL 23/08/2021 (Afternoon)

- (a) $\sec^3 \theta$ (b) $\sin^2 \theta$ (c) $\operatorname{cosec}^3 \theta$ (d) $\sec^2 \theta$

Q.527. If $\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} = 4 \sec \theta$, $0^\circ < \theta < 90^\circ$, then the value of $\cot \theta + \operatorname{cosec} \theta$ is:

SSC CGL 23/08/2021 (Evening)

- (a) $\frac{5\sqrt{3}}{3}$ (b) $3\sqrt{3}$ (c) $\sqrt{3}$ (d) $\frac{4\sqrt{3}}{3}$

Q.528. If $\sin^2 \theta = 2 \sin \theta - 1$, $0^\circ \leq \theta \leq 90^\circ$,then find the value of $\frac{1 + \operatorname{cosec} \theta}{1 - \cos \theta}$.

SSC CGL 24/08/2021 (Afternoon)

- (a) -1 (b) 1 (c) 2 (d) -2

Q.529. If $\sec 31^\circ = x$, then $\sin^2 59^\circ +$

$$\frac{1}{\sec^2 31^\circ} - \frac{1}{\sin^2 59^\circ \operatorname{cosec}^2 59^\circ}$$
 is equal to:

SSC CGL 24/08/2021 (Evening)

- (a) $\frac{x^2 - 2}{x}$ (b) $\frac{2 - x^2}{x^2}$
(c) $\frac{x^2 - 2}{x^2}$ (d) $\frac{2 - x^2}{x}$

SSC CHSL 2020 Tier - 1

Q.530. If $2 \cos \theta + \sec \theta - 2\sqrt{2} = 0$, where θ is an acute angle. Find the value of θ .

SSC CHSL 12/04/2021 (Evening)

- (a) 60° (b) 30° (c) 15° (d) 45°

Q.531. If $\cot \theta = \sqrt{2} + 1$, then $\operatorname{cosec} \theta \sec \theta = ?$

SSC CHSL 13/04/2021 (Morning)

- (a) $\frac{\sqrt{2}}{2}$ (b) $4\sqrt{2}$ (c) $\frac{\sqrt{2}}{4}$ (d) $2\sqrt{2}$

Q.532. If $0^\circ < \theta < 90^\circ$, then

$$\frac{(1 - \sin \theta)(\sec \theta + \tan \theta) \tan \theta}{(\sec \theta + \tan \theta + 1)(\cot \theta - \operatorname{cosec} \theta + 1)} = ?$$

SSC CHSL 13/04/2021 (Afternoon)

- (a) $\sin \theta$ (b) $\frac{\sin \theta}{2}$ (c) $\cos \theta$ (d) $\frac{\cos \theta}{2}$

Q.533. If $\frac{3\sqrt{3} \sec \theta + 4 \tan \theta}{3 \tan \theta + \sqrt{3} \sec \theta} = 2$, $0^\circ < \theta < 90^\circ$ then the value of $\cos \theta$ will be:

SSC CHSL 13/04/2021 (Evening)

- (a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

Q.534. If $Y = \tan 35^\circ$, then the value of $(2 \tan 55^\circ + \cot 55^\circ)$ is:

SSC CHSL 15/04/2021 (Evening)

- (a) $\frac{2 + Y^2}{Y}$ (b) $\frac{2 - Y}{Y^2}$ (c) $\frac{2}{Y^2}$ (d) $\frac{2 - Y^2}{Y}$

Q.535. If $\tan \theta + \cot \theta = 4$, then the ratio of $3(\tan^2 \theta + \cot^2 \theta)$ to $(2 \operatorname{cosec}^2 \theta \sec^2 \theta - 4)$ will be:

SSC CHSL 15/04/2021 (Evening)

- (a) 4 : 3 (b) 3 : 4 (c) 5 : 4 (d) 3 : 2

Q.536. If $3(\sec^2 \theta + \tan^2 \theta) = 5$, $0^\circ < \theta < 90^\circ$, then the value of $\operatorname{cosec} \theta$ is:

SSC CHSL 16/04/2021 (Afternoon)

- (a) $\frac{1}{\sqrt{3}}$ (b) $\frac{2\sqrt{3}}{3}$ (c) 2 (d) $\frac{1}{2}$

Q.537. If $(\cos^2 \theta - 1)(2 \sec^2 \theta) + \sec^2 \theta$ + $2 \tan^2 \theta = 2$, $0^\circ < \theta < 90^\circ$, then thevalue of $\frac{(\sec \theta + \sin \theta)}{(\operatorname{cosec} \theta - \cos \theta)}$ will be:

SSC CHSL 16/04/2021 (Afternoon)

- (a) -1 (b) -3 (c) 3 (d) 2

Q.538. $\frac{(1 + \cos \theta)(\operatorname{cosec} \theta - \cot \theta) \sec \theta}{\sin \theta (1 - \sin \theta)(\sec \theta + \tan \theta)} =$

SSC CHSL 19/04/2021 (Morning)

- (a) $\sec^2 \theta$ (b) $\sin^2 \theta$ (c) $\cos^2 \theta$ (d) $\operatorname{Cosec}^2 \theta$

Q.539. If $\sin \theta (2 \sin \theta + 3) = 2$, $0^\circ < \theta < 90^\circ$, then what is the value of $(\sec^2 \theta + \cot^2 \theta - \cos^2 \theta)$?

SSC CHSL 19/04/2021 (Morning)

- (a) $\frac{13}{3}$ (b) $\frac{31}{12}$ (c) $\frac{7}{2}$ (d) $\frac{43}{12}$

Q.540. If $\sin \theta = \frac{2\sqrt{ab}}{a+b}$, $a > b > 0$, then thevalue of $\frac{\cos \theta + 1}{\cos \theta - 1}$ will be:

SSC CHSL 19/04/2021 (Afternoon)

- (a) $-\frac{b}{a}$ (b) $-\frac{a}{b}$ (c) $\frac{a}{b}$ (d) $\frac{b}{a}$

Q.541. The value of

$$\frac{3\cos^2 27^\circ - 5 + 3\cos^2 63^\circ}{\tan^2 32^\circ + 4 - \operatorname{cosec}^2 58^\circ} + \sin 35^\circ \cos 55^\circ$$

+ $\cos 35^\circ \sin 55^\circ$ is:

SSC CHSL 19/04/2021 (Afternoon)

- (a) $-\frac{1}{4}$ (b) $-\frac{1}{3}$ (c) $\frac{1}{3}$ (d) $1\frac{2}{3}$

Q.542. $2\cos^2 \theta = 3(1 - \sin \theta)$, $0^\circ < \theta < 90^\circ$, then what is the value of $(\tan 2\theta + \operatorname{cosec} 3\theta - \sec 2\theta)$?

SSC CHSL 04/08/2021 (Evening)

- (a) $\sqrt{3} + 1$ (b) $\sqrt{3} - 1$
(c) $\frac{1 + \sqrt{3}}{\sqrt{3}}$ (d) $\frac{1 - \sqrt{3}}{\sqrt{3}}$

Q.543. Find the value of θ , if $\sec^2 \theta + (1 - \sqrt{3})\tan \theta - (1 + \sqrt{3}) = 0$, where θ is an acute angle.

SSC CHSL 05/08/2021 (Evening)

- (a) 60° (b) 30° (c) 45° (d) 15°

Q.544. If $\operatorname{cosec}^2 \theta (\cos \theta - 1)(1 + \cos \theta) = k$, then what is the value of k ?

SSC CHSL 06/08/2021 (Morning)

- (a) 1 (b) -1 (c) $\frac{1}{2}$ (d) 0

Q.545. If $\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta} = \frac{k+1}{1-k}$, $k \neq 1$, then k is equal to:

SSC CHSL 09/08/2021 (Morning)

- (a) $\sin \theta$ (b) $\operatorname{cosec} \theta$ (c) $\cos \theta$ (d) $\sec \theta$

Q.546. If $\cos^2 \theta - \sin^2 \theta - 3\cos \theta + 2 = 0$, $0^\circ < \theta < 90^\circ$, then what will be the value of $\sec \theta - \cos \theta$?

SSC CHSL 09/08/2021 (Morning)

- (a) $\frac{4}{3}$ (b) $\frac{1}{2}$ (c) $\frac{3}{2}$ (d) $\frac{2}{3}$

Q.547. If $\frac{\operatorname{cosec}^2 \theta}{\operatorname{cosec}^2 \theta - \cot^2 \theta} = \frac{13}{4}$,

$0^\circ < \theta < 90^\circ$, then the value of

$\frac{52\cos^2 \theta - 9\tan^2 \theta}{18\sec^2 \theta + 8\cot^2 \theta}$ will be:

SSC CHSL 09/08/2021 (Afternoon)

(a) $\frac{4}{11}$ (b) $\frac{8}{11}$ (c) $\frac{8}{13}$ (d) $\frac{5}{11}$

Q.548. Evaluate the following expression

$\frac{3(\cot^2 46^\circ - \sec^2 44^\circ)}{2(\sin^2 28^\circ + \sin^2 62^\circ)} + \frac{2\cos^2 60^\circ \tan^2 33^\circ \tan^2 57^\circ}{\sec^2(90^\circ - \theta) - \cot^2 \theta}$

SSC CHSL 10/08/2021 (Morning)

(a) -1 (b) 1 (c) -2 (d) 2

Q.549. For $\theta: 0^\circ < \theta < 90^\circ$, $3\sec \theta + 4\cos \theta = 4\sqrt{3}$, find the value of $(1 - \sin \theta + \cos \theta)$

SSC CHSL 10/08/2021 (Morning)

(a) $\frac{1+2\sqrt{3}}{2}$ (b) $\frac{1+\sqrt{3}}{2}$

(c) $\frac{1-\sqrt{3}}{2}$ (d) $\frac{1-2\sqrt{3}}{2}$

Q.550. If $\frac{5\cot \theta + \sqrt{3}\operatorname{cosec} \theta}{2\sqrt{3}\operatorname{cosec} \theta + 3\cot \theta} = 1$

, $0 < \theta < 90^\circ$, then the value of

$\frac{7}{2}\cot^2 \theta - \frac{3}{4}\operatorname{cosec}^2 \theta$ will be:

SSC CHSL 10/08/2021 (Afternoon)

(a) 7 (b) 2 (c) 3 (d) 5

Q.551. If $(1 + \cot^2 \theta) + (1 + (\cot^2 \theta)^{-1})$ is equal to k, then $\sqrt{k} = ?$

SSC CHSL 10/08/2021 (Evening)

(a) $\sin \theta \sec \theta$ (b) $\operatorname{cosec} \theta \cos \theta$
(c) $\operatorname{cosec} \theta \sec \theta$ (d) $\sin \theta \cos \theta$

Q.552. The value of $\frac{\sec \theta \operatorname{cosec} \theta}{2 + \tan^2 \theta + \cot^2 \theta}$ is

equal to:

SSC CHSL 11/08/2021 (Morning)

(a) $\sec \theta \operatorname{cosec} \theta$ (b) $\sec \theta \sin \theta$
(c) $\sin \theta \cos \theta$ (d) $\cos \theta \operatorname{cosec} \theta$

Q.553. If $\cot A + \operatorname{cosec} A = 2$ and A is an acute angle, then the value of

$\frac{9\tan A + 16\operatorname{cosec} A}{5\sin A + 3\tan A}$ is:

SSC CHSL 11/08/2021 (Afternoon)

(a) 3 (b) 6 (c) 4 (d) 8

Q.554. If $\cot^2 \theta + \cot^4 \theta = 2$, then the value of $2\sin^4 \theta + \sin^2 \theta$ is:

SSC CHSL 11/08/2021 (Evening)

(a) 3 (b) 5 (c) 1 (d) 2

Q.555. If $\sin \alpha + \operatorname{cosec} \alpha = \tan \frac{\pi}{3}$, then

the value of $(\sin^3 \alpha + \operatorname{cosec}^3 \alpha)$ is equal to:

SSC CHSL 11/08/2021 (Evening)

(a) $\frac{1}{2}$ (b) 1 (c) 0 (d) $\frac{3}{2}$

SSC CPO 2020 Tier - 1

Q.556. value of $\frac{\sin^2 52^\circ + 2 + \sin^2 38^\circ}{4\cos^2 43^\circ - 5 + 4\cos^2 47^\circ}$ is

SSC CPO 24/11/2020 (Evening)

(a) 3 (b) $\frac{1}{3}$ (c) -3 (d) $-\frac{1}{3}$

Q.557. If $4(\operatorname{cosec}^2 57^\circ - \tan^2 33^\circ) - \cos 90^\circ + y \times \tan^2 66^\circ \times \tan^2 24^\circ = \frac{y}{2}$

then the value of y is:

SSC CPO 25/11/2020 (Morning)

(a) 4 (b) 8 (c) -4 (d) -8

Answer Key :-

1.(b)	2.(d)	3.(b)	4.(d)
5.(b)	6.(b)	7.(a)	8.(d)
9.(a)	10.(b)	11.(d)	12.(d)
13.(c)	14.(c)	15.(c)	16.(c)
17.(a)	18.(d)	19.(a)	20.(a)
21.(d)	22.(a)	23.(a)	24.(b)
25.(a)	26.(a)	27.(c)	28.(d)
29.(a)	30.(a)	31.(b)	32.(a)
33.(b)	34.(a)	35.(d)	36.(c)
37.(b)	38.(a)	39.(a)	40.(a)
41.(a)	42.(d)	43.(a)	44.(d)
45.(c)	46.(d)	47.(c)	48.(d)
49.(d)	50.(a)	51.(c)	52.(d)
53.(a)	54.(a)	55.(b)	56.(d)
57.(c)	58.(c)	59.(d)	60.(c)
61.(d)	62.(d)	63.(c)	64.(c)
65.(b)	66.(d)	67.(c)	68.(b)
69.(b)	70.(b)	71.(a)	72.(a)
73.(b)	74.(a)	75.(a)	76.(b)
77.(b)	78.(b)	79.(c)	80.(b)
81.(c)	82.(b)	83.(b)	84.(c)
85.(c)	86.(d)	87.(d)	88.(d)
89.(c)	90.(c)	91.(a)	92.(c)
93.(a)	94.(b)	95.(d)	96.(c)
97.(d)	98.(a)	99.(d)	100.(a)
101.(a)	102.(d)	103.(a)	104.(d)
105.(a)	106.(d)	107.(c)	108.(a)
109.(c)	110.(b)	111.(d)	112.(c)
113.(c)	114.(a)	115.(b)	116.(b)
117.(a)	118.(d)	119.(b)	120.(b)
121.(b)	122.(d)	123.(a)	124.(a)
125.(c)	126.(b)	127.(b)	128.(a)

129.(a)	130.(a)	131.(b)	132.(d)
133.(b)	134.(b)	135.(a)	136.(c)
137.(d)	138.(d)	139.(d)	140.(c)
141.(c)	142.(d)	143.(a)	144.(c)
145.(b)	146.(d)	147.(c)	148.(d)
149.(c)	150.(b)	151.(b)	152.(c)
153.(b)	154.(b)	155.(a)	156.(b)
157.(d)	158.(c)	159.(c)	160.(b)
161.(c)	162.(c)	163.(a)	164.(a)
165.(c)	166.(c)	167.(c)	168.(c)
169.(b)	170.(a)	171.(a)	172.(a)
173.(b)	174.(b)	175.(b)	176.(b)
177.(a)	178.(b)	179.(c)	180.(a)
181.(a)	182.(d)	183.(a)	184.(d)
185.(c)	186.(c)	187.(d)	188.(c)
189.(b)	190.(c)	191.(d)	192.(a)
193.(c)	194.(a)	195.(b)	196.(c)
197.(d)	198.(d)	199.(c)	200.(c)
201.(d)	202.(d)	203.(b)	204.(a)
205.(b)	206.(a)	207.(a)	208.(c)
209.(c)	210.(c)	211.(d)	212.(d)
213.(a)	214.(b)	215.(b)	216.(d)
217.(b)	218.(d)	219.(c)	220.(c)
221.(d)	222.(c)	223.(a)	224.(d)
225.(c)	226.(a)	227.(b)	228.(a)
229.(c)	230.(c)	231.(a)	232.(a)
233.(b)	234.(c)	235.(b)	236.(a)
237.(c)	238.(c)	239.(c)	240.(a)
241.(b)	242.(a)	243.(a)	244.(c)
245.(b)	246.(c)	247.(d)	248.(c)
249.(d)	250.(a)	251.(d)	252.(d)
253.(c)	254.(d)	255.(d)	256.(c)
257.(b)	258.(c)	259.(b)	260.(b)
261.(a)	262.(b)	263.(d)	264.(a)
265.(a)	266.(a)	267.(c)	268.(d)
269.(a)	270.(c)	271.(b)	272.(b)
273.(a)	274.(a)	275.(a)	276.(a)
277.(a)	278.(b)	279.(c)	280.(c)
281.(a)	282.(a)	283.(b)	284.(a)
285.(b)	286.(a)	287.(c)	288.(d)
289.(b)	290.(d)	291.(d)	292.(a)
293.(c)	294.(c)	295.(d)	296.(d)
297.(c)	298.(c)	299.(b)	300.(d)
301.(d)	302.(d)	303.(c)	304.(b)
305.(b)	306.(a)	307.(d)	308.(c)
309.(a)	310.(b)	311.(c)	312.(d)

313.(d)	314.(d)	315.(c)	316.(a)
317.(d)	318.(c)	319.(d)	320.(d)
321.(a)	322.(c)	323.(a)	324.(a)
325.(c)	326.(a)	327.(d)	328.(c)
329.(d)	330.(a)	331.(b)	332.(a)
333.(c)	334.(b)	335.(c)	336.(c)
337.(a)	338.(c)	339.(b)	340.(a)
341.(a)	342.(c)	343.(c)	344.(d)
345.(a)	346.(b)	347.(c)	348.(c)
349.(a)	350.(c)	351.(d)	352.(c)
353.(c)	354.(c)	355.(c)	356.(a)
357.(c)	358.(c)	359.(a)	360.(d)
361.(d)	362.(c)	363.(d)	364.(d)
365.(b)	366.(c)	367.(b)	368.(b)
369.(d)	370.(d)	371.(d)	372.(d)
373.(d)	374.(d)	375.(b)	376.(d)
377.(d)	378.(d)	379.(d)	380.(b)
381.(c)	382.(b)	383.(d)	384.(c)
385.(c)	386.(b)	387.(c)	388.(a)
389.(c)	390.(b)	391.(c)	392.(a)
393.(d)	394.(c)	395.(b)	396.(c)
397.(c)	398.(b)	399.(a)	400.(c)
401.(c)	402.(a)	403.(d)	404.(a)
405.(b)	406.(a)	407.(d)	408.(c)
409.(b)	410.(a)	411.(d)	412.(d)
413.(b)	414.(a)	415.(b)	416.(c)
417.(a)	418.(d)	419.(d)	420.(a)
421.(d)	422.(b)	423.(a)	424.(d)
425.(d)	426.(d)	427.(b)	428.(c)
429.(c)	430.(a)	431.(c)	432.(b)
433.(c)	434.(c)	435.(b)	436.(c)
437.(a)	438.(d)	439.(b)	440.(b)
441.(a)	442.(c)	443.(b)	444.(d)
445.(c)	446.(a)	447.(b)	448.(b)
449.(a)	450.(d)	451.(a)	452.(c)
453.(a)	454.(a)	455.(c)	456.(d)
457.(c)	458.(a)	459.(d)	460.(a)
461.(c)	462.(c)	463.(b)	464.(b)
465.(b)	466.(a)	467.(b)	468.(d)
469.(a)	470.(a)	471.(b)	472.(a)
473.(c)	474.(b)	475.(b)	476.(a)
477.(d)	478.(a)	479.(d)	480.(b)
481.(c)	482.(d)	483.(b)	484.(a)
485.(b)	486.(b)	487.(a)	488.(a)
489.(c)	490.(d)	491.(c)	492.(a)
493.(c)	494.(b)	495.(c)	496.(d)

497.(d)	498.(d)	499.(a)	500.(b)
501.(d)	502.(d)	503.(d)	504.(c)
505.(d)	506.(b)	507.(d)	508.(d)
509.(b)	510.(b)	511.(b)	512.(a)
513.(a)	514.(d)	515.(b)	516.(c)
517.(d)	518.(b)	519.(d)	520.(d)
521.(a)	522.(b)	523.(d)	524.(a)
525.(a)	526.(c)	527.(c)	528.(c)
529.(b)	530.(d)	531.(d)	532.(b)
533.(d)	534.(a)	535.(d)	536.(c)
537.(c)	538.(a)	539.(d)	540.(b)
541.(c)	542.(b)	543.(a)	544.(b)
545.(a)	546.(c)	547.(b)	548.(a)
549.(b)	550.(d)	551.(c)	552.(c)
553.(c)	554.(c)	555.(c)	556.(c)
557.(d)			

Solutions :-**Sol.1.(b)**

$$\begin{aligned}
 & \tan 72^\circ - \tan 27^\circ - \tan 72^\circ \tan 27^\circ \\
 &= \tan 72^\circ - \tan 72^\circ \tan 27^\circ - \tan 27^\circ \\
 &= \tan 72^\circ (1 - \tan 27^\circ) - \tan 27^\circ \\
 &= \tan(45^\circ + 27^\circ)(1 - \tan 27^\circ) - \tan 27^\circ \\
 &= \left[\frac{\tan 45^\circ + \tan 27^\circ}{1 - \tan 45^\circ \tan 27^\circ} \right] (1 - \tan 27^\circ) - \tan 27^\circ \\
 &= \frac{(1 + \tan 27^\circ)}{(1 - \tan 27^\circ)} (1 - \tan 27^\circ) - \tan 27^\circ \\
 &= 1 + \tan 27^\circ - \tan 27^\circ = 1
 \end{aligned}$$

$$\text{Sol.2.(d)} \quad \frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta}$$

$$\begin{aligned}
 &= \frac{1 + \sin \theta - (1 - 2\sin^2 \frac{\theta}{2})}{1 + \sin \theta + 2\cos^2 \frac{\theta}{2} - 1} \\
 &= \frac{1 + \sin \theta - 1 + 2\sin^2 \frac{\theta}{2}}{\sin \theta + 2\cos^2 \frac{\theta}{2}} \\
 &= \frac{\sin \theta + 2\sin^2 \frac{\theta}{2}}{\sin \theta + 2\cos^2 \frac{\theta}{2}} \\
 &= \frac{2\sin \frac{\theta}{2} \cos \frac{\theta}{2} + 2\sin^2 \frac{\theta}{2}}{2\sin \frac{\theta}{2} \cos \frac{\theta}{2} + 2\cos^2 \frac{\theta}{2}} \\
 &= \frac{2\sin \frac{\theta}{2} (\cos \frac{\theta}{2} + \sin \frac{\theta}{2})}{2\cos \frac{\theta}{2} (\sin \frac{\theta}{2} + \cos \frac{\theta}{2})} = \tan \frac{\theta}{2}
 \end{aligned}$$

$$\begin{aligned}
 & (\cos \theta = 1 - 2\sin^2 \frac{\theta}{2} = 2\cos^2 \frac{\theta}{2} - 1) \\
 &= \frac{1 + \sin \theta - 1 + 2\sin^2 \frac{\theta}{2}}{\sin \theta + 2\cos^2 \frac{\theta}{2}} \\
 &= \frac{\sin \theta + 2\sin^2 \frac{\theta}{2}}{\sin \theta + 2\cos^2 \frac{\theta}{2}} \\
 &= \frac{2\sin \frac{\theta}{2} \cos \frac{\theta}{2} + 2\sin^2 \frac{\theta}{2}}{2\sin \frac{\theta}{2} \cos \frac{\theta}{2} + 2\cos^2 \frac{\theta}{2}} \\
 &= \frac{2\sin \frac{\theta}{2} (\cos \frac{\theta}{2} + \sin \frac{\theta}{2})}{2\cos \frac{\theta}{2} (\sin \frac{\theta}{2} + \cos \frac{\theta}{2})} = \tan \frac{\theta}{2}
 \end{aligned}$$

Sol.3.(b) Identity used:

$$\begin{aligned}
 & \sin 2A = 2\sin A \cos A \text{ and } \sin^2 A + \cos^2 A = 1 \\
 & \text{Now,} \\
 &= \frac{\cos^2 t + \sin^2 t - 2\sin t \cos t}{\cos^2 t + \sin^2 t + 2\sin t \cos t} \times
 \end{aligned}$$

$$\begin{aligned}
 & \frac{\cos(t) + \sin(t)}{\cos(t) - \sin(t)} \\
 &= \frac{(\cos t - \sin t)^2}{(\cos t + \sin t)^2} \times \frac{\cos(t) + \sin(t)}{\cos(t) - \sin(t)} \\
 &= \frac{(\cos t - \sin t)}{(\cos t + \sin t)}
 \end{aligned}$$

On dividing numerator and denominator by $\cos t$ we get,

$$= \frac{(1 - \tan t)}{(1 + \tan t)}$$

Sol.4.(d) By using identity:

$$\sin A \sin B = -\frac{1}{2} \{(\cos(A+B) - \cos(A-B))\}$$

hence,

$$\sin A \sin B = \frac{1}{2} \{(\cos(A-B) - \cos(A+B))\}$$

Sol.5.(b) Given: $\sin^3 x + \cos^3 x = \sin x \cos x$
 $\cos x$ and $\sin x = \cos x$

$$\sin^3 x + \cos^3 x = \sin x \cos x$$

$$\sin x \sin^2 x + \cos^3 x = \sin x \cos x$$

$$\cos x \sin^2 x + \cos^3 x = \sin x \cos x$$

$$\cos x (\sin^2 x + \cos^2 x) = \sin x \cos x$$

$$\cos x = \sin x \cos x$$

$$b = \sin x$$

$$\sin x = \cos x$$

on putting value of b

$$\sin x = \sin x \cos x$$

$$a = \cos x$$

Putting value of a and b

$$a^2 + b^2 = \sin^2 x + \cos^2 x = 1$$

Sol.6.(b) Given: $p \cos A = 2q \sin A$

$$\Rightarrow p = 2q \times \frac{\sin A}{\cos A} \dots (i)$$

$$2p \operatorname{cosec} A - q \sec A = 3$$

$$\Rightarrow \frac{2p}{\sin A} - \frac{q}{\cos A} = 3$$

$$\Rightarrow 2p \times \cos A - q \times \sin A = 3 \times \sin A \cos A$$

$$\Rightarrow 2 \times 2q \times \frac{\sin A}{\cos A} \times \cos A - q \times \sin A = 3 \times \sin A \cos A$$

$$\sin A \cos A \text{ [from eq. (i)]}$$

$$\Rightarrow 4q \times \sin A - q \times \sin A = 3 \times \sin A \cos A$$

$$\Rightarrow 3q \times \sin A = 3 \times \sin A \cos A \Rightarrow q = \cos A$$

Put the value of q in eq. (i)

$$\Rightarrow p = 2 \times \cos A \times \frac{\sin A}{\cos A} \Rightarrow p = 2 \sin A$$

$$\text{Now, } p^2 + 4q^2$$

$$= (2 \sin A)^2 + 4(\cos A)^2$$

$$= 4(\sin^2 A + \cos^2 A) = 4$$

Sol.7.(a) $\tan A + \sin A = x$

On squaring both side

$$\tan^2 A + \sin^2 A + 2 \tan A \sin A = x^2 \dots (i)$$

Now, $\tan A - \sin A = y$

On squaring both side

$$\tan^2 A + \sin^2 A - 2 \tan A \sin A = y^2 \dots (ii)$$

On subtracting (i) and (ii)

$$x^2 - y^2 = (\tan^2 A + \sin^2 A + 2 \tan A \sin A) -$$

$$(\tan^2 A + \sin^2 A - 2 \tan A \sin A)$$

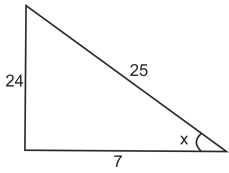
$$x^2 - y^2 = \tan^2 A + \sin^2 A + 2 \tan A \sin A -$$

$$\tan^2 A - \sin^2 A + 2 \tan A \sin A$$

$$x^2 - y^2 = 4 \tan A \sin A$$

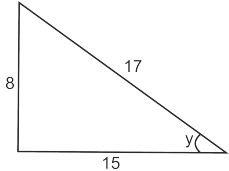
Sol.8.(d) Let $\cos^{-1}\left(\frac{7}{25}\right) = x$

$$\Rightarrow \cos x = \frac{7}{25}$$



$$\text{Then } \sin x = \frac{24}{25}$$

$$\text{and } \cos^{-1}\left(\frac{15}{17}\right) = y \Rightarrow \cos y = \frac{15}{17}$$



$$\text{Then, } \sin y = \frac{8}{17}$$

$$\begin{aligned} \text{As we know, } \sin(x-y) &= \sin x \times \cos y - \cos x \sin y \\ &= \frac{24}{25} \times \frac{15}{17} - \frac{7}{25} \times \frac{8}{17} \\ &= \frac{360 - 56}{425} = \frac{304}{425} \end{aligned}$$

$$(x-y) = \sin^{-1}\left(\frac{304}{425}\right)$$

$$\begin{aligned} \text{Then, } \cos^{-1}\left(\frac{7}{25}\right) - \cos^{-1}\left(\frac{15}{17}\right) &= \sin^{-1}\left(\frac{304}{425}\right) \end{aligned}$$

Sol.9.(a) Take $x = 45^\circ$

$$\begin{aligned} 6f(2) &= 6 \times (\sin^{p+2} x + \cos^{p+2} x) \\ &= 6 \times (\sin^4 x + \cos^4 x) \\ &= 6 \times \left(\left(\frac{1}{\sqrt{2}}\right)^4 + \left(\frac{1}{\sqrt{2}}\right)^4\right) \\ &= 6 \times \left(\frac{1}{4} + \frac{1}{4}\right) = 6 \times \frac{1}{2} = 3 \end{aligned}$$

$$\begin{aligned} 4f(4) &= 4 \times (\sin^6 x + \cos^6 x) \\ &= 4 \times ((\sin^2 x)^3 + (\cos^2 x)^3) \\ &= 4 \times \left(\left(\frac{1}{\sqrt{2}}\right)^2\right)^3 + \left(\left(\frac{1}{\sqrt{2}}\right)^2\right)^3 \\ &= 4 \times \left(\frac{1}{8} + \frac{1}{8}\right) = 4 \times \frac{1}{4} = 1 \end{aligned}$$

$$10f(0) = 10 \times (\sin^2 x + \cos^2 x) = 10 \times (1) = 10$$

$$\begin{aligned} \text{Hence, } 6f(2) - 4f(4) + 10f(0) &= 3 - 1 + 10 = 12 \end{aligned}$$

Sol.10.(b) Given :

$$\tan A = a \tan B \text{ and } \sin A = b \sin B$$

$$\text{We put the value of } A = 60^\circ \text{ and } B = 30^\circ$$

$$\tan 60^\circ = a \tan 30^\circ$$

$$\Rightarrow \sqrt{3} = a \times \frac{1}{\sqrt{3}} \Rightarrow a = 3$$

$$\sin 60^\circ = b \sin 30^\circ$$

$$\Rightarrow \frac{\sqrt{3}}{2} = b \times \frac{1}{2} \Rightarrow b = \sqrt{3}$$

$$\text{So, } \cos^2 A = (\cos 60^\circ)^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Now, put the value a and b in all options one by one only option (b) satisfied.

$$\Rightarrow \frac{b^2 - 1}{a^2 - 1} = \frac{3 - 1}{9 - 1} = \frac{2}{8} = \frac{1}{4}$$

Sol.11.(d)
$$\frac{\sin(A-B)}{\cos A \cos B} + \frac{\sin(B-C)}{\cos B \cos C}$$

$$\frac{\sin A \cos B - \cos A \sin B}{\cos A \cos B} + \frac{\sin B \cos C - \cos B \sin C}{\cos B \cos C}$$

$$\frac{\cos C(\sin A \cos B - \cos A \sin B) + \cos A(\sin B \cos C - \cos B \sin C)}{\cos A \cos B \cos C}$$

$$\frac{\sin A \cos B \cos C - \cos A \sin B \cos C - \cos A \sin B \cos C + \cos A \cos B \sin C}{\cos A \cos B \cos C}$$

$$\begin{aligned} &= \frac{\sin A \cos B \cos C - \cos A \cos B \sin C}{\cos A \cos B \cos C} \\ &= \frac{\cos B(\sin A \cos C - \cos A \sin C)}{\cos A \cos B \cos C} \\ &= \frac{(\sin A \cos C - \cos A \sin C)}{\cos A \cos C} = \frac{\sin(A-C)}{\cos A \cos C} \end{aligned}$$

Sol.12.(d)
$$\frac{3 \cos^3 \theta - 2 \cos \theta}{\sin \theta - 3 \sin^3 \theta}$$

$$\frac{\cos \theta (3 \cos^2 \theta - 2)}{\sin \theta (1 - 3 \sin^2 \theta)}$$

$$\frac{\cos \theta (3 \cos^2 \theta - 2)}{\sin \theta (1 - 3 \{1 - \cos^2 \theta\})}$$

$$\frac{\cos \theta (3 \cos^2 \theta - 2)}{\sin \theta (3 \cos^2 \theta - 2)} = \frac{\cos \theta}{\sin \theta} = \cot \theta$$

Sol.13.(c) $\cot A = \frac{7}{1} = \frac{\text{base}}{\text{perpendicular}}$

$$\text{Hypotenuse} = \sqrt{(7)^2 + (1)^2}$$

$$\text{Hypotenuse} = \sqrt{50} = 5\sqrt{2}$$

Now,

$$\begin{aligned} &5\left(\frac{7}{5\sqrt{2}}\right) + 4\left(\frac{1}{5\sqrt{2}}\right) \\ &\left(\frac{7}{5\sqrt{2}}\right)^3 + 7\left(\frac{1}{5\sqrt{2}}\right)^3 + 6\left(\frac{1}{5\sqrt{2}}\right) \\ &= \frac{\left(\frac{7}{\sqrt{2}}\right) + \left(\frac{4}{5\sqrt{2}}\right)}{\frac{343}{250\sqrt{2}} + \frac{7}{250\sqrt{2}} + \frac{6}{5\sqrt{2}}} \\ &= \frac{\left(\frac{35+4}{5\sqrt{2}}\right)}{\frac{343+7+300}{250\sqrt{2}}} = \frac{39 \times 250\sqrt{2}}{5\sqrt{2} \times 650} = 3 \end{aligned}$$

Alternate method:

$$\begin{aligned} &\frac{5 \cos A + 4 \sin A}{\cos^3 A + 7 \sin^3 A + 6 \sin A} \\ \text{Divide numerator and denominator by } \sin A & \\ &= \frac{5 \times \frac{\cos A}{\sin A} + 4 \times \frac{\sin A}{\sin A}}{\frac{\cos^3 A}{\sin A} + 7 \frac{\sin^3 A}{\sin A} + 6 \frac{\sin A}{\sin A}} \\ &= \frac{5 \times \cot A + 4 \times 1}{\cos^2 A \times \cot A + 7 \sin^2 A + 6} \\ &= \frac{5 \times 7 + 4 \times 1}{\cos^2 A \times 7 + 7 \sin^2 A + 6} \\ &= \frac{35 + 4}{7(\cos^2 A + \sin^2 A) + 6} = \frac{39}{7 + 6} = \frac{39}{13} = 3 \end{aligned}$$

Sol.14.(c)
$$\frac{[(\sin x + \sin y)(\sin y - \sin x)]}{[(\cos x + \cos y)(\cos y - \cos x)]}$$

$$= \frac{-(\sin x + \sin y)(\sin x - \sin y)}{-[(\cos x + \cos y)(\cos x - \cos y)]}$$

$$\begin{aligned} &= \frac{[(\sin^2 x - \sin^2 y)]}{[(\cos^2 x - \cos^2 y)]} \\ &= \frac{[(1 - \cos^2 x) - (1 - \cos^2 y)]}{[(\cos^2 x - \cos^2 y)]} \\ &= \frac{[1 - \cos^2 x - 1 + \cos^2 y]}{[(\cos^2 x - \cos^2 y)]} \\ &= \frac{[-\cos^2 x + \cos^2 y]}{[(\cos^2 x - \cos^2 y)]} = \frac{-(\cos^2 x - \cos^2 y)}{[(\cos^2 x - \cos^2 y)]} = -1 \end{aligned}$$

Sol.15.(c)
$$\frac{\cos 13^\circ + \sin 13^\circ}{\cos 13^\circ - \sin 13^\circ}$$

On dividing by $\cos 13^\circ$

$$\begin{aligned} &\frac{\frac{\cos 13^\circ}{\cos 13^\circ} + \frac{\sin 13^\circ}{\cos 13^\circ}}{\frac{\cos 13^\circ}{\cos 13^\circ} - \frac{\sin 13^\circ}{\cos 13^\circ}} \Rightarrow \frac{1 + \tan 13^\circ}{1 - \tan 13^\circ} \\ &\frac{\tan 45^\circ + \tan 13^\circ}{1 - \tan 45^\circ \cdot \tan 13^\circ} = \tan(45^\circ + 13^\circ) \\ &= \tan 58^\circ \end{aligned}$$

Sol.16.(c) $(\sin x + \sin y) = a$ ----- (i)

Squaring both side,

$$\sin^2 x + \sin^2 y + 2 \sin x \sin y = a^2$$
 ----- (ii)

$$(\cos x + \cos y) = b$$
 ----- (iii)

Squaring both side,

$$\cos^2 x + \cos^2 y + 2 \cos x \cos y = b^2$$
 ----- (iv)

Adding eqn (ii) & (iv) we get ;

$$(\sin^2 x + \cos^2 x) + (\sin^2 y + \cos^2 y) + 2 \sin x \sin y + 2 \cos x \cos y = a^2 + b^2$$

$$1 + 1 + 2(\sin x \sin y + \cos x \cos y) = a^2 + b^2$$

$$2(1 + \sin x \sin y + \cos x \cos y) = a^2 + b^2$$

$$(1 + \sin x \sin y + \cos x \cos y) = \frac{a^2 + b^2}{2}$$

$$\sin x \sin y + \cos x \cos y = \frac{a^2 + b^2}{2} - 1$$

$$= \frac{a^2 + b^2 - 2}{2}$$

Sol.17. (a) $\tan(t) + \cot(t) = 1$

$$\frac{\sin(t)}{\cos(t)} + \frac{\cos(t)}{\sin(t)} = 1$$

$$\frac{\sin^2(t) + \cos^2(t)}{\sin(t)\cos(t)} = 1$$

$$1 = \sin(t)\cos(t) \text{(i)}$$

$$[\sin(t) + \cos(t)]^2 = \sin^2(t) + \cos^2(t) + 2 \sin(t)\cos(t)$$

$$= \sqrt{\sin^2(t) + \cos^2(t)} + 2 \sin(t)\cos(t)$$

$$= \sqrt{1 + 2 \sin(t)\cos(t)}$$

$$= \sqrt{1 + 2 \times 1} = \sqrt{3}$$

$$\text{Then, } \frac{1}{\sin(t) + \cos(t)} = \frac{1}{\sqrt{3}}$$

Sol.18.(d) $3 \sec^4 \theta + 8 = 10 \sec^2 \theta$

$$3 \sec^4 \theta - 10 \sec^2 \theta + 8 = 0$$

$$3 \sec^4 \theta - 6 \sec^2 \theta - 4 \sec^2 \theta + 8 = 0$$

$$3 \sec^2 \theta (\sec^2 \theta - 2) - 4(\sec^2 \theta - 2) = 0$$

$$(\sec^2 \theta - 2)(3 \sec^2 \theta - 4) = 0$$

$$\sec^2 \theta = 2 \text{ and } \frac{4}{3}$$

$$\Rightarrow \sec \theta = \sqrt{2} \text{ and } \frac{2}{\sqrt{3}}$$

$$\Rightarrow \theta = 45^\circ \text{ and } 30^\circ$$

$$\text{For } \theta = 45^\circ, \tan 45^\circ = 1$$

$$\text{For } \theta = 30^\circ, \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\text{Sol.19.(a)} \cot^4 A + \cot^2 A = 1$$

$$\cot^2 A (\cot^2 A + 1) = 1$$

$$\cot A = \sin A \text{ or } \sin^2 A = \cos A$$

Now,

$$\cot^4 A - 3\cos^2 A =$$

$$\left\{ (1 - \sin^2 A)^2 - 3\sin^4 A \right\}$$

$$1 + \sin^4 A - 2\sin A - 3\sin^4 A$$

$$1 - 2(\sin^4 A + \sin^2 A) =$$

$$1 - 2(\cot^4 A + \cot^2 A) \Rightarrow 1 - 2 = -1$$

Sol.20.(a) According to the question,

$$\operatorname{cosec} \theta + \cot^3 \theta \sec \theta$$

$$\frac{1}{\sin \theta} + \frac{\cos^3 \theta}{\sin^3 \theta} \times \frac{1}{\cos \theta}$$

$$\frac{\sin^2 \theta}{\sin^3 \theta} + \frac{\cos^2 \theta}{\sin^3 \theta} = \frac{1}{\sin^3 \theta} = \operatorname{cosec}^3 \theta$$

now,

$$\cot^2 \theta = 1 - e^2 \quad (\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta)$$

$$\operatorname{cosec}^2 \theta = 2 - e^2$$

$$\operatorname{cosec} \theta = (2 - e^2)^{\frac{1}{2}}$$

(on cubing both sides)

$$\operatorname{cosec}^3 \theta = (2 - e^2)^{\frac{3}{2}}$$

Sol.21.(d) Minimum value of $a \sin^2 \theta + b$

$$\operatorname{cosec}^2 \theta = 2\sqrt{ab} \quad \{\text{when, } a > b\}$$

Hence, minimum value of $16\operatorname{cosec}^2 \theta +$

$$25\sin^2 \theta = 2\sqrt{16 \times 25} = 40$$

Sol.22.(a) $\operatorname{cosec} A + \cot A = a\sqrt{b}$

$$\left(\frac{1}{\sin A} + \frac{\cos A}{\sin A} \right)^2 = \left(\frac{1 + \cos A}{\sin A} \right)^2 = a^2 b$$

$$\frac{a^2 b - 1}{a^2 b + 1} = \frac{\left(\frac{1 + \cos A}{\sin A} \right)^2 - 1}{\left(\frac{1 + \cos A}{\sin A} \right)^2 + 1}$$

$$= \frac{1 + \cos^2 A + 2\cos A - \sin^2 A}{1 + \cos^2 A + 2\cos A + \sin^2 A}$$

$$= \frac{2\cos^2 A + 2\cos A}{2 + 2\cos A} = \frac{2\cos A (\cos A + 1)}{2 (\cos A + 1)}$$

$$= \cos A$$

Sol.23.(a) Put $\theta = 45^\circ$

$$\text{Now, } \sin \theta + \tan \theta = p \dots\dots\dots (1)$$

$$\Rightarrow \frac{1}{\sqrt{2}} + 1 = p \Rightarrow \frac{1 + \sqrt{2}}{\sqrt{2}} = p$$

$$\text{And, } \tan \theta - \sin \theta = q \dots\dots\dots (2)$$

$$\Rightarrow 1 - \frac{1}{\sqrt{2}} = q \Rightarrow \frac{\sqrt{2} - 1}{\sqrt{2}} = q$$

Now, checking through options, (a) satisfies the value of p and q.

$$p^2 - q^2 = 4\sqrt{pq} \Rightarrow \left(\frac{1 + \sqrt{2}}{\sqrt{2}} \right)^2 - \left(\frac{\sqrt{2} - 1}{\sqrt{2}} \right)^2$$

$$= 4 \sqrt{\frac{1 + \sqrt{2}}{\sqrt{2}} \times \frac{\sqrt{2} - 1}{\sqrt{2}}} \\ = \frac{(1 + \sqrt{2})^2 - (\sqrt{2} - 1)^2}{2} = 4 \times \sqrt{\frac{1}{2}}$$

$$\Rightarrow 2\sqrt{2} = 2\sqrt{2}$$

Sol.24.(b)

$$\sin \theta \cdot \cos \theta = 2\cos^3 \theta - \frac{1}{4} \cos \theta$$

$$4\sin \theta \cdot \cos \theta = 8\cos^3 \theta - \cos \theta$$

$$4\sin \theta = 8\cos^2 \theta - 1$$

$$4\sin \theta = 8 - 8\sin^2 \theta - 1$$

$$8\sin^2 \theta + 4\sin \theta - 7 = 0$$

$$\sin \theta = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-4 \pm \sqrt{4^2 - 4 \times 8 \times (-7)}}{2 \times 8}$$

$$= \frac{-4 \pm \sqrt{240}}{2 \times 8} = \frac{-1 \pm \sqrt{15}}{4}$$

Here θ is an acute angle, so

$$\Rightarrow \sin \theta = \frac{\sqrt{15} - 1}{4}$$

Sol.25.(a)

$$\text{If } A + B = 90^\circ, \text{ then } \cot A \times \cot B = 1$$

$$= 7(\cot 11^\circ \cot 13^\circ \cot 19^\circ \cot 29^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ)$$

$$= 7(\cot 11^\circ \times \cot 79^\circ)(\cot 13^\circ \times \cot 77^\circ)$$

$$(\cot 19^\circ \times \cot 71^\circ)(\cot 29^\circ \times \cot 61^\circ)$$

$$= 7(1 \times 1 \times 1 \times 1) = 7$$

Sol.26.(a) $\sin^2 A + \cos^2 A = 1$

$$\sin^4 A + \cos^4 A + 2\sin^2 A \cdot \cos^2 A = 1$$

$$\sin^4 A + \cos^4 A = 1 - 2\sin^2 A \cdot \cos^2 A$$

$$\frac{\sin^4 A + \cos^4 A}{1 - 2\sin^2 A \cos^2 A} = 1$$

Sol.27.(c) $pq = \cot^2 A - \cos^2 A$

$$= \cot^2 A \times \cos^2 A \dots (\text{identity})$$

$$\sqrt{pq} = \cot A \cdot \cos A$$

$$p^2 = \cot^2 A + \cos^2 A + 2 \times \cot A \cdot \cos A$$

$$q^2 = \cot^2 A + \cos^2 A - 2 \times \cot A \cdot \cos A$$

$$p^2 - q^2 = 4 \times \cot A \cdot \cos A$$

Now,

$$p^2 - q^2 = 4 \times \cot A \cdot \cos A = \sqrt{16pq}$$

Sol.28.(d)

$$4(\sin^6 A + \cos^6 A) - 6(\sin^4 A + \cos^4 A) + 8$$

$$[\because a^3 + b^3 = (a + b)^3 - 3ab(a + b) \text{ and } a^2 + b^2 = (a + b)^2 - 2ab]$$

$$= 4(1 - 3\sin^2 A \cdot \cos^2 A) - 6(1 - 2\sin^2 A \cdot \cos^2 A) + 8$$

$$= 4 - 12\sin^2 A \cdot \cos^2 A - 6 + 12\sin^2 A \cdot \cos^2 A$$

$$\cdot \cos^2 A + 8 = 6$$

Sol.29.(a) Concept used :

$$(1 + \cos 2\theta) = 2\cos^2 \theta$$

$$\sqrt{2 + \sqrt{2} + \sqrt{2} + 2\cos 8\theta}$$

$$= \sqrt{2 + \sqrt{2} + \sqrt{2(1 + \cos 8\theta)}}$$

$$= \sqrt{2 + \sqrt{2} + \sqrt{2 \times 2\cos^2 4\theta}}$$

$$= \sqrt{2 + \sqrt{2} + 2\cos 4\theta}$$

$$= \sqrt{2 + \sqrt{2(1 + \cos 4\theta)}}$$

$$= \sqrt{2 + \sqrt{2 \times 2\cos^2 2\theta}} = \sqrt{2 + 2\cos 2\theta}$$

$$= \sqrt{2(1 + \cos 2\theta)} = \sqrt{2 \times 2\cos^2 \theta} = 2\cos \theta$$

Short trick -

$$2\cos \left[\frac{8\theta}{2} \right] = 2\cos 2\theta$$

$$\text{Sol.30.(a)} \sqrt{\frac{1 - \sin 3\theta}{1 + \sin 3\theta}}$$

After rationalizing,

$$= \sqrt{\frac{(1 - \sin 3\theta) \times (1 - \sin 3\theta)}{(1 + \sin 3\theta) \times (1 - \sin 3\theta)}}$$

$$= \sqrt{\frac{(1 - \sin 3\theta) \times (1 - \sin 3\theta)}{(1 - \sin^2 3\theta)}}$$

$$= \sqrt{\left(\frac{1 - \sin 3\theta}{\cos 3\theta} \right)^2}$$

$$= \left(\frac{1 - \sin 3\theta}{\cos 3\theta} \right) = \sec 3\theta - \tan 3\theta$$

Sol.31.(b) $\cos^2 \theta + \sin \theta + 1 = 0$

$$1 - \sin^2 \theta + \sin \theta + 1 = 0$$

$$\sin^2 \theta - \sin \theta - 2 = 0$$

$$\sin^2 \theta - 2\sin \theta + \sin \theta - 2 = 0$$

$$\sin \theta (\sin \theta - 2) + 1(\sin \theta - 2) = 0$$

$$(\sin \theta - 2)(\sin \theta + 1) = 0$$

$$\sin \theta = -1 \text{ or } +2$$

Since $\sin \theta$ has a range of -1 to +1.

$$\text{So, } \sin \theta = -1 \text{ or } \sin \theta = \sin \frac{3\pi}{2}$$

$$\text{Comparing both sides } \Rightarrow \theta = \frac{3\pi}{2}$$

Therefore solution of $\cos^2 \theta + \sin \theta + 1$

$$= 0 \text{ lies between } \left(\frac{5\pi}{4}, \frac{7\pi}{4} \right).$$

Sol.32.(a) $(1 + \cot A + \tan A)$

$$(\sin A - \cos A) \frac{\sin A \cos A}{(\sin^3 A - \cos^3 A)}$$

$$= (1 + \cot A + \tan A)(\sin A - \cos A)$$

$$\frac{\sin A \cos A}{(\sin A - \cos A)(\sin^2 A + \cos^2 A + \sin A \cos A)}$$

$$= (1 + \cot A + \tan A) \frac{\sin A \cos A}{(1 + \sin A \cos A)}$$

Now, put $A = 45^\circ$

$$(3) \times \frac{\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}}{\left(1 + \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \right)} = 3 \times \frac{1}{2} \times \frac{2}{3} = 1$$

Sol.33.(b) $\sin m + \sin n = p$

$$\Rightarrow (\sin m + \sin n)^2 = p^2$$

$$\Rightarrow \sin^2 m + \sin^2 n + 2 \sin m \sin n = p^2$$

$$\dots\dots\dots (\text{eq. 1})$$

$$\text{And } \cos m + \cos n = q$$

$$\Rightarrow (\cos m + \cos n)^2 = q^2$$

$$\Rightarrow \cos^2 m + \cos^2 n + 2 \cos m \cos n = q^2$$

----- (eq. 2)

After adding (eq. 1 and 2)

Then,

$$\begin{aligned} & (\sin^2 m + \cos^2 m) + (\sin^2 n + \cos^2 n) + \\ & 2 (\sin m \sin n + \cos m \cos n) = p^2 + q^2 \\ \Rightarrow & 2 + 2(\sin m \sin n + \cos m \cos n) = p^2 + q^2 \\ \Rightarrow & (\sin m \sin n + \cos m \cos n) = \frac{p^2 + q^2 - 2}{2} \end{aligned}$$

Sol.34.(a)

$$b \operatorname{cosec} \theta + a \cot \theta = p$$

$$b \cot \theta + a \operatorname{cosec} \theta = q$$

$$b^2 - a^2 = p^2 - q^2$$

Sol.35.(d)

$$\begin{aligned} & \frac{\cos(\pi - A) \cdot \cot(\frac{\pi}{2} + A) \cos(-A)}{\tan(\pi + A) \tan(\frac{3\pi}{2} + A) \sin(2\pi - A)} \\ & = \frac{-\cos A \cdot (-\tan A) \cos A}{\tan A \cdot (-\cot A) \cdot (-\sin A)} = \frac{\cos^2 A}{\sin A \cot A} \\ & = \cos A \end{aligned}$$

Sol.36.(c) $\cos 2A \cos 2B + \sin^2 (A - B) - \sin^2 (A + B) = \cos 2A \cos 2B +$
 $\{ \sin(A - B) + \sin(A + B) \} \times$
 $\{ \sin(A - B) - \sin(A + B) \}$
 $= \cos 2A \cos 2B + \{ 2 \sin A \cos B \times (-2 \cos A \cdot \sin B) \}$
 $= \cos 2A \cos 2B - \sin 2A \sin 2B$
 $= \cos(2A + 2B)$

Sol.37.(b) $\frac{k - k \cot^2 30^\circ}{1 + \cot^2 30^\circ}$
 $= \frac{\sin^2 60^\circ + 4 \tan^2 45^\circ - \operatorname{cosec}^2 60^\circ}{(1 + 3)}$
 $= \frac{3(1 - 3)}{4} = \frac{3}{4} + 4 \times 1 - \frac{4}{3}$
 $k = -2 \left\{ \frac{3}{4} + 4 - \frac{4}{3} \right\} = -2 \times \left(\frac{41}{12} \right) = -6.83$

Sol.38.(a) $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$

$$\sin \theta = \sqrt{2} \cos \theta - \cos \theta$$

$$\sin \theta = \cos \theta (\sqrt{2} - 1)$$

$$\sin \theta \times \frac{1}{\sqrt{2} - 1} = \cos \theta$$

$$(\sqrt{2} + 1) \sin \theta = \cos \theta$$

Subtracting $\sin \theta$ from it

$$\sin \theta (\sqrt{2} + 1 - 1) = \cos \theta - \sin \theta$$

$$\sqrt{2} \sin \theta = \cos \theta - \sin \theta$$

Sol.39.(a) $\sin 73^\circ + \cos 137^\circ$

$$\cos 137^\circ = \cos(90^\circ + 47^\circ) = -\sin 47^\circ$$

$$\sin 73^\circ - \sin 47^\circ = 2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

$$= 2 \cos \left(\frac{73+47}{2} \right) \cdot \sin \left(\frac{73-47}{2} \right)$$

$$= 2 \cos \left(\frac{120^\circ}{2} \right) \sin \left(\frac{26^\circ}{2} \right)$$

$$= 2 \cos 60^\circ \cdot \sin 13^\circ = 2 \times \frac{1}{2} \times \sin 13^\circ = \sin 13^\circ$$

Sol.40.(a) As we know that,

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\begin{aligned} \cos^2 18^\circ &= 1 - \left(\frac{\sqrt{5}-1}{4} \right)^2 \\ &= \frac{16 - (5 + 1 - 2\sqrt{5})}{16} = \frac{10 + 2\sqrt{5}}{16} \end{aligned}$$

$$\cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}$$

Sol.41.(a) $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$
 $= \frac{2 \sin \left(\frac{4\theta+2\theta}{2} \right) \cos \left(\frac{4\theta-2\theta}{2} \right)}{2 \cos \left(\frac{4\theta+2\theta}{2} \right) \cos \left(\frac{4\theta-2\theta}{2} \right)}$
 $= \frac{\sin 3\theta \cos \theta}{\cos 3\theta \cos \theta} = \frac{\sin 3\theta}{\cos 3\theta} = \tan 3\theta$

Sol.42.(d)

$$\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ$$

$$\begin{aligned} \text{No. of terms} &= \frac{\text{Last term} - \text{1st term}}{\text{common difference}} + 1 \\ &= \frac{89 - 1}{2} + 1 = 45 \end{aligned}$$

$$\begin{aligned} \sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ \\ &= \frac{45}{2} = 22.5 \end{aligned}$$

$$\begin{aligned} \cot^2 46^\circ - \sec^2 44^\circ + (\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ) \\ &= \tan^2 44^\circ - \sec^2 44^\circ + (22.5) = -1 + 22.5 \\ &= 21.5 \end{aligned}$$

Sol.43.(a) $\sin A = 0.5$ is true for all the values of $(2\pi n + 30^\circ)$ and $(2n + 1)\pi - 30^\circ$

For the $n = 0$,

$$2\pi n + 30^\circ = 30^\circ \text{ and } (2n + 1)\pi - 30^\circ = 150^\circ$$

And for $n = 1$,

$$2\pi n + 30^\circ = 390^\circ \text{ and } (2n + 1)\pi - 30^\circ = 510^\circ$$

$\sin A = 0.5$ is true for $150^\circ, 390^\circ$ and 510°

$$\text{So, } \frac{A}{3} = 50^\circ, 130^\circ \text{ and } 170^\circ$$

Only 170° is given in the answer so it is the correct answer.

Sol.44.(d) $\cos 27^\circ = x$

$$\sin 63^\circ = \cos 27^\circ = x$$

$$\begin{aligned} \cos 63^\circ &= \sqrt{1 - x^2} \\ \Rightarrow \tan 63^\circ &= \frac{x}{\sqrt{1 - x^2}} \end{aligned}$$

Sol.45.(c) $x \sin \theta = y \cos \theta$

$$= x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$$

$$= x \sin^3 \theta + \frac{x \sin \theta}{\cos \theta} \times \cos^3 \theta = \sin \theta \cos \theta$$

$$= x \sin^3 \theta + x \sin \theta \times \cos^2 \theta = \sin \theta \cos \theta$$

$$= x \sin \theta (\sin^2 \theta + \cos^2 \theta) = \sin \theta \cos \theta$$

$$= x \sin \theta = \sin \theta \cos \theta \Rightarrow x = \cos \theta$$

$$= x \sin \theta = y \cos \theta$$

$$= \cos \theta \cdot \sin \theta = y \cos \theta \Rightarrow y = \sin \theta$$

$$\Rightarrow x^2 + y^2 = \sin^2 \theta + \cos^2 \theta = 1$$

Short trick:- In such type of questions, value of $x^2 + y^2$ is the square of constant

term before $\sin \theta \cos \theta$

Sol.46.(d) $\cos^2 \theta + \cos^4 \theta = 1$

$$\cos^4 \theta = 1 - \cos^2 \theta \Rightarrow \cos^4 \theta = \sin^2 \theta \text{ Or}$$

$$\cos^2 \theta = \sin \theta$$

$$\sin \theta + \sin^2 \theta = \cos^2 \theta + \sin^2 \theta = 1$$

Sol.47.(c) $\sin \theta - \cos \theta = \frac{7}{17}$

$$\frac{p}{h} - \frac{b}{h} = \frac{7}{17} \text{ (here, } H = 17 \text{ or its}$$

multiple) & We know,

15, 8 and 17 make right angle triangles.

Put, $p = 15$ and $b = 8$

$$\sin \theta + \cos \theta = \frac{p}{h} + \frac{b}{h} = \frac{15 + 8}{17} = \frac{23}{17}$$

Short trick :- $\sin \theta - \cos \theta = \frac{7}{17}$

$$\sin \theta + \cos \theta = \sqrt{a^2 + b^2 - m^2}$$

$$= \sqrt{1^2 + 1^2 - \left(\frac{7}{17} \right)^2} = \sqrt{\frac{529}{289}} = \frac{23}{17}$$

Sol.48.(d) sum of roots

$$= \sec \theta + \sin \theta = \frac{-b}{a} = \frac{k}{\sqrt{6}}$$

product of roots = $\sec \theta \times \sin \theta$

$$= \tan \theta = \frac{c}{a} = \frac{\sqrt{6}}{\sqrt{6}} = 1 \Rightarrow \tan \theta = 1$$

it means, $\theta = 45^\circ$

$$\sec \theta + \sin \theta = \frac{k}{\sqrt{6}} \Rightarrow \sqrt{2} + \frac{1}{\sqrt{2}} = \frac{k}{\sqrt{6}}$$

$$\frac{3}{\sqrt{2}} = \frac{k}{\sqrt{6}} \Rightarrow k = 3\sqrt{3}$$

Sol.49.(d) If $\cos A$, $\sin A$ and $\cot A$ are in geometric progression, then

$$\sin^2 A = \cos A \times \cot A$$

$$\sin^2 A = \cos A \times \frac{\cos A}{\sin A} \Rightarrow \sin^3 A = \cos^2 A$$

After dividing by $\cos^3 A$

$$\frac{\sin^3 A}{\cos^3 A} = \frac{1}{\cos A} \Rightarrow \tan^3 A = \sec A$$

$$\tan^6 A - \tan^2 A = \sec^2 A - \tan^2 A = 1$$

Sol.50.(a) $\cos 225^\circ = \cos(180 + 45)^\circ$

$$= -\cos 45^\circ = -\frac{1}{\sqrt{2}} = -\frac{1}{1.414} = -0.7071$$

Sol.51.(c) $\tan 2A = \tan[(A + B) + (A - B)]$

$$= \frac{\tan(A + B) + \tan(A - B)}{1 - \tan(A + B) \tan(A - B)}$$

$$\cos(A + B) = \frac{24}{25} \Rightarrow \tan(A + B) = \frac{7}{24}$$

$$\sin(A - B) = \frac{15}{17} \Rightarrow \tan(A - B) = \frac{15}{8}$$

$$\tan[(A + B) + (A - B)]$$

$$= \frac{\frac{7}{24} + \frac{15}{8}}{1 - \frac{7}{24} \times \frac{15}{8}} = \frac{416}{87}$$

Sol.52.(d) $\cos^2 \theta - \sin^2 \theta = \tan^2 \theta$

$$\Rightarrow \frac{\cos^2 \theta - \sin^2 \theta}{1} = \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$\Rightarrow \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} = \frac{\sin^2 \theta}{\cos^2 \theta}$$

By using componendo and Dividendo rule,

$$\Rightarrow \frac{\cos^2 \theta}{-\sin^2 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$\Rightarrow \frac{-\sin^2 \theta}{\cos^2 \theta} = \frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \theta + \cos^2 \theta}$$

$$\Rightarrow \tan^2 \theta = \cos^2 \theta - \sin^2 \theta$$

Sol.53.(a) There are a total 20 terms and the value of $\sin \theta$ can't be greater than 1. Clearly for the sum of 20 terms being 20 values of each term must be 1. So each angle must be 90°

$$(\alpha_1 + \alpha_2 + \alpha_3 \dots \alpha_{20}) = 90 \times 20 = 1800^\circ$$

Sol.54.(a) Given, $\sin(\alpha + \beta) = 1$

$$\sin(\alpha + \beta) = \sin 90^\circ \Rightarrow \alpha + \beta = 90^\circ \dots (1)$$

$$\cos(\alpha - \beta) = \frac{1}{2}$$

Similarly, $\alpha - \beta = 60^\circ \dots (2)$

Adding (1) and (2)

$$2\alpha = 150^\circ \Rightarrow \alpha = 75^\circ$$

$$\text{From eq (1)} \Rightarrow \beta = 90^\circ - 75^\circ = 15^\circ$$

Sol.55.(b) Given $\theta = 90^\circ$

$$\cot \theta \cdot \cot 2\theta \cdot \cot 3\theta \cdot \cot 4\theta \cdot \cot 5\theta \cdot \cot 6\theta \cdot \cot 7\theta \cdot \cot 8\theta \cdot \cot 9\theta$$

$$= \cot 9\theta \cdot \cot 18\theta \cdot \cot 27\theta \cdot \cot 36\theta \cdot \cot 45\theta \cdot \cot 54\theta \cdot \cot 63\theta \cdot \cot 72\theta \cdot \cot 81\theta = 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1$$

$$\dots (\cot A \cot B = 1, \text{ if } A + B = 90^\circ) = 1$$

Sol.56.(d)

$$\tan 30^\circ \cdot \tan 60^\circ \Rightarrow \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$$

$$\sin 30^\circ \Rightarrow \frac{1}{2}; \tan 45^\circ \Rightarrow 1 \Rightarrow \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Clearly $\cos 30^\circ$ is an irrational quantity.

Sol.57.(c) Values of $\cos \theta$, $\cot \theta$ and $\operatorname{cosec} \theta$ continuously decreases. Only value of $\tan \theta$ is continuously increases.

Sol.58.(c) Given,

$$\sec \theta = 4x \text{ and } \tan \theta = \frac{4}{x}$$

Squaring both sides

$$\sec^2 \theta = 16x^2 \dots (1) \text{ and } \tan^2 \theta = \frac{16}{x^2} \dots (2)$$

Subtracting (2) from (1)

$$\sec^2 \theta - \tan^2 \theta = 16x^2 - \frac{16}{x^2}$$

$$\Rightarrow 1 = 2(8x^2 - \frac{8}{x^2}) \Rightarrow 8(x^2 - \frac{1}{x^2}) = \frac{1}{2}$$

Sol.59.(d)

$$\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} = \frac{1 - r}{1 + r}$$

$$\Rightarrow \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \sqrt{\frac{\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta}}{\frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta}}}$$

$$= \frac{1 - r}{1 + r} \Rightarrow \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

$$= \frac{1 - r}{1 + r} \Rightarrow \frac{(1 - \cos \theta)}{(1 + \cos \theta)} = \frac{1 - r}{1 + r}$$

Comparing both the terms $\Rightarrow r = \cos \theta$

Sol.60.(c)

$$\frac{(\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)}{\tan^2 \theta - \sin^2 \theta} = r^3$$

$$\text{Put } \theta = 30^\circ \Rightarrow \frac{(\frac{1}{2} - 2)(\frac{\sqrt{3}}{2} - \frac{2}{\sqrt{3}})}{(\frac{1}{\sqrt{3}})^2 - (\frac{1}{2})^2} = r^3$$

$$\Rightarrow \frac{(\frac{-3}{2})(\frac{3-4}{2\sqrt{3}})}{\frac{1}{3} - \frac{1}{4}} = r^3 \Rightarrow \frac{(\frac{3}{4\sqrt{3}})}{\frac{1}{12}} = r^3$$

$$\Rightarrow r^3 = 3\sqrt{3} \Rightarrow r = \sqrt{3}$$

through options $\Rightarrow \cot \theta \Rightarrow \cot 30^\circ = \sqrt{3}$

Clearly $r = \cot \theta$

Sol.61.(d) $3 - 2\sin^2 \theta - 3\cos \theta = 0$

$$= 3 - 2(1 - \cos^2 \theta) - 3\cos \theta = 0$$

$$= 2\cos^2 \theta - 3\cos \theta + 1 = 0$$

$$= 2\cos^2 \theta - 2\cos \theta - \cos \theta + 1 = 0$$

$$= 2\cos \theta (\cos \theta - 1) - 1(\cos \theta - 1) = 0$$

$$= \cos \theta = 1 \text{ or } \frac{1}{2} \Rightarrow \text{Clearly } \theta = 60^\circ$$

$$(2\operatorname{Cosec} \theta + \tan \theta) \Rightarrow 2(\frac{2}{\sqrt{3}}) + \sqrt{3}$$

$$\Rightarrow \frac{4}{\sqrt{3}} + \sqrt{3} = \frac{7\sqrt{3}}{3}$$

Short trick:- put $\theta = 60^\circ$

$$3 - 2\sin^2 \theta - 3\cos \theta = 3 - 2 \times \frac{3}{4} - 3 \times \frac{1}{2}$$

$$= 3 - \frac{3}{2} - \frac{3}{2} = 0$$

Condition satisfied. Clearly $\theta = 60^\circ$

$$\text{So, } (2\operatorname{Cosec} \theta + \tan \theta) = 2(\frac{2}{\sqrt{3}}) + \sqrt{3}$$

$$= \frac{4}{\sqrt{3}} + \sqrt{3} = \frac{7\sqrt{3}}{3}$$

Sol.62.(d)

$$(\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta})(\frac{1}{\tan \theta + \cot \theta})$$

$$= (\frac{\sin^2 \theta + (1 + \cos \theta)^2}{\sin \theta (1 + \cos \theta)})(\frac{1}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}})$$

$$= (\frac{\sin^2 \theta + 1 + \cos^2 \theta + 2\cos \theta}{\sin \theta (1 + \cos \theta)})(\frac{1}{\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}})$$

$$= \frac{2 + 2\cos \theta}{\sin \theta (1 + \cos \theta)} (\frac{\sin \theta \cos \theta}{1})$$

$$= \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)} \times \sin \theta \cos \theta = 2\cos \theta$$

Sol.63.(c)

$$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\sin 56^\circ \cdot \sec 34^\circ + \cos 25^\circ \cdot \operatorname{cosec} 65^\circ}$$

$$= \frac{[\cos(90^\circ - 81^\circ) + \sin 81^\circ][\sec(90^\circ - 81^\circ) + \operatorname{cosec} 81^\circ]}{\sin 56^\circ \cdot \sec(90^\circ - 56^\circ) + \cos(90^\circ - 65^\circ) \cdot \operatorname{cosec} 65^\circ}$$

$$= \frac{(2\sin 81^\circ)(2\operatorname{cosec} 81^\circ)}{\sin 56^\circ \cdot \operatorname{cosec} 56^\circ + \sin 65^\circ \cdot \operatorname{cosec} 65^\circ}$$

$$= \frac{4 \times 1}{1 + 1} = 2$$

Sol.64.(c)

$$\frac{(\sec \theta + \tan \theta)(1 - \sin \theta)}{\operatorname{cosec} \theta (1 + \cos \theta)(\operatorname{cosec} \theta - \cot \theta)}$$

$$= \frac{(\frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta})(1 - \sin \theta)}{(\operatorname{cosec} \theta + \operatorname{cosec} \theta \cdot \cos \theta)(\operatorname{cosec} \theta - \cot \theta)}$$

$$= \frac{(\frac{1}{\cos \theta})(1 + \sin \theta)(1 - \sin \theta)}{(\operatorname{cosec} \theta + \cot \theta)(\operatorname{cosec} \theta - \cot \theta)}$$

$$= \frac{\frac{\cos^2 \theta}{\cos \theta}}{\operatorname{cosec}^2 \theta - \cot^2 \theta} = \cos \theta$$

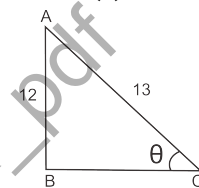
Sol.65.(b) $3\cos^2 A + 7\sin^2 A = 4$

$$= 3\cos^2 A + 7\sin^2 A = 4(\cos^2 A + \sin^2 A)$$

$$= 3\sin^2 A = \cos^2 A$$

$$= \cot^2 A = 3 \Rightarrow \cot A = \sqrt{3}$$

Sol.66.(d)



$$\operatorname{Cosec} \theta = \frac{AC}{AB} = \frac{13}{12}$$

$$AB^2 + BC^2 = AC^2 \Rightarrow BC = \sqrt{13^2 - 12^2} = 5$$

$$\sin \theta = \frac{12}{13}, \cos \theta = \frac{5}{13} \text{ and } \tan \theta = \frac{12}{5}$$

$$\sin \theta + \cos \theta - \tan \theta$$

$$= \frac{12}{13} + \frac{5}{13} - \frac{12}{5} = -\frac{71}{65}$$

Sol.67.(c) Given, $\sec 4\theta = \operatorname{Cosec}(\theta + 20^\circ)$

$$= \operatorname{Cosec}(90^\circ - 4\theta) = \operatorname{Cosec}(\theta + 20^\circ)$$

$$\text{Clearly, } (4\theta + \theta + 20^\circ) = 90^\circ$$

$$5\theta = 70^\circ \Rightarrow \theta = 14^\circ$$

Sol.68.(b) Put $\theta = 30^\circ$

$$3\sin \theta = 2\cos^2 \theta = 3 \times \frac{1}{2} = 2 \times (\frac{\sqrt{3}}{2})^2$$

$$= \frac{3}{2} = \frac{3}{2} \text{ (Condition satisfied)}$$

$$(\tan^2 \theta + \sec^2 \theta - \operatorname{cosec}^2 \theta)$$

$$= (\frac{1}{\sqrt{3}})^2 + (\frac{2}{\sqrt{3}})^2 - (2)^2$$

$$= \frac{1}{3} + \frac{4}{3} - 4 = -\frac{7}{3}$$

Sol.69.(b)

Applying componendo and dividendo

$$\frac{\tan \theta + \sin \theta}{\tan \theta - \sin \theta} = \frac{k + 1}{k - 1} \Rightarrow \frac{\tan \theta}{\sin \theta} = \frac{k}{1}$$

$$\Rightarrow \frac{\sin \theta}{\cos \theta \cdot \sin \theta} = k \Rightarrow \sec \theta = k$$

Sol.70.(b)

$$\frac{1}{\operatorname{cosec} \theta - 1} + \frac{1}{\operatorname{cosec} \theta + 1} = 2 \sec \theta$$

$$= \frac{(\operatorname{cosec} \theta + 1) + (\operatorname{cosec} \theta - 1)}{(\operatorname{cosec} \theta + 1)(\operatorname{cosec} \theta - 1)} = 2 \sec \theta$$

$$= \frac{2\operatorname{cosec} \theta}{(\operatorname{cosec}^2 \theta - 1)} = 2 \sec \theta$$

$$= \frac{2 \operatorname{cosec} \theta}{\cot^2 \theta} = 2 \sec \theta$$

$$= \frac{2 \sin^2 \theta}{\sin \theta \cdot \cos^2 \theta} = 2 \times \frac{1}{\cos \theta}$$

$$= \tan \theta = 1 \Rightarrow \theta = 45^\circ$$

$$(\cot \theta + \cos \theta) \Rightarrow 1 + \frac{1}{\sqrt{2}} = \frac{\sqrt{2} + 1}{\sqrt{2}}$$

$$= \frac{2 + \sqrt{2}}{2}$$

Short trick :- Put $\theta = 45^\circ$

$$\text{LHS} \Rightarrow \frac{1}{\operatorname{cosec} \theta - 1} + \frac{1}{\operatorname{cosec} \theta + 1}$$

$$= \frac{1}{\sqrt{2} - 1} + \frac{1}{\sqrt{2} + 1}$$

$$= \sqrt{2} + 1 + \sqrt{2} - 1 = 2\sqrt{2}$$

$$\text{RHS} = 2 \sec \theta = 2\sqrt{2}$$

$$\cot \theta + \cos \theta \Rightarrow 1 + \frac{1}{\sqrt{2}} = \frac{\sqrt{2} + 1}{\sqrt{2}}$$

$$= \frac{2 + \sqrt{2}}{2}$$

$$\text{Sol.71.(a)} \sin \theta = \frac{a}{\sqrt{a^2 + b^2}}$$

$$\cos \theta = \sqrt{1 - \left(\frac{a}{\sqrt{a^2 + b^2}}\right)^2} = \frac{b}{\sqrt{a^2 + b^2}}$$

$$\sec \theta + \tan \theta$$

$$= \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} \Rightarrow \frac{1 + \sin \theta}{\cos \theta}$$

$$= \frac{1 + \frac{a}{\sqrt{a^2 + b^2}}}{\frac{b}{\sqrt{a^2 + b^2}}} = \frac{\sqrt{a^2 + b^2} + a}{b}$$

$$\text{Sol.72.(a)} \cos^2 \theta - 3 \cos \theta + 2 = \sin^2 \theta$$

$$\Rightarrow \cos^2 \theta - 3 \cos \theta + 2 = 1 - \cos^2 \theta$$

$$= 2 \cos^2 \theta - 3 \cos \theta + 1 = 0$$

$$= 2 \cos^2 \theta - 2 \cos \theta - \cos \theta + 1 = 0$$

$$= 2 \cos \theta (\cos \theta - 1) - 1 (\cos \theta - 1) = 0$$

$$\Rightarrow \cos \theta = 1 \text{ or } \frac{1}{2}$$

Clearly $\theta = 60^\circ$, (as $0^\circ < \theta < 90^\circ$)

$$2 \operatorname{cosec} \theta + 4 \cot \theta$$

$$= 2 \left(\frac{2}{\sqrt{3}}\right) + 4 \left(\frac{1}{\sqrt{3}}\right) = \frac{8\sqrt{3}}{3}$$

$$\text{Sol.73.(b)} \frac{1}{\sec x - \tan x} - \frac{1}{\cos x}$$

$$= \sec x + \tan x - \sec x \dots \{ \because \sec x + \tan x$$

$$= \frac{1}{\sec x - \tan x} \} = \tan x$$

$$\text{Sol.74.(a)} \tan^2 \theta - 3 \sec \theta + 3 = 0$$

$$\Rightarrow \sec^2 \theta - 1 - 3 \sec \theta + 3 = 0$$

$$\Rightarrow \sec^2 \theta - 3 \sec \theta + 2 = 0$$

$$\Rightarrow \sec^2 \theta - 2 \sec \theta - \sec \theta + 2 = 0$$

$$\Rightarrow \sec \theta (\sec \theta - 2) - 1 (\sec \theta - 2) = 0$$

$$\Rightarrow (\sec \theta - 1)(\sec \theta - 2) = 0$$

$$\Rightarrow \sec \theta = 1 \text{ or } 2, \text{ Clearly } \theta = 60^\circ \text{ or } 0^\circ$$

$$\Rightarrow \text{But } \theta \neq 0, \text{ so } \theta = 60^\circ$$

$$\sin \theta + \cot \theta = \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{3}} = \frac{3 + 2}{2\sqrt{3}} = \frac{5\sqrt{3}}{6}$$

Sol.75.(a)

$$(1 + \tan^2 \theta) + (1 + (\tan^2 \theta)^{-1}) = k$$

$$\Rightarrow (1 + \tan^2 \theta) + (1 + \cot^2 \theta) = k$$

$$\Rightarrow \sec^2 \theta + \operatorname{cosec}^2 \theta = k$$

$$\Rightarrow \frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta} = k$$

$$\Rightarrow \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cdot \cos^2 \theta} = k \Rightarrow \sec^2 \theta \cdot \operatorname{cosec}^2 \theta = k$$

$$\Rightarrow \sec \theta \cdot \operatorname{cosec} \theta = \sqrt{k}$$

Short-Trick: Put $\theta = 45^\circ$

$$(1 + \tan^2 \theta) + (1 + (\tan^2 \theta)^{-1}) = k$$

$$\Rightarrow (1 + 1) + (1 + 1) = k$$

$$\Rightarrow 4 = k \Rightarrow 2 = \sqrt{k}$$

Going through options, Option A :

$$\operatorname{cosec} \theta \sec \theta = \sqrt{2} \cdot \sqrt{2} = 2$$

Clearly option (a) is the right answer.

$$\text{Sol.76.(b)} \frac{1}{\sin \theta} - \frac{\cot^2 \theta}{1 + \operatorname{cosec} \theta}$$

$$\Rightarrow \operatorname{cosec} \theta - \frac{(\operatorname{cosec}^2 \theta - 1)}{1 + \operatorname{cosec} \theta}$$

$$\Rightarrow \operatorname{cosec} \theta - \frac{(\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1)}{1 + \operatorname{cosec} \theta}$$

$$\Rightarrow \operatorname{cosec} \theta - \operatorname{cosec} \theta + 1 = 1$$

$$\text{Sol.77.(b)} \left(\frac{1}{1 + \sin^2 \theta} + \frac{1}{1 + \operatorname{cosec}^2 \theta} \right)$$

$$= \left(\frac{1}{1 + \sin^2 \theta} + \frac{1}{1 + \frac{1}{\sin^2 \theta}} \right)$$

$$\Rightarrow \left(\frac{1}{1 + \sin^2 \theta} + \frac{\sin^2 \theta}{1 + \sin^2 \theta} \right) = 1$$

$$\text{Sol.78.(b)} \frac{\cos \theta}{1 - \sin \theta} + \frac{\cos \theta}{1 + \sin \theta} = 4$$

$$= \cos \theta \left(\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} \right) = 4$$

$$= \cos \theta \left(\frac{1 + \sin \theta + 1 - \sin \theta}{1 - \sin^2 \theta} \right) = 4$$

$$\Rightarrow \cos \theta = \frac{1}{2}, \text{ Clearly } \theta = 60^\circ$$

$$\text{So, } \tan \theta + \operatorname{cosec} \theta \Rightarrow \sqrt{3} + \frac{2}{\sqrt{3}}$$

$$= \frac{3 + 2}{\sqrt{3}} = \frac{5\sqrt{3}}{3}$$

Sol.79.(c) Put $\theta = 45^\circ$,

$$\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 1}{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1}$$

$$= \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1$$

Going through the options

$$\sec \theta + \tan \theta = \sqrt{2} + 1$$

Sol.80.(b) Put $\theta = 45^\circ$

$$\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} \times \sqrt{\tan^2 \theta - \sin^2 \theta}$$

$$\Rightarrow \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2} \times \sqrt{1 - \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{2}$$

On checking all the options one by one, option (b) satisfies.

$$\sin \theta \sec^2 \theta \Rightarrow \frac{1}{\sqrt{2}} \times (\sqrt{2})^2 = \sqrt{2}$$

Sol.81.(c) $\sec \theta - \tan \theta = P$

$$\sec \theta + \tan \theta = \frac{1}{P}$$

$$2 \sec \theta = P + \frac{1}{P} \Rightarrow \cos \theta = \frac{2P}{P^2 + 1}$$

$$\cos^2 \theta = \left[\frac{2P}{P^2 + 1} \right]^2$$

$$1 - \sin^2 \theta = \frac{4P^2}{2P^2 + 1 + P^4}$$

$$\sin^2 \theta = 1 - \frac{4P^2}{2P^2 + 1 + P^4}$$

$$= \frac{2P^2 + 1 + P^4 - 4P^2}{2P^2 + 1 + P^4}$$

$$\sin \theta = \frac{1 - P^2}{P^2 + 1}, \operatorname{cosec} \theta = \frac{P^2 + 1}{1 - P^2}$$

Alternate :-

$$\sec \theta - \tan \theta = P \Rightarrow \sec \theta + \tan \theta = \frac{1}{P}$$

$$2 \sec \theta = P + \frac{1}{P}$$

$$\cos \theta = \frac{2P}{P^2 + 1} = \frac{\text{base}}{\text{hypotenuse}}$$

$$\text{Perpendicular} = 1 - P^2$$

$$\operatorname{cosec} \theta = \frac{\text{hypotenuse}}{\text{Perpendicular}} = \frac{P^2 + 1}{1 - P^2}$$

$$\text{Sol.82.(b)} \cos^2 \theta = 3 \cot^2 \theta - 3 \cos^2 \theta$$

$$\Rightarrow 4 \cos^2 \theta = 3 \cot^2 \theta \Rightarrow \sin^2 \theta = \frac{3}{4}$$

$$\Rightarrow \sin \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 60^\circ$$

$$\left(\frac{1}{2} \sec \theta + \sin \theta \right)^{-1} = \frac{1}{\left(\frac{1}{2} \times 2 + \frac{\sqrt{3}}{2} \right)}$$

$$= \frac{2}{\sqrt{3} + 2} = 2(2 - \sqrt{3})$$

$$\text{Sol.83.(b)} \left[\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} \right]^2 + 1$$

$$= \left[\frac{\sin \theta (1 - 2 \sin^2 \theta)}{\cos \theta (2 \cos^2 \theta - 1)} \right]^2 + 1$$

$$= \left[\frac{\sin \theta \cdot \cos 2\theta}{\cos \theta \cdot \cos 2\theta} \right]^2 + 1$$

$$= \tan^2 \theta + 1 = \sec^2 \theta$$

Sol.84.(c) Let $A = 45^\circ$

$$(\operatorname{cosec} A - \sin A)^2 + (\sec A - \cos A)^2 -$$

$$(\cot A - \tan A)^2$$

$$= (\operatorname{cosec} 45^\circ - \sin 45^\circ)^2 + (\sec 45^\circ - \cos 45^\circ)^2 - (\cot 45^\circ - \tan 45^\circ)^2$$

$$= \left(\sqrt{2} - \frac{1}{\sqrt{2}} \right)^2 + \left(\sqrt{2} - \frac{1}{\sqrt{2}} \right)^2 - (1 - 1)^2$$

$$= \frac{1}{2} + \frac{1}{2} - 0 = 1$$

$$\text{Sol.85.(c)} 1 + \frac{\tan^2 A}{1 + \sec A} = 1 + \frac{\sec^2 A - 1}{1 + \sec A}$$

$$= 1 + \sec A - 1 = \sec A$$

Sol.86.(d) Given,

$$(\sin^6 A + \cos^6 A + 3\sin^2 A \cdot \cos^2 A) - (\sin^4 A + \cos^4 A + 1) \operatorname{cosec}^2 A$$

Put $\angle A = 90^\circ$

$$\Rightarrow (1 + 0 + 0) - (1 + 0 + 1)1 = 1 - 2 = -1$$

Sol.87.(d) Given, $3(\sec^2 A - 1)(\operatorname{cosec} A - 1)(\operatorname{cosec} A + 1) + 4(\operatorname{cosec}^2 A - \cot^2 A)$

Using identities : $\sec^2 A - 1 = \tan^2 A$, $\operatorname{cosec}^2 A - \cot^2 A = 1$,

$$a^2 - b^2 = (a - b)(a + b)$$

$$= 3 \tan^2 A (\operatorname{Cosec}^2 A - 1^2) + 4(1)$$

$$= 3 \tan^2 A \cot^2 A + 4 \text{ (as } \operatorname{Cosec}^2 A - 1 = \cot^2 A)$$

$$= 3 \tan^2 A \times \frac{1}{\tan^2 A} + 4 \text{ (as } \cot^2 A = \frac{1}{\tan^2 A})$$

$$= 3 + 4 = 7$$

$$\text{Sol.88.(d)} \sin^2 30^\circ - \sin^2 40^\circ + \sin^2 45^\circ - \sin^2 55^\circ - \sin^2 35^\circ + \sin^2 45^\circ - \sin^2 50^\circ + \sin^2 60^\circ$$

$$= (\sin^2 30^\circ + \sin^2 60^\circ) - (\sin^2 40^\circ + \sin^2 50^\circ)$$

$$+ (\sin^2 45^\circ + \sin^2 45^\circ) - (\sin^2 55^\circ + \sin^2 35^\circ)$$

$$= 1 - 1 + 1 - 1 \text{ (}\because \sin^2 \theta + \sin^2(90^\circ - \theta) = 1)$$

$$= 0$$

$$\text{Sol.89.(c)} \cos^2 29^\circ + \cos^2 61^\circ$$

$$= \cos^2 29^\circ + \cos^2(90^\circ - 29^\circ)$$

$$= \cos^2 29^\circ + \sin^2 29^\circ = 1$$

$$\text{Sol.90.(c) Given : } \operatorname{cosec} \theta = \frac{5}{3}$$

$$\Rightarrow (\sec^2 \theta - 1) \times \cot^2 \theta \times (1 + \cot^2 \theta)$$

We know,

$$\sec^2 \theta = 1 + \tan^2 \theta, \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$$

$$\text{and } \tan^2 \theta \times \cot^2 \theta = 1$$

$$\text{Then, } (1 + \tan^2 \theta - 1) \times \cot^2 \theta \times (\operatorname{cosec}^2 \theta)$$

$$= \tan^2 \theta \times \cot^2 \theta \times \operatorname{cosec}^2 \theta$$

$$= \operatorname{cosec}^2 \theta = \left(\frac{5}{3}\right)^2 = \frac{25}{9}$$

Sol.91.(a) Identity used:

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$$

$$\frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} =$$

$$\frac{(\sin A - \cos A)[\sin^2 A + \cos^2 A + \sin A \cdot \cos A]}{\sin A - \cos A}$$

$$= \frac{(\sin A - \cos A)[1 + \sin A \cdot \cos A]}{\sin A - \cos A}$$

$$= 1 + \sin A \cdot \cos A$$

$$\text{Sol.92.(c)} \frac{\sec \alpha + \tan \alpha}{\sec \alpha - \tan \alpha} = \frac{7}{4}$$

By componendo and dividendo we get;

$$\frac{\sec \alpha}{\tan \alpha} = \frac{11}{3} \Rightarrow \sec \alpha \times \cot \alpha = \frac{11}{3}$$

$$\frac{1}{\cos \alpha} \times \frac{\cos \alpha}{\sin \alpha} = \frac{11}{3}$$

$$\frac{1}{\sin \alpha} = \frac{11}{3} \Rightarrow \operatorname{cosec} \alpha = \frac{11}{3}$$

$$\text{Sol.93.(a)} \frac{\sin A}{1 + \cos A}$$

$$= \frac{\sin A}{1 + \cos A} \times \frac{(1 - \cos A)}{(1 - \cos A)}$$

$$= \frac{\sin A(1 - \cos A)}{1 - \cos^2 A}$$

$$= \frac{\sin A(1 - \cos A)}{\sin^2 A} = \frac{1 - \cos A}{\sin A}$$

$$\text{Sol.94.(b)} \frac{7\sin \theta + 4\cos \theta}{9\sin \theta - 2\cos \theta} = \frac{5}{4}$$

$$28\sin \theta + 16\cos \theta = 45\sin \theta - 10\cos \theta$$

$$17\sin \theta = 26\cos \theta$$

$$\frac{\sin \theta}{\cos \theta} = \frac{26}{17} \Rightarrow \tan \theta = \frac{26}{17}$$

$$\text{Now, } \frac{\tan^2 \theta + 5}{\tan^2 \theta - 5} = \frac{\left(\frac{26}{17}\right)^2 + 5}{\left(\frac{26}{17}\right)^2 - 5}$$

$$= \frac{\frac{676}{289} + 5}{\frac{676}{289} - 5} = \frac{\frac{676 + 1445}{289}}{\frac{676 - 1445}{289}} = \frac{2121}{-769}$$

$$= \frac{2121}{289} \times \frac{289}{-769} = -\frac{2121}{769}$$

Sol.95.(d) Formula used :

$$\sec^2 \theta = 1 + \tan^2 \theta \text{ and } \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$$

now,

$$\frac{2 + \tan^2 2A + \cot^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A}$$

$$= \frac{1 + \tan^2 2A + 1 + \cot^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A}$$

$$= \frac{\sec^2 2A + \operatorname{cosec}^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A}$$

$$= \frac{\sec^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A} + \frac{\operatorname{cosec}^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A}$$

$$= \frac{\sec 2A}{\operatorname{cosec} 2A} + \frac{\operatorname{cosec} 2A}{\sec 2A}$$

$$= \frac{1}{\frac{\cos 2A}{\sin 2A}} + \frac{1}{\frac{\sin 2A}{\cos 2A}} = \frac{\sin 2A}{\cos 2A} + \frac{\cos 2A}{\sin 2A}$$

$$= \frac{\sin^2 2A + \cos^2 2A}{\cos 2A \cdot \sin 2A} = \frac{1}{\cos 2A \cdot \sin 2A}$$

$$= \sec 2A \cdot \operatorname{cosec} 2A$$

Sol.96.(c)

Using the given options, we have ;

$$(a) \text{ when } t = 45^\circ, \sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}},$$

$$\sin(t) \neq \cos(t) \text{ [false]}$$

$$(b) \text{ when } t = 30^\circ, \sin 30^\circ = \frac{1}{2}, \cos 30^\circ$$

$$= \frac{\sqrt{3}}{2}; \sin 30^\circ > \cos 30^\circ \text{ [false]}$$

$$(c) \text{ when } t = 30^\circ, \sin 30^\circ = \frac{1}{2}, \cos 30^\circ$$

$$= \frac{\sqrt{3}}{2}; \sin 30^\circ < \cos 30^\circ \text{ [true]}$$

$$(d) \text{ when } t = 60^\circ, \sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 60^\circ$$

$$= \frac{1}{2}; \sin 60^\circ < \cos 60^\circ \text{ [false]}$$

Clearly, we can see that only option(c) is the correct one.

Sol.97.(d) Given: $2 \tan \theta = 3$

$$\text{then } \frac{\sin \theta}{\cos \theta} = \frac{3}{2} \Rightarrow \sin \theta = \frac{3}{2} \cos \theta$$

Now,

$$\frac{3 \sin \theta - 2 \cos \theta}{3 \sin \theta + 2 \cos \theta}$$

$$= \frac{3 \times \frac{3}{2} \cos \theta - 2 \cos \theta}{3 \times \frac{3}{2} \cos \theta + 2 \cos \theta}$$

$$= \frac{\frac{9 \cos \theta - 4 \cos \theta}{2}}{\frac{9 \cos \theta + 4 \cos \theta}{2}} = \frac{5 \cos \theta}{13 \cos \theta} = \frac{5}{13}$$

Sol.98.(a) Given : $\cot \theta = \frac{4}{3}$

$$= \frac{[\sec \theta(1 + \cot^2 \theta)(\operatorname{cosec}^2 \theta - \cot^2 \theta)]}{\operatorname{cosec}^3 \theta}$$

$$= \frac{\left[\frac{1}{\cos \theta} \times \left(1 + \frac{\cos^2 \theta}{\sin^2 \theta}\right) \left(\frac{1}{\sin^2 \theta} - \frac{\cos^2 \theta}{\sin^2 \theta}\right)\right]}{\left(\frac{1}{\sin \theta}\right)^3}$$

$$= \frac{\left[\frac{1}{\cos \theta} \times \left(\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta}\right) \left(\frac{1 - \cos^2 \theta}{\sin^2 \theta}\right)\right]}{\frac{1}{\sin^3 \theta}}$$

$$\because \sin^2 \theta + \cos^2 \theta = 1, 1 - \cos^2 \theta = \sin^2 \theta$$

$$= \frac{\sin^2 \theta \times \sin^3 \theta}{\cos \theta \times \sin^4 \theta} = \frac{\sin \theta}{\cos \theta}$$

$$= \tan \theta = \frac{1}{\cot \theta} = \frac{3}{4}$$

$$\text{Sol.99.(d)} \frac{3 \sin 58^\circ}{5 \cos 32^\circ} + \frac{2 \cos 62^\circ}{5 \sin 28^\circ}$$

$$\frac{3 \sin(90^\circ - 32^\circ)}{5 \cos 32^\circ} + \frac{2 \cos(90^\circ - 28^\circ)}{5 \sin 28^\circ}$$

$$\frac{3 \cos 32^\circ}{5 \cos 32^\circ} + \frac{2 \sin 28^\circ}{5 \sin 28^\circ} = \frac{3}{5} + \frac{2}{5} = 1$$

Sol.100.(a) $7 \cos^2 \theta + 5 \sin^2 \theta - 6 = 0$

Let $\theta = 45^\circ$

$$7 \times \frac{1}{2} + 5 \times \frac{1}{2} - 6 = \frac{12}{2} - 6 = 6 - 6 = 0$$

Now,

$$1 + \sqrt{\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta}}$$

$$= 1 + \sqrt{\frac{\sec 45^\circ + \tan 45^\circ}{\sec 45^\circ - \tan 45^\circ}}$$

$$= 1 + \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1}}$$

$$\text{Now, } 1 + \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1}} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$$

$$= 1 + \sqrt{2} + 1 = \sqrt{2} + 2$$

$$\text{Sol.101.(a)} \frac{\sec 35^\circ}{\operatorname{cosec} 55^\circ} - \frac{\cos 25^\circ}{\sin 65^\circ}$$

$$\frac{\sec(90^\circ - 55^\circ)}{\operatorname{cosec} 55^\circ} - \frac{\cos(90^\circ - 65^\circ)}{\sin 65^\circ}$$

$$\frac{\operatorname{cosec} 55^\circ}{\operatorname{cosec} 55^\circ} - \frac{\sin 65^\circ}{\sin 65^\circ} = 1 - 1 = 0$$

Sol.102.(d) Concept used :

$$\operatorname{cosec}(90^\circ - \theta) = \sec \theta$$

$$\cos(90^\circ - \theta) = \sin \theta$$

Now,

$$\operatorname{cosec} 56^\circ \cos 34^\circ - \cos 59^\circ \operatorname{cosec} 31^\circ$$

$$\operatorname{cosec}(90^\circ - 34^\circ) \cdot \cos 34^\circ - \cos 59^\circ \operatorname{cosec}(90^\circ - 59^\circ)$$

$$\sec 34^\circ \cdot \cos 34^\circ - \cos 59^\circ \cdot \sec 59^\circ$$

$$1 - 1 = 0$$

$$\text{Sol.103.(a)} \left[\frac{4}{3} \tan^2 60^\circ + 3 \cos^2 30^\circ - 2 \sec^2 30^\circ - \frac{3}{4} \cot^2 60^\circ \right] \div [\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ]$$

$$\left[\frac{4}{3} \times (\sqrt{3})^2 + 3 \times \left(\frac{\sqrt{3}}{2}\right)^2 - 2 \times \left(\frac{2}{\sqrt{3}}\right)^2 - \frac{3}{4} \times \left(\frac{1}{\sqrt{3}}\right)^2 \right] \div \sin(60^\circ - 30^\circ)$$

$$\left[4 + \frac{9}{4} - \frac{8}{3} - \frac{1}{4} \right] \div \sin 30^\circ$$

$$\frac{10}{3} \div \frac{1}{2} = \frac{20}{3} = 6\frac{2}{3}$$

$$\text{Sol.104.(d)} \sin(90^\circ - \theta) = \cos \theta$$

$$\tan(90^\circ - \theta) = \cot \theta$$

$$\text{Now, } \sin 74^\circ + \tan 74^\circ$$

$$\sin(90^\circ - 16^\circ) + \tan(90^\circ - 16^\circ)$$

$$\cos 16^\circ + \cot 16^\circ$$

$$\text{Sol.105.(a)} p \sin A - \cos A = 1$$

Put $A = 90^\circ$, we have ;

$$p - 0 = 1 \Rightarrow p = 1$$

$$\text{Now, } p^2 - (1 + p^2) \cos A = 1 - (1 + 1) \cos 90^\circ = 1 - 0 = 1$$

Sol.106.(d)

$$\left(\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} \right)^3 \frac{1}{\tan \theta} - \sec^2 \theta$$

$$= \left(\frac{\sin \theta (1 - 2 \sin^2 \theta)}{\cos \theta (2 \cos^2 \theta - 1)} \right)^3 \frac{1}{\tan \theta} - \sec^2 \theta$$

$$[\because \cos 2\theta = (2 \cos^2 \theta - 1) = (1 - 2 \sin^2 \theta)]$$

$$= \left(\frac{\sin \theta \cdot \cos 2\theta}{\cos \theta \cdot \cos 2\theta} \right)^3 \times \frac{1}{\tan \theta} - \sec^2 \theta$$

$$= \left(\frac{\sin \theta}{\cos \theta} \right)^3 \times \frac{1}{\tan \theta} - \sec^2 \theta$$

$$= \tan^3 \theta \times \frac{1}{\tan \theta} - \sec^2 \theta$$

$$= \tan^2 \theta - \sec^2 \theta = -1$$

$$\text{Sol.107.(c)} 2 \operatorname{cosec}^2 \theta + 3 \cot^2 \theta = 17$$

By putting the options one by one we get, $\theta = 30^\circ$ satisfies the equation:

$$2 \operatorname{cosec}^2 \theta + 3 \cot^2 \theta = 2 \operatorname{cosec}^2 30^\circ + 3 \cot^2 30^\circ = 2 \times 4 + 3 \times 3 = 8 + 9 = 17$$

$$\text{Sol.108.(a)} \frac{1}{1 + \cos \theta} + \frac{1}{1 - \cos \theta}$$

$$\Rightarrow \frac{1 - \cos \theta + 1 + \cos \theta}{1 - \cos^2 \theta}$$

$$\Rightarrow \frac{2}{\sin^2 \theta} = 2 \operatorname{cosec}^2 \theta$$

$$\text{Sol.109.(c)} \frac{7 \sin \theta + 5 \cos \theta}{7 \sin \theta - 2 \cos \theta}$$

$$\Rightarrow \frac{7 \tan \theta + 5}{7 \tan \theta - 2} = \frac{7 \times \frac{3}{7} + 5}{7 \times \frac{3}{7} - 2}$$

$$= \frac{3 + 5}{3 - 2} = 8$$

$$\text{Sol.110.(b)} \cot^4 \phi - \cot^2 \phi = 1$$

$$\Rightarrow \cot^4 \phi = 1 + \cot^2 \phi$$

$$\Rightarrow \cot^4 \phi = \operatorname{cosec}^2 \phi \cdot (\operatorname{cosec}^2 \phi = 1 + \cot^2 \phi)$$

$$\Rightarrow 111 = \frac{1}{\sin^2 \phi}$$

$$\Rightarrow \cos^4 \phi = \sin^2 \phi \dots (i)$$

Now,

$$\cos^4 \phi + \cos^2 \phi = \sin^2 \phi + \cos^2 \phi = 1$$

$$\text{Sol.111.(d) Given:- } \tan 4\theta \cdot \tan 6\theta = 1$$

$$\Rightarrow \tan 4\theta = \frac{1}{\tan 6\theta}$$

$$\Rightarrow \tan 4\theta = \cot 6\theta$$

$$\Rightarrow \tan 4\theta = \tan(90^\circ - 6\theta)$$

$$\Rightarrow 4\theta = 90^\circ - 6\theta \Rightarrow 10\theta = 90^\circ \Rightarrow \theta = 9^\circ$$

Put the value of θ in $\cot 5\theta$

$$\cot 45^\circ = 1$$

$$\text{Sol.112.(c) Given, } 7 \cos A = 6, \cos A = \frac{6}{7}$$

$$\text{Here } B = 6, H = 7, P = \sqrt{49 - 36} = \sqrt{13}$$

$$\text{Now, } \frac{\operatorname{cosec} A + \cot A}{\operatorname{cosec} A - \cot A} = \frac{\frac{7}{\sqrt{13}} + \frac{6}{\sqrt{13}}}{\frac{7}{\sqrt{13}} - \frac{6}{\sqrt{13}}}$$

$$= \frac{13}{1} = 13$$

Sol.113.(c)

$$p(\sqrt{3} + \cot 30^\circ) = \tan^3 60^\circ - 2 \sin 60^\circ$$

$$\text{Put the value of } \cot 30^\circ = \sqrt{3}, \tan 60^\circ$$

$$= \sqrt{3} \text{ and } \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\Rightarrow p(\sqrt{3} + \sqrt{3}) = (\sqrt{3})^3 - 2 \times \frac{\sqrt{3}}{2}$$

$$\Rightarrow 2\sqrt{3}p = 3\sqrt{3} - \sqrt{3} \Rightarrow 2\sqrt{3}p = 2\sqrt{3} \Rightarrow p = 1$$

Sol.114.(a)

$$\Rightarrow \frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} = 4 \sec \theta$$

$$\Rightarrow \frac{1 + \sin \theta + 1 - \sin \theta}{1 - \sin^2 \theta} = \frac{4}{\cos \theta}$$

$$\Rightarrow \frac{2}{\cos^2 \theta} = \frac{4}{\cos \theta}$$

$$\Rightarrow \cos \theta = \frac{1}{2} = \cos 60^\circ \Rightarrow \theta = 60^\circ$$

Now,

$$(\cot \theta + \operatorname{cosec} \theta)$$

$$= \cot 60^\circ + \operatorname{cosec} 60^\circ$$

$$= \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} = \frac{3}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \sqrt{3}$$

$$\text{Sol.115.(b)} \sin 31^\circ = \alpha$$

$$\sin(90^\circ - 59^\circ) = \alpha$$

$$\cos 59^\circ = \alpha$$

$$\sin 59^\circ = \sqrt{1 - \cos^2 59^\circ} = \sqrt{1 - \alpha^2}$$

Now,

$$\cot 59^\circ = \frac{\cos 59^\circ}{\sin 59^\circ} = \frac{\alpha}{\sqrt{1 - \alpha^2}}$$

$$\text{Sol.116.(b)} \frac{\tan(x)}{\sec(x)} = \frac{1}{2}$$

$$\frac{\frac{\sin x}{\cos x}}{\frac{1}{\cos x}} = \frac{1}{2} \Rightarrow \sin x = \frac{1}{2}$$

$$\sin x = \sin 30^\circ \Rightarrow x = 30^\circ$$

Now,

$$(\sin x + \cos x)^2$$

$$\sin^2 x + \cos^2 x + 2 \sin x \cdot \cos x$$

$$1 + 2 \sin x \cdot \cos x$$

$$1 + 2 \times \sin 30^\circ \times \cos 30^\circ$$

$$1 + 2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2} \Rightarrow \frac{2 + \sqrt{3}}{2}$$

$$\text{Sol.117.(a)} \tan A = \frac{5}{7}$$

$$\text{Now, } \frac{7 \sin A - 3 \cos A}{7 \sin A + 3 \cos A} + 4$$

On dividing by $\cos A$ in fraction we get,

$$\frac{7 \tan A - 3}{7 \tan A + 3} + 4 = \frac{7 \times \frac{5}{7} - 3}{7 \times \frac{5}{7} + 3} + 4$$

$$= \frac{5 - 3}{5 + 3} + 4 = \frac{2}{8} + 4 = 4\frac{1}{4}$$

$$\text{Sol.118.(d)} \sin B = \frac{15}{17}$$

$$\text{now, } \cos B (\sec B - \tan B)$$

$$\cos B \times \frac{1}{\cos B} - \cos B \times \frac{\sin B}{\cos B}$$

$$= 1 - \sin B = 1 - \frac{15}{17} = \frac{2}{17}$$

$$\text{Sol.119.(b)} \left(\frac{\operatorname{cosec} A}{\operatorname{cosec} A - 1} + \frac{\operatorname{cosec} A}{\operatorname{cosec} A + 1} \right)$$

$$\left(\frac{\sec A}{\sec A - 1} + \frac{\sec A}{\sec A + 1} \right)^{-1}$$

$$= \frac{\frac{\operatorname{cosec}^2 A + \operatorname{cosec} A + \operatorname{cosec}^2 A - \operatorname{cosec} A}{\operatorname{cosec}^2 A - 1^2}}{\frac{\sec^2 A + \sec A + \sec^2 A - \sec A}{\sec^2 A - 1^2}}$$

$$= \frac{2 \operatorname{cosec}^2 A}{\operatorname{cosec}^2 A - 1} \times \frac{\sec^2 A - 1}{2 \sec^2 A}$$

$$= \frac{\operatorname{cosec}^2 A}{\cot^2 A} \times \frac{\tan^2 A}{\sec^2 A}$$

$$\text{Now, } A = 60^\circ$$

$$= \frac{\frac{4}{3}}{\frac{1}{3}} \times \frac{3}{4} = 4 \times \frac{3}{4} = 3$$

$$\text{Sol.120.(b)} \text{ In right } \triangle ABC, \angle B = 90^\circ$$

$$\Rightarrow \tan A = 1 \Rightarrow \tan A = \tan 45^\circ \Rightarrow A = 45^\circ$$

Now,

$$4 \sin A \cdot \cos A$$

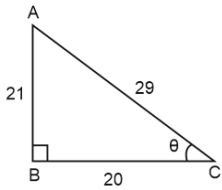
$$= 4 \times \sin 45^\circ \times \cos 45^\circ$$

$$= 4 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{4}{2} = 2$$

$$\text{Sol.121.(b) Given :}$$

$$\sec \theta = \frac{29}{20} = \frac{\text{hypotenuse}}{\text{base}}$$

By pythagorean triplet (20, 21, 29)



Then,

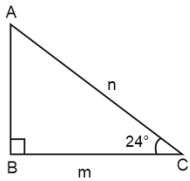
$$3 \operatorname{cosec} \theta + 3 \cot \theta$$

$$= 3 \times \frac{29}{21} + 3 \times \frac{20}{21}$$

$$= \frac{87 + 60}{21} = \frac{147}{21} = 7$$

Sol.122.(d) Given: $\cos 24^\circ = \frac{m}{n}$

$$= \frac{\text{base}}{\text{hypotenuse}}$$



By using pythagoras theorem,

$$AB^2 = AC^2 - BC^2 \Rightarrow AB^2 = n^2 - m^2$$

$$AB = \sqrt{n^2 - m^2}$$

Now,

$$(\operatorname{cosec} 24^\circ - \cos 66^\circ)$$

$$= \operatorname{cosec} 24^\circ - \cos (90^\circ - 24^\circ)$$

$$= \operatorname{cosec} 24^\circ - \sin 24^\circ$$

$$= \frac{\text{hypotenuse}}{\text{perpendicular}} - \frac{\text{perpendicular}}{\text{hypotenuse}}$$

$$= \frac{n}{\sqrt{n^2 - m^2}} - \frac{\sqrt{n^2 - m^2}}{n}$$

$$= \frac{n^2 - n^2 + m^2}{n\sqrt{n^2 - m^2}} = \frac{m^2}{n\sqrt{n^2 - m^2}}$$

Sol.123.(a)

$$\left[\frac{4 \cos(90^\circ - \theta) \sin^3(90^\circ + \theta) - 4 \sin(90^\circ + \theta) \cos^3(90^\circ - \theta)}{\cos\left(\frac{180^\circ + 8\theta}{2}\right)} \right]$$

$$= \frac{4 \sin \theta \times \cos^3 \theta - 4 \cos \theta \times \sin^3 \theta}{- \sin 4\theta}$$

$$= \frac{4 \sin \theta \cos \theta (\cos^2 \theta - \sin^2 \theta)}{- \sin 2 \times (2\theta)}$$

$$= \frac{4 \sin \theta \cos \theta (\cos^2 \theta - \sin^2 \theta)}{- 2 \sin 2\theta \cos 2\theta}$$

$$(\because \cos 2\theta = \cos^2 \theta - \sin^2 \theta)$$

$$= \frac{2 \sin \theta \cos \theta}{- \sin 2\theta} = \frac{2 \sin \theta \cos \theta}{- 2 \sin \theta \cos \theta}$$

$$= -1 (\because \sin 2\theta = 2 \sin \theta \cos \theta)$$

Sol.124.(a) $\sin A + \cos A = \frac{1}{2\sqrt{2}}$

Squaring both side, we get

$$\Rightarrow \sin^2 A + \cos^2 A + 2 \sin A \cos A = \frac{1}{8}$$

$$\Rightarrow 2 \sin A \cos A = \frac{1}{8} - 1 = \frac{-7}{8}$$

$$\Rightarrow \sin A \cos A = \frac{-7}{16}$$

Now, we know that

$$\sin^2 A + \cos^2 A = 1$$

Again Squaring both side, we get

$$\Rightarrow \sin^4 A + \cos^4 A + 2 \sin^2 A \cos^2 A = 1$$

$$\Rightarrow \sin^4 A + \cos^4 A + 2 \times \left(\frac{-49}{256}\right) = 1$$

$$\Rightarrow \sin^4 A + \cos^4 A = 1 - \frac{49}{128} = \frac{79}{128}$$

Sol.125.(c) if

$$a = x \cos \theta + y \sin \theta \text{ and } b = x \sin \theta - y \cos \theta$$

Then $a^2 + b^2 = x^2 + y^2$ (by using direct result)

Sol.126.(b) Given, $\operatorname{cosec} x$

$$= \frac{\text{hypotenuse}}{\text{perpendicular}} = \frac{5}{4}$$

By using pythagorean triplet (4, 3, 5),
base = 3

Now,

$$\frac{5 \sin^2 x + 8 \cot^2 x}{27 \tan^2 x - 125 \cos^2 x}$$

$$= \frac{5 \times \left(\frac{4}{5}\right)^2 + 8 \times \left(\frac{3}{4}\right)^2}{27 \times \left(\frac{4}{3}\right)^2 - 125 \times \left(\frac{3}{5}\right)^2}$$

$$= \frac{5 \times \frac{16}{25} + 8 \times \frac{9}{16}}{27 \times \frac{16}{9} - 125 \times \frac{9}{25}}$$

$$= \frac{\frac{16}{5} + \frac{9}{2}}{48 - 45} = \frac{32 + 45}{3 \times 10} = \frac{77}{30}$$

Sol.127.(b) $\sin \theta + \cos \theta = \sqrt{3} \cos \theta$

$$\Rightarrow \sin \theta = \sqrt{3} \cos \theta - \cos \theta$$

$$\Rightarrow \sin \theta = \cos \theta (\sqrt{3} - 1)$$

$$\Rightarrow \cot \theta = \frac{1}{\sqrt{3} - 1}$$

$$\Rightarrow \cot \theta = \frac{1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = \frac{(\sqrt{3} + 1)}{2}$$

Sol.128.(a) $\sec(t) = \frac{x}{y} = \frac{\text{Hypotenuse}}{\text{Base}}$

$$\text{Perpendicular} = \sqrt{x^2 - y^2}$$

$$\cot(t) = \frac{\text{Base}}{\text{Perpendicular}} = \frac{y}{\sqrt{x^2 - y^2}}$$

Sol.129.(a) As we know that, $\sin 45^\circ$ and

$\cos 45^\circ$ have the same value i.e. $\frac{1}{\sqrt{2}}$.

Sol.130.(a) Given: $3 \sin A + 4 \cos A = 5$

Squaring both side, we have

$$\Rightarrow (3 \sin A + 4 \cos A)^2 = 5^2$$

$$\Rightarrow 9 \sin^2 A + 16 \cos^2 A + 24 \sin A \cos A = 25$$

$$\Rightarrow 9(1 - \cos^2 A) + 16(1 - \sin^2 A) + 24 \sin A \cos A = 25$$

$$\Rightarrow 16 \sin^2 A + 9 \cos^2 A - 24 \sin A \cos A = 0$$

$$\Rightarrow (4 \sin A - 3 \cos A)^2 = 0$$

$$\Rightarrow 4 \sin A = 3 \cos A \Rightarrow \tan A = \frac{3}{4}$$

Short trick: Given: $3 \sin A + 4 \cos A = 5$

By using pythagorean triplet = (3, 4, 5)

$$\tan A = \frac{\text{Perpendicular}}{\text{Base}} = \frac{3}{4}$$

Sol.131.(b) $\frac{\tan 55^\circ}{\cot 35^\circ}$

$$= \frac{\tan 55^\circ}{\cot (90^\circ - 55^\circ)} = \frac{\tan 55^\circ}{\tan 55^\circ} = 1$$

Sol.132.(d) $\frac{\sin A}{1 + \cos A} + \cot A$

$$= \frac{\sin A}{1 + \cos A} + \frac{\cos A}{\sin A}$$

$$= \frac{\sin^2 A + \cos A + \cos^2 A}{(1 + \cos A) \cdot \sin A}$$

$$= \frac{(1 + \cos A)}{(1 + \cos A) \cdot \sin A} = \frac{1}{\sin A} = \operatorname{cosec} A$$

Sol.133.(b) $\tan \theta = \frac{8}{15} = \frac{\text{perpendicular}}{\text{base}}$

By pythagorean triplet = (8, 15, 17)

Now,

$$\frac{\sqrt{1 - \sin \theta}}{\sqrt{1 + \sin \theta}} = \frac{\sqrt{1 - \frac{8}{17}}}{\sqrt{1 + \frac{8}{17}}} = \frac{\sqrt{\frac{17 - 8}{17}}}{\sqrt{\frac{17 + 8}{17}}}$$

$$= \sqrt{\frac{9}{25}} = \frac{3}{5}$$

Sol.134.(b) $\sqrt{\frac{\sec \theta + 1}{\sec \theta - 1}} + \sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}}$

$$= \sqrt{\frac{\sec \theta + 1}{\sec \theta - 1}} \times \frac{\sec \theta + 1}{\sec \theta + 1} +$$

$$\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} \times \frac{\sec \theta - 1}{\sec \theta - 1}$$

$$= \frac{\sec \theta + 1}{\tan \theta} + \frac{\sec \theta - 1}{\tan \theta}$$

$$= \frac{2 \sec \theta}{\tan \theta} = 2 \operatorname{cosec} \theta$$

Sol.135.(a) $\sqrt{\frac{9 \sin^2 26^\circ + 9 \sin^2 64^\circ + 36}{32 - 7 \cos^2 32^\circ - 7 \cos^2 58^\circ}}$

$$= \sqrt{\frac{9 \cos^2 64^\circ + 9 \sin^2 64^\circ + 36}{32 - (7 \sin^2 58^\circ + 7 \cos^2 58^\circ)}}$$

$$= \sqrt{\frac{9 + 36}{32 - 7}} = \sqrt{\frac{45}{25}} = \frac{3\sqrt{5}}{5}$$

Sol.136.(c) $\sin x + \operatorname{cosec} x = 2$

$$\text{Put } x = 90^\circ \Rightarrow 1 + 1 = 2$$

$$\Rightarrow 2 = 2 \text{ (LHS = RHS)}$$

$$\text{Now, } \sin^7 x + \operatorname{cosec}^7 x = 1^7 + 1^7 = 2$$

Sol.137.(d) $\sin A + \cos A = \sqrt{3}$

Squaring on both side, we get

$$\Rightarrow 1 + 2 \sin A \cos A = 3$$

$$\Rightarrow \sin A \cos A = 1$$

$$\tan A + \cot A + 2 \sin A \cos A$$

$$= \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} + 2 \sin A \cos A$$

$$= \frac{\sin^2 A + \cos^2 A}{\cos A \sin A} + 2 \sin A \cos A$$

$$= 1 + 2 = 3$$

Sol.138.(d) $3\sin^2\theta + 5\cos^2\theta = 4$
 $\Rightarrow 3(\sin^2\theta + \cos^2\theta) + 2\cos^2\theta = 4$
 $\Rightarrow 2\cos^2\theta = 1 \Rightarrow \cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$
 $\therefore \tan\theta = \tan 45^\circ = 1$

Sol.139.(d) Maximum value of $a \sin x + b \cos x = \sqrt{a^2 + b^2}$
 maximum value of $(2 \sin \theta + 3 \cos \theta) = \sqrt{4 + 9} = \sqrt{13}$

Sol.140.(c) $[1 - \sin^2 32^\circ + \frac{1}{1 + \tan^2 58^\circ}]$
 $= \cos^2 32^\circ + \frac{1}{\sec^2 58^\circ}$
 $[\because 1 - \sin^2 A = \cos^2 A, 1 + \tan^2 A = \sec^2 A]$
 $= \cos^2 32^\circ + \cos^2 58^\circ$
 $= \cos^2 32^\circ + \sin^2 32^\circ = 1$

Sol.141.(c) Given, $\tan \theta = \frac{5}{8}$
 $\frac{(1 + \cos \theta)(1 - \cos \theta)}{(1 + \sin \theta)(1 - \sin \theta)}$
 $\frac{1 - \cos^2 \theta}{1 - \sin^2 \theta} = \frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta = (\frac{5}{8})^2 = \frac{25}{64}$

Sol.142.(d) $\tan \theta + \cot \theta = 2$
 Taking $\theta = 45^\circ$ then satisfies the equation,
 LHS = $\tan 45^\circ + \cot 45^\circ = 1 + 1 = 2 = \text{RHS}$
 So, the value of θ will be 45°

Sol. 143.(a) $\sqrt{\frac{1 - \cos A}{1 + \cos A}} + \sqrt{\frac{1 + \cos A}{1 - \cos A}}$
 $= \sqrt{\frac{1 - \cos A}{1 + \cos A}} \times \frac{1 - \cos A}{1 - \cos A} + \sqrt{\frac{1 + \cos A}{1 - \cos A}} \times \frac{1 + \cos A}{1 + \cos A}$
 $= \sqrt{\frac{(1 - \cos A)^2}{1 - \cos^2 A}} + \sqrt{\frac{(1 + \cos A)^2}{1 - \cos^2 A}}$
 $= \sqrt{\frac{(1 - \cos A)^2}{\sin^2 A}} + \sqrt{\frac{(1 + \cos A)^2}{\sin^2 A}}$
 $= \frac{1 - \cos A}{\sin A} + \frac{1 + \cos A}{\sin A}$
 $= \frac{1 - \cos A + 1 + \cos A}{\sin A} = 2 \operatorname{cosec} A$

Sol.144.(c) $\sqrt{3} \cos \theta + \sin \theta = 1$
 Putting value of $\theta = 90^\circ$ satisfies the given equation,
 LHS = $\sqrt{3} \cos \theta + \sin \theta$
 $= \sqrt{3} \times \cos 90^\circ + \sin 90^\circ$
 $= \sqrt{3} \times 0 + 1 = 1 = \text{RHS}$

Sol.145.(b) According to question,
 $x^2 + y^2 + z^2$
 $= (r \cos t)^2 + (r \sin t \cdot \cos u)^2 + (r \sin t \cdot \sin u)^2$
 $= r^2 \cos^2 t + r^2 \cos^2 u \cdot \sin^2 t + r^2 \sin^2 t \cdot \sin^2 u$
 $= r^2 \cos^2 t + r^2 \sin^2 t (\cos^2 u + \sin^2 u)$
 $= r^2 (\cos^2 t + \sin^2 t) = r^2$

Sol.146.(d) $4 \tan \theta - 3 = 0$

$\Rightarrow 4 \tan \theta = 3 \Rightarrow \tan \theta = \frac{3}{4}$

Now,

$\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \frac{1 - 1 + 2\sin^2 \theta}{1 + 2\cos^2 \theta - 1} = \frac{2\sin^2 \theta}{2\cos^2 \theta}$
 $= \tan^2 \theta = (\frac{3}{4})^2 = \frac{9}{16}$

Sol.147.(c) $\sin A - \cos A = \frac{\sqrt{3} - \sqrt{2}}{2}$

Taking square both side,

$(\sin A - \cos A)^2 = (\frac{\sqrt{3} - \sqrt{2}}{2})^2$
 $\sin^2 A + \cos^2 A - 2\sin A \cdot \cos A = \frac{3 + 2 - 2\sqrt{6}}{4}$

$1 - 2\sin A \cdot \cos A = \frac{5 - 2\sqrt{6}}{4}$

$2\sin A \cdot \cos A = 1 - \frac{5 - 2\sqrt{6}}{4}$

$2\sin A \cdot \cos A = \frac{4 - 5 + 2\sqrt{6}}{4}$

$\sin A \cdot \cos A = \frac{-1 + 2\sqrt{6}}{8} = \frac{2\sqrt{6} - 1}{8}$

Sol.148.(d) $\cos A + \cos^2 A = 1$

$\Rightarrow \cos A = 1 - \cos^2 A \Rightarrow \cos A = \sin^2 A$

Then,

$\sin^2 A + \sin^4 A = \sin^2 A + \cos^2 A = 1$

Sol.149.(c) $\frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\sin \theta}$

$= \frac{1}{\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta}} - \frac{1}{\sin \theta}$
 $= \frac{\sin \theta}{1 - \cos \theta} - \frac{1}{\sin \theta} = \frac{\sin^2 \theta - 1 + \cos \theta}{(1 - \cos \theta) \sin \theta}$
 $= \frac{-(1 - \sin^2 \theta) + \cos \theta}{(1 - \cos \theta) \sin \theta} = \frac{-\cos^2 \theta + \cos \theta}{(1 - \cos \theta) \sin \theta}$
 $= \frac{\cos \theta}{\sin \theta} = \cot \theta$

Sol.150.(b) $\frac{\sin^2 \theta - 2 \sin^4 \theta}{2 \cos^4 \theta + \cos^2 \theta}$

$= \frac{(\sin 45^\circ)^2 - 2(\sin 45^\circ)^4}{2(\cos 45^\circ)^4 + (\cos 45^\circ)^2}$
 $= \frac{(\frac{1}{\sqrt{2}})^2 - 2(\frac{1}{\sqrt{2}})^4}{2(\frac{1}{\sqrt{2}})^4 + (\frac{1}{\sqrt{2}})^2} = \frac{\frac{1}{2} - 2 \times \frac{1}{4}}{2 \times \frac{1}{4} + \frac{1}{2}}$
 $= \frac{0}{1} = 0$

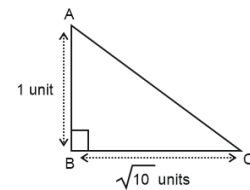
Sol.151.(b) According to the question,

$\sin^2 t = \frac{1}{2} \times \tan^2 t$

$\sin^2 t = \frac{1}{2} \times \frac{\sin^2 t}{\cos^2 t} \Rightarrow \cos^2 t = \frac{1}{2}$

$\cos t = \frac{1}{\sqrt{2}} = \cos 45^\circ \Rightarrow t = 45^\circ$

Sol.152.(c) $\tan A = \frac{1}{\sqrt{10}}$



By using pythagoras theorem,

$AB^2 + BC^2 = AC^2$

$AC^2 = 1^2 + (\sqrt{10})^2 \Rightarrow AC^2 = 1 + 10$

$AC = \sqrt{11}$ units

Now,

$\sin A + \operatorname{cosec} A$

$\frac{1}{\sqrt{11}} + \frac{\sqrt{11}}{1} \Rightarrow \frac{1 + 11}{\sqrt{11}} = \frac{12}{\sqrt{11}}$

Sol.153.(b) $\sec \theta = \sqrt{2}$

$\Rightarrow \sec \theta = \sec 45^\circ \Rightarrow \theta = 45^\circ$

$\therefore \operatorname{cosec} 45^\circ = \sqrt{2}$

Sol.154.(b) $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$

$\Rightarrow 2 \sin^2 \theta + 3 \sin \theta - 2 = 0$

if $\theta = 30^\circ$

$\Rightarrow 2 \times \frac{1}{4} + 3 \times \frac{1}{2} - 2 = 2 \Rightarrow 2 = 2$

$\therefore \theta = 30^\circ$

Sol.155.(a) We know that

$\sin \theta = \cos(90^\circ - \theta)$

$\sin 3A = \cos(90^\circ - 3A)$

$\Rightarrow \cos(A - 26^\circ) = \cos(90^\circ - 3A)$

$\Rightarrow A - 26^\circ = 90^\circ - 3A \Rightarrow 4A = 116^\circ$

$\Rightarrow A = 29^\circ$

Sol.156.(b)

$(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2$

By value putting method, Let $A = 45^\circ$

$= (\frac{1}{\sqrt{2}} + \sqrt{2})^2 + (\frac{1}{\sqrt{2}} + \sqrt{2})^2$

$= 2(\frac{1}{\sqrt{2}} + \sqrt{2})^2 = 2 \times \frac{9}{2} = 9$

By checking options one by one, Option (b) is satisfy

i.e. $7 + \tan^2 A + \cot^2 A = 7 + 1 + 1 = 9$

Sol.157.(d) Given : $3 \tan A = 4$

$\tan A = \frac{4}{3} = \frac{P}{B}$

$\therefore H = 5$ (3, 4, 5 is a triplet)

Now, $4 \sin A + 3 \cos A = 4 \times \frac{4}{5} + 3 \times \frac{3}{5}$

$= \frac{16}{5} + \frac{9}{5} = 5$

Sol.158.(c) $10 \sin^2 \theta + 6 \cos^2 \theta = 7$

$4 \sin^2 \theta + 6 \sin^2 \theta + 6 \cos^2 \theta = 7$

$4 \sin^2 \theta + 6(\sin^2 \theta + \cos^2 \theta) = 7$

$4 \sin^2 \theta = 1$

$\sin^2 \theta = \frac{1}{4} \Rightarrow \sin \theta = \frac{1}{2}$

$\sin \theta = \sin 30^\circ \dots (\sin 30^\circ = \frac{1}{2}) \Rightarrow \theta = 30^\circ$

Now, $\tan 30^\circ = \frac{1}{\sqrt{3}}$

$$\text{Sol.159.(c)} \frac{11\sqrt{2} \cos 3A + 10 \sin 2A}{7\sqrt{3} \sin 4A}$$

$$\text{Put } A = 15^\circ$$

$$\frac{11\sqrt{2} \cos 3 \times 15^\circ + 10 \sin 2 \times 15^\circ}{7\sqrt{3} \sin 4 \times 15^\circ}$$

$$\frac{11\sqrt{2} \cos 45^\circ + 10 \sin 30^\circ}{7\sqrt{3} \sin 60^\circ}$$

$$\frac{11\sqrt{2} \times \frac{1}{\sqrt{2}} + 10 \times \frac{1}{2}}{7\sqrt{3} \times \frac{\sqrt{3}}{2}} \Rightarrow \frac{11 + 5}{\frac{21}{2}}$$

$$= \frac{32}{21}$$

$$\text{Sol.160.(b)} 2 \frac{\tan 54^\circ}{\cot 36^\circ} - \frac{\cot 41^\circ}{\tan 49^\circ}$$

$$2 \frac{\tan 54^\circ}{\cot 36^\circ} - \frac{\cot 41^\circ}{\tan 49^\circ} = 2 - 1 = 1$$

$$\text{Sol.161.(c)} (\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)$$

$$(\tan \theta + \cot \theta)$$

$$\left(\frac{1}{\sin \theta} - \sin \theta\right) \left(\frac{1}{\cos \theta} - \cos \theta\right) \left(\frac{1}{\cot \theta} + \cot \theta\right)$$

$$\left(\frac{1 - \sin^2 \theta}{\sin \theta}\right) \left(\frac{1 - \cos^2 \theta}{\cos \theta}\right) \left(\frac{1 + \cot^2 \theta}{\cot \theta}\right)$$

$$\left(\frac{\cos^2 \theta}{\sin \theta}\right) \left(\frac{\sin^2 \theta}{\cos \theta}\right) \left(\frac{\operatorname{cosec}^2 \theta}{\cot \theta}\right)$$

$$(\cos \theta)(\sin \theta) \left(\frac{\frac{1}{\sin^2 \theta}}{\frac{\cos \theta}{\sin \theta}}\right)$$

$$(\cos \theta)(\sin \theta) \left(\frac{1}{\sin \theta \cos \theta}\right) = 1$$

$$\text{Sol.162.(c)} \Rightarrow m + 5$$

$$= \sec^2 A \cdot \operatorname{cosec}^2 A (1 - \cos^2 A)(1 - \sin^2 A)$$

$$+ 5$$

$$\Rightarrow m + 5 = \sec^2 A \cdot \operatorname{cosec}^2 A \cdot \sin^2 A \cdot \cos^2 A$$

$$+ 5$$

$$\Rightarrow m + 5 = \frac{1}{\cos^2 A} \cdot \frac{1}{\sin^2 A} \cdot \sin^2 A \cdot \cos^2 A +$$

$$5 \Rightarrow m = 1$$

$$\text{Sol.163.(a)} \cot^2 \theta - 2 \cos^2 \theta = 0$$

$$\cot^2 \theta = 2 \cos^2 \theta \Rightarrow \frac{\cos^2 \theta}{\sin^2 \theta} = 2 \cos^2 \theta$$

$$\sin^2 \theta = \frac{1}{2} \Rightarrow \sin \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

$$\text{Sol.164.(a)} x = p \sec A \cos B$$

$$\frac{x}{p} = \sec A \cos B \dots (i)$$

$$y = q \sec A \sin B$$

$$\frac{y}{q} = \sec A \sin B \dots (ii)$$

$$z = r \tan A$$

$$\frac{z}{r} = \frac{\sin A}{\cos A} \dots (iii)$$

On squaring eq. (i), (ii) and (iii) and then adding (i), (ii) and subtracting (iii) we get,

$$\frac{x^2}{p^2} + \frac{y^2}{q^2} - \frac{z^2}{r^2}$$

$$= \sec^2 A \cos^2 B + \sec^2 A \sin^2 B - \frac{\sin^2 A}{\cos^2 A}$$

$$= \sec^2 A (\cos^2 B + \sin^2 B) - \frac{\sin^2 A}{\cos^2 A}$$

$$= \frac{1}{\cos^2 A} - \frac{\sin^2 A}{\cos^2 A} = \frac{1 - \sin^2 A}{\cos^2 A} = \frac{\cos^2 A}{\cos^2 A} = 1$$

Short trick: Put $A = B = 0^\circ$

then

$$x = p \sec A \cos B$$

$$\frac{x}{p} = 1 \dots (i)$$

$$y = q \sec A \sin B$$

$$\frac{y}{q} = 0 \dots (ii)$$

$$z = r \tan A$$

$$\frac{z}{r} = 0 \dots (iii)$$

hence,

$$\frac{x}{p} + \frac{y}{q} - \frac{z}{r} = 1 + 0 - 0 = 1$$

$$\text{Sol.165.(c)} 3 \sin 15^\circ - 4 \sin^3 15^\circ$$

$$= \sin 3A = \sin 3 \times 15^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\text{Sol.166.(c)} \sec \theta + \tan \theta = x \dots (i)$$

$$\text{and, } \sec \theta - \tan \theta = \frac{1}{x} \dots (ii)$$

On adding (i) and (ii) we get,

$$2 \sec \theta = x + \frac{1}{x}$$

$$\sec \theta = \frac{x^2 + 1}{2x} = \frac{\text{hypotenuse}}{\text{base}}$$

$$\text{then, perpendicular} = x^2 - 1$$

$$\sin \theta = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{x^2 - 1}{x^2 + 1}$$

$$\text{Sol.167.(c)} \sin(48^\circ + k) = \cos 13^\circ$$

$$\sin(48^\circ + k) = \cos(90^\circ - 77^\circ)$$

$$\sin(48^\circ + k) = \sin 77^\circ$$

$$(48^\circ + k) = 77^\circ$$

$$k = 77^\circ - 48^\circ = 29^\circ$$

$$\text{Sol.168.(c)}$$

$$\frac{5}{1 + \cot^2 \theta} + \frac{3}{1 + \tan^2 \theta} + 2 \cos^2 \theta$$

$$\frac{5}{\operatorname{cosec}^2 \theta} + \frac{3}{\sec^2 \theta} + 2 \cos^2 \theta$$

$$5 \sin^2 \theta + 3 \cos^2 \theta + 2 \cos^2 \theta$$

$$5 \sin^2 \theta + 5 \cos^2 \theta \Rightarrow 5(\sin^2 \theta + \cos^2 \theta) = 5$$

$$\text{Sol.169.(b)} \tan A + \cot A = 2$$

$$\text{Put } A = 45^\circ$$

$$\text{LHS} = \tan 45^\circ + \cot 45^\circ = 1 + 1 = \text{RHS}$$

Now,

$$7 \tan^8 45^\circ - 6 \cot^8 45^\circ + 8 \sec^2 45^\circ$$

$$7 \times (1)^8 - 6 \times (1)^8 + 8 \times (\sqrt{2})^2$$

$$\Rightarrow 7 - 6 + 16 = 17$$

$$\text{Sol.170.(a)} \Rightarrow p = \frac{\sin A}{1 + \cos A}$$

$$\Rightarrow p = \frac{\sin A}{1 + \cos A} \times \frac{1 - \cos A}{1 - \cos A}$$

$$\Rightarrow p = \frac{\sin A(1 - \cos A)}{1 - \cos^2 A}$$

$$\Rightarrow p = \frac{\sin A(1 - \cos A)}{\sin^2 A}$$

$$\Rightarrow p = \frac{(1 - \cos A)}{\sin A} \Rightarrow \frac{\sin A}{(1 - \cos A)} = \frac{1}{p}$$

$$\text{Sol.171.(a)} 8 \cot \theta = 7$$

$$\cot \theta = \frac{7}{8} = \frac{\text{Base}}{\text{Perpendicular}}$$

$$\text{Hence, Hypotenuse} = \sqrt{113}$$

$$\frac{1 + \sin \theta}{\cos \theta} = \frac{1 + \frac{\text{perpendicular}}{\text{Hypotenuse}}}{\frac{\text{Base}}{\text{Hypotenuse}}}$$

$$= \frac{1 + \frac{8}{\sqrt{113}}}{\frac{7}{\sqrt{113}}} = \frac{\frac{\sqrt{113} + 8}{\sqrt{113}}}{\frac{7}{\sqrt{113}}} = \frac{\sqrt{113} + 8}{7}$$

$$\text{Sol.172.(a)} \frac{\sin^2 C + \cos^2 C}{1 + \cot^2 C}$$

$$= \frac{1}{1 + \frac{1}{\tan^2 C}} = \frac{1}{1 + \frac{1}{3}} = \frac{1}{\frac{4}{3}} = \frac{3}{4}$$

$$\text{Sol.173.(b)} \sin^4 \theta + \cos^4 \theta$$

$$(\sin^2 \theta + \cos^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta$$

$$1 - 2 \sin^2 \theta \cos^2 \theta$$

$$\text{As } 2 \sin^2 \theta \cos^2 \theta \geq 0$$

$$\text{hence, } \sin^4 \theta + \cos^4 \theta \leq 1$$

$$\text{Sol.174.(b)} x = a \sec \theta \cos \theta$$

On squaring both side

$$\frac{x^2}{a^2} = \sec^2 \theta \cos^2 \theta \dots (i)$$

$$y = b \sec \theta \sin \theta$$

On squaring both side

$$\frac{y^2}{b^2} = \sec^2 \theta \sin^2 \theta \dots (ii)$$

$$z = c \tan \theta$$

On squaring both side

$$\frac{z^2}{c^2} = \tan^2 \theta \dots (iii)$$

On adding (i), (ii) and then subtracting (iii)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2}$$

$$= \sec^2 \theta \cos^2 \theta + \sec^2 \theta \sin^2 \theta - \tan^2 \theta$$

$$= \sec^2 \theta (\cos^2 \theta + \sin^2 \theta) - \tan^2 \theta$$

$$= \sec^2 \theta - \tan^2 \theta = 1$$

$$\text{Sol.175.(b)} \cot \theta = \frac{\sqrt{11}}{1} = \frac{\text{base}}{\text{perpendicular}}$$

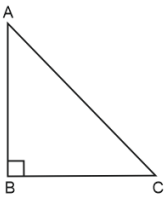
$$\text{hypotenuse}(h) = \sqrt{(\sqrt{11})^2 + 1^2} = \sqrt{12}$$

Now,

$$\frac{\operatorname{cosec}^2 \theta - \sec^2 \theta}{\operatorname{cosec}^2 \theta + \sec^2 \theta}$$

$$\left(\frac{h}{p}\right)^2 - \left(\frac{h}{b}\right)^2 \Rightarrow \frac{\left(\frac{\sqrt{12}}{1}\right)^2 - \left(\frac{\sqrt{12}}{\sqrt{11}}\right)^2}{\left(\frac{h}{p}\right)^2 + \left(\frac{h}{b}\right)^2} \Rightarrow \frac{\left(\frac{\sqrt{12}}{1}\right)^2 + \left(\frac{\sqrt{12}}{\sqrt{11}}\right)^2}$$

$$\frac{12 - \frac{12}{11}}{12 + \frac{12}{11}} = \frac{132 - 12}{132 + 12} = \frac{120}{144} = \frac{5}{6}$$

Sol.176.(b)

$$\operatorname{cosec} A = \frac{AC}{BC} = \frac{2\sqrt{2}}{1}$$

$$AB = \sqrt{(2\sqrt{2})^2 - (1)^2} = \sqrt{8 - 1} = \sqrt{7}$$

$$\sin A \cdot \cos C + \cos A \cdot \sin C$$

$$= \frac{BC}{AC} \times \frac{BC}{AC} + \frac{AB}{AC} \times \frac{AB}{AC}$$

$$= \frac{1}{2\sqrt{2}} \times \frac{1}{2\sqrt{2}} + \frac{\sqrt{7}}{2\sqrt{2}} \times \frac{\sqrt{7}}{2\sqrt{2}}$$

$$= \frac{1}{8} + \frac{7}{8} = 1$$

Sol.177.(a) $(2xy : x^2 + y^2 : x^2 - y^2)$ are pythagorean triplet

$$\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{2xy}{x^2 + y^2}$$

$$\text{So, } \tan \theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{2xy}{x^2 - y^2}$$

Sol.178.(b)

$$12 \sin^2 A + 2 \cos^2 A = 7 \text{-----(i)}$$

$$12(1 - \cos^2 A) + 2 \cos^2 A = 7$$

$$12 - 12 \cos^2 A + 2 \cos^2 A = 7$$

$$-10 \cos^2 A = -5$$

$$\cos^2 A = \frac{1}{2} \text{-----(ii)}$$

$$\cos A = \pm \frac{1}{\sqrt{2}}$$

From equation (i) and (ii)

$$12 \sin^2 A + 2 \times \frac{1}{2} = 7 \Rightarrow 12 \sin^2 A = 6$$

$$\sin^2 A = \frac{1}{2} \Rightarrow \sin A = \pm \frac{1}{\sqrt{2}}$$

$$\text{Now, } \cot A = \frac{\cos A}{\sin A} = \frac{\pm \frac{1}{\sqrt{2}}}{\pm \frac{1}{\sqrt{2}}} = \pm 1$$

Sol.179.(c) As we know ,

$$\{\cos(x+y) = \cos x \cdot \cos y - \sin x \cdot \sin y\}$$

$$\{\sin(x-y) = \sin x \cdot \cos y - \cos x \cdot \sin y\}$$

$$\{\tan(2z) = \frac{2 \tan z}{1 - \tan^2 z}\}$$

$$\text{So, } \cos(x+y) - \sin(x-y) + \tan(2z)$$

$$= \cos x \cdot \cos y - \sin x \cdot \sin y - (\sin x \cdot \cos y -$$

$$\cos x \cdot \sin y) + \frac{2 \tan z}{1 - \tan^2 z}$$

$$= \cos x \cdot \cos y - \sin x \cdot \sin y - \sin x \cdot \cos y +$$

$$\cos x \cdot \sin y + \frac{2 \tan z}{1 - \tan^2 z}$$

$$\text{Sol.180.(a)} \sin \theta = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{15}{17}$$

$$\text{Base} = \sqrt{(17)^2 - (15)^2}$$

$$= \sqrt{289 - 225} = \sqrt{64} = 8 \text{ unit}$$

$$\text{now, } \cot \theta = \frac{\text{base}}{\text{perpendicular}} = \frac{8}{15}$$

$$\sec \theta = \frac{\text{base}}{\text{hypotenuse}} = \frac{17}{8}$$

$$\frac{15 \cot \theta + 17 \sin \theta}{8 \tan \theta + 16 \sec \theta}$$

$$= \frac{15 \cot \theta + 17 \sin \theta}{8 \times \frac{1}{\cot \theta} + 16 \sec \theta}$$

$$= \frac{15 \times \frac{8}{15} + 17 \times \frac{15}{17}}{8 \times \frac{15}{8} + 16 \times \frac{17}{8}} = \frac{8 + 15}{15 + 34} = \frac{23}{49}$$

Sol.181.(a) According to the question,

$$\tan \theta + \cot \theta = 2$$

$$\text{Let } \theta = 45^\circ \text{ then,}$$

$$1 + 1 = 2 \text{ (satisfied)}$$

Now, put the value of $\theta = 45^\circ$ we get,

$$\Rightarrow \tan^3 \theta + \cot^3 \theta + 6 \tan^3 \theta \cot^2 \theta$$

$$\Rightarrow 1 + 1 + 6 = 8$$

$$\text{Sol.182.(d)} \sqrt{13} \sin \theta = 2$$

$$\sin \theta = \frac{2}{\sqrt{13}} = \frac{p}{h}$$

$$b = \sqrt{(\sqrt{13})^2 - (2)^2} = 3$$

According to the question,

$$\frac{3 \times \frac{2}{3} + \sqrt{13} \times \frac{2}{\sqrt{13}}}{\sqrt{13} \times \frac{3}{\sqrt{13}} - 3 \times \frac{2}{3}} = \frac{2+2}{3-2} = 4$$

Sol.183.(a)On putting $\theta = 45^\circ$ then we get,

$$\Rightarrow (\cot \theta + \tan \theta)(\operatorname{cosec} \theta - \sin \theta)(\cos \theta - \sec \theta)$$

$$\Rightarrow (1+1)(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}})(\frac{1}{\sqrt{2}} - \sqrt{2})$$

$$\Rightarrow (2)(\frac{1}{\sqrt{2}})(\frac{-1}{\sqrt{2}}) \Rightarrow -1$$

Sol.184.(d) Put $\theta = 0^\circ$ we get,

$$\sin(30+0) - \cos(60-0) \Rightarrow \frac{1}{2} - \frac{1}{2} = 0$$

$$\text{Sol.185.(c)} \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta}$$

Given that $\alpha + \beta = 45^\circ$

$$\text{Then, } \tan 45^\circ = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta} = 1$$

$$\tan \alpha + \tan \beta = 1 - \tan \alpha \cdot \tan \beta$$

$$\tan \alpha + \tan \beta + \tan \alpha \cdot \tan \beta = 1 \text{-----(i)}$$

$$\text{Now, } (\tan \alpha + 1)(\tan \beta + 1) = 2x$$

$$\Rightarrow \tan \alpha + \tan \beta + \tan \alpha \cdot \tan \beta + 1 = 2x$$

Putting the value from equation (i) we get;

$$1 + 1 = 2x \Rightarrow x = 1$$

Sol.186.(c) According to the question,

$$\tan 570^\circ = \tan(180^\circ \times 3 + 30^\circ)$$

[$\therefore$ the value of $\tan \theta$ in 3rd quadrant is positive]

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

Sol.187.(d)

$$\frac{1}{\sin^4(90^\circ - 2\alpha)} + \frac{1}{\cos^2(90^\circ - 2\alpha) - 1}$$

$$\frac{1}{\cos^4 2\alpha} - \frac{1}{1 - \sin^2 2\alpha} (\because \sin(90^\circ - \theta) = \cos \theta)$$

$$\frac{1}{\cos^4 2\alpha} - \frac{1}{\cos^2 2\alpha}$$

$$\frac{1}{\cos^2 2\alpha} \left(\frac{1}{\cos^2 2\alpha} - 1 \right)$$

$$\frac{1}{\cos^2 2\alpha} \left(\frac{1 - \cos^2 2\alpha}{\cos^2 2\alpha} \right)$$

$$\frac{1}{\cos^2 2\alpha} \left(\frac{\sin^2 2\alpha}{\cos^2 2\alpha} \right) = \sec^2 2\alpha \tan^2 2\alpha$$

Sol.188.(c) $\cot \theta + \tan \theta = 2$ Let $\theta = 45^\circ$

$$\cot 45^\circ + \tan 45^\circ = 2$$

$$1 + 1 = 2 \Rightarrow 2 = 2 \text{ (LHS = RHS)}$$

now, $\tan^{12} \theta + \cot^{12} \theta + 2 \tan^5 \theta \cot^7 \theta$

$$\Rightarrow (\tan 45^\circ)^{12} + (\cot 45^\circ)^{12} + 2 (\tan 45^\circ)^5$$

$$(\cot 45^\circ)^7 \Rightarrow 1 + 1 + 2(1)(1) = 4$$

$$\text{Sol.189.(b)} \cos \theta = \frac{\text{base}}{\text{hypotenuse}} = \frac{24}{25}$$

$$\text{Perpendicular} = \sqrt{(25)^2 - (24)^2}$$

$$= \sqrt{625 - 576} = \sqrt{49} = 7$$

Now, $\cot(90^\circ - \theta) = \tan \theta$

$$= \frac{\text{perpendicular}}{\text{base}} = \frac{7}{24}$$

$$\text{Sol.190.(c)} \left[\frac{\operatorname{cosec} 31^\circ - \sec 59^\circ}{(\sin 18^\circ \div \cos 72^\circ)} \times \frac{\tan 26^\circ}{\cot 64^\circ} \right]$$

$$+ \left[\frac{\tan 30^\circ + \tan 15^\circ}{(1 - \tan 30^\circ \tan 15^\circ)} \right]$$

$$= \left[\frac{\operatorname{cosec} 31^\circ - \sec(90^\circ - 31^\circ)}{(\sin 18^\circ \div \cos 72^\circ)} \times \frac{\tan(90^\circ - 64^\circ)}{\cot 64^\circ} \right]$$

$$+ \left[\frac{\tan 30^\circ + \tan 15^\circ}{(1 - \tan 30^\circ \tan 15^\circ)} \right]$$

$$= \left[\frac{\operatorname{cosec} 31^\circ - \operatorname{cosec} 31^\circ}{(\sin 18^\circ \div \cos 72^\circ)} \times \frac{\cot 64^\circ}{\cot 64^\circ} \right] + \tan(30^\circ + 15^\circ)$$

$$= \left[\frac{0}{(\sin 18^\circ \div \cos 72^\circ)} \times \frac{\cot 64^\circ}{\cot 64^\circ} \right] + \tan 45^\circ$$

$$= 0 + 1 = 1$$

$$\text{Sol.191.(d)} \cot A = \frac{5}{12} = \frac{b}{p}$$

$$h = \sqrt{12^2 + 5^2} \Rightarrow h = 13$$

According to the question,

$$\Rightarrow \frac{\sqrt{\sec A - \operatorname{cosec} A}}{\sqrt{\sec A + \operatorname{cosec} A}} - \sqrt{\frac{17}{7}}$$

$$\Rightarrow \frac{\sqrt{\frac{13}{5} - \frac{13}{12}}}{\sqrt{\frac{13}{5} + \frac{13}{12}}} - \sqrt{\frac{17}{7}}$$

$$\Rightarrow \frac{\sqrt{\frac{156 - 65}{60}}}{\sqrt{\frac{156 + 65}{60}}} - \sqrt{\frac{17}{7}}$$

$$\Rightarrow \sqrt{\frac{91}{221}} - \sqrt{\frac{17}{7}} \Rightarrow \frac{\sqrt{13 \times 7}}{\sqrt{13 \times 17}} - \frac{\sqrt{17}}{\sqrt{7}}$$

$$\Rightarrow \frac{\sqrt{7}}{\sqrt{17}} - \frac{\sqrt{17}}{\sqrt{7}} \Rightarrow -\frac{10}{\sqrt{119}}$$

Sol.192.(a) $\tan(\theta) = \frac{\text{perpendicular}}{\text{base}} = \frac{3}{7}$

Hypotenuse = $\sqrt{(3)^2 + (7)^2} = \sqrt{58}$

$\operatorname{cosec}(\theta) = \frac{\text{hypotenuse}}{\text{perpendicular}} = \frac{\sqrt{58}}{3}$

Sol.193.(c) $\sin \theta = \frac{11}{12} = \frac{p}{h}$

$b = \sqrt{(12)^2 - (11)^2}$

$b = \sqrt{144 - 121} = \sqrt{23}$

According to the question,

$\Rightarrow (\tan \theta + \sec \theta) = \frac{p}{b} + \frac{h}{b} = \frac{p+h}{b}$

$\Rightarrow \frac{11+12}{\sqrt{23}} = \frac{23}{\sqrt{23}} = \sqrt{23}$

Sol.194.(a) $\cos^4 \theta - \sin^4 \theta = \frac{4}{5}$

$(\cos^2 \theta)^2 - (\sin^2 \theta)^2 = \frac{4}{5}$

$(\cos^2 \theta + \sin^2 \theta)(\cos^2 \theta - \sin^2 \theta) = \frac{4}{5}$

$(\cos^2 \theta - \sin^2 \theta) = \frac{4}{5}$

$\cos 2\theta = \frac{4}{5}$ [$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$]

$\cos^2 2\theta = \frac{16}{25}$

Now, $\sin^2 2\theta = 1 - \cos^2 2\theta = 1 - \frac{16}{25}$

$\sin^2 2\theta = \frac{9}{25} \Rightarrow \sin 2\theta = \frac{3}{5}$

Now, $\sin 4\theta = 2 \sin 2\theta \cos 2\theta$

Putting the value of $\sin 2\theta$ and $\cos 2\theta$

$\sin 4\theta = 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$

Sol.195.(b) $\frac{P \cos \alpha - Q \sin \alpha}{P \cos \alpha + Q \sin \alpha}$

Divide numerator and denominator by $\sin \alpha$

$$\frac{P \times \frac{\cos \alpha}{\sin \alpha} - Q \times \frac{\sin \alpha}{\sin \alpha}}{P \times \frac{\cos \alpha}{\sin \alpha} + Q \times \frac{\sin \alpha}{\sin \alpha}}$$

$$= \frac{P \times \frac{P}{Q} - Q}{P \times \frac{P}{Q} + Q} = \frac{P^2 - Q^2}{P^2 + Q^2}$$

So, $\frac{P \cos \alpha - Q \sin \alpha}{P \cos \alpha + Q \sin \alpha} = \frac{P^2 - Q^2}{P^2 + Q^2} + 3$

$$= \frac{P^2 - Q^2}{P^2 + Q^2} - \frac{P^2 - Q^2}{P^2 + Q^2} + 3 = 3$$

Sol.196.(c) $\sin \theta = \frac{7}{25} = \frac{\text{perpendicular}}{\text{hypotenuse}}$

base = $\sqrt{(25)^2 - (7)^2}$

base = 24

$\cos \theta = \frac{24}{25}$, $\tan \theta = \frac{7}{24}$

Sol.197.(d) Concept :- maximum value

of $A \cos x + B \sin x = \sqrt{A^2 + B^2}$

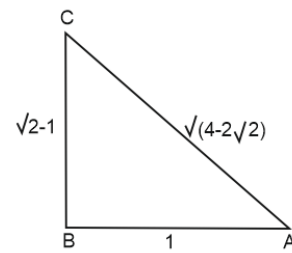
Maximum value $(7 \cos A + 24 \sin A)$

$= \sqrt{(7)^2 + (24)^2} = \sqrt{625} = 25$

So, maximum value of

$7 \cos A + 24 \sin A + 32 = 25 + 32 = 57$

Sol.198.(d)



$AC = \sqrt{(\sqrt{2}-1)^2 + 1^2} = \sqrt{3-2\sqrt{2}+1}$

$= \sqrt{4-2\sqrt{2}}$

$\sin A \cos A + \tan A$

$= \frac{\sqrt{2}-1}{\sqrt{4-2\sqrt{2}}} \times \frac{1}{\sqrt{4-2\sqrt{2}}} + \sqrt{2}-1$

$= \frac{\sqrt{2}-1}{4-2\sqrt{2}} + \sqrt{2}-1$

$= \frac{\sqrt{2}-1}{2\sqrt{2}(\sqrt{2}-1)} + \sqrt{2}-1$

$= \frac{1}{2\sqrt{2}} + \sqrt{2}-1 = \frac{\sqrt{2}}{4} + \sqrt{2}-1$

$= \frac{\sqrt{2}+4\sqrt{2}-4}{4} = \frac{5\sqrt{2}-4}{4}$

Sol.199.(c) $4 \sin^2 \theta - 4\sqrt{3} \sin \theta + 3 = 0$

$4 \sin^2 \theta - 2\sqrt{3} \sin \theta - 2\sqrt{3} \sin \theta + 3 = 0$

$2 \sin \theta (2 \sin \theta - \sqrt{3}) - \sqrt{3} (2 \sin \theta - \sqrt{3}) = 0$

$(2 \sin \theta - \sqrt{3})(2 \sin \theta - \sqrt{3}) = 0$

$(2 \sin \theta - \sqrt{3})^2 = 0$

$2 \sin \theta = \sqrt{3} \Rightarrow \sin \theta = \frac{\sqrt{3}}{2}$

$\sin \theta = \sin 60^\circ$ So, $\theta = 60^\circ$

Sol.200.(c)

$\left(\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} \right) \left(\frac{1}{\tan \theta + \cot \theta} \right)$

$= \frac{\sin^2 \theta + (1 + \cos \theta)^2}{(1 + \cos \theta) \sin \theta} \times \left(\frac{1}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}} \right)$

$= \frac{\sin^2 \theta + 1 + \cos^2 \theta + 2 \cos \theta}{(1 + \cos \theta) \sin \theta} \times \frac{1}{\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}}$

$= \frac{2(1 + \cos \theta)}{(1 + \cos \theta) \sin \theta} \times \cos \theta \sin \theta = 2 \cos \theta$

Sol.201.(d) $\frac{\sin 60^\circ - \cos 60^\circ}{\sin 60^\circ + \cos 60^\circ}$

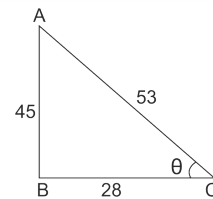
$\frac{\frac{\sqrt{3}}{2} - \frac{1}{2}}{\frac{\sqrt{3}}{2} + \frac{1}{2}} = \frac{\frac{\sqrt{3}-1}{2}}{\frac{\sqrt{3}+1}{2}}$

$= \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$

$= \frac{(\sqrt{3}-1)^2}{(\sqrt{3})^2 - (1)^2} = \frac{(\sqrt{3})^2 + (1)^2 - 2\sqrt{3}}{3-1}$

$= \frac{3+1-2\sqrt{3}}{2} = \frac{4-2\sqrt{3}}{2} = 2-\sqrt{3}$

Sol.202.(d) According to question, we have given triangle :



$AC = \sqrt{45^2 + 28^2} = \sqrt{2809} = 53$

Now, $\sin \theta - \cos \theta = \frac{45}{53} - \frac{28}{53} = \frac{17}{53}$

Sol.203.(b) $\sec 6A = \operatorname{cosec} (A - 29^\circ)$

$\operatorname{cosec} (90^\circ - 6A) = \operatorname{cosec} (A - 29^\circ)$

$\Rightarrow 90^\circ - 6A = A - 29^\circ$

$7A = 119 \Rightarrow A = \frac{119}{7} = 17^\circ$

Sol.204.(a)

Formula used : $\sin^2 A + \cos^2 A = 1$

$\sec^2 A - \tan^2 A = 1$

$\operatorname{cosec}^2 A - \cot^2 A = 1$

Now, $(1 + \tan^2 A) + (1 + \cot^2 A)$

$= \sec^2 A + \operatorname{cosec}^2 A$

$\frac{1}{\cos^2 A} + \frac{1}{\sin^2 A} = \frac{\sin^2 A + \cos^2 A}{\sin^2 A \cos^2 A}$

$= \frac{1}{\sin^2 A \cos^2 A}$

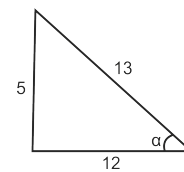
Now, solving option (a) we get ;

$\frac{1}{\sin^2 A - \sin^4 A} = \frac{1}{\sin^2 A (1 - \sin^2 A)}$

$= \frac{1}{\sin^2 A \cos^2 A}$

So, the correct option is (a).

Sol.205.(b) According to question,



$\cos \alpha \cdot \operatorname{cosec} \alpha \cdot \cot \alpha = \cot^2 \alpha = \left(\frac{12}{5} \right)^2$

$= \frac{144}{25}$

Sol.206.(a)

$(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 3(\sin^2 A + \cos^2 A)$

$\left(1 + \frac{\cos A}{\sin A} - \frac{1}{\sin A} \right) \left(1 + \frac{\sin A}{\cos A} + \frac{1}{\cos A} \right) - 3 \times 1$

$= \left(\frac{\sin A + \cos A - 1}{\sin A} \right) \left(\frac{\cos A + \sin A + 1}{\cos A} \right) - 3$

$= \frac{(\sin A + \cos A)^2 - 1}{\sin A \cos A} - 3$

$= \frac{\sin^2 A + \cos^2 A + 2 \sin A \cos A - 1}{\sin A \cos A} - 3$

$= \frac{1 + 2 \sin A \cos A - 1}{\sin A \cos A} - 3 = 2 - 3 = -1$

Sol.207.(a) Given:

$(19 \sin \theta + 6 \cot \theta \sin \theta)$

$= (19 \sin \theta + 6 \cos \theta)$

We know that,

$$y = (A \sin \theta + B \cos \theta) \text{ where, } A > B$$

$$y_{\max} = \sqrt{A^2 + B^2}$$

So,

Maximum value of $(19 \sin \theta + 6 \cos \theta)$

$$= \sqrt{19^2 + 6^2} = \sqrt{361 + 36} \Rightarrow \sqrt{397}$$

Sol.208.(c) Given: $\tan \theta = \frac{3}{10}$

$$\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)} = \frac{(1)^2 - (\sin \theta)^2}{(1)^2 - (\cos \theta)^2}$$

$$\frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\tan^2 \theta} = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

Sol.209.(c) $x = \sin A + \cos A \dots$ (i)

On squaring both side

$$x^2 = \sin^2 A + \cos^2 A + 2 \sin A \cdot \cos A$$

$$x^2 = 1 + 2 \sin A \cdot \cos A$$

$$\frac{x^2 - 1}{2} = \sin A \cdot \cos A \dots$$
 (ii)

Now,

$$y = \sec A + \operatorname{cosec} A$$

$$y = \frac{\sin A + \cos A}{\sin A \cos A}$$

$$y = \frac{x}{\frac{x^2 - 1}{2}} \text{ from (i) and (ii)}$$

$$y = \frac{2x}{x^2 - 1} \Rightarrow (x^2 - 1)y = 2x$$

$$(x^2 y - y) = 2x \Rightarrow x^2 y = 2x + y$$

Short trick : Put $A = 45^\circ$

Sol.210.(c) $2 \cos^2 A + 3 \cos A - 2 = 0$

$$\Rightarrow 2 \cos^2 A + 4 \cos A - \cos A - 2 = 0$$

$$\Rightarrow 2 \cos A (\cos A + 2) - 1 (\cos A + 2) = 0$$

$$\Rightarrow (\cos A + 2)(2 \cos A - 1) = 0$$

$$\Rightarrow (\cos A + 2) = 0 \text{ and } (2 \cos A - 1) = 0$$

$$\Rightarrow \cos A = -2 \text{ and } \cos A = \frac{1}{2}$$

$$\cos A = \frac{1}{2} \Rightarrow \cos A = \cos 60^\circ \Rightarrow A = 60^\circ$$

Sol.211.(d) $\frac{1 - \cot^2 \theta}{\tan^2 \theta - 1}$

$$\frac{1 - \frac{\cos^2 \theta}{\sin^2 \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta} - 1} \Rightarrow \frac{\frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\cos^2 \theta}}$$

$$= \frac{\cos^2 \theta}{\sin^2 \theta} = \cot^2 \theta$$

Sol.212.(d) $\frac{(\sec A + \tan A)}{\sec A - \tan A} = 2 \frac{51}{79}$

$$\frac{(\sec A + \tan A)}{\sec A - \tan A} = \frac{209}{79}$$

By applying componendo and dividendo rule

$$\frac{\sec A}{\tan A} = \frac{209 + 79}{209 - 79}$$

$$\frac{\sec A}{\tan A} = \frac{288}{130} \Rightarrow \frac{\frac{1}{\cos A}}{\frac{\sin A}{\cos A}} = \frac{288}{130}$$

$$\frac{1}{\sin A} = \frac{144}{65} \Rightarrow \sin A = \frac{65}{144}$$

Sol.213.(a) $4 \sin^2 \theta = 3(1 + \cos \theta)$

$$4(1 - \cos^2 \theta) = 3(1 + \cos \theta)$$

$$4(1 - \cos \theta)(1 + \cos \theta) = 3(1 + \cos \theta)$$

$$(1 - \cos \theta) = \frac{3}{4}$$

$$\cos \theta = 1 - \frac{3}{4} = \frac{1}{4} = \frac{\text{base}}{\text{hypotenuse}}$$

$$\text{Perpendicular} = \sqrt{15}$$

Now,

$$\sqrt{15} \tan \theta + \frac{4}{\sqrt{15}} \sin \theta + 2 \sec \theta$$

$$\sqrt{15} \times \frac{\text{Perpendicular}}{\text{base}} + \frac{4}{\sqrt{15}} \times$$

$$\frac{\text{Perpendicular}}{\text{hypotenuse}} + 2 \times \frac{\text{hypotenuse}}{\text{base}}$$

$$\sqrt{15} \times \frac{\sqrt{15}}{1} + \frac{4}{\sqrt{15}} \times \frac{\sqrt{15}}{4} + 2 \times \frac{4}{1}$$

$$15 + 1 + 8 = 24$$

Sol.214.(b) $\tan \theta + \cot \theta = 6$

On squaring both side,

$$(\tan \theta + \cot \theta)^2 = 36$$

$$\tan^2 \theta + \cot^2 \theta + 2 \tan \theta \cdot \cot \theta = 36$$

$$\text{We know that, } \tan \theta \cdot \cot \theta = 1$$

Hence,

$$\tan^2 \theta + \cot^2 \theta = 36 - 2 = 34$$

Sol.215.(b) $(1 - \cos^2 A) \sec^2 A$

Using identity $\therefore \sin^2 A = 1 - \cos^2 A$ and

$$\sec^2 A = \frac{1}{\cos^2 A}$$

$$\text{then, } \sin^2 A \times \frac{1}{\cos^2 A} = \tan^2 A$$

Sol.216.(d) $\sin x + \cos x = \sqrt{2} \sin x$

On squaring both side,

$$(\sin x + \cos x)^2 = (\sqrt{2} \sin x)^2$$

$$\sin^2 x + \cos^2 x + 2 \sin x \cdot \cos x = 2 \sin^2 x$$

$$= 1 + 2 \sin x \cdot \cos x = 2 \sin^2 x$$

$$= 2 \sin x \cdot \cos x = 2 \sin^2 x - 1 \dots$$
 (i)

$$\text{Now, } (\sin x - \cos x)^2$$

$$= \sin^2 x + \cos^2 x - 2 \sin x \cdot \cos x$$

$$= 1 - 2 \sin x \cdot \cos x \dots \text{ from (i)}$$

$$= 1 - (2 \sin^2 x - 1) = 1 - 2 \sin^2 x + 1$$

$$= 2 - 2 \sin^2 x = 2(1 - \sin^2 x)$$

$$(\sin x - \cos x)^2 = 2 \cos^2 x$$

$$\sin x - \cos x = \sqrt{2} \cos x$$

Sol.217.(b) In $\triangle ABC$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\frac{\angle A}{2} + \frac{\angle B + \angle C}{2} = 90^\circ$$

$$\text{Now, } \left\{ \tan \frac{A}{2} + \operatorname{cosec} \frac{B+C}{2} \right\} \left\{ \tan \frac{A}{2} - \operatorname{cosec} \frac{B+C}{2} \right\}$$

$$\left\{ \tan \frac{A}{2} + \operatorname{cosec} (90^\circ - \frac{A}{2}) \right\} \left\{ \tan \frac{A}{2} - \operatorname{cosec} (90^\circ - \frac{A}{2}) \right\}$$

$$\left\{ \tan \frac{A}{2} + \sec \frac{A}{2} \right\} \left\{ \tan \frac{A}{2} - \sec \frac{A}{2} \right\}$$

$$\left\{ \tan \frac{A}{2} + \sec \frac{A}{2} \right\} \left\{ \tan \frac{A}{2} - \sec \frac{A}{2} \right\}$$

$$\left\{ \tan \frac{A}{2} + \sec \frac{A}{2} \right\} \left\{ \tan \frac{A}{2} - \sec \frac{A}{2} \right\}$$

$$\tan^2 \frac{A}{2} - \sec^2 \frac{A}{2} = -1$$

Sol.218.(d) $= \frac{\sin^3 A + \cos^3 A}{\sin A + \cos A} =$

$$\frac{(\sin A + \cos A)(\sin^2 A + \cos^2 A - \sin A \cos A)}{\sin A + \cos A}$$

$$= (1 - \sin A \cos A)$$

Sol.219.(c) $\sin(x + y) = 1$

$$x + y = 90^\circ \dots (\text{as } \sin 90^\circ = 1) \dots$$
 (i)

$$\cos(x - y) = \frac{\sqrt{3}}{2}$$

$$x - y = 30^\circ \dots (\text{as } \cos 30^\circ = \frac{\sqrt{3}}{2}) \dots$$
 (ii)

From eq. (i) and (ii)

$$x = \frac{120^\circ}{2} = 60^\circ \text{ and } y = 30^\circ$$

Sol.220.(c) $\left(\frac{2 \cos^3 \theta - \cos \theta}{\sin \theta - 2 \sin^3 \theta} \right)^2 + 1$

$$= \left[\frac{\cos \theta (2 \cos^2 \theta - 1)}{\sin \theta (1 - 2 \sin^2 \theta)} \right]^2 + 1$$

$$(\because \cos 2A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A)$$

$$= \cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$$

Sol.221.(d) $\sin^4 \theta + \cos^4 \theta = 2 \sin^2 \theta \cos^2 \theta$

$$\sin^4 \theta + \cos^4 \theta - 2 \sin^2 \theta \cos^2 \theta = 0$$

$$(\sin^2 \theta - \cos^2 \theta)^2 = 0$$

Taking square root

$$\sin^2 \theta - \cos^2 \theta = 0 \Rightarrow \sin^2 \theta = \cos^2 \theta$$

$$1 = \frac{\cos^2 \theta}{\sin^2 \theta} \Rightarrow 1 = \cot^2 \theta \Rightarrow \cot \theta = 1$$

Sol.222.(c) $\tan^4 A + \tan^2 A$

$$\tan^2 A (\tan^2 A + 1)$$

$$(\sec^2 A - 1)(\sec^2 A) \Rightarrow \sec^4 A - \sec^2 A$$

Sol.223.(a)

$$[\tan \theta + \sec \theta + 1][\cot \theta - \operatorname{cosec} \theta + 1]$$

$$\left[\frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta} + 1 \right] \left[\frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta} + 1 \right]$$

$$\left[\frac{(\sin \theta + \cos \theta) + 1}{\cos \theta} \right] \left[\frac{(\sin \theta + \cos \theta) - 1}{\sin \theta} \right]$$

$$\left[\frac{(\cos \theta + \sin \theta)^2 - 1^2}{\cos \theta \cdot \sin \theta} \right]$$

$$\left[\frac{\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta - 1}{\cos \theta \cdot \sin \theta} \right]$$

$$\left[\frac{1 - 1 + 2 \sin \theta \cos \theta}{\cos \theta \cdot \sin \theta} \right] = \left[\frac{2 \sin \theta \cos \theta}{\cos \theta \cdot \sin \theta} \right] = 2$$

Sol.224.(d)

$$(\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)$$

$$(\tan \theta + \cot \theta)$$

$$\left(\frac{1}{\sin \theta} - \sin \theta \right) \left(\frac{1}{\cos \theta} - \cos \theta \right) \left(\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right)$$

$$\left(\frac{1 - \sin^2 \theta}{\sin \theta} \right) \left(\frac{1 - \cos^2 \theta}{\cos \theta} \right) \left(\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \cdot \sin \theta} \right)$$

$$\left(\frac{\cos^2 \theta}{\sin \theta} \right) \left(\frac{\sin^2 \theta}{\cos \theta} \right) \left(\frac{1}{\cos \theta \cdot \sin \theta} \right)$$

$$\left(\frac{\cos^2 \theta}{\sin \theta}\right)\left(\frac{\sin^2 \theta}{\cos \theta}\right)\left(\frac{1}{\cos \theta \sin \theta}\right) = 1$$

Short trick :-

Put $\theta = 45^\circ$

Sol.225.(c)

$$\begin{aligned} & (1 + \tan \theta + \sec \theta) \times (1 + \cot \theta - \operatorname{cosec} \theta) \\ &= \left(1 + \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta}\right) \times \left(1 + \frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta}\right) \\ &= \left(\frac{\sin \theta + \cos \theta + 1}{\cos \theta}\right) \times \left(\frac{\sin \theta + \cos \theta - 1}{\sin \theta}\right) \\ &= \frac{(\sin \theta + \cos \theta)^2 - (1)^2}{\cos \theta \cdot \sin \theta} \\ &= \frac{(\sin^2 \theta + \cos^2 \theta + 2 \cos \theta \cdot \sin \theta - 1)}{\cos \theta \cdot \sin \theta} \\ &= \frac{(1 + 2 \cos \theta \cdot \sin \theta - 1)}{\cos \theta \cdot \sin \theta} = \frac{2 \cos \theta \cdot \sin \theta}{\cos \theta \cdot \sin \theta} \\ &= 2 \end{aligned}$$

Short trick -

Put $\theta = 45^\circ$

Sol.226.(a) Formula used :- $\tan(A+B)$

$$= \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

$$\text{Then, } \frac{\tan 71^\circ + \tan 19^\circ}{1 - \tan 71^\circ \tan 19^\circ}$$

$$= \tan(71^\circ + 19^\circ) = \tan(90^\circ) = \text{not defined}$$

Sol.227.(b) $\sin(x+y) = \cos\{3(x+y)\}$

When, $\sin a = \cos b$ then $a + b = 90^\circ$

Hence,

$$(x+y) + 3(x+y) = 90^\circ$$

$$4(x+y) = 90^\circ \Rightarrow 2(x+y) = 45^\circ$$

Now,

$$\cot 2(x+y) = \cot(45^\circ) = \cot 45^\circ = 1$$

Sol.228.(a)

Identity used :- $\sin 2A = 2 \sin A \cos A$ and

$$\cos 2A = 1 - 2 \sin^2 A$$

$$\text{Now, } \frac{2 \cos 15^\circ \sin 15^\circ}{\cos^2 15^\circ - \sin^2 15^\circ}$$

$$\begin{aligned} &= \frac{\sin 2(15^\circ)}{1 - \sin^2 15^\circ - \sin^2 15^\circ} = \frac{\sin 30^\circ}{1 - 2 \sin^2 15^\circ} \\ &= \frac{\sin 30^\circ}{\cos 2(15^\circ)} = \frac{\sin 30^\circ}{\cos 30^\circ} = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}} \end{aligned}$$

$$\text{Sol.229.(c)} \quad \frac{17}{\sec^2 \theta} + 10 \sin^2 \theta + \frac{7}{1 + \cot^2 \theta}$$

$$= 17 \cos^2 \theta + 10 \sin^2 \theta + \frac{7}{1 + \frac{\cos^2 \theta}{\sin^2 \theta}}$$

$$= 17 \cos^2 \theta + 10 \sin^2 \theta + \frac{7 \sin^2 \theta}{\sin^2 \theta + \cos^2 \theta}$$

$$= 17 \cos^2 \theta + 10 \sin^2 \theta + 7 \sin^2 \theta$$

$$= 17(\cos^2 \theta + \sin^2 \theta) \Rightarrow 17$$

Sol.230.(c) $\sin^2 15^\circ + \sin^2 25^\circ + \sin^2 35^\circ +$

$$\sin^2 45^\circ + \sin^2 55^\circ + \sin^2 65^\circ + \sin^2 75^\circ$$

$$= \sin^2(90^\circ - 75^\circ) + \sin^2(90^\circ - 65^\circ) +$$

$$\sin^2(90^\circ - 55^\circ) + \sin^2 45^\circ + \sin^2 55^\circ +$$

$$\sin^2 65^\circ + \sin^2 75^\circ$$

$$= \cos^2 75^\circ + \cos^2 65^\circ + \cos^2 55^\circ + \sin^2 45^\circ$$

$$+ \sin^2 55^\circ + \sin^2 65^\circ + \sin^2 75^\circ$$

$$= \cos^2 75^\circ + \sin^2 75^\circ + \cos^2 65^\circ + \sin^2 65^\circ$$

$$+ \cos^2 55^\circ + \sin^2 55^\circ + \sin^2 45^\circ$$

$$= 1 + 1 + 1 + \frac{1}{2} \Rightarrow 3 + \frac{1}{2} = \frac{7}{2}$$

$$\text{Sol.231.(a)} \quad 7\left(\frac{\operatorname{cosec} 24^\circ}{\sec 66^\circ}\right)^3 + 8\left(\frac{\cot 37^\circ}{\tan 53^\circ}\right)^4 -$$

$$\left(2\frac{\sec 14^\circ}{\operatorname{cosec} 76^\circ}\right)^2 + \left(-3\frac{\tan 82^\circ}{\cot 8^\circ}\right)^3$$

$$\Rightarrow 7\left(\frac{\operatorname{cosec} 24^\circ}{\sec(90-24^\circ)}\right)^3 + 8\left(\frac{\cot 37^\circ}{\tan(90-37^\circ)}\right)^4$$

$$- \left(2\frac{\sec 14^\circ}{\operatorname{cosec}(90-14^\circ)}\right)^2 + \left(-3\frac{\tan 82^\circ}{\cot(90-82^\circ)}\right)^3$$

$$\Rightarrow 7\left(\frac{\operatorname{cosec} 24^\circ}{\operatorname{cosec} 24^\circ}\right)^3 + 8\left(\frac{\cot 37^\circ}{\cot 37^\circ}\right)^4 - \left(2\frac{\sec 14^\circ}{\sec 14^\circ}\right)^2 + \left(-3\frac{\tan 82^\circ}{\tan 82^\circ}\right)^3$$

$$\Rightarrow 7(1)^3 + 8(1)^4 - (2)^2 + (-3)^3$$

$$\Rightarrow 7 + 8 - 4 - 27 = -16$$

Sol.232.(a) Given, $8 \tan A = 5$

$$\Rightarrow \tan A = \frac{5}{8}$$

$$\text{Now, } \frac{8 \sin A - 7 \cos A}{8 \sin A + 11 \cos A}$$

On dividing numerator and denominator by $\cos A$, we get

$$\Rightarrow \frac{8 \frac{\sin A}{\cos A} - 7}{8 \frac{\sin A}{\cos A} + 11} = \frac{8 \tan A - 7}{8 \tan A + 11}$$

$$\Rightarrow \frac{8 \times \frac{5}{8} - 7}{8 \times \frac{5}{8} + 11} = \frac{5 - 7}{5 + 11} = \frac{-2}{16} = \frac{-1}{8}$$

Sol.233.(b) Let $A = 45^\circ$

(a) LHS = $\cos^2 A \cot^2 A$

$$= \cos^2 45^\circ \cot^2 45^\circ = \frac{1}{2}$$

$$\text{RHS} = \cot^2 A - \cos^2 A$$

$$= \cot^2 45^\circ - \cos^2 45^\circ = 1 - \frac{1}{2} = \frac{1}{2}$$

$$(b) \frac{1 + \tan^2 A}{\tan^2 A} = \sin^2 A$$

$$\text{LHS} = \frac{1 + \tan^2 A}{\tan^2 A} = \frac{1 + 1}{1} = 2$$

$$\text{RHS} = \sin^2 A = \sin^2 45^\circ = \frac{1}{2}$$

Hence LHS $\neq$ RHS

$$\Rightarrow \frac{1 + \tan^2 A}{\tan^2 A} = \sin^2 A \text{ (not the correct formula).}$$

Sol.234.(c) Given expression

$$\frac{5 \cos^2 120^\circ + 4 \sec^2 30^\circ - \tan^2 135^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$$

$$\Rightarrow \frac{5\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) + 4\left(\frac{2}{\sqrt{3}}\right)\left(\frac{2}{\sqrt{3}}\right) - (-1)(-1)}{\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)}$$

$$\Rightarrow \frac{\frac{5}{4} + \frac{16}{3} - 1}{\frac{1}{4} + \frac{3}{4}} = \frac{\frac{67}{12}}{\frac{4}{4}} = \frac{67}{12}$$

Sol.235.(b)

$$\begin{aligned} & \left(\frac{\sin 23^\circ \cos 67^\circ + \cos 23^\circ \sin 67^\circ}{\operatorname{cosec}^2 15^\circ - \tan^2 75^\circ}\right)^3 \\ & \Rightarrow \left(\frac{\sin 23^\circ \cos(90-23^\circ) + \cos 23^\circ \sin(90-23^\circ)}{\operatorname{cosec}^2 15^\circ - \tan^2(90-15^\circ)}\right)^3 \\ & \Rightarrow \left(\frac{\sin 23^\circ \sin 23^\circ + \cos 23^\circ \cos 23^\circ}{\operatorname{cosec}^2 15^\circ - \cot^2 15^\circ}\right)^3 \\ & \Rightarrow \left(\frac{\sin^2 23^\circ + \cos^2 23^\circ}{1}\right)^3 \text{ (since } \operatorname{cosec}^2 \theta - \cot^2 \theta = 1) \end{aligned}$$

$$\cot^2 \theta = 1 \text{ and } \sin^2 \theta + \cos^2 \theta = 1$$

$$= 1^3 = 1$$

Sol.236.(a) We know

$$\operatorname{Cosec}^2 A - \cot^2 A = 1$$

$$\Rightarrow \operatorname{Cosec}^2 A - \left(\frac{2p}{p^2 - 1}\right)^2 = 1$$

$$\Rightarrow \operatorname{Cosec}^2 A = 1 + \left(\frac{2p}{p^2 - 1}\right)^2$$

$$\Rightarrow \operatorname{Cosec}^2 A = \frac{p^4 + 1 - 2p^2 + 4p^2}{(p^2 - 1)^2}$$

$$\Rightarrow \operatorname{Cosec}^2 A = \frac{p^4 + 1 + 2p^2}{(p^2 - 1)^2}$$

$$\Rightarrow \operatorname{Cosec}^2 A = \frac{(p^2 + 1)^2}{(p^2 - 1)^2}$$

$$\Rightarrow \operatorname{Cosec} A = \frac{p^2 + 1}{p^2 - 1}$$

Sol.237.(c)

$$\operatorname{cosec} \theta \left[\frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta} \right] - 2 \cot^2 \theta$$

$$= \operatorname{cosec} \theta \left[\frac{(1 + \cos \theta)^2 + \sin^2 \theta}{(\sin \theta)(1 + \cos \theta)} \right] - 2 \cot^2 \theta$$

$$= \operatorname{cosec} \theta \left[\frac{1 + \cos^2 \theta + 2 \cos \theta + \sin^2 \theta}{(\sin \theta)(1 + \cos \theta)} \right] - 2 \cot^2 \theta$$

$$= \operatorname{cosec} \theta \left[\frac{1 + 1 + 2 \cos \theta}{(\sin \theta)(1 + \cos \theta)} \right] - 2 \cot^2 \theta$$

$$(\because \sin^2 \theta + \cos^2 \theta = 1)$$

$$= \operatorname{cosec} \theta \times \frac{2}{\sin \theta} - 2 \cot^2 \theta$$

$$= 2 \operatorname{cosec}^2 \theta - 2 \cot^2 \theta = 2$$

$$(\because \operatorname{cosec}^2 \theta - \cot^2 \theta = 1)$$

Sol.238.(c) $(\sin A + \cos A) = \frac{4}{3}$

Squaring both side, we have;

$$\sin^2 A + \cos^2 A + 2 \sin A \cos A = \frac{16}{9}$$

$$1 + 2 \sin A \cos A = \frac{16}{9} \text{ (As, } \sin^2 A + \cos^2 A = 1)$$

$$2 \sin A \cos A = \frac{16}{9} - 1 = \frac{7}{9}$$

$$\Rightarrow \sin A \cos A = \frac{7}{9 \times 2} = \frac{7}{18}$$

$$\tan A + \cot A = \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}$$

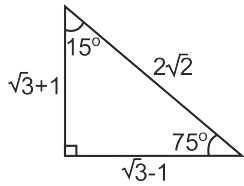
$$= \frac{\sin^2 A + \cos^2 A}{\cos A \sin A} = \frac{1}{\sin A \cos A} = \frac{1}{\frac{7}{18}} = \frac{18}{7}$$

Sol.239.(c) $\sin^2 \theta + \cos^2 \theta - 1 = 0$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Clearly we can see that there are infinitely many values which satisfy the given expression.

Sol.240.(a)



$$\sin 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$\sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$\begin{aligned}\sin 75^\circ + \sin 15^\circ &= \frac{\sqrt{3}+1}{2\sqrt{2}} + \frac{\sqrt{3}-1}{2\sqrt{2}} \\ &= \frac{\sqrt{3}+1+\sqrt{3}-1}{2\sqrt{2}} = \frac{2\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{3} \times \sqrt{2}}{\sqrt{2} \times \sqrt{2}} = \frac{\sqrt{6}}{2}\end{aligned}$$

Sol.241.(b) As we know,

$$\sin(90^\circ - \theta) = \cos \theta$$

$$\cos(90^\circ - \theta) = \sin \theta$$

$$\begin{aligned}\frac{\cos 25^\circ - \sin 65^\circ}{\cos 25^\circ + \sin 65^\circ} &= \frac{\sin 65^\circ - \sin 65^\circ}{\sin 65^\circ + \sin 65^\circ} \\ &= \frac{0}{2\sin 65^\circ} = 0\end{aligned}$$

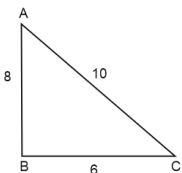
Sol.242.(a) $\sin(5x - 40^\circ) = \cos(5y + 40^\circ)$

$$\cos[90^\circ - (5x - 40^\circ)] = \cos(5y + 40^\circ)$$

$$130^\circ - 5x = 5y + 40^\circ \Rightarrow 5y + 5x = 90^\circ$$

$$x + y = 18^\circ$$

Sol.243.(a) According to the question, we have given triangle;



$$\tan^2 A = \left(\frac{6}{8}\right)^2 = \frac{36}{64} = \frac{9}{16}$$

$$\cos^2 C = \left(\frac{6}{10}\right)^2 = \frac{36}{100} = \frac{9}{25}$$

$$\begin{aligned}\tan^2 A + \cos^2 C &= \frac{9}{16} + \frac{9}{25} = 9 \left(\frac{25+16}{400} \right) \\ &= 9 \times \frac{41}{400} = \frac{369}{400}\end{aligned}$$

Sol.244.(c) $\sin \theta + \frac{1}{\sin \theta} = \sqrt{5}$

Then, $\sin^3 \theta + \frac{1}{\sin^3 \theta} = \left(\sin \theta + \frac{1}{\sin \theta}\right)^3 -$

$$3\left(\sin \theta + \frac{1}{\sin \theta}\right)$$

$$= (\sqrt{5})^3 - 3\sqrt{5} = 5\sqrt{5} - 3\sqrt{5} = 2\sqrt{5}$$

Sol.245.(b) $\cos 2\theta = 0$

$$\cos 2\theta = \cos 90^\circ \Rightarrow 2\theta = 90^\circ$$

$$\theta = \left(\frac{90^\circ}{2}\right) = 45^\circ$$

$$\sin(75^\circ - \theta) = \sin(75^\circ - 45^\circ) = \sin 30^\circ = \frac{1}{2}$$

Sol.246.(c) $P \sin 60^\circ = Q \operatorname{cosec} 45^\circ$

$$P \times \frac{\sqrt{3}}{2} = Q \times \sqrt{2} \Rightarrow \frac{P}{Q} = \frac{2\sqrt{2}}{\sqrt{3}}$$

$$\frac{P^2 + Q^2}{P^2 - Q^2} = \frac{(2\sqrt{2})^2 + (\sqrt{3})^2}{(2\sqrt{2})^2 - (\sqrt{3})^2} = \frac{8+3}{8-3} = \frac{11}{5}$$

Sol.247.(d)

$$\tan^2 \theta + \tan^4 \theta = \tan^2 \theta (1 + \tan^2 \theta)$$

$$= \tan^2 \theta \cdot \sec^2 \theta = (\sec^2 \theta - 1) \cdot \sec^2 \theta$$

$$= \sec^4 \theta - \sec^2 \theta$$

Sol.248.(c) $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$

Squaring both sides we get,

$$\Rightarrow (\cos \theta - \sin \theta)^2 = 2 \sin^2 \theta$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta - 2 \cos \theta \sin \theta = 2 \sin^2 \theta$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta - 2 \sin^2 \theta = 2 \cos \theta \sin \theta$$

$$\Rightarrow \cos^2 \theta - \sin^2 \theta = 2 \cos \theta \sin \theta$$

$$\Rightarrow (\cos \theta - \sin \theta)(\cos \theta + \sin \theta)$$

$$= 2 \cos \theta \sin \theta$$

$$\Rightarrow \sqrt{2} \sin \theta (\cos \theta + \sin \theta) = 2 \cos \theta \sin \theta$$

$$\Rightarrow (\cos \theta + \sin \theta) = \sqrt{2} \cos \theta$$

Sol.249.(d) According to the question,

$$(1 + \sin^4 A - \cos^4 A) \operatorname{cosec}^2 A$$

On Putting $A = 90^\circ$

$$(1 + 1^4 - 0^4) 1^2 = 2$$

Sol.250.(a) According to the question,

$$\cos A = \frac{\text{base}}{\text{hypotenuse}} = \frac{15}{17}$$

So the perpendicular = 8 (8, 15, 17 pythagorean triples)

$$\cot(90^\circ - A) = \tan A$$

$$\tan A = \frac{\text{perpendicular}}{\text{base}} = \frac{8}{15}$$

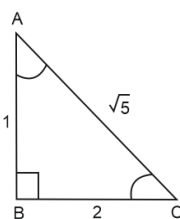
Sol.251.(d)

If $\sin A = \cos B$ then, $A + B = 90^\circ$

$$\sin 2A = \cos 15^\circ$$

$$\Rightarrow 2A + 15^\circ = 90^\circ \Rightarrow A = 37.5^\circ$$

Sol.252.(d)



$$\cos A + \tan C = \frac{1}{\sqrt{5}} + \frac{1}{2} = \frac{2+\sqrt{5}}{2\sqrt{5}}$$

Sol.253.(c) According to the question, $\sec A - \tan A = p$

$$\sec A + \tan A = \frac{1}{p} \Rightarrow \sec A = \frac{p^2 + 1}{2p}$$

Sol.254.(d) $\sqrt{\frac{1-\tan A}{1+\tan A}}$

$$= \sqrt{\frac{1 - \frac{\sin A}{\cos A}}{1 + \frac{\sin A}{\cos A}}} = \sqrt{\frac{\frac{\cos A - \sin A}{\cos A}}{\frac{\cos A + \sin A}{\cos A}}}$$

$$= \sqrt{\frac{\cos A - \sin A}{\cos A + \sin A} \times \frac{\cos A - \sin A}{\cos A - \sin A}}$$

$$= \sqrt{\frac{(\cos A - \sin A)^2}{\cos^2 A - \sin^2 A}}$$

$$= \sqrt{\frac{1 - 2 \sin A \cos A}{\cos^2 A - \sin^2 A}} = \sqrt{\frac{1 - \sin 2A}{\cos 2A}}$$

Sol.255.(d)

$$\sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sqrt{\frac{1 + \sin A}{1 - \sin A} \times \frac{1 + \sin A}{1 + \sin A}}$$

$$= \sqrt{\frac{(1 + \sin A)^2}{(1 - \sin A)(1 + \sin A)}}$$

$$= \frac{(1 + \sin A)}{\sqrt{1 - \sin^2 A}} = \frac{(1 + \sin A)}{\cos A} = \sec A + \tan A$$

Sol.256.(c) $\sin A = \frac{2}{3} = \frac{\text{perpendicular}}{\text{hypotenuse}}$

Then, base = $\sqrt{9 - 4} = \sqrt{5}$

Now, $(7 - \tan A)(3 + \cos A)$

$$= \left(7 - \frac{2}{\sqrt{5}}\right) \left(3 + \frac{\sqrt{5}}{3}\right)$$

$$= \left(21 - \frac{2}{3}\right) + \left(\frac{7\sqrt{5}}{3} - \frac{6}{\sqrt{5}}\right)$$

$$= \frac{61}{3} + \frac{17}{3\sqrt{5}}$$

Sol.257.(b) $\sin \beta = \frac{1}{3}$

Then, $(\sec \beta - \tan \beta)^2$

$$= \left(\frac{1}{\cos \beta} - \frac{\sin \beta}{\cos \beta}\right)^2 = \left(\frac{1 - \sin \beta}{\cos \beta}\right)^2$$

$$= \frac{(1 - \sin \beta)^2}{\cos^2 \beta} = \frac{(1 - \sin \beta)^2}{1 - \sin^2 \beta} = \frac{(1 - \frac{1}{3})^2}{1 - \frac{1}{9}}$$

$$= \frac{1}{2}$$

Sol.258.(c) $\frac{\sin^2 39^\circ + \sin^2(90^\circ - 39^\circ)}{\cos^2 35^\circ + \cos^2(90^\circ - 35^\circ)} +$

$$3 \tan 25^\circ \tan 65^\circ$$

$$= \frac{\sin^2 39^\circ + \cos^2 39^\circ}{\cos^2 35^\circ + \sin^2 35^\circ} + 3 \tan 25^\circ \times \frac{1}{\tan 25^\circ}$$

$$= 1 + 3 = 4$$

Sol.259.(b) According to the question,

$$\tan \theta + \cot \theta = 2$$

On Putting $\theta = 45^\circ$

$$1 + 1 = 2 \text{ (satisfied)}$$

Now,

$$2 \tan^{25} \theta + 3 \cot^{20} \theta + 5 \tan^{30} \theta \cot^{15} \theta$$

$$2 \times 1 + 3 \times 1 + 5 \times 1 \times 1 = 10$$

Sol.260.(b) According to the question,

$$\sin(50^\circ + A) - \cos(40^\circ - A)$$

Let $A = 10^\circ$

$$\text{Then } \sin 60^\circ - \cos 30^\circ = \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} = 0$$

Sol.261.(a) According to the question,

$$\sin \theta + \cos \theta = \sqrt{2} \cos \theta$$

$$\sin \theta = \cos \theta (\sqrt{2} - 1)$$

$$\text{now, } \frac{\sin\theta - \cos\theta}{\sin\theta} = \frac{\sin\theta}{\sin\theta} - \frac{\cos\theta}{\sin\theta}$$

$$1 - \frac{\cos\theta}{\cos\theta(\sqrt{2}-1)} \\ = 1 - \frac{1}{\sqrt{2}-1} \times \frac{(\sqrt{2}+1)}{(\sqrt{2}+1)} = -\sqrt{2}$$

$$\text{Sol.262.(b)} \quad \frac{\cos\theta}{\sec\theta-1} + \frac{\cos\theta}{\sec\theta+1} \\ = \cos\theta \left(\frac{\sec\theta+1+\sec\theta-1}{\tan^2\theta} \right) \\ = \cos\theta \left(\frac{2\sec\theta}{\tan^2\theta} \right) = 2\cot^2\theta$$

$$\text{Sol.263.(d)} \quad \sin\theta - \cos\theta = 0$$

$$\sin\theta = \cos\theta \Rightarrow \theta = 45^\circ$$

$$\text{Now, } \sin(\pi - \theta) + \sin\left(\frac{\pi}{2} - \theta\right)$$

$$= \sin(\theta) + \cos(\theta) = \sin(45^\circ) + \cos(45^\circ)$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

$$\text{Sol.264.(a)} \quad \cos\theta + \sec\theta = \sqrt{3} \\ \cos^3\theta + \sec^3\theta = (\sqrt{3})^3 - 3(\sqrt{3}) = 0$$

$$\text{Sol.265.(a)}$$

Pythagoras triplets :- (16, 63, 65)

$$\cos A = \frac{63}{65} \Rightarrow \sin A = \frac{16}{65}$$

$$\text{Now, } \tan A + \cot A = \frac{1}{\sin A \cos A}$$

$$= \frac{65 \times 65}{16 \times 63} = 4.19$$

$$\text{Sol.266.(a)} \quad \frac{\cos\theta}{(1+\sin\theta)} + \frac{\cos\theta}{(1-\sin\theta)} = 4$$

$$\frac{2}{(\cos^2\theta)} = \frac{4}{\cos\theta} \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

$$\text{Sol.267.(c)} \quad \sin\alpha + \cos\alpha = \frac{2}{\sqrt{3}}$$

$$1 + 2\sin\alpha \cos\alpha = \frac{4}{3}$$

$$\sin 2\alpha = \frac{4}{3} - 1 = \frac{1}{3}$$

$$\text{Now, } \tan\alpha + \cot\alpha = \frac{2}{\sin 2\alpha} = 6$$

$$\text{Sol.268.(d)} \quad (\operatorname{cosec}^2 B - 1)(\sec^2 B - 1) \\ (\cot^2 B)(\tan^2 B) = 1$$

$$\text{Sol.269.(a)} \quad \sin\theta + \cos\theta = \frac{1}{29} \Rightarrow 2\sin\theta$$

$$\cos\theta = \left(\frac{1}{29}\right)^2 - 1 = \left(\frac{-840}{841}\right)$$

$$\sin\theta - \cos\theta = \sqrt{1 - \left(\frac{-840}{841}\right)^2}$$

$$= \frac{41}{29}$$

$$\text{Now, } \frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta} = \frac{1}{29} \times \frac{29}{41} = \frac{1}{41}$$

$$\text{Sol.270.(c)}$$

$$\frac{(1 + \tan^2 A)}{\operatorname{cosec}^2 A \times \tan A} = \frac{(\sec^2 A)}{\operatorname{cosec}^2 A \times \tan A}$$

$$= \frac{(\sin^2 A)}{\cos^2 A \times \tan A} = \tan A$$

$$\text{Sol.271.(b)} \quad \cos^4 A - \sin^4 A \\ = (\cos^2 A - \sin^2 A)(\cos^2 A + \sin^2 A) \\ = 1 - 2\sin^2 A$$

$$\text{Sol.272.(b)} \quad \operatorname{cosec}\theta = \frac{29}{22} \\ \cot^2\theta = \operatorname{cosec}^2\theta - 1 = \frac{841}{484} - 1 = \frac{357}{484}$$

$$\text{Sol.273.(a)}$$

$$\sin A \left(1 + \frac{\sin A}{\cos A}\right) + \cos A \left(1 + \frac{\cos A}{\sin A}\right) \\ = \sin A + \frac{\sin^2 A}{\cos A} + \cos A + \frac{\cos^2 A}{\sin A} \\ = \frac{\sin^2 A + \cos^2 A}{\cos A} + \frac{\sin^2 A + \cos^2 A}{\sin A} \\ = \frac{1}{\cos A} + \frac{1}{\sin A} = \sec A + \operatorname{cosec} A$$

$$\text{Sol.274.(a)} \quad \frac{3 - \sin^2 A + \cos^2 A}{2 + 2\cos^2 A} \\ = \frac{\sin^2 A + \cos^2 A + 2 - \sin^2 A + \cos^2 A}{2 + 2\cos^2 A} \\ = \frac{\cos^2 A + 2 + \cos^2 A}{2 + 2\cos^2 A} = 1$$

$$\text{Sol.275.(a)} \quad \sec\theta + \tan\theta = \sqrt{3} \\ \sec\theta - \tan\theta = \frac{1}{\sqrt{3}} \\ \frac{\sqrt{3} + \frac{1}{\sqrt{3}}}{2} = \frac{2}{\sqrt{3}} \Rightarrow \theta = 30^\circ \\ \text{Now, } \sin\theta + \cos\theta = \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1 + \sqrt{3}}{2}$$

$$\text{Sol.276.(a)} \quad 2\tan^2 A + 4\cos^4 A = 3$$

$$\text{Put } A = 45^\circ$$

$$2\tan^2 45^\circ + 4\cos^4 45^\circ \\ = 2 \times 1 + 4 \times \left(\frac{1}{\sqrt{2}}\right)^4 = 2 + 1 = 3$$

$$\text{Sol.277.(a)} \quad \cot^2 A - \cos^2 A = \\ \cot^2 A \cdot \cos^2 A \dots (\text{identity}) \\ a^2 = \cot^2 A + \cos^2 A + 2 \cdot \cot A \cdot \cos A \\ b^2 = \cot^2 A + \cos^2 A - 2 \cdot \cot A \cdot \cos A \\ ab = \cot^2 A - \cos^2 A \\ \text{Now, } \\ a^2 - b^2 - 4\sqrt{ab} = 4 \cdot \cot A \cdot \cos A - 4 \\ \sqrt{\cot^2 A - \cos^2 A} \\ = 4 \cdot \cot A \cdot \cos A - 4 \cdot \cot A \cdot \cos A = 0$$

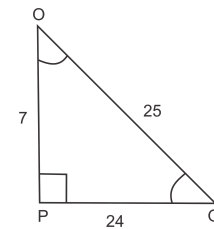
$$\text{Sol.278.(b)} \quad 1 - \frac{\tan^2 \phi}{\sec^2 \phi} = \frac{\sec^2 \phi - \tan^2 \phi}{\sec^2 \phi} \\ = \frac{1}{\sec^2 \phi} = \cos^2 \phi = \frac{\sin^2 \phi \cdot \cos^2 \phi}{\sin^2 \phi} \\ = \sin^2 \phi \cdot \cot^2 \phi$$

$$\text{Sol.279.(c)} \quad 1 + \cot^2 B = \operatorname{cosec}^2 B$$

$$\cot^2 B - \operatorname{cosec}^2 B = -1$$

$$\text{Sol.280.(c)}$$

Pythagoras triplets:- (7, 24, 25)



$$\sin Q + \cos Q = \frac{7}{25} + \frac{24}{25} = \frac{31}{25}$$

$$\text{Sol.281.(a)} \quad \frac{2 \sec A}{\sec^2 A - 1} = \frac{2 \cot^2 A}{\cos A} \\ = \frac{2 \cos A}{\sin^2 A} = \frac{2 \cot A}{\sin A} = 2 \operatorname{cosec} A \cdot \cot A$$

$$\text{Sol.282.(a)} \quad \frac{\cos^2 A}{(1+\sin A)} = \frac{1-\sin A}{(1+\sin A)} \\ = \frac{(1+\sin A)(1-\sin A)}{(1+\sin A)} = (1-\sin A)$$

$$\text{Sol.283.(b)}$$

If $A + B = 90^\circ$ then $\sin^2 A + \sin^2 B = 1$

$$\sin^2 30^\circ + \sin^2 40^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \\ \sin^2 35^\circ + \sin^2 45^\circ + \sin^2 50^\circ + \sin^2 60^\circ \\ = \sin^2 30^\circ + \sin^2 60^\circ + \sin^2 40^\circ + \sin^2 50^\circ \\ + \sin^2 45^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 35^\circ \\ = 1 + 1 + 1 + 1 = 4$$

$$\text{Sol.284.(a)}$$

If $A + B = 90^\circ$, Then $\sin^2 A + \sin^2 B = 1$

and $\cos^2 A + \cos^2 B = 1$

$$\frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos^2 53^\circ \\ \sin 37^\circ \\ = \frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos^2 53^\circ \\ = \frac{1}{1} + 1 = 2$$

$$\text{Sol.285.(b)} \quad \frac{1}{\sin\theta} - \frac{\cot^2 \theta}{1 + \operatorname{cosec}\theta} \\ = \operatorname{cosec}\theta - \frac{(\operatorname{cosec}\theta - 1)(\operatorname{cosec}\theta + 1)}{1 + \operatorname{cosec}\theta} \\ = \operatorname{cosec}\theta - \operatorname{cosec}\theta + 1 = 1$$

$$\text{Sol.286.(a)}$$

$$\frac{\sin 58^\circ}{\cos 32^\circ} + \frac{\sin 55^\circ \cdot \sec 35^\circ}{\tan 5^\circ \cdot \tan 45^\circ \cdot \tan 85^\circ} \\ = \frac{\cos 32^\circ}{\cos 32^\circ} + \frac{\sin 55^\circ \cdot \operatorname{cosec} 55^\circ}{\tan 5^\circ \cdot \tan 85^\circ} \\ = 1 + \frac{1}{\tan 5^\circ \cdot \cot 5^\circ} = 1 + 1 = 2$$

$$\text{Sol.287.(c)} \quad \sec\theta = \frac{50}{23} \Rightarrow \sec^2 \theta = \frac{2500}{529} \\ \tan^2 \theta = \sec^2 \theta - 1 = \frac{2500 - 529}{529} \\ = \frac{1971}{529} = 3 \frac{384}{529}$$

$$\text{Sol.288.(d)} \frac{\cos^2 25^\circ - \sin^2 65^\circ}{\cos^2 25^\circ + \sin^2 65^\circ} =$$

$$\frac{\cos^2 25^\circ - \cos^2 25^\circ}{\cos^2 25^\circ + \sin^2 65^\circ} = 0$$

$$\text{Sol.289.(b)} \frac{\cot^3 A - 1}{\cot A - 1}$$

$$= \frac{(\cot A - 1)(\cot^2 A + \cot A + 1)}{\cot A - 1}$$

$$= (\cot^2 A + \cot A + 1)$$

$$\text{Sol.290.(d)} \text{ If } \sec \theta + \tan \theta = a$$

$$\text{Then, } \sec \theta - \tan \theta = \frac{1}{a}$$

on solving above equation, we get

$$\sec \theta = \frac{a + \frac{1}{a}}{2} = \frac{a^2 + 1}{2a}$$

$$\text{Sol.291.(d)} \frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} + 2$$

$$= \frac{1 - \sin A}{\cos A} + \frac{1 + \sin A}{\cos A} + 2$$

$$= \frac{1 - \sin A + 1 + \sin A}{\cos A} + 2$$

$$= 2 \sec A + 2 = 2(1 + \sec A)$$

$$\text{Sol.292.(a)} \sec \theta + \tan \theta = \frac{1}{\sqrt{3}}$$

$$\sec \theta - \tan \theta = \sqrt{3}$$

$$\sec \theta = \frac{\frac{1}{\sqrt{3}} + \sqrt{3}}{2} = \frac{2}{\sqrt{3}} \Rightarrow \cos \theta = \frac{\sqrt{3}}{2}$$

Therefore, $\theta = 30^\circ$

$$\text{Now, } \cot \theta + \cos \theta = \sqrt{3} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

$$\text{Sol.293.(c)} \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} = \frac{4}{5}$$

$$\frac{\sin \theta - \cos \theta + \sin \theta + \cos \theta}{\sin \theta - \cos \theta - \sin \theta - \cos \theta} = \frac{4 + 5}{4 - 5}$$

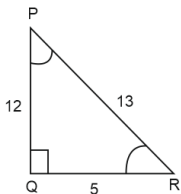
$$\frac{2 \sin \theta}{-2 \cos \theta} = \frac{9}{-1} \Rightarrow \cot \theta = \frac{1}{9}$$

$$\operatorname{cosec}^2 \theta = 1 + \left(\frac{1}{9}\right)^2 = \left(\frac{82}{81}\right)$$

$$\text{Now, } \frac{\operatorname{cosec}^2 \theta}{2 - \operatorname{cosec}^2 \theta} = \frac{\frac{82}{81}}{2 - \frac{82}{81}}$$

$$= \frac{82}{80} = \frac{41}{40}$$

$$\text{Sol.294.(c)}$$



$$\tan P + \cot R = \frac{5}{12} + \frac{5}{12} = \frac{10}{12}$$

$$\text{Sol.295.(d)} \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}} = \frac{\frac{5}{6}}{1 - \frac{1}{6}} = 1$$

$$\Rightarrow A + B = 45^\circ$$

$$\text{Sol.296.(d)} \tan A = \frac{12}{5}$$

$$\frac{5(1 - \cos^2 A)}{6(1 - \sin^2 A)} = \frac{5(\sin^2 A)}{6(\cos^2 A)} = \frac{5}{6} \times \tan^2 A$$

$$= \frac{5}{6} \times \frac{144}{25} = \frac{24}{5}$$

$$\text{Sol.297.(c)} \tan \theta \cdot \tan 2\theta = 1$$

Put $\theta = 30^\circ$, Then

$$\tan 30^\circ \cdot \tan 60^\circ = \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$$

....(satisfied)

$$\cot 5\theta = \cot(150^\circ) = \cot(90^\circ + 60^\circ)$$

$$= -\tan 60^\circ = -\sqrt{3}$$

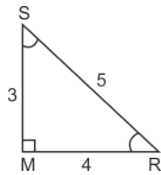
$$\text{Sol.298.(c)} 4 \sin^2 A - 3 = 0$$

Multiply by $\sin A$, both sides

$$4 \sin^3 A - 3 \sin A = 0 \text{ or}$$

$$3 \sin A - 4 \sin^3 A = 0$$

$$\text{Sol.299.(b)} \text{ Pythagoras triplets :- } (3, 4, 5)$$



$$\sec R = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{5}{4}$$

$$\text{Sol.300.(d)} 2(\tan^2 A \cdot \cos^2 A + \cos^2 A)$$

$$= 2 \cos^2 A (\tan^2 A + 1)$$

$$= 2 \cos^2 A (\sec^2 A) = 2$$

$$\text{Sol.301.(d)}$$

$$\tan(A + B - C) = \frac{1}{\sqrt{3}} = \tan 30^\circ$$

$$\Rightarrow (A + B - C) = 30^\circ \text{e.q. (1)}$$

$$\cos(B + C - A) = \frac{1}{2} = \cos 60^\circ$$

$$\Rightarrow (B + C - A) = 60^\circ \text{e.q. (2)}$$

$$\sin(C + A - B) = \frac{1}{\sqrt{2}} = \sin 45^\circ$$

$$\Rightarrow (C + A - B) = 45^\circ \text{e.q. (3)}$$

Adding e.q. (1), e.q. (2) and e.q. (3), we get

$$A + B + C = 30^\circ + 60^\circ + 45^\circ = 135^\circ$$

$$\text{Sol.302.(d)} \frac{(1 + \sec \phi)}{\sec \phi} \times (1 - \cos \phi)$$

$$= (1 + \cos \phi)(1 - \cos \phi)$$

$$= (1 - \cos^2 \phi) = \sin^2 \phi$$

$$\text{Sol.303.(c)} \sec C + \sin C \cdot \sec C$$

$$= \sec C + \tan C = \sec 45^\circ + \tan 45^\circ$$

$$= \sqrt{2} + 1$$

$$\text{Sol.304.(b)} 4 \sin \theta + 5 \cos \theta = 3$$

$$4 \cos \theta - 5 \sin \theta = x$$

$$\text{-----}$$

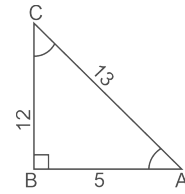
$$(4)^2 + (5)^2 = (3)^2 + (x)^2$$

$$x = \sqrt{16 + 25 - 9} = \sqrt{32} = 4\sqrt{2}$$

$$\text{Sol.305.(b)}$$

Pythagorean triplets:- (5, 12, 13)

$$\tan A = \frac{\text{Perpendicular}}{\text{Base}} = \frac{12}{5}$$



$$\operatorname{cosec} A = \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{13}{12}$$

$$\sec A = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{13}{5}$$

$$\text{Sol.306.(a)} \cos \theta = \sin \theta \Rightarrow \theta = 45^\circ$$

$$\text{Now, } \sin^8 \theta + \cos^8 \theta = \left(\frac{1}{\sqrt{2}}\right)^8 + \left(\frac{1}{\sqrt{2}}\right)^8$$

$$= \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$$

$$\text{Sol.307.(d)} \frac{3(\sin^4 Z - \cos^4 Z + 1)}{\sin^2 Z}$$

$$= \frac{3[(\sin^2 Z + \cos^2 Z)(\sin^2 Z - \cos^2 Z) + 1]}{\sin^2 Z} \because$$

$$(\sin^2 Z + \cos^2 Z = 1)$$

$$= \frac{3(\sin^2 Z - \cos^2 Z + 1)}{\sin^2 Z}$$

$$= \frac{3(\sin^2 Z + \sin^2 Z)}{\sin^2 Z} \because (1 - \cos^2 Z = \sin^2 Z)$$

$$= \frac{3 \times 2 \sin^2 Z}{\sin^2 Z} = 6$$

Short trick : Put $Z = 90^\circ$

$$\frac{3(\sin^4 Z - \cos^4 Z + 1)}{\sin^2 Z} = \frac{3(1 - 0 + 1)}{1} = 6$$

$$\text{Sol.308.(c)}$$

$$\sin 47^\circ \cos 33^\circ - \cos 43^\circ \sin 57^\circ$$

$$= \sin(90^\circ - 43^\circ) \cos 33^\circ - \cos 43^\circ$$

$$\sin(90^\circ - 33^\circ)$$

$$= \cos 43^\circ \cos 33^\circ - \cos 43^\circ \cos 33^\circ = 0$$

$$\text{Sol.309.(a)} \frac{1 + \sin A}{\cos A} + \frac{\cos A}{1 + \sin A}$$

$$= \frac{(1 + \sin A)^2 + (\cos A)^2}{(1 + \sin A) \cos A}$$

$$= \frac{1 + \sin^2 A + 2 \sin A + \cos^2 A}{(1 + \sin A) \cos A}$$

$$= \frac{2 + 2 \sin A}{(1 + \sin A) \cos A}$$

$$= \frac{2(1 + \sin A)}{(1 + \sin A) \cos A} = 2 \sec A$$

$$\text{Sol.310.(b)} (1 + \tan^2 A)(1 + \cot^2 A)$$

$$(\sec^2 A)(\operatorname{cosec}^2 A) = \frac{1}{\cos^2 A} \times \frac{1}{\sin^2 A}$$

$$= \frac{1}{\sin^2 A (1 - \sin^2 A)}$$

Sol.311.(c) According to the question,

$$\frac{\cos B}{\sin B} - \frac{\sin B}{\cos B} \times (\sin B \cdot \cos B)$$

$$= \frac{(\cos B)^2 - (\sin B)^2}{\sin B \cdot \cos B} \times (\sin B \cdot \cos B)$$

$$= 2\cos^2 B - 1$$

Sol.312.(d) Given:- $\cot^2 \theta - \operatorname{cosec}^2 \theta = ?$

$$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$-(\cot^2 \theta - \operatorname{cosec}^2 \theta) = 1$$

$$\cot^2 \theta - \operatorname{cosec}^2 \theta = -1$$

Sol.313.(d) According to the question,

$$(\sin^4 \theta + \cos^4 \theta) = (\sin^2 \theta + \cos^2 \theta)^2 - 2\sin^2 \theta \cdot \cos^2 \theta$$

$$\text{We know that } (\sin^2 \theta + \cos^2 \theta) = 1$$

$$1^2 - 2 \times \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$

Sol.314.(d) According to the question,

$$\sqrt{\frac{1 + \cos P \times (1 + \cos P)}{1 - \cos P \times (1 + \cos P)}} = \frac{1 + \cos P}{\sin P}$$

$$= \operatorname{cosec} P + \cot P$$

Sol.315.(c) $\tan(50^\circ + \theta) - \cot(40^\circ - \theta)$

$$= \tan(50^\circ + \theta) - \cot[90^\circ - (50^\circ + \theta)] \{ \cot(90^\circ - \theta) = \tan \theta \}$$

$$= \tan(50^\circ + \theta) - \tan(50^\circ + \theta) = 0$$

Sol.316.(a) $(1 - \sin^2 A) \operatorname{cosec}^2 A$

$$= (\cos^2 A) \operatorname{cosec}^2 A = \frac{\cos^2 A}{\sin^2 A} = \cot^2 A$$

Sol.317.(d)

$$\tan 15^\circ = 2 - \sqrt{3} \Rightarrow \cot 15^\circ = 2 + \sqrt{3}$$

$$\tan 15^\circ \cot 75^\circ + \tan 75^\circ \cot 15^\circ$$

$$= \tan 15^\circ \cot(90^\circ - 15^\circ) + \tan(90^\circ - 15^\circ) \cot 15^\circ$$

$$= \tan 15^\circ \tan 15^\circ + \cot 15^\circ \cot 15^\circ$$

$$= (\tan 15^\circ)^2 + (\cot 15^\circ)^2$$

$$= (2 - \sqrt{3})^2 + (2 + \sqrt{3})^2$$

$$= 4 + 3 - 4\sqrt{3} + 4 + 3 + 4\sqrt{3} = 14$$

Sol.318.(c) $\frac{3\sin\theta - \cos\theta}{\cos\theta + \sin\theta} = 1$

$$2\sin\theta = 2\cos\theta \Rightarrow \frac{\cos\theta}{\sin\theta} = \frac{2}{2} \Rightarrow \cot\theta = 1$$

Sol.319.(d) Formula used : $\cos(A - B)$

$$= \cos A \cdot \cos B + \sin A \cdot \sin B$$

$$\cos(36^\circ + A) \cdot \cos(36^\circ - A) + \cos(54^\circ + A) \cdot \cos(54^\circ - A)$$

$$\Rightarrow \cos(36^\circ + A) \cdot \cos(36^\circ - A) + \cos[90^\circ - (36^\circ - A)] \cdot \cos[90^\circ - (36^\circ + A)]$$

$$\Rightarrow \cos(36^\circ + A) \cdot \cos(36^\circ - A) + \sin(36^\circ + A) \cdot \sin(36^\circ - A)$$

$$\Rightarrow \cos[36^\circ + A - (36^\circ - A)] \Rightarrow \cos(2A)$$

Sol.320.(d)

$$\sin 23^\circ = \frac{a}{b} = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

$$\text{Base} = \sqrt{b^2 - a^2}$$

$$\sec 23^\circ - \sin 67^\circ = \sec 23^\circ - \cos 23^\circ$$

$$= \frac{b}{\sqrt{b^2 - a^2}} - \frac{\sqrt{b^2 - a^2}}{b}$$

$$= \frac{b^2 - b^2 + a^2}{b\sqrt{b^2 - a^2}} = \frac{a^2}{b\sqrt{b^2 - a^2}}$$

Sol.321.(a) $\frac{\cos \beta}{\sec \alpha} = 15$ and $\frac{\sin \beta}{\sec \alpha} = 16$

$$\frac{\cos \beta}{\sec \alpha} \times \frac{\sec \alpha}{\sin \beta} = \frac{15}{16}$$

$$\Rightarrow \cot \beta = \frac{15}{16} = \frac{\text{Base}}{\text{Perpendicular}}$$

$$\text{Hypotenuse} = \sqrt{225 + 256} = \sqrt{481}$$

$$\sin^2 \beta = \left(\frac{\text{Perpendicular}}{\text{hypotenuse}}\right)^2$$

$$= \left(\frac{16}{\sqrt{481}}\right)^2 = \frac{256}{481}$$

Sol.322.(c) $\frac{\sin^3 \alpha + \cos^3 \alpha}{\sin \alpha + \cos \alpha}$

$$= \frac{(\sin \alpha + \cos \alpha)(\sin^2 \alpha + \cos^2 \alpha - \sin \alpha \cos \alpha)}{\sin \alpha + \cos \alpha}$$

$$= 1 - \sin \alpha \cos \alpha$$

Sol.323.(a) According to question,

$$\tan \frac{\pi}{6} + \sec \frac{\pi}{6} = x \Rightarrow \tan 30^\circ + \sec 30^\circ = x$$

$$\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} \Rightarrow x = \sqrt{3}$$

Sol.324.(a) x lies in third quadrant In third quadrant only $\tan \theta$ and $\cot \theta$ are positive

$$\text{Now, } x = \cos^{-1}\left(-\frac{1}{2}\right) = 60^\circ$$

$$\text{Then, } \tan x = \tan(60^\circ) = \sqrt{3}$$

Sol.325.(c) According to the question,

$$\cos 75^\circ = \cos(45^\circ + 30^\circ)$$

$$= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4}$$

Sol.326.(a) According to the question,

$$\frac{\sin^4 \theta + \cos^4 \theta}{1 - 2\sin^2 \theta \cdot \cos^2 \theta}$$

$$= \frac{(\sin^2 \theta + \cos^2 \theta)^2 - 2\sin^2 \theta \cos^2 \theta}{1 - 2\sin^2 \theta \cos^2 \theta}$$

$$= \frac{1 - 2\sin^2 \theta \cos^2 \theta}{1 - 2\sin^2 \theta \cos^2 \theta} = 1$$

Sol.327.(d) $\left(\frac{\sin(x-y)}{\sin(x+y)}\right) \left(\frac{\tan x + \tan y}{\tan x - \tan y}\right)$

$$= \left(\frac{\sin x \cos y - \cos x \sin y}{\sin x \cos y + \cos x \sin y}\right) \left(\frac{\tan x + \tan y}{\tan x - \tan y}\right)$$

$$= \left(\frac{\frac{\sin x \cos y - \cos x \sin y}{\cos x \cos y}}{\frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y}}\right) \left(\frac{\tan x + \tan y}{\tan x - \tan y}\right)$$

$$= \left(\frac{\tan x - \tan y}{\tan x + \tan y}\right) \left(\frac{\tan x + \tan y}{\tan x - \tan y}\right) = 1$$

Sol.328.(c) $\sin\left(\frac{A}{2}\right) = \sqrt{\frac{(1 - \cos A)}{2}}$

$$\text{And, } \cos\left(\frac{A}{2}\right) = \sqrt{\frac{(1 + \cos A)}{2}}$$

$$\text{Given, } \tan \frac{A}{2} = x$$

$$= \sqrt{\frac{(1 - \cos A)}{2}} \div \sqrt{\frac{(1 + \cos A)}{2}}$$

$$= \sqrt{\frac{(1 - \cos A)}{2}} \times \sqrt{\frac{2}{(1 + \cos A)}}$$

$$= \sqrt{\frac{(1 - \cos A)}{(1 + \cos A)}}$$

Formula :- $\tan \frac{A}{2} = \sqrt{\frac{(1 - \cos A)}{(1 + \cos A)}}$

Sol.329.(d)

$$\left(\frac{1}{\sin \theta} + \frac{1}{\tan \theta}\right) \left(\frac{1}{\sin \theta} - \frac{1}{\tan \theta}\right)$$

$$= \left(\frac{1}{\sin^2 \theta} - \frac{1}{\tan^2 \theta}\right) \Rightarrow (\operatorname{cosec}^2 \theta - \cot^2 \theta) = 1$$

Sol.330.(a) $\sin 2x + 2 \sin 4x + \sin 6x$

$$2 \sin \left(\frac{6x+2x}{2}\right) \cos \left(\frac{6x-2x}{2}\right) + 2 \sin 4x$$

$$= 2 \sin 4x \cos 2x + 2 \sin 4x$$

$$= 2 \sin 4x (\cos 2x + 1)$$

$$= 2 \sin 4x \{2 \cos^2 x - 1 + 1\}$$

$$= 2 \sin 4x (2 \cos^2 x) = 4 \cos^2 x \sin 4x$$

Sol.331.(b) In third quadrant, only value of $\tan \theta$ and $\cot \theta$ is positive.

$$\text{Given that, } \sin A = -\frac{3}{5}$$

$$= \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

Using pythagoras theorem,

$$H^2 = P^2 + B^2 \Rightarrow B^2 = 25 - 9 = 16$$

$$B = \sqrt{16} = \pm 4$$

$$\text{Now, } \sec A = \frac{\text{Hypotenuse}}{\text{Base}} = -\frac{5}{4}$$

Sol.332.(a) $12(\sin^4 \theta + \cos^4 \theta) +$

$$18(\sin^6 \theta + \cos^6 \theta) + 78\sin^2 \theta \cos^2 \theta$$

$$\text{Put } \theta = 0^\circ$$

$$= 12(0 + 1) + 18(0 + 1) + 78 \times 0 \times 1 = 30$$

Sol.333.(c) Let $A = 45^\circ$ and $B = 30^\circ$

According to question,

$$\tan(45^\circ - 30^\circ) = \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ}$$

$$\tan 15^\circ = \frac{1 - \frac{1}{\sqrt{3}}}{1 + 1 \times \frac{1}{\sqrt{3}}} = \frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$$

$$= \frac{3+1-2\sqrt{3}}{3-1} = 2 - \sqrt{3}$$

Sol.334.(b) $\frac{\sin x - \cos x}{\sin x + \cos x} = \frac{2}{5}$

$$5 \sin x - 5 \cos x = 2 \sin x + 2 \cos x$$

$$3 \sin x = 7 \cos x$$

$$\frac{\cos x}{\sin x} = \frac{3}{7} \Rightarrow \cot x = \frac{3}{7} \Rightarrow \cot^2 x = \frac{9}{49}$$

$$\text{Now, } \frac{1 + \cot^2 x}{1 - \cot^2 x} = \frac{1 + \frac{9}{49}}{1 - \frac{9}{49}} = \frac{\frac{49+9}{49}}{\frac{49-9}{49}}$$

$$= \frac{58}{49} \times \frac{49}{40} = \frac{58}{40} = 1.45$$

Sol.335.(c) $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2$

$$= \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta + \sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta$$

$$= 2(\sin^2 \theta + \cos^2 \theta) = 2 \times 1 = 2$$

Sol.336.(c) $\left(\frac{\cos A}{1 - \sin A}\right) + \left(\frac{\cos A}{1 + \sin A}\right) = 4$

$$\cos A \left(\frac{1 + \sin A + 1 - \sin A}{(1 - \sin A)(1 + \sin A)}\right) = 4$$

$$\cos A \left(\frac{2}{1 - \sin^2 A}\right) = 4$$

$$\cos A \left(\frac{2}{\cos^2 A}\right) = 4$$

$$\cos A = \frac{1}{2} \Rightarrow A = 60^\circ$$

Sol.337.(a) $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$

$$\cot 15^\circ = \cot(45^\circ - 30^\circ)$$

$$= \frac{\cot 45^\circ \cot 30^\circ + 1}{\cot 30^\circ - \cot 45^\circ}$$

$$= \frac{1 \times \sqrt{3} + 1}{\sqrt{3} - 1} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{4 + 2\sqrt{3}}{3 - 1} = 2 + \sqrt{3}$$

Sol.338.(c) $P \cos \alpha = 3 \Rightarrow \sec \alpha = \frac{P}{3}$

taking square both sides,

$$\sec^2 \alpha = \frac{P^2}{9} \quad \text{---(1)}$$

$$\text{Now, } 4 \tan \alpha = Q \Rightarrow \tan \alpha = \frac{Q}{4}$$

taking square both sides

$$\tan^2 \alpha = \frac{Q^2}{16} \quad \text{---(2)}$$

from equation (1) and (2), we get

$$\sec^2 \alpha - \tan^2 \alpha = \frac{P^2}{9} - \frac{Q^2}{16}$$

$$\therefore \frac{P^2}{9} - \frac{Q^2}{16} = 1$$

Sol.339.(b) $\sin \theta + \sin^2 \theta = 1$

$$\sin \theta = 1 - \sin^2 \theta$$

$$\sin \theta = \cos^2 \theta \Rightarrow \sin^2 \theta = \cos^4 \theta$$

Now,

$$\cos^2 \theta + \cos^4 \theta = \cos^2 \theta + \sin^2 \theta = 1$$

Sol.340.(a) $(a - b)^2 + (a + b)^2 = 2(a^2 + b^2)$

$$(\cos x - \sin x)^2 + (\cos x + \sin x)^2$$

$$= 2(\cos^2 x + \sin^2 x) = 2$$

Sol.341.(a) $\frac{\sec^2(90^\circ - \theta) - \cot^2 \theta}{2(\sin^2 35^\circ + \sin^2 55^\circ)}$

$$\frac{\csc^2(\theta) - \cot^2 \theta}{2(\sin^2 35^\circ + \cos^2 35^\circ)} = \frac{1}{2} = 0.5$$

Sol.342.(c) $\operatorname{cosec} 2910^\circ = \operatorname{cosec}(360^\circ \times 8 + 30^\circ) = \operatorname{cosec} 30^\circ = 2$

$$\sec 4260^\circ = \sec(360^\circ \times 11 + 300^\circ)$$

$$= \sec 300^\circ = \sec(360^\circ - 60^\circ) = \sec 60^\circ = 2$$

$$\tan 2565^\circ = \tan(360^\circ \times 7 + 45^\circ)$$

$$= \tan(45^\circ) = 1$$

$$\cot 1755^\circ = \cot(360^\circ \times 4 + 315^\circ)$$

$$= \cot(315^\circ) = \cot(270^\circ + 45^\circ)$$

$$= -\tan 45^\circ = -1$$

Now,

$$\operatorname{cosec} 2910^\circ + \sec 4260^\circ + \tan 2565^\circ + \cot 1755^\circ \Rightarrow 2 + 2 + 1 - 1 = 4$$

Sol.343.(c) For the maximum value of

$$\sin^{12} \theta + \cos^{20} \theta, \text{ Put } \theta = 90^\circ,$$

$$\sin^{12}(90^\circ) + \cos^{20}(90^\circ) = 1 + 0 = 1$$

Sol.344.(d) $\sin(-405^\circ) = -\sin(360^\circ + 45^\circ)$

$$= -\sin 45^\circ = \frac{-1}{\sqrt{2}}$$

Sol.345.(a) $\cos 2\theta = (2\cos^2 \theta - 1)$

$$= (1 - 2\sin^2 \theta)$$

$$\text{Now, } \frac{\sin \theta - 2\sin^3 \theta}{2\cos^3 \theta - \cos \theta} = \frac{\sin \theta(1 - 2\sin^2 \theta)}{\cos \theta(2\cos^2 \theta - 1)}$$

$$= \tan \theta$$

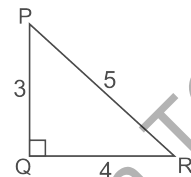
Sol.346.(b) $\frac{1 + \sin t}{4 - 4 \sin t} - \frac{1 - \sin t}{4 + 4 \sin t}$

$$= \frac{1}{4} \left[\frac{1 + \sin t}{(1 - \sin t)} - \frac{1 - \sin t}{(1 + \sin t)} \right]$$

$$= \frac{1}{4} \left[\frac{(1 + \sin t)^2 - (1 - \sin t)^2}{(1 - \sin t)(1 + \sin t)} \right]$$

$$= \frac{1}{4} \left[\frac{4 \sin t}{\cos^2 t} \right] = \frac{\sin t}{\cos t} \times \frac{1}{\cos t} = \tan t \sec t$$

Sol.347.(c)



$$\sin P = \frac{4}{5}, \sin Q = 90^\circ = 1 \text{ And, } \sin R = \frac{3}{5}$$

$$\text{Hence, } \sin P + \sin Q + \sin R$$

$$= \frac{4}{5} + 1 + \frac{3}{5} = \frac{12}{5}$$

Sol.348.(c) Given, $\cot \theta = \frac{3}{2}$

$$= \frac{\text{Base}}{\text{Perpendicular}}$$

By using pythagoras theorem, we get

$$\text{Hypotenuse} = \sqrt{13}$$

$$\text{Now, } \frac{\sqrt{13} \sin \theta - 3 \tan \theta}{3 \tan \theta + \sqrt{13} \cos \theta}$$

$$= \frac{\sqrt{13} \times \frac{2}{\sqrt{13}} - 3 \times \frac{2}{3}}{3 \times \frac{2}{3} + \sqrt{13} \times \frac{3}{\sqrt{13}}} = 0$$

Sol.349.(a)

$$\tan(45^\circ - 30^\circ) = \frac{\tan 45^\circ - \tan 30^\circ}{(1 + \tan 45^\circ \tan 30^\circ)}$$

$$= \frac{1 - \frac{1}{\sqrt{3}}}{(1 + 1 \times \frac{1}{\sqrt{3}})} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} = 0.268$$

Short trick :-

$$\tan 15^\circ = (2 - \sqrt{3}) = 2 - 1.732 = 0.268$$

Sol.350.(c) If $\operatorname{Cosec} \theta + \cot \theta = m,$

$$\text{Then } \operatorname{Cosec} \theta - \cot \theta = \frac{1}{m}$$

$$\text{Now, } \frac{m^2 - 1}{m^2 + 1} = \frac{m - \frac{1}{m}}{m + \frac{1}{m}}$$

$$= \frac{\operatorname{Cosec} \theta + \cot \theta - (\operatorname{Cosec} \theta - \cot \theta)}{\operatorname{Cosec} \theta + \cot \theta + (\operatorname{Cosec} \theta - \cot \theta)}$$

$$= \frac{2 \cot \theta}{2 \operatorname{Cosec} \theta} = \cos \theta$$

Sol.351.(d) $\sin \theta + \operatorname{cosec} \theta = \sqrt{5}$

$$\sin \theta + \frac{1}{\sin \theta} = \sqrt{5} \text{ Then, } \sin^3 \theta + \frac{1}{\sin^3 \theta}$$

$$= (\sqrt{5})^3 - 3 \times \sqrt{5} = 2\sqrt{5}$$

Sol.352.(c)

$$\text{Given, } \cot \theta = \frac{b}{a} = \frac{\text{Base}}{\text{Perpendicular}}$$

By using pythagoras theorem,

$$\text{Hypotenuse} = \sqrt{b^2 + a^2}$$

$$\text{Now, } \frac{b \cos \theta - a \sin \theta}{b \cos \theta + a \sin \theta}$$

$$= \frac{\frac{b \times b}{\sqrt{b^2 + a^2}} - \frac{a \times a}{\sqrt{b^2 + a^2}}}{\frac{b \times b}{\sqrt{b^2 + a^2}} + \frac{a \times a}{\sqrt{b^2 + a^2}}} = \frac{b^2 - a^2}{b^2 + a^2}$$

Sol.353.(c) Let $\theta = 45^\circ$

According to question,

$$\frac{1 + \sin^4 \theta + \cos^4 \theta}{\cos^2 \theta + \sin^4 \theta} = \frac{1 + \sin^4 45^\circ + \cos^4 45^\circ}{\cos^2 45^\circ + \sin^4 45^\circ}$$

$$= \frac{1 + \left(\frac{1}{\sqrt{2}}\right)^4 + \left(\frac{1}{\sqrt{2}}\right)^4}{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^4}$$

$$= \frac{1 + \frac{1}{4} + \frac{1}{4}}{\frac{1}{2} + \frac{1}{4}} = \frac{\frac{6}{4}}{\frac{3}{4}} = 2$$

Sol.354.(c) $\operatorname{cosec}(\alpha + \beta)$

$$= \frac{\sec \alpha \times \sec \beta \times \operatorname{cosec} \alpha \times \operatorname{cosec} \beta}{\sec \alpha \times \operatorname{cosec} \beta + \operatorname{cosec} \alpha \times \sec \beta}$$

$$\operatorname{cosec}(45^\circ + 30^\circ)$$

$$= \frac{\sec 45^\circ \times \sec 30^\circ \times \operatorname{cosec} 45^\circ \times \operatorname{cosec} 30^\circ}{\sec 45^\circ \times \operatorname{cosec} 30^\circ + \operatorname{cosec} 45^\circ \times \sec 30^\circ}$$

$$= \frac{\sqrt{2} \times \frac{2}{\sqrt{3}} \times \sqrt{2} \times 2}{\sqrt{2} \times 2 + \sqrt{2} \times \frac{2}{\sqrt{3}}}$$

$$= \frac{\sqrt{2} \times \frac{2}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{2\sqrt{2}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3} + 1}$$

$$= \frac{2\sqrt{2}}{\sqrt{3} + 1} \times \frac{\sqrt{3} - 1}{\sqrt{3} - 1}$$

$$= \sqrt{2}(\sqrt{3} - 1) = \sqrt{6} - \sqrt{2}$$

Sol.355.(c) $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$

$$= \sqrt{\frac{(1 + \cos \theta) \times (1 + \cos \theta)}{(1 - \cos \theta) \times (1 + \cos \theta)}}$$

$$= \sqrt{\frac{(1 + \cos \theta)^2}{1 - \cos^2 \theta}} = \sqrt{\frac{(1 + \cos \theta)^2}{\sin^2 \theta}}$$

$$= \frac{1 + \cos \theta}{\sin \theta} = \operatorname{cosec} \theta + \cot \theta$$

Sol.356.(a) $\frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta}$

$$= \frac{\cos^2 \theta}{\sin \theta} = \cos \theta \cdot \cot \theta$$

Sol.357.(c) $\frac{21 \cos A + 3 \sin A}{3 \cos A + 4 \sin A} = 2$

$$= 21 \cos A + 3 \sin A = 6 \cos A + 8 \sin A$$

$$= 15 \cos A = 5 \sin A$$

$$\Rightarrow \cot A = \frac{5}{15} = \frac{1}{3}$$

Sol.358.(c) $\tan 4384^\circ + \cot 6814^\circ$

$$= \tan (360^\circ \times 12 + 64^\circ) + \cot (360^\circ \times 18 + 334^\circ) = \tan 64^\circ + \cot (270^\circ + 64^\circ)$$

$$= \tan 64^\circ - \tan 64^\circ = 0$$

Sol.359.(a)

$$10\sqrt{2} \sin 45^\circ - 7\sqrt{2} \cos 45^\circ + 9 \tan 45^\circ$$

$$10 - 7 + 9 = 12$$

Sol.360.(d)

$$\sin 600^\circ \cos 750^\circ + \sin 150^\circ \cos 240^\circ$$

$$= \sin (180^\circ \times 3 + 60^\circ) \cos (180^\circ \times 4 + 30^\circ)$$

$$+ \sin (180^\circ - 30^\circ) \cos (180^\circ + 60^\circ)$$

$$= -\sin 60^\circ \cos 30^\circ - \sin 30^\circ \cos 60^\circ$$

$$= -\sin (60^\circ + 30^\circ) = -1$$

Sol.361.(d) $\sin 2\theta = 2 \sin \theta \cos \theta$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\frac{\sin \theta}{1 + \cos \theta} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = \tan \left(\frac{\theta}{2} \right)$$

$$\Rightarrow \tan \left(\frac{\theta}{2} \right) = \frac{\sin \theta}{1 + \cos \theta}$$

Short trick : put $\theta = 30^\circ$

Sol.362.(c)

$$3x = \sec A \Rightarrow 9x^2 = \sec^2 A \dots\dots (1)$$

$$\frac{3}{x} = \tan A \Rightarrow \frac{9}{x^2} = \tan^2 A \dots\dots (2)$$

Subtracting eqn.(2) from (1), we get

$$9x^2 - \frac{9}{x^2} = 1 \Rightarrow 9(x^2 - \frac{1}{x^2}) = 1$$

Sol.363.(d) $\tan^4 \theta + \tan^2 \theta = 1$

$$\tan^2 \theta (\tan^2 \theta + 1) = 1 \Rightarrow \tan^2 \theta \times \sec^2 \theta = 1$$

$$\frac{\sin^2 \theta}{\cos^2 \theta} \times \frac{1}{\cos^2 \theta} = 1 \Rightarrow \sin^2 \theta = \cos^4 \theta$$

Now, $11 (\cos^4 \theta + \cos^2 \theta)$

$$= 11 (\sin^2 \theta + \cos^2 \theta) = 11$$

Sol.364.(d) $\frac{32}{\sec^2 \theta} - \frac{20}{1 + \tan^2 \theta} + 12 \sin^2 \theta$

Put, $\theta = 0^\circ$ Then

$$\frac{32}{\sec^2 0^\circ} - \frac{20}{1 + \tan^2 0^\circ} + 12 \sin^2 0^\circ$$

$$= 32 - 20 + 0 = 12$$

Sol.365.(b) $\sin \theta + \cos \theta = \frac{\sqrt{11}}{3}$

$$\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = \frac{11}{9}$$

$$2 \sin \theta \cos \theta = \frac{11}{9} - 1 = \frac{2}{9}$$

$$\sin \theta \cos \theta = \frac{1}{9}$$

Now, $\tan \theta + \cot \theta = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = 1 \times \frac{9}{1} = 9$$

Sol.366.(c) $1 + \sin^2 \theta = a \cos^2 \theta$

$$\frac{1}{\cos^2 \theta} + \frac{\sin^2 \theta}{\cos^2 \theta} = a$$

$$\sec^2 \theta + \tan^2 \theta = a$$

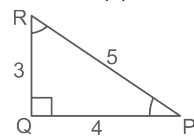
and $\sec^2 \theta - \tan^2 \theta = 1$ (identity)

Now, $\sec^4 \theta - \tan^4 \theta$

$$= (\sec^2 \theta + \tan^2 \theta)(\sec^2 \theta - \tan^2 \theta)$$

$$\sec^4 \theta - \tan^4 \theta = a \times 1 = a$$

Sol.367.(b)



Pythagoras triplets : - (3, 4, 5)

$$\sin P = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{3}{5}$$

base = 4 So, $\angle Q = 90^\circ = \frac{\pi}{2}$

Sol.368.(b) $\operatorname{Cosec} 41^\circ - \sec 49^\circ$

$$= \operatorname{Cosec} 41^\circ - \sec (90^\circ - 41^\circ)$$

$$= \operatorname{Cosec} 41^\circ - \operatorname{Cosec} 41^\circ = 0$$

Sol.369.(d) $\sin x + \sin^2 x = 1$

$$\sin x = \cos^2 x \Rightarrow \cos^4 x = \sin^2 x$$

Now,

$$\cos^8 x + 2 \cos^6 x + \cos^4 x = (\cos^4 x + \cos^2 x)^2$$

$$= (\sin^2 x + \cos^2 x)^2 = 1$$

Sol.370.(d) $\left(\frac{1 + \sec^2 A}{1 + \cos^2 A} \right) \left(\frac{1 + \sin^2 A}{1 + \operatorname{cosec}^2 A} \right)$

$$= \left(\frac{1 + \frac{1}{\cos^2 A}}{1 + \cos^2 A} \right) \left(\frac{1 + \frac{1}{\operatorname{cosec}^2 A}}{1 + \operatorname{cosec}^2 A} \right)$$

$$\left(\frac{1}{\cos^2 A} \right) \left(\frac{1}{\operatorname{cosec}^2 A} \right) = \tan^2 A$$

Sol.371.(d) $\frac{\sqrt{1 - \sin A}}{\sqrt{1 + \sin A}}$

$$= \frac{\sqrt{1 - \sin A}}{\sqrt{1 + \sin A}} \times \frac{\sqrt{1 - \sin A}}{\sqrt{1 - \sin A}}$$

$$= \frac{(1 - \sin A)}{\sqrt{(1 + \sin A)} \sqrt{(1 - \sin A)}}$$

$$= \frac{(1 - \sin A)}{\sqrt{1 - \sin^2 A}} = \frac{(1 - \sin A)}{\cos A}$$

$$= \sec A - \tan A$$

Sol.372.(d)

$$(1 + \tan \theta - \sec \theta)(1 + \cot \theta + \operatorname{cosec} \theta)$$

$$= \left(1 + \frac{\sin \theta}{\cos \theta} - \frac{1}{\cos \theta} \right) \left(1 + \frac{\cos \theta}{\sin \theta} + \frac{1}{\sin \theta} \right)$$

$$= \left(1 + \frac{\sin \theta - 1}{\cos \theta} \right) \left(1 + \frac{\cos \theta + 1}{\sin \theta} \right)$$

$$= \left(\frac{\cos \theta + \sin \theta - 1}{\cos \theta} \right) \left(\frac{\cos \theta + \sin \theta + 1}{\sin \theta} \right)$$

$$= \frac{(\cos \theta + \sin \theta)^2 - 1^2}{\sin \theta \cos \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta - 1}{\sin \theta \cos \theta}$$

$$= \frac{2 \sin \theta \cos \theta}{\sin \theta \cos \theta} = 2$$

Sol.373.(d) $\frac{\cos(50^\circ + A) - \sin(40^\circ - A)}{\cos 40^\circ \cdot \sec 40^\circ}$

$$= \frac{\cos(50^\circ + A) - \sin(40^\circ - A)}{\cos 40^\circ \times \frac{1}{\cos 40^\circ}}$$

Let $A = 40^\circ$ then,

$$\frac{\cos(50^\circ + 40^\circ) - \sin(40^\circ - 40^\circ)}{1}$$

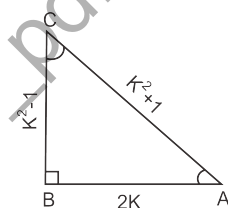
$$= \frac{\cos(90^\circ)}{1} = 0$$

Sol.374.(d) $\sec A + \tan A = K$ E.q.(1),

$$\sec A - \tan A = \frac{1}{K}$$
 E.q.(2)

On adding e.q.(1) and e.q.(2), we get

$$2 \sec A = \frac{K^2 + 1}{K} \Rightarrow \cos A = \frac{2K}{K^2 + 1}$$



Now, $\sin A = \frac{K^2 - 1}{K^2 + 1}$

Sol.375.(b) $\frac{\operatorname{cosec} 90^\circ}{(1 - \cos^2 60^\circ)}$

$$= \frac{1}{(1 - \frac{1}{4})} = \frac{1}{(\frac{3}{4})} = \frac{4}{3}$$

Sol.376.(d)

$$\sqrt{\frac{\operatorname{cosec} A + 1}{\operatorname{cosec} A - 1}} + \sqrt{\frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1}}$$

$$= \frac{\sqrt{\operatorname{cosec} A + 1}}{\sqrt{\operatorname{cosec} A - 1}} + \frac{\sqrt{\operatorname{cosec} A - 1}}{\sqrt{\operatorname{cosec} A + 1}}$$

$$= \frac{(\operatorname{cosec} A + 1) + (\operatorname{cosec} A - 1)}{\sqrt{\operatorname{cosec}^2 A - 1}}$$

$$= \frac{(2 \operatorname{cosec} A)}{\sqrt{\cot^2 A}} = \frac{(2 \operatorname{cosec} A)}{\cot A}$$

$$= \frac{2(\frac{1}{\sin A})}{\frac{\cos A}{\sin A}} = 2 \sec A$$

Sol.377.(d) Putting $\theta = 45^\circ$

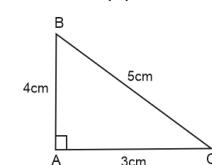
$$m = \sec \theta - \tan \theta = \sqrt{2} - 1$$

$$n = \operatorname{cosec} \theta + \cot \theta = \sqrt{2} + 1$$

Putting the value of m and n in $m + n(m - 1)$

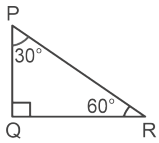
$$\sqrt{2} - 1 + (\sqrt{2} + 1)(\sqrt{2} - 1 - 1) = -1$$

Sol.378.(d)



$$\cos B = \frac{4}{5} \text{ and } \cot C = \frac{3}{4}$$

$$\cos B + \cot C \Rightarrow \frac{4}{5} + \frac{3}{4} = \frac{31}{20}$$

Sol.379.(d)

$$\cos(30^\circ + 60^\circ) = \cos 90^\circ = 0$$

$$\text{Sol.380.(b)} \quad \sqrt{3} \tan^2 \theta - 4 \tan \theta + \sqrt{3} = 0$$

$$= \sqrt{3} \tan^2 \theta - \tan \theta - 3 \tan \theta + \sqrt{3} = 0$$

$$= \tan \theta (\sqrt{3} \tan \theta - 1) - \sqrt{3} (\sqrt{3} \tan \theta - 1) = 0$$

$$= (\sqrt{3} \tan \theta - 1)(\tan \theta - \sqrt{3})$$

$$\Rightarrow \tan \theta = \left(-\frac{1}{\sqrt{3}} \text{ and } \sqrt{3}\right) \text{ [here both value}$$

exists] $\Rightarrow \theta = 30^\circ$ and 60° Then, by putting $\theta = 30^\circ$

$$\tan^2 \theta + \cot^2 \theta \Rightarrow \frac{1}{3} + 3 = \frac{10}{3}$$

$$\text{Sol.381.(c)} \quad 6 \cot \theta = 5 \Rightarrow \cot \theta = \frac{5}{6}$$

$$\text{Now, } \frac{(6 \cos \theta + \sin \theta)}{(6 \cos \theta - 4 \sin \theta)} = \frac{(6 \cot \theta + 1)}{(6 \cot \theta - 4)}$$

$$= \frac{(6 \times \frac{5}{6} + 1)}{(6 \times \frac{5}{6} - 4)} = 6$$

Sol.382.(b) $\operatorname{cosec} \theta - \sin \theta = 6$ Then, formula:- $(x+y)^2 = (x-y)^2 + 4xy$

$$\operatorname{cosec} \theta + \sin \theta = \sqrt{(6^2) + 4} = \sqrt{40}$$

$$\text{Sol.383.(d)} \quad \sqrt{2} \sec^2 \theta - 4 \sec \theta + 2\sqrt{2} = 0$$

$$\sqrt{2} \sec^2 \theta - 2 \sec \theta - 2 \sec \theta + 2\sqrt{2} = 0$$

$$\sqrt{2} \sec \theta (\sec \theta - \sqrt{2}) - 2(\sec \theta - \sqrt{2}) = 0$$

$$\sec \theta = \sqrt{2}, \sec \theta = \frac{2}{\sqrt{2}} = \frac{\sqrt{2}}{1} = \frac{H}{B}$$

So, $P = 1$

$$\sin^2 \theta + \tan^2 \theta = \frac{1}{2} + 1 = \frac{3}{2}$$

Short trick :-

$$\sqrt{2} \sec^2 \theta - 4 \sec \theta + 2\sqrt{2} = 0$$

Put $\theta = 45^\circ$

$$\text{So value of } \sin^2 \theta + \tan^2 \theta = \frac{1}{2} + 1 = \frac{3}{2}$$

$$\text{Sol.384.(c)} \quad \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} + \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta}$$

$$= \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} \times \frac{\sin \theta + \cos \theta}{\sin \theta + \cos \theta} +$$

$$\frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} \times \frac{\sin \theta - \cos \theta}{\sin \theta - \cos \theta}$$

$$= \frac{(\sin \theta + \cos \theta)^2}{\sin^2 \theta - \cos^2 \theta} + \frac{(\sin \theta - \cos \theta)^2}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta + \sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{2}{(\sin^2 \theta - \cos^2 \theta)}$$

$$\text{Sol.385.(c)} \quad \frac{1 - \sin A}{\cos A} + \frac{\cos A}{1 - \sin A}$$

$$= \frac{(1 - \sin A)^2 + \cos^2 A}{(\cos A)(1 - \sin A)}$$

$$= \frac{(1)^2 + \sin^2 A - 2 \sin A + \cos^2 A}{(\cos A)(1 - \sin A)}$$

$$= \frac{2(1 - \sin A)}{(\cos A)(1 - \sin A)} = 2 \sec A$$

Sol.386.(b) $\sec x + \tan x = 5$

$$\text{Then, } \sec x - \tan x = \frac{1}{5}$$

$$\operatorname{cosec} y - \cot y = \frac{1}{3}$$

$$\text{then, } \operatorname{cosec} y + \cot y = 3$$

Now,

$$(\sec x + \operatorname{cosec} y) - (\tan x - \cot y)$$

$$= (\sec x - \tan x + \operatorname{cosec} y + \cot y)$$

$$= \frac{1}{5} + 3 = \frac{16}{5} = 3.2$$

Sol.387.(c) Put the value of $\theta = 45^\circ$,

$$A = a\sqrt{2} + b \text{ and } B = a + b\sqrt{2}$$

$$\text{Then, } A^2 = 2a^2 + b^2 + 2\sqrt{2}ab$$

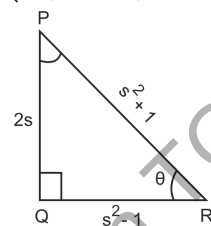
$$B^2 = a^2 + 2b^2 + 2\sqrt{2}ab$$

$$A^2 - B^2 = 2a^2 + b^2 + 2\sqrt{2}ab - a^2 - 2b^2 - 2\sqrt{2}ab$$

$$ab \Rightarrow A^2 - B^2 = (a^2 - b^2)$$

Sol.388.(a) Pythagoras triplets :-

$$(2s, s^2 + 1, s^2 - 1)$$



$$\operatorname{cosec} \theta + \cot \theta = s$$

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{s}$$

$$\operatorname{Cosec} \theta = \frac{s^2 + 1}{2s} \Rightarrow \sin \theta = \frac{2s}{s^2 + 1}$$

$$\text{So that, } \cos \theta = \frac{s^2 - 1}{s^2 + 1}$$

Sol.389.(c) $x = 30^\circ$ (put)

$$\sec x + \tan x = \sqrt{3}$$

$$\Rightarrow \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} = \sqrt{3} \text{ (satisfied)}$$

According to the question,

$$\tan x = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

Sol.390.(b)Complementary angle of $\tan = \cot$

$$\text{Sol.391.(c)} \quad \sec \theta + \tan \theta = 2 \text{ ----- (eq. 1)}$$

$$\text{As we know } \sec^2 \theta - \tan^2 \theta = 1$$

$$\text{Then, } (\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$$

$$\Rightarrow (\sec \theta - \tan \theta) = \frac{1}{2} \text{ ----- (eq. 2)}$$

By (eq. 1 and 2)

$$2 \sec \theta = 2 + \frac{1}{2}$$

$$\Rightarrow \sec \theta = \frac{\text{hypotenuse}}{\text{base}} = \frac{5}{4}$$

$$\text{then, perpendicular} = \sqrt{(5)^2 - (4)^2} = 3$$

$$\text{Therefore, } \cot \theta = \frac{\text{base}}{\text{perpendicular}} = \frac{4}{3}$$

Sol.392.(a) $\tan 10^\circ \tan 80^\circ + \tan 20^\circ \tan$

$$70^\circ + \tan 30^\circ \tan 60^\circ + \tan 40^\circ \tan 50^\circ$$

$$= \tan 10^\circ \tan(90^\circ - 10^\circ) + \tan 20^\circ \tan$$

$$(90^\circ - 20^\circ) + \tan 30^\circ \tan(90^\circ - 30^\circ) + \tan$$

$$40^\circ \tan(90^\circ - 40^\circ)$$

$$= \tan 10^\circ \cot 10^\circ + \tan 20^\circ \cot 20^\circ + \tan$$

$$30^\circ \cot 30^\circ + \tan 40^\circ \cot 40^\circ$$

$$= 1 + 1 + 1 + 1 = 4$$

Sol.393.(d) Let $\theta = 45^\circ$

$$\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1} = \frac{1 + \sqrt{2} - 1}{1 - \sqrt{2} + 1} = \frac{\sqrt{2}}{2 - \sqrt{2}}$$

$$\frac{\sqrt{2}}{\sqrt{2}(\sqrt{2} - 1)} \text{ (multiply by } (\sqrt{2} + 1) \text{ in}$$

numerator and denominator)

$$\frac{(\sqrt{2} + 1)}{1} = (\sqrt{2} + 1)$$

According to the question,

In option d, put $\theta = 45^\circ$

$$(\cot \theta + \operatorname{cosec} \theta) \Rightarrow (1 + \sqrt{2}) \text{ (satisfied)}$$

Sol.394.(c) $6 \sec \theta = 10$

$$\sec \theta = \frac{10}{6} = \frac{5}{3} = \frac{\text{Hypotenuse}}{\text{Base}}$$

Using Pythagoras theorem;

$$\text{Perpendicular} = \sqrt{5^2 - 3^2} = 4,$$

$$\operatorname{cosec} \theta = \frac{5}{4} \text{ and } \cot \theta = \frac{3}{4}$$

$$\frac{(5 \operatorname{cosec} \theta - 3 \cot \theta)}{(4 \cos \theta + 3 \sin \theta)}$$

$$= \frac{(5 \times \frac{5}{4} - 3 \times \frac{3}{4})}{(4 \times \frac{3}{5} + 3 \times \frac{4}{5})} = \frac{(\frac{25}{4} - \frac{9}{4})}{(\frac{12}{5} + \frac{12}{5})}$$

$$= \frac{(\frac{25-9}{4})}{(\frac{12+12}{5})} = \frac{20}{24} = \frac{5}{6}$$

Sol.395.(b)

$$\operatorname{cosec} A(1 - \cos A)(\operatorname{cosec} A + \cot A)$$

$$\Rightarrow \frac{1}{\sin A}(1 - \cos A)\left(\frac{1 + \cos A}{\sin A}\right)$$

$$\Rightarrow \frac{(1 - \cos^2 A)}{\sin^2 A} \Rightarrow \frac{(1 - \cos^2 A)}{(1 - \cos^2 A)} = 1$$

$$\text{Sol.396.(c)} \quad \frac{(2 \cos^3 A - \cos A)}{(\sin A - 2 \sin^3 A)} = 1$$

$$\Rightarrow \frac{\cos A(2 \cos^2 A - 1)}{\sin A(1 - 2 \sin^2 A)} = 1 \Rightarrow \frac{\cos A \cdot \cos 2A}{\sin A \cdot \cos 2A}$$

$$= 1 \Rightarrow \cot A = 1 \Rightarrow A = 45^\circ$$

Sol.397.(c)

$$(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2$$

Putting the value of $\theta = 45^\circ$

$$\left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)^2 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)^2 = 9$$

By putting value of $\theta = 45^\circ$ in the option (c)we get, $7 + \tan^2 \theta + \cot^2 \theta = 7 + 1 + 1 = 9$.

Sol.398.(b)

$$\tan^2 \theta + \cot^2 \theta - \sec^2 \theta \cdot \operatorname{cosec}^2 \theta$$

$$\text{Put } \theta = 45^\circ$$

$$\tan^2 45^\circ + \cot^2 45^\circ - \sec^2 45^\circ \cdot \operatorname{cosec}^2 45^\circ \\ \Rightarrow 1 + 1 - 2 \times 2 \Rightarrow -2$$

Sol.399.(a) Formula :-

$$\sin a - \sin b = 2 \sin\left(\frac{a-b}{2}\right) \cos\left(\frac{a+b}{2}\right)$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

Sol.400.(c) In third quadrant value of $\tan \theta$ and $\cot \theta$ is +ve.

$$\tan 240^\circ = \tan (180^\circ + 60^\circ)$$

$$= \tan 60^\circ = \sqrt{3}$$

Sol.401.(c) $\tan^2 \theta = 1 - a^2$

Adding 1 both side,

$$1 + \tan^2 \theta = 1 - a^2 + 1$$

$$\sec^2 \theta = 2 - a^2$$

$$\sec \theta = \sqrt{(2 - a^2)} \quad \text{..... (eq.1)}$$

Now, $\sec \theta + \tan^3 \theta \times \operatorname{cosec} \theta$

$$= \frac{1}{\cos \theta} + \frac{\sin^3 \theta}{\cos^3 \theta} \times \frac{1}{\sin \theta}$$

$$= \frac{\cos^2 \theta + \sin^2 \theta}{\cos^3 \theta} = \sec^3 \theta \quad \text{.....(eq.2)}$$

Putting the value of (eq.1) in (eq.2), we get, $\sec^3 \theta = (2 - a^2)^{\frac{3}{2}}$

Sol.402.(a) Formula :-

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

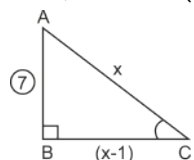
$$\sin(A + B) = \cos(A + B) \quad \text{.....(given)}$$

Clearly value of $\sin \theta$ and $\cos \theta$ are equal at 45° , $(A + B) = 45^\circ \Rightarrow A = 45^\circ - B$

Now, $\tan A = \tan(45^\circ - B)$

$$= \frac{\tan 45^\circ - \tan B}{1 + \tan 45^\circ \tan B} = \frac{1 - \tan B}{1 + \tan B}$$

Sol.403.(d) Let, the length of side AC = x cm, then BC = (x - 1)cm



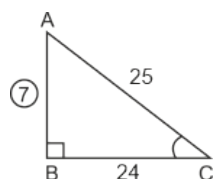
Now, using pythagoras theorem :-

$$AC^2 = AB^2 + BC^2 \Rightarrow x^2 = 7^2 + (x - 1)^2$$

$$x^2 = 49 + x^2 + 1 - 2x$$

$$2x = 50 \Rightarrow x = 25 \text{ cm}$$

$$\text{Now, } \sin C = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{7}{25}$$

Short Trick:-

Using pythagoras triplets : (7, 24, 25)

AB = 7 cm

And AC - BC = 25 - 24 = 1 cm (Satisfy the given condition)

$$\text{Then, } \sin C = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{7}{25}$$

Sol.404.(a) Formula :-

$$\frac{(1 - \cos \theta)}{\sin \theta} = \frac{\sin \theta}{(1 + \cos \theta)}$$

$$\frac{(1 - \cos \theta)}{\sin \theta} = \frac{1}{5} \quad \text{.....(given)}$$

$$\text{Then, } \frac{(1 + \cos \theta)}{\sin \theta} = 5$$

$$\text{Sol.405.(b)} \quad \frac{\cos A}{1 + \tan A} - \frac{\sin A}{1 + \cot A}$$

$$= \frac{\cos A}{1 + \frac{\sin A}{\cos A}} - \frac{\sin A}{1 + \frac{\cos A}{\sin A}}$$

$$= \frac{\cos^2 A}{\cos A + \sin A} - \frac{\sin^2 A}{\sin A + \cos A}$$

$$= \frac{\cos^2 A - \sin^2 A}{\cos A + \sin A}$$

$$= \frac{(\cos A + \sin A)(\cos A - \sin A)}{(\cos A + \sin A)}$$

$$= (\cos A - \sin A)$$

Sol.406.(a)

$$\frac{1}{1 + \cos(90^\circ - \theta)} + \frac{1}{1 - \cos(90^\circ - \theta)}$$

$$= \frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta}$$

$$= \frac{(1 - \sin \theta) + (1 + \sin \theta)}{1 - \sin^2 \theta}$$

$$= \frac{2}{\cos^2 \theta} \Rightarrow 2 \sec^2 \theta$$

Sol.407.(d) We know that,

$$\cos^2 \theta = (1 - \sin^2 \theta)$$

$$\sin^2 \theta + \cos^2 \theta = 1 \quad \text{.....eq.(1)}$$

On squaring eq.(1), we get

$$(\sin^2 \theta + \cos^2 \theta)^2 = (1)^2$$

$$\sin^4 \theta + \cos^4 \theta + 2 \sin^2 \theta \cos^2 \theta = 1$$

$$\sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \theta \cos^2 \theta$$

$$\sin^4 407^\circ + \cos^4 \theta = 1 - 2 \sin^2 \theta (1 - \sin^2 \theta)$$

$$\sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \theta + 2 \sin^4 \theta$$

$$\text{Sol.408.(c)} \quad \frac{\cos x - \sqrt{3} \sin x}{2}$$

$$= \frac{1}{2} \times \cos x - \frac{\sqrt{3}}{2} \times \sin x$$

$$= \sin 30^\circ \cos x - \cos 30^\circ \sin x$$

$$= \sin(30^\circ - x) = \sin[90^\circ - (60^\circ + x)]$$

$$= \cos(60^\circ + x) = \cos\left(\frac{\pi}{3} + x\right)$$

Sol.409.(b)

$$\sqrt{\frac{1 + \cos A}{1 - \cos A}} = \sqrt{\frac{1 + \cos A}{1 - \cos A}} \times \frac{1 + \cos A}{1 + \cos A}$$

$$= \sqrt{\frac{(1 + \cos A)^2}{1 - \cos^2 A}} = \frac{1 + \cos A}{\sin A}$$

$$= \operatorname{cosec} A + \cot A$$

Sol.410.(a)

$$\sin \theta + \cos \theta = \frac{\sqrt{3} - 1}{2\sqrt{2}} \quad \text{.....eq.(1)}$$

Squaring eq.(1), we get

$$\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta$$

$$= \left(\frac{\sqrt{3} - 1}{2\sqrt{2}}\right)^2 = \frac{4 - 2\sqrt{3}}{8}$$

$$2 \sin \theta \cos \theta = \frac{4 - 2\sqrt{3}}{8} - 1$$

$$= \frac{(4 - 2\sqrt{3}) - 8}{8} = \frac{-(2 + \sqrt{3})}{4}$$

$$\text{Now, } \tan \theta + \cot \theta = \frac{2}{2 \sin \theta \cos \theta}$$

$$= \frac{-8}{(2 + \sqrt{3})} \times \frac{(2 - \sqrt{3})}{(2 - \sqrt{3})} = 8(\sqrt{3} - 2)$$

Sol.411.(d)

If $A + B = 90^\circ$, then $\sec A = \operatorname{cosec} B$

$$\sec 2\theta = \operatorname{cosec}(\theta - 36^\circ)$$

$$\text{Now, } 2\theta + (\theta - 36^\circ) = 90^\circ \Rightarrow \theta = 42^\circ$$

Sol.412.(d) Formula :-

$$\tan 2x = \frac{2 \tan x}{(1 - \tan^2 x)}$$

$$\cos x \cdot \operatorname{cosec} x - \sin x \cdot \sec x$$

$$= \cot x - \tan x = \frac{1}{\tan x} - \frac{\tan^2 x}{\tan x}$$

$$= \frac{2 \times (1 - \tan^2 x)}{2 \times \tan x} = \frac{2 \times 1}{\tan 2x} = 2 \cot 2x$$

Sol.413.(b) $\cot^2 A - \cos^2 A$

$$= \frac{\cos^2 A}{\sin^2 A} - \cos^2 A = \cos^2 A (\operatorname{cosec}^2 A - 1)$$

$$= \cos^2 A \cdot \cot^2 A$$

Sol.414.(a) $b \cdot \sin \theta = a$

$$\Rightarrow \sin \theta = \frac{a}{b} = \frac{\text{Perpendicular}}{\text{hypotenuse}}$$

by, using pythagoras theorem, we get

$$\text{Base} = \sqrt{b^2 - a^2} \quad \text{Now,}$$

$$\sec \theta + \tan \theta = \frac{b}{\sqrt{b^2 - a^2}} + \frac{a}{\sqrt{b^2 - a^2}}$$

$$= \frac{b + a}{\sqrt{b^2 - a^2}} = \frac{\sqrt{b+a} \times \sqrt{b+a}}{\sqrt{b+a} \times \sqrt{b-a}} = \sqrt{\frac{b+a}{b-a}}$$

Sol.415.(b) Formula :-

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

$$\sin 10^\circ - \frac{4}{3} \sin^3 10^\circ = \frac{3 \sin 10^\circ - 4 \sin^3 10^\circ}{3}$$

$$= \frac{\sin 30^\circ}{3} = \frac{1}{6}$$

Sol.416.(c) (i) $\cos(-\theta) = \cos \theta$

$$(ii) \cos(2n\pi - \theta) = \cos \theta$$

$$\text{Now, } \cos\left(\frac{-17\pi}{3}\right) = \cos\left(\frac{17\pi}{3}\right)$$

$$= \cos\left(6\pi - \frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

Sol.417.(a) Formula :-

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$(\sin A + \cos A)(1 - \sin A \cos A)$$

$$= (\sin A + \cos A)$$

$$(\sin^2 A + \cos^2 A - \sin A \cos A)$$

$$= \sin^3 A + \cos^3 A$$

Sol.418.(d) $\sec \theta - 2 \cos \theta = \frac{7}{2}$

$$= \sec \theta - \frac{2}{\sec \theta} = \frac{7}{2}$$

$$= \sec^2 \theta - 2 = \frac{7}{2} \sec \theta$$

$$= 2 \sec^2 \theta - 7 \sec \theta - 4 = 0$$

$$= 2 \sec^2 \theta - 8 \sec \theta + \sec \theta - 4 = 0$$

$$= (\sec \theta - 4)(2 \sec \theta + 1) = 0$$

$$= \sec \theta = 4$$

Sol.419.(d) $\sec 3\theta = \operatorname{cosec} (4\theta - 15^\circ)$
 $\Rightarrow \operatorname{cosec} (90^\circ - 3\theta) = \operatorname{cosec} (4\theta - 15^\circ)$
 $\Rightarrow 90^\circ - 3\theta = 4\theta - 15^\circ$
 $\Rightarrow 7\theta = 90^\circ + 15^\circ$
 $\Rightarrow 7\theta = 105^\circ \Rightarrow \theta = 15^\circ$
 Then, $\tan 3\theta = \tan 45^\circ = 1$

Sol.420.(a) In a right angled triangle, right angled at A,
 $= \angle A = \angle B + \angle C = 90^\circ$
 $= \angle A - \angle B - \angle C = 0^\circ$
 $\tan(A - B - C) = \tan 0^\circ = 0$
 $\tan(A - B - C)$

$$= \frac{\tan A - \tan B - \tan C - \tan A \tan B \tan C}{1 + \tan A \tan B - \tan B \tan C + \tan C \tan A}$$

$$\tan A - \tan B - \tan C - \tan A \tan B \tan C = 0$$

$$\tan A - \tan B - \tan C = \tan A \tan B \tan C$$

Sol.421.(d)

$$1 + \tan^2 \alpha = 4 + Q^2 \text{ (adding 1 both sides)}$$

$$\sec^2 \alpha = 4 + Q^2$$

$$\cos \alpha = (4 + Q^2)^{-\frac{1}{2}}$$

then, $\sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha$

$$= \frac{1}{\cos \alpha} + \frac{\sin^3 \alpha}{\cos^3 \alpha \times \sin \alpha}$$

$$= \frac{1}{\cos \alpha} + \frac{\sin^2 \alpha}{\cos^2 \alpha \times \cos \alpha}$$

$$= \frac{1 + \tan^2 \alpha}{\cos \alpha} = \frac{(4 + Q^2)}{(4 + Q^2)^{-1/2}} = (4 + Q^2)^{\frac{3}{2}}$$

Short Trick:- Put, $= 60^\circ$, then,

$$\tan^2 60^\circ = 3 + Q^2 \Rightarrow Q = 0$$

Now, $\sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha = 2 + 3\sqrt{3} \times \frac{2}{\sqrt{3}} = 8$ Clearly, option (d) satisfy the

given equation

$$\Rightarrow (4 + Q^2)^{3/2} = (4 + 0^2)^{3/2}$$

$$= (2^2)^{3/2} = 2^3 = 8$$

Sol.422.(b) $1 + \sin \theta = m \cos \theta$

$$\frac{1 + \sin \theta}{\cos \theta} = m \Rightarrow \frac{(1 + \sin \theta)^2}{\cos^2 \theta} = m^2$$

$$\frac{(1 + \sin \theta)^2}{(1 + \sin \theta)(1 - \sin \theta)} = m^2$$

$$\frac{(1 + \sin \theta)}{(1 - \sin \theta)} = m^2$$

By using componendo and dividendo,

we get $\frac{(1 + \sin \theta + 1 - \sin \theta)}{(1 + \sin \theta - 1 + \sin \theta)} = \frac{m^2 + 1}{m^2 - 1}$

$$\frac{(2)}{(2 \sin \theta)} = \frac{m^2 + 1}{m^2 - 1} \Rightarrow \sin \theta = \frac{m^2 - 1}{m^2 + 1}$$

Sol.423.(a) Put $\theta = 90^\circ$

$$5(\sin^4 \theta + \cos^4 \theta) + 3(\sin^6 \theta + \cos^6 \theta) + 19 \sin^2 \theta \cos^2 \theta$$

$$= 5 \times 1 + 3 \times 1 + 19 \times 0 = 8$$

Sol.424.(d) Formula: $\cos 2A = 1 - 2 \sin^2 A$
 Sin Y = x then, $\cos 2Y = 1 - 2x^2$

Sol.425.(d) Formula:- $\tan(A + B + C)$

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

In a right angled triangle, right angled at C,

$$= \angle C = \angle A + \angle B = 90^\circ$$

$$= \angle A + \angle B - \angle C = 0^\circ$$

$$\tan(A + B - C) = \tan 0^\circ = 0$$

$$\Rightarrow \tan(A + B - C)$$

$$= \frac{\tan A + \tan B - \tan C + \tan A \tan B \tan C}{1 - \tan A \tan B + \tan B \tan C + \tan C \tan A}$$

$$\Rightarrow \tan A + \tan B - \tan C + \tan A \tan B \tan C = 0$$

$$\Rightarrow \tan A + \tan B - \tan C = -\tan A \tan B \tan C$$

$$\Rightarrow \tan C - \tan A - \tan B = \tan A \tan B \tan C$$

Sol.426.(d) $x = 8(\sin \theta + \cos \theta)$ and $y = 9$
 $(\sin \theta - \cos \theta),$

$$\Rightarrow \frac{x^2}{8^2} + \frac{y^2}{9^2}$$

$$= \frac{64(\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta)}{64}$$

$$+ \frac{81(\sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta)}{81}$$

$$= 1 + 2 \sin \theta \cos \theta + 1 - 2 \sin \theta \cos \theta = 2$$

Sol.427.(b) $2 \cos 135^\circ \sin 15^\circ$

$$= [\sin(135^\circ + 15^\circ) - \sin(135^\circ - 15^\circ)]$$

$$= \sin 150^\circ - \sin 120^\circ$$

$$= \cos 60^\circ - \cos 30^\circ = \frac{1}{2} - \frac{\sqrt{3}}{2} = \frac{1 - \sqrt{3}}{2}$$

Sol.428.(c) $\sec x + \cos x =$

$$\sqrt{(\sec x - \cos x)^2 + 4 \times \sec x \cos x}$$

$$= \sqrt{4^2 + 4} = \sqrt{20} = 2\sqrt{5}$$

$$\frac{(1 + \cos^2 x)}{\cos x} = \frac{1}{\cos x} + \frac{\cos^2 x}{\cos x}$$

$$= \sec x + \cos x = 2\sqrt{5}$$

Sol.429.(c) $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$

$$= \frac{\sin A \sin B}{\cos A \cos B} + \frac{\cos x}{\cos A \cos B} = 1$$

$$= \frac{\sin A \sin B + \cos x}{\cos A \cos B} = 1$$

$$= \sin A \sin B + \cos x = \cos A \cos B$$

$$= \cos x = \cos A \cos B - \sin A \sin B$$

$$= \cos x = \cos(A + B) \Rightarrow x = A + B$$

Sol.430.(a) $\cos^3 60^\circ - \cos^3 240^\circ - \cos^3 360^\circ$

$$= \cos^3 60^\circ - \cos^3 (180^\circ + 60^\circ) - 1$$

$$= \cos^3 60^\circ + \cos^3 60^\circ - 1$$

$$= \left(\frac{1}{2}\right)^3 \times 2 - 1 = \frac{1}{4} - 1 = \frac{-3}{4}$$

Sol.431.(c) $5 \sin^2 A + 3 \cos^2 A = 4$

$$\Rightarrow 2 \sin^2 A + 3 \sin^2 A + 3 \cos^2 A = 4$$

$$\Rightarrow 2 \sin^2 A + 3(\sin^2 A + \cos^2 A) = 4$$

$$\Rightarrow 2 \sin^2 A + 3 = 4 \Rightarrow 2 \sin^2 A = 1$$

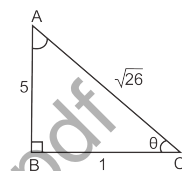
$$\sin^2 A = \frac{1}{2} \Rightarrow \sin A = \frac{1}{\sqrt{2}}$$

$$\Rightarrow A = 45^\circ$$

then, $\tan A = \tan 45^\circ = 1$

Sol.432.(b) $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = \frac{3}{2}$

$$\frac{\sin \theta}{\cos \theta} = \frac{3 + 2}{3 - 2} = \frac{5}{1} \Rightarrow \tan \theta = \frac{5}{1}$$



$$\sin^4 \theta - \cos^4 \theta = \left(\frac{5}{\sqrt{26}}\right)^4 - \left(\frac{1}{\sqrt{26}}\right)^4$$

$$= \frac{625}{676} - \frac{1}{676} = \frac{624}{676} = \frac{12}{13}$$

Sol.433.(c)

$$\operatorname{cosec}^2 \theta + \sec^2 \theta = \frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta} = \frac{1}{\frac{2}{9}} = \frac{9}{2}$$

Sol.434.(c) Formula :- $(\cos 2x = 2 \cos^2 x - 1)$

$$2 \frac{\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$$

$$\Rightarrow 2 \frac{\cos^2 x - \frac{1}{\cos^2 x}}{\frac{\sin^2 x}{\cos^2 x}} = 2 \frac{\cos^4 x - 1}{\sin^2 x}$$

$$= 2 \frac{(\cos^2 x - 1)(\cos^2 x + 1)}{1 - \cos^2 x}$$

$$= -2(\cos^2 x + 1) = -2 \cos^2 x - 2$$

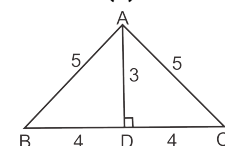
$$= -(\cos 2x + 1) - 2 = a + b \cos 2x$$

$$-3 - \cos 2x = a + b \cos 2x$$

Comparing both side, we get,

$$a = -3 \text{ and } b = -1$$

Sol.435.(b)



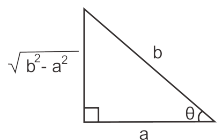
Since, $\triangle ABC$ is an isosceles triangle then, AD(median) divides BC in the ratio of 1 : 1.

By the given figure,

$$\tan C = \frac{3}{4}, \cot B = \frac{4}{3}$$

$$\text{So, } \tan C - \cot B = \frac{3}{4} - \frac{4}{3} = \frac{9 - 16}{12} = \frac{-7}{12}$$

Sol.436.(c) $b \cos \theta = a \Rightarrow \cos \theta = \frac{a}{b}$

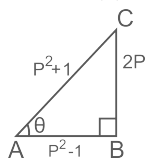


$p = \sqrt{b^2 - a^2}$ (using pythagoras theorem)

$\operatorname{cosec} \theta + \cot \theta = \frac{b}{\sqrt{b^2 - a^2}} + \frac{a}{\sqrt{b^2 - a^2}}$

$= \frac{b+a}{(\sqrt{b-a})(\sqrt{b+a})} = \sqrt{\frac{b+a}{b-a}}$

Sol.437.(a)



$\operatorname{cosec} \theta + \cot \theta = p$ ----- (1)

$\operatorname{cosec} \theta - \cot \theta = \frac{1}{p}$ ----- (2)

(As $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$)

Adding eq. (1) and (2) we get :

$\operatorname{cosec} \theta = \frac{p^2+1}{2p}$. then, $\cot \theta = \frac{p^2-1}{2p}$

Now, $\frac{\frac{p^2-1}{2p}}{\frac{p^2+1}{2p}} = \frac{p^2-1}{p^2+1} = \frac{\cot \theta}{\operatorname{cosec} \theta} = \cos \theta$

Sol.438.(d) $\sin^2 \theta - 3 \sin \theta + 2 = 0$

$\sin^2 \theta - 2 \sin \theta - \sin \theta + 2 = 0$

$\sin \theta (\sin \theta - 2) - 1(\sin \theta - 2) = 0$

$(\sin \theta - 1)(\sin \theta - 2) = 0 \Rightarrow \sin \theta = 1$ ($\sin \theta \neq 2$)

$\therefore (-1 \leq \sin \theta \leq 1)$

So, $\sin \theta = 1$ (As, $0^\circ \leq \theta \leq 90^\circ$) $\Rightarrow \theta = 90^\circ$

Sol.439.(b) $\tan 3\theta \cdot \tan 7\theta = 1$

$\tan 3\theta = \frac{1}{\tan 7\theta} = \cot 7\theta$

$\tan 3\theta = \tan(90^\circ - 7\theta)$

$3\theta = 90^\circ - 7\theta \Rightarrow 10\theta = 90^\circ$

$\Rightarrow \theta = \frac{90}{10} = 9^\circ$

So, $\cot 150 = \cot 135^\circ = -1$

Sol.440.(b) $2 \sin \theta + 2 \sin^2 \theta = 2$

$\sin \theta + \sin^2 \theta = 1 \Rightarrow \sin \theta = 1 - \sin^2 \theta$

$\Rightarrow \sin \theta = \cos^2 \theta$

So, $2 \cos^4 \theta + 2 \cos^2 \theta = 2(\cos^4 \theta + \cos^2 \theta)$

$= 2(\sin^2 \theta + \cos^2 \theta) = 2 \times 1 = 2$

Sol.441.(a) $\cos \theta + \cos^2 \theta = 1$

$\cos \theta = 1 - \cos^2 \theta \Rightarrow \cos \theta = \sin^2 \theta$

So, $\sqrt{\sin^4 \theta + \cos^2 \theta} = \sqrt{\cos^2 \theta + \cos^2 \theta}$

$= \sqrt{2 \cos^2 \theta} = \sqrt{2} \cos \theta$

Sol.442.(c) $\sec^2 A + \tan^2 A = 3$

$1 + \tan^2 A + \tan^2 A = 3$

$1 + 2 \tan^2 A = 3 \Rightarrow 2 \tan^2 A = 2$

$\tan^2 A = 1 \Rightarrow \frac{1}{\cot^2 A} = 1$

$\cot^2 A = 1 \Rightarrow \cot A = 1$

Sol.443.(b) $\operatorname{cosec} 15^\circ \sec 15^\circ$

$= \frac{1}{\sin 15^\circ \cos 15^\circ} = \frac{2}{2 \sin 15^\circ \cos 15^\circ}$

$= \frac{2}{\sin 30^\circ} = \frac{2}{\frac{1}{2}} = 4$

Sol.444.(d) $\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A}$

$= \frac{\cos A}{1 - \frac{\sin A}{\cos A}} + \frac{\sin A}{1 - \frac{\cos A}{\sin A}}$

$= \frac{\cos^2 A}{\cos A - \sin A} - \frac{\sin^2 A}{\cos A - \sin A}$

$= \frac{\cos^2 A - \sin^2 A}{\cos A - \sin A}$

$= \frac{(\cos A + \sin A)(\cos A - \sin A)}{\cos A - \sin A}$

$= \sin A + \cos A$

Sol.445.(c) $\tan 40^\circ = \cot 50^\circ = a$

So, $\frac{\tan 320^\circ - \tan 310^\circ}{1 + \tan 320^\circ \tan 310^\circ}$

$= \frac{\tan(360^\circ - 40^\circ) - \tan(360^\circ - 50^\circ)}{1 + \tan(360^\circ - 40^\circ) \tan(360^\circ - 50^\circ)}$

$= \frac{-\tan 40^\circ + \tan 50^\circ}{1 + \tan 40^\circ \tan 50^\circ}$

$= \frac{-a + \frac{1}{a}}{1 + a \times \frac{1}{a}} = \frac{1 - a^2}{2a}$

Sol.446.(a) Formula :-

$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$\tan 2\alpha = \tan\{(\alpha + \beta) + (\alpha - \beta)\}$

$= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta) \tan(\alpha - \beta)} = \frac{a+b}{1-ab}$

Sol.447.(b) Short trick :

Putting $\theta = 45^\circ$, we have ,

$x = \frac{2 \sin 45^\circ}{1 + \cos 45^\circ + \sin 45^\circ} = \frac{\sqrt{2}}{1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}$

$= \frac{2}{\sqrt{2} + 2} = \frac{\sqrt{2}}{1 + \sqrt{2}}$

Now,

$\frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta} = \frac{1 - \cos 45^\circ + \sin 45^\circ}{1 + \sin 45^\circ}$

$= \frac{1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} = \frac{\sqrt{2} - 1 + 1}{\sqrt{2} + 1}$

$= \frac{\sqrt{2}}{\sqrt{2} + 1} = x$

Sol.448.(b) $\left\{ \left(\frac{\sec \theta - 1}{\sec \theta + 1} \right) \right\}^n = \operatorname{cosec} \theta - \cot \theta$

$\left\{ \frac{(\sec \theta - 1)(\sec \theta - 1)}{(\sec \theta + 1)(\sec \theta - 1)} \right\}^n = \operatorname{cosec} \theta - \cot \theta$

$\left\{ \frac{(\sec \theta - 1)}{\tan \theta} \right\}^{2n} = \operatorname{cosec} \theta - \cot \theta$

$\{ \operatorname{cosec} \theta - \cot \theta \}^{2n} = \operatorname{cosec} \theta - \cot \theta$

Comparing both side, we get

$2n = 1 \Rightarrow n = \frac{1}{2} = 0.5$

Sol.449.(a)

$(2 \cos^2 \theta - 1) \left[\frac{1 + \tan \theta}{1 - \tan \theta} + \frac{1 - \tan \theta}{1 + \tan \theta} \right]$

$= \cos 2\theta \left[\frac{(1 + \tan \theta)^2 + (1 - \tan \theta)^2}{1 - \tan^2 \theta} \right]$

$(\because \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1)$

$= \cos 2\theta \left[\frac{2(1 + \tan^2 \theta)}{1 - \tan^2 \theta} \right]$

$= \cos 2\theta \left[\frac{2(\cos^2 \theta + \sin^2 \theta)}{\cos^2 \theta - \sin^2 \theta} \right]$

$= \cos 2\theta \times \frac{2}{\cos 2\theta} = 2$

Sol.450.(d) $\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta}$

Multiplying by 2 in numerator and denominator,

$\frac{2 \sin 8\theta \cos \theta - 2 \sin 6\theta \cos 3\theta}{2 \cos 2\theta \cos \theta - 2 \sin 3\theta \sin 4\theta}$

$= \frac{\sin 9\theta + \sin 7\theta - (\sin 9\theta + \sin 3\theta)}{\cos 3\theta + \cos \theta - (\cos 7\theta - \cos 7\theta)}$

$= \frac{\sin 7\theta - \sin 3\theta}{\cos 3\theta + \cos 7\theta} = \frac{2 \cos 5\theta \cdot \sin 2\theta}{2 \cos 5\theta \cdot \cos 2\theta} = \tan 2\theta$

Sol.451.(a)

$(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 1$

Putting $A = 45^\circ$, we have :

$(1 + \cot 45^\circ - \operatorname{cosec} 45^\circ)(1 + \tan 45^\circ + \sec 45^\circ) - 1$

$= (2 - \sqrt{2})(2 + \sqrt{2}) - 1 \Rightarrow 4 - 2 - 1 = 1$

Sol.452.(c) As we know that,

$\cos^2 \theta + \sin^2 \theta = 1$

Then, $\cos A = \sqrt{1 - \left(\frac{1}{3}\right)^2} = \sqrt{\frac{8}{9}}$

$= -\frac{2\sqrt{2}}{3}$ and $\cos B = \sqrt{1 - \left(\frac{1}{5}\right)^2}$

$= \sqrt{\frac{24}{25}} = -\frac{2\sqrt{6}}{5}$

($\because$ A and B are in second quadrant)

$\cos(A - B) = \cos A \cdot \cos B + \sin A \cdot \sin B$

$= \left(-\frac{2\sqrt{2}}{3}\right) \times \left(-\frac{2\sqrt{6}}{5}\right) + \frac{1}{3} \times \frac{1}{5}$

$= \frac{8\sqrt{3} + 1}{15}$

Sol.453.(a)

$\frac{\sin(A+B)}{\sin A \cos B} = \frac{\sin A \cos B + \cos A \sin B}{\sin A \cos B}$

$= 1 + \frac{\cos A \sin B}{\sin A \cos B} = 1 + \cot A \tan B$

Sol.454.(a) $\cos^2 45^\circ + \cos^2 135^\circ + \cos^2 225^\circ + \cos^2 315^\circ$

$= \left(\frac{1}{\sqrt{2}}\right)^2 + \cos^2(90^\circ + 45^\circ) + \cos^2(270^\circ -$

$45^\circ) + \cos^2(360^\circ - 45^\circ)$

$= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 +$

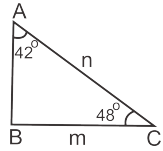
$\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 2$

Sol.455.(c) Formula used :

$$\cos 2\theta = 2\cos^2\theta - 1 \text{ or } 1 - 2\sin^2\theta$$

$$\text{Now, } \frac{2\cos^3\theta - \cos\theta}{\sin\theta - 2\sin^3\theta} = \frac{\cos\theta(2\cos^2\theta - 1)}{\sin\theta(1 - 2\sin^2\theta)}$$

$$= \frac{\cos\theta \cdot \cos 2\theta}{\sin\theta \cdot \cos 2\theta} = \cot\theta$$

Sol.456.(d)

$$\sec 48^\circ = \frac{n}{m} \text{ and } \cot 42^\circ = \frac{\sqrt{n^2 - m^2}}{m}$$

$$\text{So, } \sec 48^\circ - \cot 42^\circ = \frac{n}{m} - \frac{\sqrt{n^2 - m^2}}{m}$$

$$= \frac{n - \sqrt{n^2 - m^2}}{m}$$

$$\text{Sol.457.(c) } \sec^2\theta \sqrt{1 - \sin^2\theta}$$

$$= \sec^2\theta \times \sqrt{\cos^2\theta} = \sec^2\theta \times \cos\theta = \sec\theta$$

Sol.458.(a) As we know, that

$$\operatorname{cosec}^2 A - \cot^2 A = 1$$

$$= (\operatorname{cosec} A + \cot A)(\operatorname{cosec} A - \cot A) = 1$$

$$= m \times (\operatorname{cosec} A - \cot A) = 1$$

$$\Rightarrow (\operatorname{cosec} A - \cot A) = \frac{1}{m}$$

$$\text{Sol.459.(d) } (\operatorname{cosec} A + \cot A)(1 - \cos A)$$

$$= \left(\frac{1}{\sin A} + \frac{\cos A}{\sin A}\right)(1 - \cos A)$$

$$= \frac{1 + \cos A}{\sin A} \times (1 - \cos A)$$

$$= \frac{(1 - \cos^2 A)}{\sin A} = \frac{\sin^2 A}{\sin A} = \sin A$$

$$\text{Sol.460.(a) } \cos A = \sin^2 A,$$

$$a \sin^{12} A + b \sin^{10} A + c \sin^8 A + \sin^6 A = 1$$

$$= a \cos^6 A + b \cos^5 A + c \cos^4 A + \cos^3 A = 1 \dots \dots (1)$$

$$\text{We know that, } (\cos A + \cos^2 A)^3 =$$

$$= \cos^3 A + \cos^6 A + 3\cos^3 A(\cos A + \cos^2 A)$$

$$= \cos^6 A + 3\cos^5 A + 3\cos^4 A + \cos^3 A$$

$$= 1 \dots \dots (2)$$

from eq. (2) with (1),

$$a = 1, b = 3 \text{ and } c = 3$$

$$\text{Then, } a + b + c = 7$$

$$\text{Sol.461.(c) } 4\cos\theta + 3\sin\theta = x \dots \dots (1)$$

$$4\sin\theta - 3\cos\theta = y \dots \dots (2)$$

Squaring eq. (1) and (2), we get

$$16\cos^2\theta + 9\sin^2\theta + 24\sin\theta \cdot \cos\theta$$

$$= x^2 \dots \dots (3)$$

$$16\sin^2\theta + 9\cos^2\theta - 24\sin\theta \cdot \cos\theta$$

$$= y^2 \dots \dots (4)$$

Adding eq. (3) and (4),

$$16(\sin^2\theta + \cos^2\theta) + 9(\sin^2\theta + \cos^2\theta)$$

$$= x^2 + y^2$$

$$\text{then } x^2 + y^2 = 16 + 9 = 25$$

$$\text{Sol.462.(c) } \frac{\cos^4\theta - \sin^4\theta}{\sin^2\theta}$$

$$= \frac{(\cos^2\theta - \sin^2\theta)(\sin^2\theta + \cos^2\theta)}{\sin^2\theta}$$

$$= \cot^2\theta - 1$$

Sol.463.(b)

$$\frac{\sin A}{\cot A + \operatorname{cosec} A} - \frac{\sin A}{\cot A - \operatorname{cosec} A} + 1$$

$$= \frac{\sin A \cot A - 1 - \sin A \cot A - 1}{\cot^2 A - \operatorname{cosec}^2 A} + 1$$

$$= \frac{-2}{-1} + 1 = 3$$

$$\text{Sol.464.(b) Put } \theta = 45^\circ$$

$$1 + \sin^2\theta - 3\sin\theta \cos\theta = 0$$

$$\Rightarrow 1 + \frac{1}{2} - \frac{3}{2} = 0$$

$$\text{Now, } \cot\theta = 1$$

$$\text{Sol.465.(b) } \cos^4\alpha - \sin^4\alpha = \frac{5}{6}$$

$$\Rightarrow (\cos^2\alpha - \sin^2\alpha)(\cos^2\alpha + \sin^2\alpha) = \frac{5}{6}$$

$$\Rightarrow \cos^2\alpha - (1 - \cos^2\alpha) = \frac{5}{6}$$

$$\Rightarrow \cos^2\alpha + \cos^2\alpha - 1 = \frac{5}{6}$$

$$\text{then, } 2\cos^2\alpha - 1 = \frac{5}{6}$$

Sol.466.(a)

$$\left[\frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta}\right] = \frac{\cos^2\theta + \sin^2\theta}{\sin\theta \times \cos\theta}$$

$$= \frac{1}{\sin\theta \times \cos\theta} = \operatorname{cosec}\theta \cdot \sec\theta$$

Sol.467.(b)

$$\frac{\cos 50^\circ}{\sin 40^\circ} + \frac{3\operatorname{cosec} 80^\circ}{\sec 10^\circ} - 2\cos 50^\circ \cdot \operatorname{cosec} 40^\circ$$

$$= \frac{\sin 40^\circ}{\sin 40^\circ} + \frac{3\sec 10^\circ}{\sec 10^\circ} - 2\sin 40^\circ \cdot \operatorname{cosec} 40^\circ$$

$$= 1 + 3 - 2\left(\sin 40^\circ \times \frac{1}{\sin 40^\circ}\right)$$

$$= 1 + 3 - 2(1) = 4 - 2 = 2$$

$$\text{Sol.468.(d) } \sqrt{3}\tan\theta = 3\sin\theta$$

$$\Rightarrow \frac{\tan\theta}{\sin\theta} = \frac{3}{\sqrt{3}} = \sqrt{3} \Rightarrow \frac{1}{\cos\theta} = \sqrt{3}$$

$$\Rightarrow \cos^2\theta = \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$

As per trigonometric identities, we have

$$\sin^2\theta + \cos^2\theta = 1$$

$$\Rightarrow \sin^2\theta = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\text{So, } \sin^2\theta - \cos^2\theta = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$$

$$\text{Sol.469.(a) } \frac{1}{\tan\theta} + \tan\theta = \frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta}$$

$$= \frac{\cos^2\theta + \sin^2\theta}{\sin\theta \times \cos\theta} = \frac{1}{\sin\theta \times \cos\theta}$$

$$= \frac{\sec\theta \times \operatorname{cosec}\theta}{1} = \sec\theta \times \operatorname{cosec}\theta$$

Sol.470.(a)

$$[(\sin\theta \tan\theta + \cos\theta)^2 - 1] = \sin^2\theta$$

$$\tan^2\theta + \cos^2\theta + 2\sin\theta \cdot \tan\theta \cdot \cos\theta - 1$$

$$= \sin^2\theta \tan^2\theta + \cos^2\theta + 2\sin^2\theta - 1$$

$$= \sin^2\theta \tan^2\theta + \cos^2\theta + \sin^2\theta + \sin^2\theta - 1$$

$$= \sin^2\theta(1 + \tan^2\theta) + (\sin^2\theta + \cos^2\theta) - 1$$

$$= \sin^2\theta \cdot \sec^2\theta + 1 - 1 = \tan^2\theta$$

Short Trick:- Put $\theta = 45^\circ$

$$[(\sin\theta \tan\theta + \cos\theta)^2 - 1]$$

$$\left[\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)^2 - 1\right] = 1$$

Only option (a) satisfies the given condition. $\Rightarrow \tan^2\theta = 1$ **Sol.471.(b)**

$$\sin^2 \frac{2\pi}{3} + \cos^2 \frac{5\pi}{6} - \tan^2 \frac{3\pi}{4}$$

$$= \sin^2 120^\circ + \cos^2 150^\circ - \tan^2 135^\circ$$

$$= \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{-\sqrt{3}}{2}\right)^2 - 1$$

$$= \frac{3}{4} + \frac{3}{4} - 1 = \frac{6}{4} - 1 = \frac{1}{2}$$

Sol.472.(a)

$$\frac{2\sin\theta}{\sqrt{4 - 4\sin^2\theta}} = \frac{2\sin\theta}{\sqrt{4(1 - \sin^2\theta)}}$$

$$= \frac{2\sin\theta}{\sqrt{4\cos^2\theta}} = \frac{2\sin\theta}{2\cos\theta} = \tan\theta$$

Sol.473.(c)

$$\frac{1}{\cos x} \sqrt{\frac{\cos(\pi + x) \cos(-x)}{\sin(\pi - x) \cos\left(\frac{\pi}{2} + x\right)}}$$

$$= \frac{1}{\cos x} \sqrt{\frac{-\cos x \cos x}{\sin x (-\sin x)}}$$

$$= \frac{1}{\cos x} \times \frac{\cos x}{\sin x} = \frac{1}{\sin x} = \operatorname{cosec} x$$

Sol.474.(b) $\cos\theta = \sin(2\theta)$

$$\sin(90^\circ - \theta) = \sin(2\theta) \Rightarrow 90^\circ - \theta = 2\theta$$

$$\Rightarrow 90^\circ = 3\theta \Rightarrow \theta = 30^\circ. \text{ Now,}$$

$$(\cos^4 30^\circ + \cos^3 30^\circ + \cos 30^\circ) + (\sin^4 30^\circ + \sin^3 30^\circ + \sin 30^\circ) + (\sin^2 30^\circ + \cos^2 30^\circ)$$

$$= \left(\frac{9}{16} + \frac{3\sqrt{3}}{8} + \frac{\sqrt{3}}{2}\right) + \left(\frac{1}{16} + \frac{1}{8} + \frac{1}{2}\right) + 1$$

$$= \left(\frac{9 + 14\sqrt{3}}{16}\right) + \frac{11}{16} + 1$$

$$= \frac{9 + 14\sqrt{3} + 11 + 16}{16} = \frac{36 + 14\sqrt{3}}{16}$$

$$= \frac{2(18 + 7\sqrt{3})}{16} = \frac{18 + 7\sqrt{3}}{8}$$

Sol.475.(b)

$$\frac{x(1 - \operatorname{cosec}^2 30^\circ)}{1 + \operatorname{cosec}^2 30^\circ} = \left(\frac{1}{2}\right)^2 + (1 \times 4) - 2^2$$

$$\frac{x(1 - 4)}{1 + 4} = \frac{1}{4} + 4 - 4$$

$$x \times \frac{-3}{5} = \frac{1}{4} \Rightarrow x = \frac{1}{4} \times \frac{5}{-3} = -\frac{5}{12}$$

Sol.476.(a) Given that $14\sec\theta = x$

$$\sec\theta - \cos\theta = 14$$

$$\sec \theta - \frac{1}{\sec \theta} = 14 \Rightarrow \frac{\sec^2 \theta - 1}{\sec \theta} = 14$$

[$14 \sec \theta = x$, given]

$$14 \sec \theta = \sec^2 \theta - 1 = \tan^2 \theta$$

$$14 \sec \theta = x = \tan^2 \theta$$

Sol.477.(d) $(1 + \tan 10^\circ)(1 + \tan 35^\circ)$

We know,

If $A + B = 45^\circ$ or 225° then, $(1 + \tan A)$

$$(1 + \tan B) = 2$$

$$(1 + \tan 10^\circ)(1 + \tan 35^\circ) = 2$$

$$\begin{aligned} \text{Sol.478.(a)} \quad & \frac{\sin 40^\circ + \sin 20^\circ}{\cos 40^\circ + \cos 20^\circ} \\ &= \frac{2 \sin \left(\frac{40^\circ + 20^\circ}{2} \right) \cos \left(\frac{40^\circ - 20^\circ}{2} \right)}{2 \cos \left(\frac{40^\circ + 20^\circ}{2} \right) \cos \left(\frac{40^\circ - 20^\circ}{2} \right)} \\ &= \frac{\sin 30^\circ \cos \theta}{\cos 30^\circ \cos \theta} = \frac{\sin 30^\circ}{\cos 30^\circ} = \tan 30^\circ \end{aligned}$$

Sol.479.(d) $\sin x - 3 \cos x = \sqrt{3} \cos x$

$$\sin x = \sqrt{3} \cos x + 3 \cos x$$

Dividing both side by $\cos x$. we get

$$\tan x = \sqrt{3} + 3$$

$$\cot x = \frac{1}{\sqrt{3} + 3} = \frac{\sqrt{3} - 3}{3 - 9}$$

$$= -\frac{\sqrt{3} - 3}{6} = \frac{3 - \sqrt{3}}{6}$$

Sol.480.(b) As per trigonometric identity,

$$\text{we have } \sin 3x = 3 \sin x - 4 \sin^3 x$$

$$\text{Now, } (4 \sin^3 x - 3 \sin x + \sin 3x)$$

$$= 4 \sin^3 x - 3 \sin x + 3 \sin x - 4 \sin^3 x = 0$$

Sol.481.(c)

$$\cos^2(90^\circ - \theta) - \left[\frac{\cos(90^\circ - \theta) \cos \theta}{\cot \theta} \right]$$

$$= \sin^2 \theta - \left[\frac{\sin \theta \cos \theta}{\cot \theta} \right]$$

$$= \sin^2 \theta - [\sin \theta \cos \theta \times \frac{\sin \theta}{\cos \theta}]$$

$$\Rightarrow \sin^2 \theta - \sin^2 \theta = 0$$

$$\text{Sol.482.(d)} \quad \sin 2\theta \left(\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right)$$

$$\Rightarrow 2 \sin \theta \cos \theta \left(\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta} \right)$$

$$\Rightarrow 2 \sin \theta \cos \theta \times \frac{1}{\sin \theta \cos \theta} = 2$$

Sol.483.(b) Formula used:-

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

Putting $A = 75^\circ$, $B = 15^\circ$. We have ,

$$2 \times \cos 75^\circ \cos 15^\circ = \cos(75^\circ + 15^\circ)$$

$$+ \cos(75^\circ - 15^\circ)$$

$$2 \times \cos 75^\circ \cos 15^\circ = \cos 90^\circ + \cos 60^\circ$$

$$\cos 75^\circ \cos 15^\circ = \frac{1}{2} \left(0 + \frac{1}{2} \right)$$

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Sol.484.(a) $1 + \tan \theta = \sqrt{3}$

$$\tan \theta = \sqrt{3} - 1$$

$$\sqrt{3} \cot \theta - 1 = \sqrt{3} \times \frac{1}{\sqrt{3} - 1} - 1$$

$$\Rightarrow \frac{\sqrt{3}(\sqrt{3} + 1)}{2} - 1 = \frac{3 + \sqrt{3} - 2}{2}$$

$$\Rightarrow \frac{\sqrt{3} + 1}{2}$$

$$\text{Sol.485.(b)} \quad \operatorname{cosec}^2 \theta + \cot^2 \theta = \frac{5}{3}$$

$$\Rightarrow \frac{1}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{5}{3}$$

$$\Rightarrow \frac{1 + 1 - \sin^2 \theta}{\sin^2 \theta} = \frac{5}{3} \Rightarrow \frac{2 - \sin^2 \theta}{\sin^2 \theta} = \frac{5}{3}$$

$$\Rightarrow 6 - 3 \sin^2 \theta = 5 \sin^2 \theta$$

$$\Rightarrow 6 = 8 \sin^2 \theta \Rightarrow \sin^2 \theta = \frac{6}{8} = \frac{3}{4}$$

$$\Rightarrow \sin \theta = \frac{\sqrt{3}}{2} = \sin 60^\circ \Rightarrow \theta = 60^\circ$$

$$\cot 2\theta = \cot(2 \times 60^\circ) = \cot 120^\circ = -\frac{1}{\sqrt{3}}$$

Short trick:- Put $\theta = 60^\circ$

$$\operatorname{cosec}^2 \theta + \cot^2 \theta = \left(\frac{2}{\sqrt{3}} \right)^2 + \left(\frac{1}{\sqrt{3}} \right)^2 = \frac{5}{3}$$

LHS = RHS . Now,

$$\cot 2\theta = \cot(2 \times 60^\circ) = \cot 120^\circ = -\frac{1}{\sqrt{3}}$$

Sol.486.(b) $(\operatorname{cosec} \theta - \cot \theta)^2$

$$= \left(\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} \right)^2$$

$$= \frac{(1 - \cos \theta)^2}{\sin^2 \theta} = \frac{(1 - \cos \theta)(1 - \cos \theta)}{1 - \cos^2 \theta}$$

$$= \frac{(1 - \cos \theta)(1 - \cos \theta)}{(1 - \cos \theta)(1 + \cos \theta)} = \frac{(1 - \cos \theta)}{(1 + \cos \theta)}$$

Sol.487.(a) Put $x = 45^\circ$

$$\tan \frac{45^\circ}{2} = \frac{1 - \cos 45^\circ}{\sin 45^\circ}$$

$$\tan 22.5^\circ = \frac{1 - \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} \Rightarrow \tan 22.5^\circ = \sqrt{2} - 1$$

Sol.488.(a) $4 \sin^2 \theta = 3(1 + \cos \theta)$

$$4 \sin^2 \theta = 3 + 3 \cos \theta$$

$$4(1 - \cos^2 \theta) = 3 + 3 \cos \theta$$

$$4 - 4 \cos^2 \theta = 3 + 3 \cos \theta$$

$$-4 \cos^2 \theta - 3 \cos \theta + 1 = 0$$

$$4 \cos^2 \theta + 3 \cos \theta - 1 = 0$$

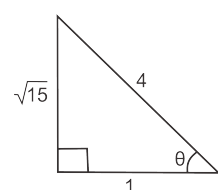
$$4 \cos^2 \theta + 4 \cos \theta - \cos \theta - 1 = 0$$

$$4 \cos \theta (\cos \theta + 1) - 1(\cos \theta + 1) = 0$$

$$(\cos \theta + 1)(4 \cos \theta - 1) = 0$$

$\cos \theta = -1$ not possible as $0^\circ < \theta < 90^\circ$, so

$$\cos \theta = \frac{1}{4}$$



$$2 \tan \theta + 4 \sin \theta - \sec \theta$$

$$= 2 \left(\frac{\sqrt{15}}{1} \right) + 4 \left(\frac{\sqrt{15}}{4} \right) - \frac{4}{1}$$

$$= 2\sqrt{15} + \sqrt{15} - 4 = 3\sqrt{15} - 4$$

Sol.489.(c) $3 \sec^2 \theta + \tan \theta - 7 = 0$

$$3(1 + \tan^2 \theta) + \tan \theta - 7 = 0$$

$$3 + 3 \tan^2 \theta + \tan \theta - 7 = 0$$

$$3 \tan^2 \theta + \tan \theta - 4 = 0$$

$$3 \tan^2 \theta + 4 \tan \theta - 3 \tan \theta - 4 = 0$$

$$3 \tan^2 \theta - 3 \tan \theta + 4 \tan \theta - 4 = 0$$

$$3 \tan \theta (\tan \theta - 1) + 4 (\tan \theta - 1) = 0$$

$$\tan \theta = -\frac{4}{3} \text{ not possible as } 0^\circ < \theta < 90^\circ, \tan \theta = 1, \text{ hence } \theta = 45^\circ$$

$$90^\circ, \tan \theta = 1, \text{ hence } \theta = 45^\circ$$

$$\text{So, } \frac{2 \sin \theta + 3 \cos \theta}{\operatorname{cosec} \theta + \sec \theta}$$

$$= \frac{2 \sin 45^\circ + 3 \cos 45^\circ}{\operatorname{cosec} 45^\circ + \sec 45^\circ}$$

$$= \frac{2 \left(\frac{1}{\sqrt{2}} \right) + 3 \left(\frac{1}{\sqrt{2}} \right)}{\sqrt{2} + \sqrt{2}} = \frac{\frac{5}{\sqrt{2}}}{2\sqrt{2}} = \frac{5}{4}$$

Alternate :- Put $\theta = 45^\circ$

$$3 \sec^2 \theta + \tan \theta - 7 = 3 \times (\sqrt{2})^2 + 1 - 7 = 6 + 1 - 7 = 0 \text{ (LHS = RHS)}$$

$$\text{So, } \frac{2 \sin \theta + 3 \cos \theta}{\operatorname{cosec} \theta + \sec \theta}$$

$$= \frac{2 \sin 45^\circ + 3 \cos 45^\circ}{\operatorname{cosec} 45^\circ + \sec 45^\circ}$$

$$= \frac{2 \left(\frac{1}{\sqrt{2}} \right) + 3 \left(\frac{1}{\sqrt{2}} \right)}{\sqrt{2} + \sqrt{2}} = \frac{\frac{5}{\sqrt{2}}}{2\sqrt{2}} = \frac{5}{4}$$

Sol.490.(d) Putting $\theta = 45^\circ$

$$1 + \sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$$

$$= 1 + \sqrt{\frac{\cot 45^\circ + \cos 45^\circ}{\cot 45^\circ - \cos 45^\circ}}$$

$$= 1 + \sqrt{\frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}} = 1 + \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1}}$$

$$= 1 + \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}}$$

$$= 1 + \sqrt{\frac{(\sqrt{2} + 1)^2}{1}} = 1 + \sqrt{2} + 1 = 2 + \sqrt{2}$$

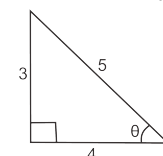
Putting $\theta = 45^\circ$ in options, we find that option (d) gets satisfied.

$$1 + \sec \theta + \tan \theta$$

$$= 1 + \sec 45^\circ + \tan 45^\circ$$

$$= 1 + \sqrt{2} + 1 \Rightarrow 2 + \sqrt{2}$$

Sol.491.(c) In this question, we will put the value of pythagoras triplet.



$$\left(\frac{1 - \cot \theta}{1 - \tan \theta} \right)^2 - 1 = \left(\frac{1 - \frac{4}{3}}{1 - \frac{3}{4}} \right)^2 - 1$$

$$\Rightarrow \left(\frac{-\frac{1}{3}}{\frac{1}{4}} \right)^2 - 1 = \left(\frac{-4}{3} \right)^2 - 1$$

$$\Rightarrow \frac{16}{9} - 1 = \frac{7}{9}$$

Putting given triplets in options, we find that option (c) gets satisfied.

$$\cot^2 \theta - 1 = \left(\frac{4}{3}\right)^2 - 1 = \frac{16}{9} - 1 = \frac{7}{9}$$

Sol.492.(a) Let $\theta = 45^\circ$, now

$$\begin{aligned} & (\cos^6 \theta + \sin^6 \theta - 1) \\ & (\tan^2 \theta + \cot^2 \theta + 2) + 3 \\ & = \left[\left(\frac{1}{\sqrt{2}}\right)^6 + \left(\frac{1}{\sqrt{2}}\right)^6 - 1\right][1^2 + 1^2 + 2] + 3 \\ & = \left[\frac{1}{8} + \frac{1}{8} - 1\right][4] + 3 = -3 + 3 = 0 \end{aligned}$$

Sol.493.(c) $\sin^2 \theta - \cos^2 \theta - 3\sin \theta + 2 = 0$

$$\sin^2 \theta - (1 - \sin^2 \theta) - 3\sin \theta + 2 = 0$$

$$\sin^2 \theta - 1 + \sin^2 \theta - 3\sin \theta + 2 = 0$$

$$2\sin^2 \theta - 3\sin \theta + 1 = 0$$

$$2\sin^2 \theta - 2\sin \theta - \sin \theta + 1 = 0$$

$$2\sin \theta (\sin \theta - 1) - 1(\sin \theta - 1) = 0$$

$$(\sin \theta - 1)(2\sin \theta - 1) = 0$$

$$\sin \theta = 1 \text{ not possible, so } \sin \theta = \frac{1}{2},$$

$$\theta = 30^\circ \Rightarrow \text{So, } 1 + \sec \theta + \tan \theta$$

$$= 1 + \sec 30^\circ + \tan 30^\circ$$

$$= 1 + \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} = 1 + \frac{3}{\sqrt{3}} = 1 + \sqrt{3}$$

Alternate :- put $\theta = 30^\circ, 60^\circ$ one by one, we find that $\theta = 30^\circ$ satisfies the given terms.

$$\sin^2 \theta - \cos^2 \theta - 3\sin \theta + 2$$

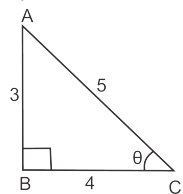
$$= \left(\frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 - 3 \times \frac{1}{2} + 2$$

$$= \frac{1}{4} - \frac{3}{4} - \frac{3}{2} + 2 = 0 \text{ (LHS = RHS)}$$

$$\text{So, } 1 + \sec \theta + \tan \theta \Rightarrow 1 + \sec 30^\circ + \tan 30^\circ$$

$$= 1 + \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} = 1 + \frac{3}{\sqrt{3}} = 1 + \sqrt{3}$$

Sol.494.(b) In this question, we will use pythagorean triplets of 3, 4, 5.



$$\sec \theta = \frac{5}{4}, \tan \theta = \frac{3}{4}, \sin \theta = \frac{3}{5} \text{ and}$$

$$\cos \theta = \frac{4}{5}$$

$$(\sec \theta - \tan \theta)^2 (1 + \sin \theta)^2 \div \cos^2 \theta$$

$$= \left(\frac{5}{4} - \frac{3}{4}\right)^2 \left(1 + \frac{3}{5}\right)^2 \div \left(\frac{4}{5}\right)^2$$

$$= \left(\frac{1}{2}\right)^2 \times \left(\frac{8}{5}\right)^2 \div \left(\frac{4}{5}\right)^2$$

$$= \frac{1}{4} \times \frac{64}{25} \div \frac{16}{25} = \frac{1}{4} \times \frac{64}{25} \times \frac{25}{16} = 1$$

Sol.495.(c) $2\sin^2 \theta + 3\cos \theta = 3$

$$2(1 - \cos^2 \theta) + 3\cos \theta = 3$$

$$2 - 2\cos^2 \theta + 3\cos \theta = 3$$

$$-2\cos^2 \theta + 3\cos \theta - 1 = 0$$

$$2\cos^2 \theta - 3\cos \theta + 1 = 0$$

$$2\cos^2 \theta - 2\cos \theta - \cos \theta + 1 = 0$$

$$2\cos \theta (\cos \theta - 1) - 1(\cos \theta - 1) = 0$$

$$(\cos \theta - 1)(2\cos \theta - 1) = 0$$

$$\cos \theta = 1 \text{ not possible as } 0^\circ < \theta < 90^\circ,$$

$$\text{so } \cos \theta = \frac{1}{2}, \theta = 60^\circ$$

$$\text{So, } \sec^2 \theta + \cot^2 \theta = \sec^2 60^\circ + \cot^2 60^\circ$$

$$= (2)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 = 4 + \frac{1}{3} = 4\frac{1}{3}$$

Alternate method :- put $\theta = 60^\circ$

$$2\sin^2 \theta + 3\cos \theta = 2 \times \left(\frac{\sqrt{3}}{2}\right)^2 + 3 \times \frac{1}{2}$$

$$= 2 \times \frac{3}{4} + \frac{3}{2} = \frac{3}{2} + \frac{3}{2} = 3 \text{ i.e. LHS = RHS}$$

$$\text{So, } \sec^2 \theta + \cot^2 \theta = \sec^2 60^\circ + \cot^2 60^\circ$$

$$= (2)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 = 4 + \frac{1}{3} = 4\frac{1}{3}$$

Sol.496.(d) Put $A = 135^\circ$

$$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} - \sin A$$

$$= \frac{-\frac{1}{\sqrt{2}}}{1 - (-1)} + \frac{\frac{1}{\sqrt{2}}}{1 - (-1)} - \frac{1}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$$

$$\text{Only } \cos A \text{ gives } -\frac{1}{\sqrt{2}} \text{ at } A = 135^\circ.$$

so $\cos A$ is answer

Sol.497.(d)

$$\operatorname{cosec}^4 A (1 - \cos^4 A) - 2\cot^2 A - 1$$

Put $A = 45^\circ$ in equation because all option gives different values at $A = 45^\circ$

$$(\sqrt{2})^4 \left\{1 - \left(\frac{1}{\sqrt{2}}\right)^4\right\} - 2 \times (1)^2 - 1$$

$$= 4 \times \frac{3}{4} - 3 = 3 - 3 = 0$$

Sol.498.(d) $6\tan A(\tan A + 1) = 5 - \tan A$,

$$6\tan^2 A + 6\tan A + \tan A - 5 = 0$$

$$6\tan^2 A + 7\tan A - 5 = 0$$

$$6\tan^2 A + 10\tan A - 3\tan A - 5 = 0$$

$$2\tan A(3\tan A + 5) - 1(3\tan A + 5) = 0$$

$$(2\tan A - 1)(3\tan A + 5) = 0$$

$$\tan A = \frac{1}{2}, -\frac{5}{3}$$

$$\text{Taking } \tan A = \frac{1}{2} \dots (0 < A < \frac{\pi}{2})$$

$$\sin A = \frac{1}{\sqrt{5}}, \cos A = \frac{2}{\sqrt{5}}$$

$$\sin A + \cos A = \frac{1}{\sqrt{5}} + \frac{2}{\sqrt{5}} = \frac{3}{\sqrt{5}}$$

Sol.499.(a)

$$\left(\frac{1 - \cot \theta}{1 - \tan \theta}\right)^2 + 1 \Rightarrow \left(\frac{1 - \cot \theta}{1 - \frac{1}{\cot \theta}}\right)^2 + 1$$

$$= \left(\frac{1 - \cot \theta}{\cot \theta - 1}\right)^2 \times \cot^2 \theta + 1$$

$$= (-1)^2 \times \cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$$

$$\Rightarrow \cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$$

$$\text{Sol.500.(b)} \frac{\sin^2 \theta - 3\sin \theta + 2}{\cos^2 \theta} = 1$$

$$\sin^2 \theta - 3\sin \theta + 2 = \cos^2 \theta$$

$$\sin^2 \theta - 3\sin \theta + 2 = 1 - \sin^2 \theta$$

$$2\sin^2 \theta - 3\sin \theta + 1 = 0$$

$$2\sin^2 \theta - 2\sin \theta - \sin \theta + 1 = 0$$

$$2\sin \theta (\sin \theta - 1) - 1(\sin \theta - 1) = 0$$

$$(2\sin \theta - 1)(\sin \theta - 1) = 0$$

$$\sin \theta = \frac{1}{2}, \theta = 30^\circ, 90^\circ$$

90° is not possible because $\cos \theta$ cannot be 0. $\Rightarrow \cos 2\theta - \sin 3\theta + \operatorname{cosec} 2\theta$

$$\cos 60^\circ - \sin 90^\circ + \operatorname{cosec} 60^\circ$$

$$= \frac{1}{2} - 1 + \frac{2}{\sqrt{3}} = \frac{-3 + 4\sqrt{3}}{6}$$

Sol.501.(d)

$$\frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}{\tan \theta + \cot \theta}$$

$$= \frac{1 - \sin^2 \theta + \cos \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}}}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}}$$

$$= \frac{\cos^2 \theta + \cos \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta \times \sin^2 \theta}}}{\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \times \sin \theta}}$$

$$= \frac{\cos \theta (1 + \cos \theta)}{\sin \theta (1 + \cos \theta)} \times \frac{1}{\frac{\cos \theta \times \sin \theta}{\cos \theta \times \sin \theta}}$$

$$= \frac{\cos \theta}{\sin \theta} = \cot \theta$$

Sol.502.(d)

$$1 + 2\tan^2 \theta + 2\sin \theta \sec^2 \theta = \frac{a}{b}$$

$$1 + \tan^2 \theta + \tan^2 \theta + 2\sin \theta \sec^2 \theta = \frac{a}{b}$$

$$\sec^2 \theta + \tan^2 \theta + 2\sin \theta \sec^2 \theta = \frac{a}{b}$$

$$\sec^2 \theta + \sec^2 \theta - 1 + 2\sin \theta \sec^2 \theta = \frac{a}{b}$$

$$\frac{2\sec^2 \theta - 1 + 2\sin \theta \sec^2 \theta}{1} = \frac{a}{b}$$

Using componendo and dividendo,

$$\frac{2\sec^2 \theta - 1 + 2\sin \theta \sec^2 \theta + 1}{2\sec^2 \theta - 1 + 2\sin \theta \sec^2 \theta - 1} = \frac{a + b}{a - b}$$

$$\text{So, } \frac{a + b}{a - b} = \frac{2\sec^2 \theta + 2\sin \theta \sec^2 \theta}{2\sec^2 \theta - 2 + 2\sin \theta \sec^2 \theta}$$

$$= \frac{\frac{2}{\cos^2 \theta} + \frac{2\sin \theta}{\cos^2 \theta}}{\frac{2}{\cos^2 \theta} - 2 + \frac{2\sin \theta}{\cos^2 \theta}}$$

$$= \frac{2 + 2\sin \theta}{2 - 2\cos^2 \theta + 2\sin \theta}$$

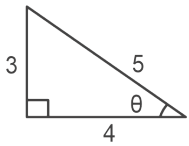
$$\Rightarrow \frac{2 + 2\sin \theta}{2\sin^2 \theta + 2\sin \theta} = \frac{2(1 + \sin \theta)}{2\sin \theta (1 + \sin \theta)}$$

$$\Rightarrow \frac{1}{\sin \theta} = \operatorname{cosec} \theta$$

Sol.503.(d)

$$\frac{(1 + \sec \theta \operatorname{cosec} \theta)^2 (\sec \theta - \tan \theta)^2 (1 + \sin \theta)}{(\sin \theta + \sec \theta)^2 + (\cos \theta + \operatorname{cosec} \theta)^2}$$

.....(1)



Put $\tan\theta = \frac{3}{4}$, $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$ and so

on.

Putting respective values in (1), we get

$$\begin{aligned} & \frac{(1 + \sec\theta \csc\theta)^2 (\sec\theta - \tan\theta)^2 (1 + \sin\theta)}{(\sin\theta + \sec\theta)^2 + (\cos\theta + \csc\theta)^2} \\ &= \frac{(1 + \frac{5}{4} \times \frac{5}{3})^2 (\frac{5}{4} - \frac{3}{4})^2 (1 + \frac{3}{5})}{(\frac{3}{5} + \frac{5}{4})^2 + (\frac{4}{5} + \frac{5}{3})^2} \\ &= \frac{(1 + \frac{25}{12})^2 \times (\frac{1}{4}) \times (\frac{8}{5})}{(\frac{12+25}{20})^2 + (\frac{12+25}{15})^2} \\ &= \frac{\frac{1}{12 \times 12} \times \frac{1}{4} \times \frac{8}{5}}{\frac{1}{400} + \frac{1}{225}} = \frac{\frac{1}{12 \times 6} \times \frac{1}{5}}{\frac{625}{400 \times 225}} = \frac{2}{5} \end{aligned}$$

Among all the given options, only option (d) satisfies the above obtained value.

Sol.504.(c) $\left(\frac{\tan^3\theta}{\sec^2\theta} + \frac{\cot^3\theta}{\csc^2\theta} + 2\sin\theta\cos\theta\right) \div (1 + \csc^2\theta + \tan^2\theta)$

$$\begin{aligned} &= \left(\frac{\sin^3\theta}{\cos^3\theta \times \sec^2\theta} + \frac{\cos^3\theta}{\sin^3\theta \times \csc^2\theta} + 2\sin\theta\cos\theta\right) \div (\sec^2\theta + \csc^2\theta) \\ &= \left(\frac{\sin^3\theta}{\cos\theta} + \frac{\cos^3\theta}{\sin\theta} + 2\sin\theta\cos\theta\right) \div \left(\frac{1}{\sin^2\theta} + \frac{1}{\cos^2\theta}\right) \\ &= \left(\frac{\sin^4\theta + \cos^4\theta + 2\sin^2\theta\cos^2\theta}{\sin\theta\cos\theta}\right) \div \left(\frac{\cos^2\theta + \sin^2\theta}{\sin^2\theta\cos^2\theta}\right) \\ &= \frac{(\sin^2\theta + \cos^2\theta)^2}{\sin\theta\cos\theta} \div \left(\frac{1}{\sin^2\theta\cos^2\theta}\right) \\ &= \frac{1}{\sin\theta\cos\theta} \div \left(\frac{1}{\sin^2\theta\cos^2\theta}\right) = \sin\theta\cos\theta \end{aligned}$$

Sol.505.(d) $\frac{\tan^6\theta - \sec^6\theta + 3\sec^2\theta \tan^2\theta}{\tan^2\theta + \cot^2\theta + 2}$

Putting $\theta = 45^\circ$ in above expression, we get

$$\begin{aligned} & \frac{\tan^6 45^\circ - \sec^6 45^\circ + 3\sec^2 45^\circ \tan^2 45^\circ}{\tan^2 45^\circ + \cot^2 45^\circ + 2} \\ &= \frac{(1)^6 - (\sqrt{2})^6 + 3(\sqrt{2})^2 (1)^2}{(1)^2 + (1)^2 + 2} \\ &= \frac{1 - 8 + 6}{4} = \frac{-1}{4} \end{aligned}$$

If we put $\theta = 45^\circ$ in given options, then only option (d) satisfies the above value.

Sol.506.(b)

$$\begin{aligned} & \frac{\cos^4\theta - \sin^4\theta + 2\sin^2\theta + 3}{(\csc\theta + \cot\theta + 1)(\csc\theta - \cot\theta + 1) - 2} \\ &= \frac{\cos^2\theta - \sin^2\theta + 2\sin^2\theta + 3}{(\csc\theta + 1)^2 - \cot^2 - 2} \end{aligned}$$

$$\begin{aligned} &= \frac{\cos^2\theta + \sin^2\theta + 3}{\csc^2\theta + 1 + 2\csc\theta - \cot^2\theta - 2} \\ &= \frac{1 + 3}{1 + 1 + 2\csc\theta - 2} = \frac{4}{2\csc\theta} = 2\sin\theta \end{aligned}$$

Sol.507.(d)

$$\begin{aligned} & \frac{(1 - \sin\theta + \cos\theta)^2 (1 - \cos\theta) \sec^3\theta \csc^2\theta}{(\sec\theta - \tan\theta)(\tan\theta + \cot\theta)} \\ \text{Putting } \theta &= 45^\circ, \\ & \frac{(1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}})^2 (1 - \frac{1}{\sqrt{2}}) (\sqrt{2})^3 (\sqrt{2})^2}{(\sqrt{2} - 1)(1 + 1)} \\ &= \frac{(1)^2 (\frac{\sqrt{2}-1}{\sqrt{2}}) 4\sqrt{2}}{(\sqrt{2} - 1)(1 + 1)} = \frac{4}{2} = 2 \end{aligned}$$

Put $\theta = 45^\circ$ in options

So, option (d) satisfies. $\Rightarrow 2 \tan 45^\circ = 2$

Sol.508.(d) $\frac{\sin^2\theta}{\cos^2\theta - 3\cos\theta + 2} = 1$

$$\Rightarrow \sin^2\theta = \cos^2\theta - 3\cos\theta + 2$$

$$\Rightarrow 1 - \cos^2\theta = \cos^2\theta - 3\cos\theta + 2$$

$$\Rightarrow 2\cos^2\theta - 3\cos\theta + 1 = 0$$

$$\Rightarrow 2\cos^2\theta - 2\cos\theta - \cos\theta + 1 = 0$$

$$\Rightarrow 2\cos\theta (\cos\theta - 1) - 1 (\cos\theta - 1) = 0$$

$$\Rightarrow (2\cos\theta - 1) (\cos\theta - 1) = 0$$

$$\text{For } \cos\theta = \frac{1}{2}, \theta = 60^\circ \text{ and for}$$

$$\cos\theta = 1, \theta = 0^\circ,$$

as value of θ lies in first quadrant which is not possible in case of $\cos\theta = 1$,

so value of $\cos\theta = \frac{1}{2}$ and $\theta = 60^\circ$ is

admissible and correct.

Putting $\theta = 60^\circ$, in $\frac{\tan^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}}{\tan\theta + \sin\theta}$

we get, $\frac{\tan^2 60^\circ + \sin^2 60^\circ}{\tan 60^\circ + \sin 60^\circ}$

$$\begin{aligned} &= \frac{\tan^2 30^\circ + \sin^2 30^\circ}{\tan 60^\circ + \sin 60^\circ} \\ &= \frac{(\frac{1}{\sqrt{3}})^2 + (\frac{1}{2})^2}{\sqrt{3} + \frac{\sqrt{3}}{2}} = \frac{\frac{1}{3} + \frac{1}{4}}{\frac{3\sqrt{3}}{2}} = \frac{\frac{7}{12}}{\frac{3\sqrt{3}}{2}} \end{aligned}$$

$$= \frac{14}{36\sqrt{3}} = \frac{7}{18\sqrt{3}} = \frac{7}{18\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

(multiplying numerator and denominator

$$\text{by } \sqrt{3}) = \frac{7\sqrt{3}}{54}$$

Sol.509.(b)

$$\frac{(1 - 2\sin^2\theta \cos^2\theta)(\cot\theta + 1) \cos\theta}{(\sin^4\theta + \cos^4\theta)(1 + \tan\theta) \csc\theta} - 1$$

Put $\theta = 45^\circ$

$$\begin{aligned} & \frac{\{(1 - 2(\frac{1}{\sqrt{2}})^2 (\frac{1}{\sqrt{2}})^2)\} (1 + 1) \frac{1}{\sqrt{2}}}{[(\frac{1}{\sqrt{2}})^4 + (\frac{1}{\sqrt{2}})^4] (1 + 1) \sqrt{2}} - 1 \\ &= \frac{(1 - \frac{1}{2})(1 + 1) \frac{1}{\sqrt{2}}}{[\frac{1}{4} + \frac{1}{4}] 2\sqrt{2}} - 1 = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2} \times 2\sqrt{2}} - 1 \end{aligned}$$

$$= \frac{1}{\sqrt{2} \times \sqrt{2}} - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$$

on putting $\theta = 45^\circ$ in given options, then only option (b) satisfies.

$$\Rightarrow -\sin^2\theta = -(\frac{1}{\sqrt{2}})^2 = -\frac{1}{2}$$

Sol.510.(b) As, $(\cos^2\theta + \sin^2\theta) = 1$

So,

$$\begin{aligned} & (\cos^2 x + \sin^2 x) + (\cos^2 2x + \sin^2 2x) + \\ & (\cos^2 3x + \sin^2 3x) + \dots + \\ & (\cos^2 30x + \sin^2 30x) \\ &= 1 + 1 + 1 + \dots + 1 = 1 \times 30 = 30 \end{aligned}$$

$$\text{So, required average} = \frac{30}{60} = 0.5$$

Sol.511.(b)

$$\begin{aligned} &= \frac{\csc\theta}{\csc\theta - 1} + \frac{\csc\theta}{\csc\theta + 1} - \tan^2\theta \\ &= \frac{\csc^2\theta + \csc\theta + \csc^2\theta - \csc\theta}{\csc^2\theta - 1} - \tan^2\theta \\ &= \frac{2\csc^2\theta}{\csc^2\theta - 1} - \tan^2\theta = \frac{2\csc^2\theta}{\cot^2\theta} - \tan^2\theta \\ &= 2\sec^2\theta - \tan^2\theta = \sec^2\theta + \sec^2\theta - \tan^2\theta \\ &= \sec^2\theta + 1 \end{aligned}$$

Sol.512.(a) $2\cos^2\theta = 3\sin\theta$

$$2(1 - \sin^2\theta) = 3\sin\theta \Rightarrow 2 - 2\sin^2\theta = 3\sin\theta$$

$$2\sin^2\theta + 3\sin\theta - 2 = 0 \Rightarrow \sin\theta = -2 \text{ or } 0.5$$

So, $\theta = 30^\circ$

$$(\sec^2\theta - \tan^2\theta + \cos^2\theta)$$

$$\text{Putting } \theta = 30^\circ \Rightarrow \frac{4}{3} - \frac{1}{3} + \frac{3}{4} = \frac{7}{4}$$

Sol.513.(a) $\frac{\tan(45^\circ - \alpha)}{\cot(45^\circ + \alpha)}$

$$= \frac{(\cos 19^\circ + \sin 71^\circ)(\sec 19^\circ + \csc 71^\circ)}{\tan 12^\circ \cdot \tan 24^\circ \cdot \tan 66^\circ \cdot \tan 78^\circ}$$

Let $\alpha = 0^\circ \Rightarrow \frac{\tan(45^\circ)}{\cot(45^\circ)}$

$$= \frac{(\cos 19^\circ + \cos 19^\circ)(\sec 19^\circ + \sec 19^\circ)}{\tan 12^\circ \cdot \tan 24^\circ \cdot \cot 24^\circ \cdot \cot 12^\circ}$$

$$1 - \frac{2\cos 19^\circ \cdot 2\sec 19^\circ}{1} \Rightarrow 1 - \frac{2 \times 2}{1}$$

$$\Rightarrow 1 - 4 = -3$$

Sol.514.(d) $1 + 2\tan^2\theta + 2\sin\theta \sec^2\theta$

$$= \sec^2\theta + \tan^2\theta + 2\sin\theta \frac{\sec\theta}{\cos\theta}$$

$$= \sec^2\theta + \tan^2\theta + 2\sec\theta \tan\theta$$

$$= (\sec\theta + \tan\theta)^2$$

$$= \left(\frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta}\right)^2 = \left(\frac{1 + \sin\theta}{\cos\theta}\right)^2$$

$$= \frac{(1 + \sin\theta)^2}{\cos^2\theta} = \frac{(1 + \sin\theta)^2}{(1 + \sin\theta)(1 - \sin\theta)}$$

$$= \frac{1 + \sin\theta}{1 - \sin\theta}$$

Sol.515.(b) $\sin\left(\frac{2A+B}{2}\right) = \frac{\sqrt{3}}{2} = \sin 60^\circ$

$$\Rightarrow 2A + B = 120^\circ \dots (\text{eq. 1})$$

$$\cos\left(\frac{2A-B}{2}\right) = \frac{\sqrt{3}}{2} = \cos 30^\circ$$

$$\Rightarrow 2A - B = 60^\circ \dots (\text{eq. 2})$$

On solving eq. (1) and (2) we get,

$A = 45^\circ$ and $B = 30^\circ$

$$\sin[3(A - B)] = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

Sol.516.(c)

$$\begin{aligned} & (\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}) \left(\frac{\sin \theta (1 + \cos \theta)}{1 + \cos \theta - \sin^2 \theta} \right) \\ & \left(\sqrt{\frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}} \right) \left(\frac{\sin \theta (1 + \cos \theta)}{1 + \cos \theta - 1 + \cos^2 \theta} \right) \\ & = \frac{1}{\sin \theta \cdot \cos \theta} \times \frac{\sin \theta}{\cos \theta} = \sec^2 \theta \end{aligned}$$

$$\text{Sol.517.(d)} \quad \frac{1}{\cos \theta} + \frac{1}{\tan \theta - \sec \theta}$$

$$= \frac{1}{\cos \theta} - \frac{(\tan \theta - \sec \theta)(\tan \theta + \sec \theta)}{\tan \theta - \sec \theta}$$

$$= \sec \theta - \tan \theta - \sec \theta = -\tan \theta$$

$$\text{Sol.518.(b)} \quad 4 \sin^2(2x - 10)^\circ = 3$$

$$\sin^2(2x - 10)^\circ = 3/4 \Rightarrow \sin(2x - 10)^\circ = \frac{\sqrt{3}}{2}$$

$$(2x - 10)^\circ = 60^\circ \Rightarrow x = 35^\circ$$

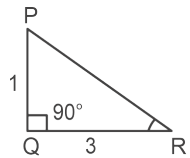
$$\frac{\sin^4(x - 5)^\circ + \cos^4(x - 5)^\circ}{1 - 2 \sin^2(3x - 15)^\circ \cos^2(3x - 15)^\circ}$$

$$\begin{aligned} & \Rightarrow \frac{\sin^4 30^\circ + \cos^4 30^\circ}{1 - 2 \sin^2 90^\circ \cos^2 90^\circ} \\ & = \frac{\left(\frac{1}{2}\right)^4 + \left(\frac{\sqrt{3}}{2}\right)^4}{1 - 2 \times 1 \times 0} = \frac{\frac{1}{16} + \frac{9}{16}}{1} = \frac{5}{8} \end{aligned}$$

$$\text{Sol.519.(d)} \quad \frac{\sqrt{2} \tan(60^\circ - \theta) \tan(30^\circ + \theta)}{\sin^2(45^\circ + \theta) + \sin^2(45^\circ - \theta)}$$

$$\text{Put } \theta = 0^\circ \Rightarrow \frac{\sqrt{2} \tan(60^\circ) \tan(30^\circ)}{\sin^2(45^\circ) + \sin^2(45^\circ)}$$

$$\Rightarrow \frac{\sqrt{2} \times \sqrt{3} \times \frac{1}{\sqrt{3}}}{\frac{1}{2} + \frac{1}{2}} = \sqrt{2}$$

Sol.520.(d)

$$PR = \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$\frac{\sec P(\cos R + \sin P)}{\operatorname{cosec} R(\sin R - \operatorname{cosec} P)}$$

$$\begin{aligned} & = \frac{\frac{\sqrt{10}}{1} \left(\frac{3}{\sqrt{10}} + \frac{3}{\sqrt{10}} \right)}{\frac{\sqrt{10}}{1} \left(\frac{1}{\sqrt{10}} - \frac{\sqrt{10}}{3} \right)} = \frac{\frac{6\sqrt{10}}{\sqrt{10}}}{\frac{3 - 10}{3}} = \frac{6}{-7/3} \\ & = -\frac{18}{7} \end{aligned}$$

Sol.521.(a)

$$(\sec \theta + \tan \theta)^2 + \frac{1 + \operatorname{cosec} \theta}{1 - \operatorname{cosec} \theta}$$

$$= (\sec \theta + \tan \theta)^2 + \frac{(1 + \operatorname{cosec} \theta)^2}{1 - \operatorname{cosec}^2 \theta}$$

$$\Rightarrow \sec^2 \theta + \tan^2 \theta + 2 \sec \theta \times \tan \theta +$$

$$\frac{1 + \operatorname{cosec}^2 \theta + 2 \operatorname{cosec} \theta}{- \cot^2 \theta}$$

$$\Rightarrow \sec^2 \theta + \tan^2 \theta + 2 \sec \theta \times \tan \theta - \tan^2 \theta - \sec^2 \theta - 2 \sec \theta \times \tan \theta = 0$$

$$\text{Sol.522.(b)} \quad \sin^6 \theta + \cos^6 \theta = \frac{1}{3}$$

$$\begin{aligned} & (\sin^2 \theta + \cos^2 \theta) [(\sin^2 \theta + \cos^2 \theta)^2 - 3 \sin^2 \theta \cdot \cos^2 \theta] = \frac{1}{3} \\ & [1 - 3 \sin^2 \theta \cdot \cos^2 \theta] = \frac{1}{3} \end{aligned}$$

$$\Rightarrow 3 \sin^2 \theta \cdot \cos^2 \theta = \frac{2}{3}$$

$$\Rightarrow 3 \sin^2 \theta \cdot \cos^2 \theta = \frac{2}{3}$$

$$\sin^2 \theta \cdot \cos^2 \theta = \frac{2}{9} \Rightarrow \sin \theta \cdot \cos \theta = \frac{\sqrt{2}}{3}$$

$$\text{Sol.523.(d)} \quad 2 \cos^2 \theta - 5 \cos \theta + 2 = 0$$

$$\text{Put } \theta = 60^\circ$$

$$= 2 \cos^2 60^\circ - 5 \cos 60^\circ + 2 = 0$$

$$= 2 \times \frac{1}{4} - \frac{5}{2} + 2 = 0 \Rightarrow 0 = 0, \text{ satisfy then}$$

$$(\sec \theta + \tan \theta) = \sec 60^\circ + \tan 60^\circ$$

$$= 2 + \sqrt{3}$$

$$\text{Sol.524.(a)} \quad \sec^4 \theta (1 - \sin^4 \theta) - 2 \tan^2 \theta$$

$$= \sec^4 \theta \cos^2 \theta (1 + \sin^2 \theta) - 2 \tan^2 \theta$$

$$= \sec^2 \theta + \sec^2 \theta \cdot \sin^2 \theta - \tan^2 \theta - \tan^2 \theta$$

$$= 1 + 0 = 1$$

$$\text{Sol.525.(a)} \quad \sin^2 \theta - \cos^2 \theta - 3 \sin \theta + 2 = 0$$

$$\text{Put } \theta = 30^\circ \Rightarrow \frac{1}{4} - \frac{3}{4} - \frac{3}{2} + 2 = 0$$

$$\Rightarrow \frac{9}{4} - \frac{9}{4} = 0$$

$$\Rightarrow 0 = 0 \text{ (condition satisfied). So,}$$

$$\frac{1}{\sqrt{\sec \theta - \tan \theta}} = \frac{1}{\sqrt{\frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}}}} = \sqrt[4]{3}$$

$$\text{Sol.526.(c)} \quad \sqrt{\frac{\sec^2 \theta + \operatorname{cosec}^2 \theta}{\tan^2 \theta - \sin^2 \theta}}$$

$$= \sqrt{\frac{\frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}}{\sin^2 \theta \left(\frac{1}{\cos^2 \theta} - 1 \right)}}$$

$$= \sqrt{\frac{\frac{1}{\sin^2 \theta \cdot \cos^2 \theta}}{\frac{\sin^2 \theta \cdot \sin^2 \theta}{\cos^2 \theta}}} = \sqrt{\operatorname{cosec}^6 \theta} = \operatorname{cosec}^3 \theta$$

Short trick :

$$\text{Put } \theta = 30^\circ \Rightarrow \sqrt{\frac{\sec^2 30^\circ + \operatorname{cosec}^2 30^\circ}{\tan^2 30^\circ - \sin^2 30^\circ}} = 8$$

$$\text{Check option (c)} \Rightarrow \operatorname{Cosec}^3 30^\circ = 8$$

$$\text{Sol.527.(c)} \quad \frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} = 4 \sec \theta$$

$$\frac{2}{1 - \sin^2 \theta} = 4 \sec \theta \Rightarrow \frac{2}{\cos^2 \theta} = \frac{4}{\cos \theta}$$

$$\sec \theta = 2, \theta = 60^\circ, \text{ then}$$

$$\cot 60^\circ + \operatorname{cosec} 60^\circ = \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} = \sqrt{3}$$

$$\text{Sol.528.(c)} \quad \sin^2 \theta = 2 \sin \theta - 1,$$

$$\sin^2 \theta - 2 \sin \theta + 1 = 0 \Rightarrow (\sin \theta - 1)^2 = 0$$

$$\sin \theta = 1 \text{ so, } \theta = 90^\circ$$

$$\frac{1 + \operatorname{cosec} \theta}{1 - \cos \theta} = \frac{1 + 1}{1 - 0} = 2$$

Sol.529.(b)

$$\sin^2 59^\circ + \frac{1}{\sec^2 31^\circ} - \frac{1}{\sin^2 59^\circ \operatorname{cosec}^2 59^\circ}$$

$$\Rightarrow \cos^2 31^\circ + \frac{1}{x^2} - \frac{1}{1}$$

$$\Rightarrow \frac{1}{x^2} + \frac{1}{x^2} - 1 = \frac{2}{x^2} - 1 = \frac{2 - x^2}{x^2}$$

$$\text{Sol.530.(d)} \quad 2 \cos \theta + \sec \theta - 2\sqrt{2} = 0$$

$$\text{Put } \theta = 45^\circ$$

$$2 \cos 45^\circ + \sec 45^\circ - 2\sqrt{2} = 0$$

$$= 2 \frac{1}{\sqrt{2}} + \sqrt{2} - 2\sqrt{2} = 0$$

$$= \sqrt{2} + \sqrt{2} - 2\sqrt{2} = 0 \Rightarrow 0 = 0, \text{ satisfy}$$

Sol.531.(d) As we know that

$$\cot \theta = \frac{B}{P} = \frac{\sqrt{2} + 1}{1}$$

Applying pythagoras theorem

$$H^2 = B^2 + P^2 = 2 + 1 + 2\sqrt{2} + 1 = 4 + 2\sqrt{2}$$

$$\operatorname{Cosec} \theta \sec \theta = \frac{H^2}{B \times P} = \frac{4 + 2\sqrt{2}}{\sqrt{2} + 1} = 2\sqrt{2}$$

Sol.532.(b) Put $\theta = 30^\circ$ in

$$\frac{(1 - \sin \theta)(\sec \theta + \tan \theta) \tan \theta}{(\sec \theta + \tan \theta + 1)(\cot \theta - \operatorname{cosec} \theta + 1)}$$

$$= \frac{(1 - \frac{1}{2})(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}) \frac{1}{\sqrt{3}}}{(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} + 1)(\sqrt{3} - 2 + 1)}$$

$$= \frac{(\frac{1}{2})(\frac{3}{\sqrt{3}}) \frac{1}{\sqrt{3}}}{(\frac{3}{\sqrt{3}} + 1)(\sqrt{3} - 1)}$$

$$= \frac{(\frac{1}{2})}{(\sqrt{3} + 1)(\sqrt{3} - 1)} = \frac{1}{4} = \frac{\sin \theta}{2}$$

$$\text{Sol.533.(d)} \quad \frac{3\sqrt{3} \sec \theta + 4 \tan \theta}{3 \tan \theta + \sqrt{3} \sec \theta} = 2$$

$$3\sqrt{3} \sec \theta + 4 \tan \theta = 2 \times (3 \tan \theta + \sqrt{3} \sec \theta)$$

$$3\sqrt{3} \sec \theta + 4 \tan \theta = 6 \tan \theta + 2\sqrt{3} \sec \theta$$

$$\sqrt{3} \sec \theta = 2 \tan \theta$$

$$\sin \theta = \frac{\sqrt{3}}{2}, \theta = 60^\circ \Rightarrow \cos 60^\circ = \frac{1}{2}$$

$$\text{Sol.534.(a)} \quad Y = \tan 35^\circ$$

$$Y = \cot 55^\circ (\tan A = \cot B, \text{ if } A + B = 90^\circ)$$

$$\text{Now, } 2 \tan 55^\circ + \cot 55^\circ$$

$$= \frac{2 + \cot^2 55^\circ}{\cot 55^\circ} = \frac{2 + Y^2}{Y}$$

$$\text{Sol.535.(d)} \quad \tan \theta + \cot \theta = 4$$

Squaring both sides

$$\tan^2 \theta + \cot^2 \theta = 14 \dots (1)$$

$$\text{Also, } \tan \theta + \cot \theta = 4$$

$$\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = 4 \Rightarrow \operatorname{cosec} \theta \sec \theta = 4$$

$$\operatorname{cosec}^2 \theta \sec^2 \theta = 16 \dots (2)$$

On putting value from eq. 1 and 2

$$\frac{3(\tan^2 \theta + \cot^2 \theta)}{(2 \operatorname{cosec}^2 \theta \sec^2 \theta - 4)} = \frac{3 \times 14}{2 \times 16 - 4}$$

$$= \frac{42}{28} = 3 : 2$$

$$\text{Sol.536.(c)} \quad 3(\sec^2 \theta + \tan^2 \theta) = 5$$

$$\theta = 30^\circ \text{ (by hit and trial)} \Rightarrow \operatorname{Cosec} 30^\circ = 2$$

Sol.537.(c)

$$\begin{aligned}
 (\cos^2 \theta - 1)(2\sec^2 \theta) + \sec^2 \theta + 2\tan^2 \theta &= 2 \\
 \Rightarrow (2 - 2\sec^2 \theta) + \sec^2 \theta + 2\tan^2 \theta &= 2 \\
 \Rightarrow -2\tan^2 \theta + \sec^2 \theta + 2\tan^2 \theta &= 2 \\
 \Rightarrow \sec^2 \theta &= 2 \Rightarrow \theta = 45^\circ
 \end{aligned}$$

$$\frac{(\sec \theta + \sin \theta)}{(\operatorname{cosec} \theta - \cos \theta)} = \frac{\sqrt{2} + \frac{1}{\sqrt{2}}}{\sqrt{2} - \frac{1}{\sqrt{2}}} = 3$$

Sol.538.(a)

$$\begin{aligned}
 &\frac{(1 + \cos \theta)(\operatorname{cosec} \theta - \cot \theta)\sec \theta}{\sin \theta(1 - \sin \theta)(\sec \theta + \tan \theta)} \\
 &= \frac{(1 + \cos \theta)(1 - \cos \theta) \frac{1}{\sin \theta \cos \theta}}{\sin \theta(1 - \sin \theta)(1 + \sin \theta) \frac{1}{\cos \theta}} \\
 &= \frac{\frac{\sin \theta}{\cos \theta}}{\sin \theta \cos \theta} = \sec^2 \theta
 \end{aligned}$$

Sol.539.(d) $\sin \theta(2\sin \theta + 3) = 2$

$$\theta = 30^\circ \quad (\text{by hit and trial})$$

$$\begin{aligned}
 \sec^2 \theta + \cot^2 \theta - \cos^2 \theta \\
 = \sec^2 30^\circ + \cot^2 30^\circ - \cos^2 30^\circ \\
 = \frac{4}{3} + 3 - \frac{3}{4} = \frac{43}{12}
 \end{aligned}$$

Sol.540.(b) $\sin \theta = \frac{2\sqrt{ab}}{a+b}$

$$\text{Let } a = 2, b = 1$$

$$\text{Then } \sin \theta = \frac{2\sqrt{2}}{3}, \cos \theta = \frac{1}{3}$$

$$\text{Now, } \frac{\cos \theta + 1}{\cos \theta - 1} = \frac{\frac{1}{3} + 1}{\frac{1}{3} - 1} = \frac{-2}{1}$$

When the value of a and b are put in all options, option (b) is the correct answer.

Sol.541.(c) $\frac{3\cos^2 27^\circ - 5 + 3\cos^2 63^\circ}{\tan^2 32^\circ + 4 - \operatorname{cosec}^2 58^\circ}$

$$+ \sin 35^\circ \cos 55^\circ + \cos 35^\circ \sin 55^\circ$$

We know that $\sin A = \cos B$, if

$$A + B = 90^\circ$$

$$\text{So, } \frac{3\cos^2 27^\circ - 5 + 3\sin^2 27^\circ}{\cot^2 58^\circ + 4 - \operatorname{cosec}^2 58^\circ}$$

$$+ \sin 35^\circ \sin 35^\circ + \cos 35^\circ \cos 35^\circ = \frac{3-5}{-1+4} + 1 = \frac{1}{3}$$

Sol.542.(b) $2\cos^2 \theta = 3(1 - \sin \theta)$

$$\text{Put } \theta = 30^\circ$$

$$\Rightarrow 2\cos^2 30^\circ = 3(1 - \sin 30^\circ) \Rightarrow \frac{3}{2} = \frac{3}{2}$$

$$\theta = 30^\circ \text{ satisfies the given equation.}$$

$$(\tan 2\theta + \operatorname{cosec} 3\theta - \sec 2\theta) = \sqrt{3} + 1 - 2 = \sqrt{3} - 1$$

Sol.543.(a)

$$\sec^2 \theta + (1 - \sqrt{3})\tan \theta - (1 + \sqrt{3}) = 0,$$

$$1 + \tan^2 \theta + \tan \theta - \sqrt{3}\tan \theta - 1 - \sqrt{3} = 0$$

$$\tan \theta(\tan \theta + 1) - \sqrt{3}(\tan \theta + 1) = 0$$

$$(\tan \theta + 1)(\tan \theta - \sqrt{3}) = 0$$

$$\tan \theta = -1, \sqrt{3} \Rightarrow \theta = 60^\circ$$

Sol.544.(b)

$$\operatorname{cosec}^2 \theta (\cos \theta - 1)(1 + \cos \theta) = k,$$

$$\operatorname{cosec}^2 \theta \{-(1 - \cos \theta)(1 + \cos \theta)\}$$

$$-\operatorname{cosec}^2 \theta \times \sin^2 \theta = k \Rightarrow k = -1$$

Sol.545.(a)

$$\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta} = \frac{k+1}{1-k}$$

(By using componendo and dividendo)

$$\frac{\cot \theta}{\cos \theta} = \frac{2}{2k} \Rightarrow k = \sin \theta$$

Sol.546.(c) $\cos^2 \theta - \sin^2 \theta - 3\cos \theta + 2 = 0,$

$$\text{Put } \theta = 60^\circ$$

$$\cos^2 60^\circ - \sin^2 60^\circ - 3\cos 60^\circ + 2 = 0,$$

$$\frac{1}{4} - \frac{3}{4} - \frac{3}{2} + 2 = 0$$

$$\Rightarrow 0 = 0 \text{ (LHS = RHS)}$$

$$\sec \theta - \cos \theta = 2 - \frac{1}{2} = \frac{3}{2}$$

Sol.547.(b) $\frac{\operatorname{cosec}^2 \theta}{\operatorname{cosec}^2 \theta - \cot^2 \theta} = \frac{13}{4}$

$$\Rightarrow \operatorname{cosec}^2 \theta = \frac{13}{4} (\operatorname{cosec}^2 \theta - \cot^2 \theta = 1)$$

$$\sin \theta = \frac{2}{\sqrt{13}} = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

By pythagoras theorem

$$\text{Base} = \sqrt{\text{Hypotenuse}^2 - \text{Perpendicular}^2} \\ \Rightarrow \text{Base} = 3$$

$$\cos \theta = \frac{3}{\sqrt{13}}, \sec \theta = \frac{\sqrt{13}}{3},$$

$$\tan \theta = \frac{2}{3}, \cot \theta = \frac{3}{2}$$

$$\frac{52\cos^2 \theta - 9\tan^2 \theta}{18\sec^2 \theta + 8\cot^2 \theta} = \frac{52 \times \frac{9}{13} - 9 \times \frac{4}{9}}{18 \times \frac{13}{9} + 8 \times \frac{9}{4}}$$

$$= \frac{36 - 4}{26 + 18} = \frac{32}{44} = \frac{8}{11}$$

Sol.548.(a) $\frac{3(\cot^2 46^\circ - \sec^2 44^\circ)}{2(\sin^2 28^\circ + \sin^2 62^\circ)} +$

$$\frac{2\cos^2 60^\circ \tan^2 33^\circ \tan^2 57^\circ}{\sec^2 (90^\circ - \theta) - \cot^2 \theta}$$

$$\Rightarrow \frac{3(\tan^2 44^\circ - \sec^2 44^\circ)}{2(\sin^2 28^\circ + \cos^2 28^\circ)} +$$

$$\frac{2\cos^2 60^\circ \tan^2 33^\circ \cot^2 33^\circ}{\operatorname{cosec}^2 \theta - \cot^2 \theta} \Rightarrow \frac{-3}{2} + \frac{1}{2} = -1$$

Sol.549.(b) $3\sec \theta + 4\cos \theta = 4\sqrt{3},$

$$\text{Put } \theta = 30^\circ$$

$$3 \times \frac{2}{\sqrt{3}} + 4 \times \frac{\sqrt{3}}{2} = 4\sqrt{3} \text{ (Equation satisfies)}$$

$$(1 - \sin \theta + \cos \theta) = (1 - \sin 30^\circ +$$

$$\cos 30^\circ) = \frac{1 + \sqrt{3}}{2}$$

Sol.550.(d) $\frac{5\cot \theta + \sqrt{3}\operatorname{cosec} \theta}{2\sqrt{3}\operatorname{cosec} \theta + 3\cot \theta} = 1$

$$\Rightarrow 5\cos \theta + \sqrt{3} = 2\sqrt{3} + 3\cos \theta$$

$$2\cos \theta = \sqrt{3} \Rightarrow \cos \theta = \frac{\sqrt{3}}{2}, \theta = 30^\circ$$

$$\Rightarrow \frac{\frac{7}{2}\cot^2 \theta - \frac{3}{4}\operatorname{cosec}^2 \theta}{4\sin^2 \theta + \frac{3}{2}\tan^2 \theta}$$

$$= \frac{\frac{7}{2} \times 3 - \frac{3}{4} \times 4}{4 \times \frac{1}{4} + \frac{3}{2} \times \frac{1}{3}} = 5$$

Sol.551.(c) $(1 + \cot^2 \theta) + (1 + (\cot^2 \theta)^{-1})$

$$\Rightarrow \operatorname{cosec}^2 \theta + (1 + \tan^2 \theta)$$

$$\Rightarrow \operatorname{cosec}^2 \theta + \sec^2 \theta \Rightarrow \frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta}$$

$$\Rightarrow \operatorname{cosec}^2 \theta \cdot \sec^2 \theta$$

$$k = \operatorname{cosec}^2 \theta \cdot \sec^2 \theta \Rightarrow \sqrt{k} = \operatorname{cosec} \theta \cdot \sec \theta$$

Sol.552.(c) $\frac{\sec \theta \operatorname{cosec} \theta}{2 + \tan^2 \theta + \cot^2 \theta}$

$$= \frac{\sec \theta \operatorname{cosec} \theta}{\sec^2 \theta + \operatorname{cosec}^2 \theta} = \frac{\sec \theta \operatorname{cosec} \theta}{\frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta}}$$

$$= \sec \theta \cdot \operatorname{cosec} \theta \sin^2 \theta \cos^2 \theta = \sin \theta \cos \theta$$

Sol.553.(c) $\cot A + \operatorname{cosec} A = 2 \text{ ---- (1)}$

$$\operatorname{cosec}^2 A - \cot^2 A = 1$$

$$(\operatorname{cosec} A + \cot A)(\operatorname{cosec} A - \cot A) = 1$$

$$(\operatorname{cosec} A - \cot A) = \frac{1}{2} \text{ ---- (2)}$$

Adding (1) and (2)

$$2\operatorname{cosec} A = \frac{5}{2} \Rightarrow \operatorname{cosec} A = \frac{5}{4}$$

$$\sin A = \frac{4}{5}; \tan A = \frac{4}{3}$$

$$\frac{9\tan A + 16\operatorname{cosec} A}{5\sin A + 3\tan A} = \frac{9 \times \frac{4}{3} + 16 \times \frac{5}{4}}{5 \times \frac{4}{5} + 3 \times \frac{4}{3}} = 4$$

Sol.554.(c) $\cot^2 \theta + \cot^4 \theta = 2$

$$\cot^2 \theta(1 + \cot^2 \theta) = 2$$

$$\cot^2 \theta = 2\sin^2 \theta \Rightarrow \cot^4 \theta = 4\sin^4 \theta$$

$$2\sin^4 \theta + \sin^2 \theta = \frac{\cot^4 \theta}{2} + \frac{\cot^2 \theta}{2}$$

$$= \frac{\cot^2 \theta + \cot^4 \theta}{2} = 1$$

Short-Trick: put $\theta = 45^\circ$

Sol.555.(c) $\sin \alpha + \operatorname{cosec} \alpha = \tan \frac{\pi}{3}$

$$\sin \alpha + \operatorname{cosec} \alpha = \sqrt{3}$$

$$\sin^3 \alpha + \operatorname{cosec}^3 \alpha = a^3 - 3a$$

$$= (\sqrt{3})^3 - 3(\sqrt{3}) = 0$$

Sol.556.(c) $\frac{\sin^2 52^\circ + 2 + \sin^2 38^\circ}{4\cos^2 43^\circ - 5 + 4\cos^2 47^\circ}$

$$\frac{\sin^2 52^\circ + 2 + \cos^2 52^\circ}{4\cos^2 43^\circ - 5 + 4\sin^2 43^\circ}$$

$$\frac{1 + 2}{4\cos^2 43^\circ + 4\sin^2 43^\circ - 5} = \frac{3}{-1} = -3$$

Sol.557.(d)

$$4(\operatorname{cosec}^2 57^\circ - \tan^2 33^\circ) - \cos 90^\circ$$

$$+ y \times \tan^2 66^\circ \times \tan^2 24^\circ = \frac{y}{2},$$

We know $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$

And $\tan A \tan B = 1$ if $(A + B = 90^\circ)$

So, $4(1 + \cot^2 57^\circ - \cot^2 57^\circ) - \cos 90^\circ +$

$$y \times 1 = \frac{y}{2} \Rightarrow 4 + y = \frac{y}{2}$$

$$\Rightarrow \frac{y}{2} = -4 \Rightarrow y = -8$$